
ALGORITHM THEORETIC BASIS DOCUMENT (ATBD) FOR NEXT GENERATION MODTRAN®

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ABSTRACT

MODTRAN6 is validated software designed to compute spectral transmittance, radiances and fluxes for a user-specified atmospheric state. Pressures, temperatures and constituent densities are defined via vertical profiles above the locally spherical ground, which is characterized most generally by a surface spectral bi-directional reflectance distribution function (BRDF). Although the core radiative transfer approach is its band model, MODTRAN also includes correlated- k and line-by-line algorithms. Scattering is computed most accurately using the DISORT discrete ordinate method. This ATBD describes the full suite of MODTRAN6 radiative transfer algorithms.

INTRODUCTION

The Next Generation MODTRAN (MODTRAN6) Algorithm Theoretical Basis Document (ATBD) provides an up-to-date description of the model's radiative transfer. The Air Force Research Laboratory (AFRL) contract to Spectral Sciences, Inc. (SSI) that lead to the first delivery of MODTRAN began over 30 years ago. That effort called for the integration of a 2-parameter, temperature dependent, 5 cm^{-1} molecular band model [Robertson, *et al.*, 1981] into LOWTRAN6 [Kneizys *et al.*, 1983] as an alternative to the 1-parameter, 20 cm^{-1} spectral resolution band models of Pierluissi and Maragoudakis [1986]. The decision was made to increase the spectral resolution of the delivered band model to 1 cm^{-1} , and MODTRAN was born. The final report, entitled "MODTRAN™: A Moderate-Resolution Model for LOWTRAN 7," [Berk *et al.*, 1989] described the initial MODTRAN statistical band model, which was largely based on the earlier work of Robertson, *et al.*, [1981].

The initial delivery of MODTRAN band model provided the basic framework for radiative transfer model of today. Many aspects of the original algorithm have been retained. MODTRAN continues to partition molecular absorption in each spectral bin into three contributions:

1. lines centered within the bin,
2. the tails of lines centered in neighboring spectral bins, and
3. continua from distant lines.

Line center transmittances are still computed from the Plass [1958] expression for randomly distributed, identical lines. Temperature dependent absorption coefficient and line density band model parameters are still used to characterize line strength distributions. Molecular transmittances are still computed assuming random correlation of the absorption from the contributing molecules. Finally, inhomogeneous paths can still be modeled using Curtis-Godson [van de Hulst, 1945; Curtis, 1952; Godson, 1953] path averaging.

Even though many aspects of the original band model have been retained, there have been a large number of critically important improvements. In general, these improvements to the band model can be viewed as methods for better characterizing the statistical representation of line strength distributions and for computing transmittances from these distributions. In particular, the introduction of the finer 0.1 cm^{-1} spectral resolution band model (MODTRAN5 [Berk *et al.*, 2006]) would not have been possible without these upgrades. Some highlights include the

- spectral curve fitting of line tail contributions using the ratio of pressure and temperature dependent quadratic polynomials,
- redefining the line strength distribution as arises from two groups of randomly distributed identical lines based on two pairs of absorption coefficient and line spacing band model parameters, and
- inclusion of an accurate prescription for calculation of finite bin Voigt transmittances [Berk, 2013].

In addition to the upgrades made to the transmittance algorithm, MODTRAN methods for computing spectral radiances and fluxes have seen major improvements over the years. The original delivery of MODTRAN included a 2-stream algorithm for calculating thermal and solar multiple scatter [Meador and Weaver, 1980; Isaacs *et al.*, 1986; Isaacs *et al.*, 1987]. MODTRAN2 [Anderson *et al.*, 1993] introduced N-stream multiple scattering via the discrete ordinate method of DISORT [Stamnes *et al.*, 1988; Stamnes *et al.*, 2000]. The processing capabilities of computers at that time greatly restricted use of the DISORT option. As computers got faster and use of DISORT became more commonplace, two major problems with the DISORT integration into MODTRAN were uncovered. First, MODTRAN, being a band model, does not obey Beer's Law, i.e. product of spectral band segment transmittances for a multiple segmented path is not equal to total path transmittance. The problem is that the DISORT method relies on an

adherence to Beer's Law. A second difficulty with the original integration of DISORT into MODTRAN is a geometry incompatibility: MODTRAN models a spherical atmosphere above the local Earth radius while DISORT is a plane-parallel atmosphere model.

MODTRAN4 [Bernstein *et al.*, 1996] introduced Beer's Law into MODTRAN by developing a statistical Correlated- k (C_k) algorithm. C_k methods recast the spectral band transmittance calculation into a weighted set of monochromatic problems, each satisfying Beer's Law. Traditionally, the generation of k (absorption coefficient) distributions is computationally intensive, requiring calculation of high spectral resolution absorption data. Instead, the MODTRAN statistical approach determines the k -distributions directly from the spectral bin band model parameters, interpolating into pre-computed and compact k -distribution tables. The C_k option enables a first principles calculation of scattered radiances using DISORT.

To eliminate the geometry incompatibility, MODTRAN6 introduces "spherical DISORT". Plane-parallel DISORT generates line-of-sight (LOS) radiant intensities from a set of altitude (optical depth) levels to the ground if the view angle faces downward or to space if the view angle faces upward. DISORT has been generalized for MODTRAN6 to output segment *source* radiant intensities defined for each of MODTRAN's spherical refractive geometry LOS segments. These source intensities are multiplied by MODTRAN foreground refractive path transmittances and summed to compute spherical atmosphere LOS radiances. The calculation of the segment source radiant intensities is complicated because DISORT solves a *scaled* radiative transfer equation (RTE) in which scattering phase functions are represented as the sum of a forward direction δ -function and a Legendre expansion [Wiscombe, 1977]. The *unscaled* radiant intensities, required by MODTRAN, are determined from the *scaled* values by adding a forward scattered contribution that must be determined via back to front integration along the LOS. Finally, MODTRAN subtracts out the DISORT path thermal emission and single scatter solar contributions, replacing them with the more accurate MODTRAN values.

MODTRAN6 also introduces a MODTRAN line-by-line (LBL) radiative transfer algorithm. This option facilitates inclusion of scattering into one's LBL calculations and simplifies validation of the band model and C_k algorithms. Unlike other LBL models, MODTRAN solves the LBL RTE within 0.1 cm^{-1} bins, stitching together these distinct calculations to generate high resolution spectra over the full requested bandpass. The pre-computed spectral fits of line tail contributions, developed for the band model, are generalized for use in the MODTRAN LBL algorithm. This data greatly reduces the number of molecular transitions that must be explicitly modeled at each spectral point.

The following chapters describe in detail the MODTRAN radiative transfer algorithms. *Caveats regarding the MODTRAN implementation or a convention it uses, not directly related to the MODTRAN radiative transfer theory, are highlighted in brown italics. Readers focused on learning the radiative transfer theory, can skip over these sections without loss of continuity.* Chapter 1 lays the foundation by defining the monochromatic LOS radiance from the integral RTE. Chapter 2 illustrates that the radiance expression defined in term of LOS transmittance integrals can be employed to compute band model radiances, without relying on Beer's Law. The cornerstone of band model algorithms is the calculation for spectral band transmittance; the MODTRAN approach is described in Chapter 3. Chapter 4 describes MODTRAN's statistical C_k algorithm including the method for defining k -distributions. The DISORT scattering algorithm and its integration into MODTRAN6 is detailed in Chapter 5. For the most part, the MODTRAN atmosphere is defined via vertical profiles, leading to spherical shell layering. The one exception is an option to model local chemical (gas) clouds. The approach used is described in Chapter 6. The MODTRAN6 LBL algorithm is formulated and validated in Chapter 7. A number of appendices provide further details on MODTRAN methods.

1. LINE-OF-SIGHT SPECTRAL RADIANCE

Calculation of in-band radiances via the line-by-line (LBL) approach is a computationally intensive. At each modeled spectral point within the bandpass, molecular and particulate absorption and scattering coefficients for the series of pressure and temperature pairs encountered in the atmosphere must be generated. Contributions from molecular transitions centered far from the spectral point under consideration must be summed. Once the extinction data is defined, the radiative transfer equation (RTE) can be solved for the individual spectral point. Because of the fine structure of molecular absorption, especially at higher altitudes (lower pressures), a very finely spaced spectral grid is required. A spectral step size of between 0.01 and 0.00001 cm⁻¹ is common, depending upon scenario and accuracy requirements.

The MODTRAN solution to this problem is to use a narrow *statistical* band model. In this approach, the RTE is solved for a relatively narrow spectral band in a single step. A number of physical and statistical approximations are made. The quality of the MODTRAN results depends largely upon the validity of these approximations for any specific scenario. Over the past 30+ years of MODTRAN development, much effort has been focused on fine tuning the model to improve the accuracy of the approach. The MODTRAN methodology, highlighting current approximations and the rationale for their application, is presented in this document.

The LBL radiative transfer equations are described in this first chapter to lay the foundation for the MODTRAN band model (Chapters 2 and 3) and for the LBL algorithm introduced into MODTRAN6 in 2015 to facilitate band model validation (Chapter 7).

1.1 Line-of-Sight Spectral Radiance

The integral form of the radiation transport equation defines the line-of-sight (LOS) *monochromatic* radiant intensity $I_0(\Omega_0; \nu)$ at spectral frequency ν along a refracted path from the sensor (path length $\ell = 0$) and in the sensor view direction, $\Omega_0 \equiv (\mu_0, \phi_0)$:

$$I_0(\Omega_0; \nu) = \exp[-\tau_\ell(\nu)] I_\ell(\Omega_\ell; \nu) + \int_0^{\tau_\ell(\nu)} \exp[-\tau_\ell(\nu)] J_\ell(\Omega_\ell; \nu) d\tau_\ell(\nu) \quad (1.1)$$

MODTRAN Angles: *MODTRAN uses the convention that the directions are defined by the reverse of the photon's trajectory. So, μ_0 is the cosine of the sensor zenith angle at the sensor and ϕ_0 is the sensor azimuth angle in degrees East of North. Nadir viewing corresponds $\mu = -1$ and the azimuth angle is 270° when the sensor faces West. The angles along the LOS, $\Omega_\ell \equiv (\mu_\ell, \phi_\ell)$, vary with path length ℓ due to the Earth's curvature and refractive bending. Similarly, the solar angles $\Omega_\ell^{\text{sun}} \equiv (\mu_\ell^{\text{sun}}, \phi_\ell^{\text{sun}})$ a distance ℓ from a sensor along a LOS will include a positive μ_ℓ^{sun} if the sun's elevation at ℓ is positive and a 45° azimuth angle if the sun is located North-East. The convention for lunar angles is completely analogous.*

MODTRAN Input Angles: *MODTRAN provides multiple options for inputting the LOS angles. The sensor zenith angle μ_0 is input variable OBSZEN. Alternatively, one can specify the zenith angle from the end of the LOS, $\ell = L$, back towards the sensor, $-\mu_L$; this input is BCKZEN. There are also multiple options, based on the IPARM input, for defining the solar/lunar position and path azimuth with IPARM equal to 0, 1, 10 or 11 for the ephemeris inputs:*

- *Set IPARM to 2 to define $\mu_0^{\text{sun}} = \text{PARM2}$ and $\phi_0^{\text{sun}} - \phi_0 = \text{PARM1}$;*
- *Set IPARM to 12 to define $\mu_L^{\text{sun}} = \text{PARM2}$ and $\phi_L^{\text{sun}} - \phi_L = \text{PARM1}$;*
- *Set IPARM to 0 or 1 to define $\phi_0 = \text{TRUEAZ}$; or*
- *Set IPARM to 10 or 11 to define $-\phi_L = \text{TRUEAZ}$*

For IPARM equal to 0, 1, 10 or 11, solar/lunar position is defined by latitude, longitude, date and time inputs.

Eq. (1.1) states that the observed spectral radiant intensity at the sensor is the sum of two terms. The first is a boundary term equal to the product of the spectral radiant intensity, $I_\ell(\Omega_\ell; \nu)$, directed towards the sensor at a path length distance ℓ along the LOS and of the spectral transmittance, $\exp[-\tau_\ell(\nu)]$, that results from the extinction in the foreground path. The second term is the path radiance, equal to the LOS integral over the source radiation $J_{\ell'}(\Omega_{\ell'}; \nu)$ directed towards the sensor and attenuated by the foreground transmittance $\exp[-\tau_{\ell'}(\nu)]$.

The path spectral transmittances in Eq. (1.1) fall-off exponentially with the dimensionless extinction optical depth, $\tau_\ell(\nu)$. The optical depth terms themselves are path integrals over extinction coefficients, $b_{\ell'}(\nu)$, with dimensions of reciprocal length:

$$\tau_\ell(\nu) \equiv \int_0^\ell b_{\ell'}(\nu) d\ell' \quad . \quad (1.2)$$

The extinction coefficients consists of absorption $b_\ell^{abs}(\nu)$ and scattering $b_\ell^{sct}(\nu)$ components, which are computed as sums over atmospheric molecular and particulate species contributions $b_{\ell,i}(\nu)$:

$$b_\ell(\nu) \equiv b_\ell^{abs}(\nu) + b_\ell^{sct}(\nu) \equiv \sum_{i \text{ species}} \left[b_{\ell,i}^{abs}(\nu) + b_{\ell,i}^{sct}(\nu) \right] \equiv \sum_{i \text{ species}} \left[\kappa_{\ell,i}^{abs}(\nu) + \kappa_{\ell,i}^{sct}(\nu) \right] \rho_i(\ell) \quad . \quad (1.3)$$

The individual species absorption and scattering coefficients are themselves the product of a cross-section $\kappa_{\ell,i}(\nu)$ [with dimensions of area] and a density $\rho_i(\ell)$ [number per unit volume]. Absorption and scattering optical depths are defined by

$$\tau_\ell^{abs}(\nu) \equiv \int_0^\ell b_{\ell'}^{abs}(\nu) d\ell' \quad \text{and} \quad \tau_\ell^{sct}(\nu) \equiv \int_0^\ell b_{\ell'}^{sct}(\nu) d\ell' \quad . \quad (1.4)$$

Differentiating Eqs. (1.2) and (1.4) with respect to path length and solving for the differential path length, $d\ell$, produces the following useful identities:

$$d\ell = \frac{d\tau_\ell(\nu)}{b_\ell(\nu)} = \frac{d\tau_\ell^{abs}(\nu)}{b_\ell^{abs}(\nu)} = \frac{d\tau_\ell^{sct}(\nu)}{b_\ell^{sct}(\nu)} \quad . \quad (1.5)$$

The MODTRAN atmospheric source term $J_\ell(\Omega_\ell; \nu)$ consists of three distinct components: local thermal emission $J_\ell^{em}(\Omega_\ell; \nu)$, directly transmitted single scattered solar (or lunar) irradiance $J_\ell^{ss}(\Omega_\ell; \nu)$, and diffuse radiation scattered into the LOS $J_\ell^{ms}(\Omega_\ell; \nu)$:

$$\begin{aligned} J_\ell(\Omega_\ell; \nu) &= J_\ell^{em}(\Omega_\ell; \nu) + J_\ell^{ss}(\Omega_\ell; \nu) + J_\ell^{ms}(\Omega_\ell; \nu) \\ &= \frac{b_\ell^{abs}(\nu)}{b_\ell(\nu)} B(T_\ell; \nu) \quad \text{local thermal emission} \\ &\quad + \frac{b_\ell^{sct}(\nu)}{b_\ell(\nu)} p_\ell(\Omega_\ell, \Omega_\ell^{sun}; \nu) \exp[-\tau_\ell^{sun}(\nu)] F^{sun}(\nu) \quad \text{single scatter solar} \\ &\quad + \frac{b_\ell^{sct}(\nu)}{b_\ell(\nu)} \int_{4\pi} p_\ell(\Omega_\ell, \Omega'_\ell; \nu) I_\nu(\Omega'_\ell; \nu) d\Omega'_\ell \quad \text{multiple scatter} \quad . \end{aligned} \quad (1.6)$$

The first term, the thermal emission, is the product of the Planck blackbody function $B(T_\ell; \nu)$ at local temperature T_ℓ and the local emissivity, which is represented as a ratio of absorption to extinction coefficients.¹

The latter two terms in Eq. (1.6) are the source terms contributing to path scattered radiance. The single scatter solar radiation is the product of the top-of-atmosphere (TOA) solar irradiance $F^{sun}(\nu)$, the scattering point to sun transmittance $\exp[-\tau_\ell^{sun}(\nu)]$, the *effective* scattering phase function $p_\ell(\Omega_\ell, \Omega_\ell^{sun}; \nu)$, and the single scattering albedo, equal to the ratio of scattering to extinction coefficients.

Sun as a Point Source: *MODTRAN models the sun as a point source at Ω_ℓ^{sun} , even though its radial diameter is slightly more than half a degree (the radial diameter varies with earth to sun distance or, equivalently, day of year). In reality, the solar irradiance varies across the solar disk. Unless one is modeling a direct view of the sun, the point source approximation is very accurate.*

TOA Solar Irradiance: *MODTRAN provides multiple TOA solar irradiance curves from which to choose in its DATA/ directory. The set of 6 curves (low, med1, med2, high1, high2 and peak) recently provided by Fontenla, et al. [2015] vary with solar activity. The Fontenla med2 curve, Figure 1.1, is used as the default within MODTRAN6. Earlier versions of MODTRAN relied on data generated from Kurucz [1992; 1995], either directly from his calculations (kurucz1995.dat) or scaled in spectral regions to match integrated measurement data (kurucz1997.dat) [Acharya et al., 1998]. The TOA solar irradiances are corrected for sun-earth distance based on the day-of-year input. The maximum occurs at day 1 (3.4% above the mean sun-earth distance value) and the minimum, at day 182 (3.3% below the mean sun-earth distance value) {add figure}.*

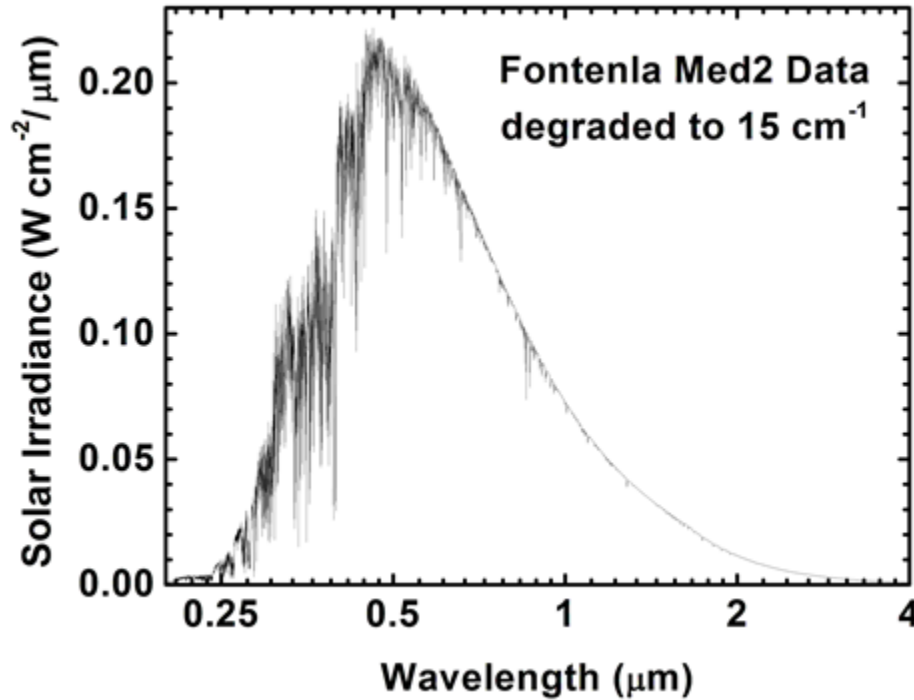


Figure 1.1. Fontenla *et al.* (2015) “med2” Top-Of-Atmosphere Solar Irradiances, 15 cm⁻¹ FWHM

¹ This expression for thermal emission assumes local thermodynamic equilibrium (LTE) conditions - an assumption, valid in the lower terrestrial atmosphere, that molecular collisions occur in sufficient frequency to control upper state population densities.

Lunar Source Function: When lunar calculations are performed, the TOA solar irradiance is multiplied by the product of 2.04472×10^{-7} , a phase angle correction between 0 and 100%, Figure 1.2a, and a spectral lunar albedo that varies from 0.001 at 0.1 μm to 0.4 at 5.0 μm and beyond, Figure 1.2b. MODTRAN does not model lunar thermal emission. The lunar modeling occurs in module **source_mod.f90**.

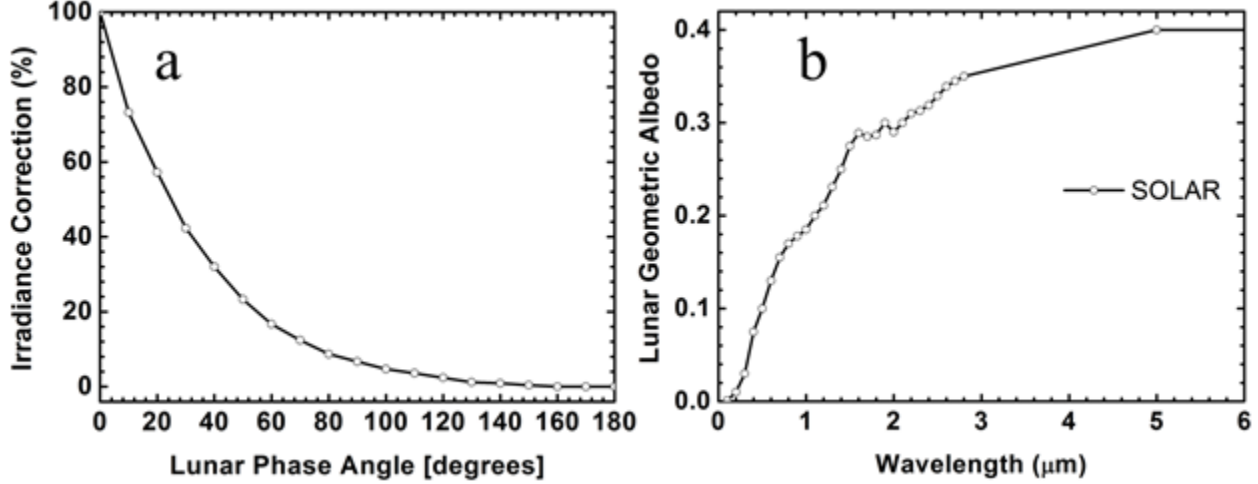


Figure 1.2. (a) Lunar Irradiance Correction and (b) Lunar Geometric Albedo

The last term in Eq. (1.6), the multiply scattered radiation, is an integral over all directions (4π steradians) of the incoming diffuse radiation scattered into the LOS. The expression “multiple scatter” is somewhat of a misnomer since it does include thermal emission single scatter contributions. The dependence of the multiple scattering on the path radiance significantly complicates the radiation transport equation, coupling the source and radiant intensity terms. MODTRAN provides three options for solving these coupled equations:

1. 2-stream (upper and lower hemisphere) Isaacs’ model [Isaacs *et al.*, 1986; 1987];
2. The high-fidelity DISORT discrete ordinate algorithm [Chandrasekhar 1960; Stamnes *et al.*, 1988; 2000; Chapter 7]; and
3. Isaacs’ model scaled to DISORT at a handful of spectral window frequencies.

The *effective* phase function that appears in both scattered radiance source terms of Eq. (1.6) is a scattering coefficient weighted average of individual species i scattering phase functions $p_{\ell,i}(\Omega_\ell, \Omega'; \nu)$:

$$p_\ell(\Omega_\ell, \Omega'; \nu) \equiv \frac{\sum_{i \text{ species}} b_{\ell,i}^{sct}(\nu) p_{\ell,i}(\Omega_\ell, \Omega'; \nu)}{b_\ell^{sct}(\nu)} \quad (1.7)$$

These phase functions are all unit normalized, with dimensions of reciprocal steradians (1/sr). It is somewhat more common to define phase functions as dimensionless with a 4π sr normalization, but MODTRAN uses the unit normalization to eliminate constant factors from the source function equation.

Substituting the source term of Eq. (1.6) into the integral RTE, the spectral radiant intensity of Eq. (1.1) can be rewritten as integrals over absorption, $\tau_\ell^{abs}(\nu)$, and scattering, $\tau_\ell^{sct}(\nu)$, optical depth:

$$\begin{aligned}
I_0(\Omega_0; \nu) &= \exp[-\tau_\ell(\nu)] I_\ell(\Omega_\ell; \nu) + \int_0^{\tau_\ell(\nu)} \exp[-\tau_{\ell'}(\ell')] \left[\frac{b_{\ell'}^{abs}(\nu)}{b_{\ell'}(\nu)} B(T_{\ell'}, \nu) \right. \\
&\quad \left. + \frac{b_{\ell'}^{sct}(\nu)}{b_{\ell'}(\nu)} p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}^{sun}; \nu) \exp[-\tau_{\ell'}^{sun}(\nu)] F^{sun}(\nu) \right. \\
&\quad \left. + \frac{b_{\ell'}^{sct}(\nu)}{b_{\ell'}(\nu)} \int_{4\pi} p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}'; \nu) I_{\ell'}(\Omega_{\ell'}'; \nu) d\Omega_{\ell'}' \right] d\tau_{\ell'}(\nu) \\
&= \exp[-\tau_\ell(\nu)] I_\ell(\Omega_\ell; \nu) + \int_0^{\tau_\ell^{abs}(\nu)} \exp[-\tau_{\ell'}(\nu)] B(T_{\ell'}, \nu) d\tau_{\ell'}^{abs}(\nu) \\
&\quad + \int_0^{\tau_\ell^{sct}(\nu)} \exp[-\tau_{\ell'}(\nu)] \left[p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}^{sun}; \nu) \exp[-\tau_{\ell'}^{sun}(\ell')] F^{sun}(\nu) \right. \\
&\quad \left. + \int_{4\pi} I_{\ell'}(\Omega_{\ell'}'; \nu) p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}'; \nu) d\Omega_{\ell'}' \right] d\tau_{\ell'}^{sct}(\nu) .
\end{aligned} \tag{1.8}$$

The identities of Eq. (1.5) are used to convert the integrals over extinction optical depth, $\tau_\ell(\nu)$, in the first line of Eq. (1.8) to integrals over absorption and scattering optical depth, $\tau_{\ell'}^{abs}(\nu)$ and $\tau_{\ell'}^{sct}(\nu)$, respectively.

For band model theory, the key atmospheric attenuation quantity is the transmittance itself, not optical depth. The RTE can be re-expressed in terms of transmittance, $t_\ell(\nu) = \exp[-\tau_\ell(\nu)]$. The following expressions define the relationship between differential optical depth and differential transmittance:

$$\begin{aligned}
t_\ell^{abs}(\nu) &= \exp[-\tau_\ell^{abs}(\nu)] \Rightarrow dt_\ell^{abs}(\nu) = -d\tau_\ell^{abs}(\nu) \exp[-\tau_\ell^{abs}(\nu)] = -t_\ell^{abs}(\nu) d\tau_\ell^{abs}(\nu) \\
t_\ell^{sct}(\nu) &= \exp[-\tau_\ell^{sct}(\nu)] \Rightarrow dt_\ell^{sct}(\nu) = -d\tau_\ell^{sct}(\nu) \exp[-\tau_\ell^{sct}(\nu)] = -t_\ell^{sct}(\nu) d\tau_\ell^{sct}(\nu) \\
\Rightarrow -t_\ell^{sct}(\nu) dt_\ell^{abs}(\nu) &= \exp[-\tau_\ell(\nu)] d\tau_\ell^{abs}(\nu) \quad \text{and} \quad -t_\ell^{abs}(\nu) dt_\ell^{sct}(\nu) = \exp[-\tau_\ell(\nu)] d\tau_\ell^{sct}(\nu)
\end{aligned} \tag{1.9}$$

The final expressions from Eq. (1.9) can be directly inserted into Eq. (1.8) to obtain the transmittance dependent RTE:

$$\begin{aligned}
I_0(\Omega_0; \nu) &= t_\ell(\nu) I_\ell(\Omega_\ell; \nu) + \int_{t_\ell^{abs}(\nu)}^1 t_{\ell'}^{sct}(\nu) B(T_{\ell'}, \nu) dt_{\ell'}^{abs}(\nu) \\
&\quad + \int_{t_\ell^{sct}(\nu)}^1 t_{\ell'}^{abs}(\nu) \left[p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}^{sun}; \nu) t_{\ell'}^{sun}(\nu) F^{sun}(\nu) + \int_{4\pi} I_{\ell'}(\Omega_{\ell'}'; \nu) p_{\ell'}(\Omega_{\ell'}, \Omega_{\ell'}'; \nu) d\Omega_{\ell'}' \right] dt_{\ell'}^{sct}(\nu) .
\end{aligned} \tag{1.10}$$

Within MODTRAN, the boundary radiant intensity term $I_\ell(\Omega_\ell; \nu)$ for the full LOS path is typically zero unless the path terminates at the Earth or other surface.² The surface radiance includes both emissive and reflective components:

² There is an option to include grey body emission with no reflected source at the path final altitude. Additional options for modeling the surface term will be considered in future distributions. This is particularly pertinent given the new approach used to integrate DISORT into MODTRAN6 in which the boundary term effects the path scattered radiance; currently this boundary term is set to zero for paths terminating in the atmosphere.

$$\begin{aligned}
I_s(\Omega_s; \nu) &= \varepsilon_s(\Omega_s; \nu) B(T_g, \nu) && \{ \text{Surface emission} \\
&+ \mu_s^{sun} f_s(\Omega_s, \Omega_s^{sun}; \nu) t_s^{sun}(\nu) F^{sun}(\nu) && \left\{ \begin{array}{l} \text{Directly - transmitted} \\ \text{reflected solar irradiance} \end{array} \right. \\
&+ \int_{2\pi} I_s(\Omega'_s; \nu) f_s(\Omega_s, \Omega'_s; \nu) \mu'_s d\Omega'_s && \{ \text{Surface reflected downward flux}
\end{aligned} \tag{1.11}$$

The variables in this equation are defined at a surface (e.g., the ground) a distance $\ell = s$ from the sensor along the LOS. The cosine of the zenith angle, μ'_s , in the *Surface reflected downward flux* integral is defined with respect to the surface normal hemisphere, Ω'_s ; $\varepsilon_s(\Omega_s; \nu)$ is the dimensionless monochromatic directional emissivity for photons leaving the surface in direction Ω_s ; and $f_s(\Omega_s, \Omega'_s; \nu)$ is the monochromatic surface bi-direction reflectance distribution function (BRDF) in units of reciprocal steradians. The BRDF defines the reflected radiance resulting from surface illumination; the Helmholtz reciprocity principle dictates that it be a symmetric function of the incoming and outgoing angles with $f_s(\Omega_s, \Omega'_s; \nu) = f_s(-\Omega', -\Omega_s; \nu)$; and it is normalized to the surface albedo $\alpha_s(\nu)$:

$$\alpha_s(\nu) = \frac{1}{\pi} \int_{2\pi} \int_{2\pi} f_s(\Omega_s, \Omega'_s; \nu) \mu'_s d\Omega'_s \mu_s d\Omega_s \tag{1.12}$$

The directional emissivity, the complement of the directional reflectivity $\rho_s(\Omega_s; \nu)$, dictates the angular distribution of the surface thermal emission, and is computed directly from the BRDF by the following relationship:

$$\varepsilon_s(\Omega_s; \nu) \equiv 1 - \rho_s(\Omega_s; \nu) = 1 - \int_{2\pi} f_s(\Omega_s, \Omega'_s; \nu) \mu'_s d\Omega'_s \tag{1.13}$$

The Earth viewing radiance is obtained by substituting the surface radiance term, Eq. (1.11), into the RTE, Eq. (1.10):

$$\begin{aligned}
I_0(\Omega; \nu) &= t_\ell(\nu) I_\ell(\Omega; \nu) + \int_{t_\ell^{abs}(\nu)}^1 t_{\ell'}^{sct}(\nu) B(T_{\ell'}, \nu) dt_{\ell'}^{abs}(\nu) \\
&+ \int_{t_\ell^{sct}(\nu)}^1 t_{\ell'}^{abs}(\nu) \left\{ t_{\ell'}^{sun}(\nu) F^{sun}(\nu) p_{\ell'}(\Omega, \Omega^{sun}; \nu) \right. \\
&\quad \left. + \int_{4\pi} I_{\ell'}(\Omega'; \nu) p_{\ell'}(\Omega, \Omega'; \nu) d\Omega' \right\} dt_{\ell'}^{sct}(\nu) ; \\
I_\ell(\Omega; \nu) &= \begin{cases} 0 & \text{if } \ell \text{ is at the top of atmosphere (TOA)} \\ \varepsilon_s(\Omega; \nu) B(T_g, \nu) + \int_{2\pi} \left[t_s^{sun}(\nu) \frac{F^{sun}(\nu)}{\pi} \delta(\Omega - \Omega^{sun}) + I_s(\Omega'; \nu) \right] f_s(\Omega, \Omega'; \nu) \mu'_s d\Omega' & \text{if } \ell = s \end{cases} .
\end{aligned} \tag{1.14}$$

In writing this final set of equations, the explicit dependence of the solid angle Ω on path length ℓ has been suppressed for notational simplicity and the Dirac delta function $\delta(\Omega - \Omega^{sun})$ has been introduced to move the surface direct solar illumination term inside the angular integration. This form properly treats the case in which the sun is in shadow. If the path termination point ℓ is neither the TOA nor a surface, then the atmosphere beyond ℓ must be modeled to determine $I_\ell(\Omega; \nu)$; of course, one is left to wonder why a path would be terminated prematurely at ℓ if one were not planning to insert an object at that distance.

2. LINE-OF-SIGHT MODTRAN BAND MODEL RADIANCE

The MODTRAN statistic band model computes radiative transfer quantities integrated over *narrow* spectral bins. Thus, the MODTRAN radiant intensity measured at a sensor, $\langle I_0(\Omega) \rangle$, is defined by:

$$\langle I_0(\Omega) \rangle \equiv \frac{1}{\Delta\nu} \int_{\Delta\nu} I_0(\Omega; \nu) d\nu \quad . \quad (2.1)$$

As shown here, band model quantities are identified by angle brackets; no spectral band index is included for notational simplicity. The challenge of the band model is to efficiently and accurately predict the in-band attenuation arising from the large number of molecular transitions that contribute to the total absorption in each spectral bin. The width $\Delta\nu$ of the MODTRAN bins (0.1, 1.0, 5.0 or 15.0 cm^{-1}) is chosen sufficiently narrow to insure at most a modest spectral variation from the extinction sources other than molecular absorption; these sources include particulate (cloud or aerosol) scattering and absorption and also Rayleigh scattering. This definition enables all the *continua* data to be modeled via *effective* band values, generally defined at the spectral bin center and denoted by an over-line (e.g., a band averaged value of u would be denoted $\bar{u}$).

Continua Data: *The ground consists of solids and liquids. Employing lower spectral resolution surface data is justified because condensed matter does not exhibit the fine spectral structure of molecular gas absorption. Similarly, even though scattering data can be highly structure for a collection of particles of a specific size and shape; terrestrial aerosols and clouds consists of particulates of various sizes, shapes and orientations. These distributions smooth out fine spectral structure. Rayleigh scattering optical properties also vary smoothly. The variation of the Planck blackbody function at atmospheric temperatures (~ 180 to 330 K) for the MODTRAN band model widths is modest for the spectral range over which thermal emission is significant.*

The band model complement to the monochromatic LOS radiance expression is obtained by spectrally integrating Eq. (1.14):

$$\begin{aligned} \langle I_0(\Omega) \rangle = & \langle t_\ell I_\ell(\Omega) \rangle + \int_{\langle t_\ell^{abs} \rangle}^1 \overline{t_{\ell'}^{sct} B(T_{\ell'})} d\langle t_{\ell'}^{abs} \rangle \\ & + \int_{\overline{t_\ell^{sct}}}^1 \left[\left\langle t_{0 \rightarrow \ell' \rightarrow \text{sun}}^{abs} \right\rangle \overline{t_{\ell' \rightarrow \text{sun}}^{sct} p_{\ell'}(\Omega, \Omega^{\text{sun}}) F^{\text{sun}}} + \int_{4\pi} \left\langle t_{\ell'}^{abs} I_{\ell'}(\Omega') \right\rangle \overline{p_{\ell'}(\Omega, \Omega')} d\Omega' \right] d\overline{t_{\ell'}^{sct}} \quad . \end{aligned} \quad (2.2)$$

If the LOS terminates at a surface, $\ell = s$, the boundary term has the following form:

$$\begin{aligned} \langle t_s I_s(\Omega) \rangle = & \left\langle t_s^{abs} \right\rangle \overline{t_s^{sct} \mathcal{E}_s(\Omega) B(T_s)} + \left\langle t_{0 \rightarrow s \rightarrow \text{sun}}^{abs} \right\rangle \overline{t_s^{sct} t_{s \rightarrow \text{sun}}^{sct} f_s(\Omega, \Omega^{\text{sun}}) F^{\text{sun}}} \\ & + \int_{2\pi} \left\langle t_s^{abs} I_s(\Omega') \right\rangle \overline{t_s^{sct} f_s(\Omega, \Omega') \mu' d\Omega'} \quad . \end{aligned} \quad (2.3)$$

In these equations, the convention is that a single path length subscript (ℓ , ℓ' or s) denotes a direct LOS from the sensor; arrows ($\rightarrow$) are introduced into the subscript for direct solar illumination paths ($\ell \rightarrow \text{sun}$) or sensor to scattering-point to sun L-shaped paths ($0 \rightarrow \ell \rightarrow \text{sun}$). Remember, this manuscript always references paths by their reverse photon direction. There are 3 band model spectral integrals in these equations:

- (1) $\langle t_{\ell'}^{abs} \rangle$, the sensor LOS attenuation due to absorption;

- (2) $\langle t_{0 \rightarrow \ell' \rightarrow \text{sun}}^{abs} \rangle$, the L-shaped single scatter solar path attenuation due to absorption; and
- (3) $\langle t_{\ell'}^{abs} I_{\ell'}(\Omega') \rangle$ the incoming radiation scattered towards the sensor at ℓ' along the LOS and attenuated by foreground absorption.

The absorption spectral integrals include both molecular and particulate contributions. Although MODTRAN models the particulate absorption via spectral bin central values, i.e., as a continuum source,

$$\left\langle t_{\ell'}^{abs} \right\rangle \equiv \overline{t_{\ell'}^{part-abs}} \left\langle t_{\ell'}^{mol-abs} \right\rangle, \quad (2.4)$$

this partitioning is not explicitly highlighted in Eqs. (2.2) and (2.3). The reader is tasked to remember that $t_{\ell'}^{part-abs}$ is not a band model quantity even though it is embedded in the angle bracket terms. Other quantities computed at the spectral bin center are the path transmittance due to scattering $t_{\ell'}^{sct}$, the blackbody emission $B(T_{\ell'})$, the scattering-point to sun transmittance due to scattering $t_{\ell' \rightarrow \text{sun}}^{sct}$, the scattering phase function $p_{\ell'}(\Omega, \Omega')$, the *imaged-pixel* directional emissivity $\epsilon_s(\Omega)$, and the *imaged-pixel* BRDF $f_s(\Omega, \Omega')$. Although the TOA solar irradiance F^{sun} is also included under the over-line, its band model values are pre-computed by integrating high resolution solar spectral data and dividing by the spectral bin width.

Band Model TOA Solar Irradiance: *One could argue that the band model bin TOA solar irradiance F^{sun} should be coupled into the L-shaped path band model absorption calculations, $\langle t_{0 \rightarrow \ell' \rightarrow \text{sun}}^{abs} F^{sun} \rangle$. After all, the TOA solar irradiance is spectrally highly structured due the Fraunhofer line absorption in the solar atmosphere. MODTRAN models this spectral structure as being randomly correlated with terrestrial absorption, a very reasonable assumption.*

MODTRAN typically models the ground as a spectral Lambertian (angularly-independent) surface with $f_s(\Omega, \Omega') = \alpha_s/\pi$. Under these conditions, the angular distribution of the downwelling (surface impingent) radiance, $I_s(\Omega')$, is decoupled from the foreground molecular absorption. Instead, the surface reflected downward radiance equals the product of the hemisphere-integrated downward flux over π steradians, $\langle F_s^{\downarrow}/\pi \rangle$, and the surface albedo:

$$\int_{2\pi} \left\langle t_s^{abs} I_s(\Omega') \right\rangle \overline{t_s^{sct} f_s(\Omega, \Omega') \mu' d\Omega'} = \overline{t_s^{sct}} \left\langle t_s^{abs} \int_{2\pi} I_s(\Omega') \alpha_s/\pi \mu' d\Omega' \right\rangle = \overline{\alpha_s t_s^{sct}} \left\langle t_s^{abs} F_s^{\downarrow}/\pi \right\rangle. \quad (2.5)$$

MODTRAN does include a handful of terrain BRDF models [Walthall, 1985; Hapke, 1981; Hapke 1986; Rahman *et al.*, 1993; Roujean *et al.*, 1992; Wanner *et al.*, 1995; Wanner *et al.*, 1997; Lucht *et al.*, 1999; Pinty and Verstraete, 1991] as well as an ocean BRDF model [Ross *et al.*, 2005; Ross and Dion, 2007]. When one of these is chosen, the DISORT discrete ordinate scattering model defines its lower boundary, the ground, based on the specified BRDF to incorporate it into the solution for the spectral radiance, $I_v(\Omega; \ell)$. However, MODTRAN does not model the angular coupling between the down-welling radiance and the surface BRDF when computing the LOS boundary term. Instead angular averages are used:

$$\begin{aligned} \int_{2\pi} \left\langle t_s^{abs} I_s(\Omega') \right\rangle \overline{t_s^{sct} f_s(\Omega, \Omega') \mu' d\Omega'} &\approx \overline{t_s^{sct}} \int_{2\pi} \overline{f_s(\Omega, \Omega') \mu' d\Omega'} \left\langle \frac{t_s^{abs}}{\pi} \int_{2\pi} I_s(\Omega') \mu' d\Omega' \right\rangle \\ &= \overline{\rho_s(\Omega) t_s^{sct}} \left\langle t_s^{abs} F_s^{\downarrow}/\pi \right\rangle. \end{aligned} \quad (2.6)$$

In the thermal dominated spectral regions, this angular averaging will generally not introduce significant errors since the downwelling radiance is fairly isotropic. The approximation is more suspect in the solar

dominated spectral region where the diffuse radiance predominantly originates from the solar direction and reflectances are strongly peaked in the hot spot direction for terrains, and at specular and grazing angles over calm water bodies.

DISORT BRDF coupling: *MODTRAN can be upgraded to model this angular coupling of the surface BRDF and the downwelling radiances. One reason this had not previously been implemented is that MODTRAN provides an adjacency option. An “area-averaged” BRDF is used within DISORT to model distant surface reflections, but the $f_s(\Omega, \Omega')$ in Eq. (2.6) is the “imaged-pixel” BRDF. Only recently (2015), was the DISORT expansion of the “imaged-pixel” BRDF defined and integrated into the DISORT module to compute the angularly coupled reflected downwelling radiance. Now that this term is calculated, it should be incorporated into the calculation of the transmitted reflected downwelling radiance.*

MODTRAN invokes a second approximation in its band model calculation of Eq. (2.2) radiances. Band model transmittances do not obey Beer’s Law, which dictates that the *spectral* transmittance of a multi-segmented path equals the product of the individual segment transmittances:

$$t_{0 \rightarrow \ell' \rightarrow \ell}(\nu) = t_{0 \rightarrow \ell'}(\nu) t_{\ell' \rightarrow \ell}(\nu) \quad . \quad (2.7)$$

Equivalently, segment spectral extinction optical depths are additive, $\tau_{0 \rightarrow \ell' \rightarrow \ell}(\nu) = \tau_{0 \rightarrow \ell'}(\nu) + \tau_{\ell' \rightarrow \ell}(\nu)$. This relationship breaks down for band model in-band transmittances:

$$\frac{1}{\Delta \nu} \int_{\Delta \nu} t_{0 \rightarrow \ell' \rightarrow \ell}(\nu) d\nu = \langle t_{0 \rightarrow \ell' \rightarrow \ell} \rangle \neq \langle t_{0 \rightarrow \ell'} \rangle \langle t_{\ell' \rightarrow \ell} \rangle = \left[\frac{1}{\Delta \nu} \int_{\Delta \nu} t_{0 \rightarrow \ell'}(\nu) d\nu \right] \times \left[\frac{1}{\Delta \nu} \int_{\Delta \nu} t_{\ell' \rightarrow \ell}(\nu) d\nu \right] \quad . \quad (2.8)$$

Basically, the spectral integral of the product of transmittances does not necessarily equal the product of the spectral integrals. Figure 2.1 illustrates the problem. In this example, the transmission is zero in the center half of the spectral bins for segments A and B; the remaining half, the edge regions, are fully transparent. The band model transmittance of each segment is one-half. Applying Beer’s Law, the transmittance for the combined path is also be one-half; if in-band segment transmittances were multiplicative, the value would be one-fourth!

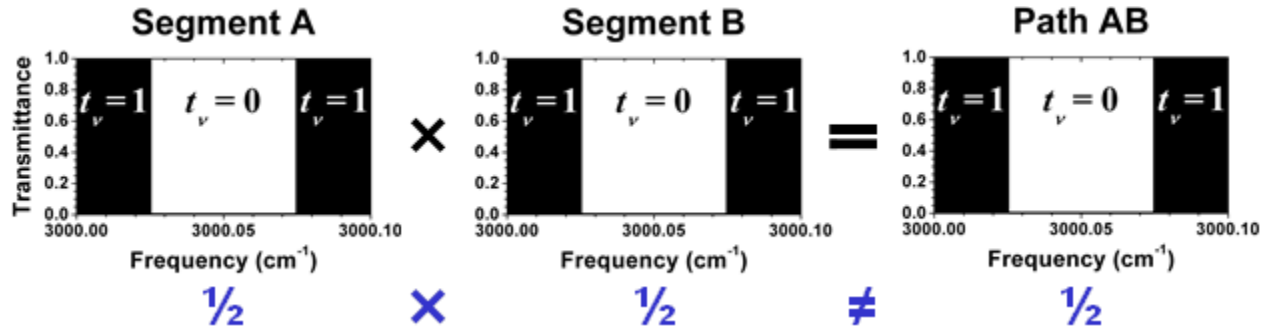


Figure 2.1. Calculation Illustrating that In-Band Transmittances do not Obey Beer’s Law

The plane parallel atmosphere discrete ordinate solution to the RTE is derived assuming additive optical depths. For that reason, the DISORT generated values for the in-band radiances $\langle I_\ell(\Omega') \rangle$ are not rigorously defined. Furthermore, the band model does not spectrally couple the absorption attenuated band model radiance terms, $\langle t_\ell^{abs} I_\ell(\Omega') \rangle$; instead, these are approximated by a band model product: $\langle t_\ell^{abs} I_\ell(\Omega') \rangle \approx \langle t_\ell^{abs} \rangle \times \langle I_\ell(\Omega') \rangle$. In practice, however, the resulting band model LOS radiances, $\langle I_0(\Omega) \rangle$, are sufficiently accurate for many applications, within a few percent of the exact solution. This may seem surprising, but the reality is that the bandpass integral of spectral products generally do not deviate far from the product of the in-band quantities. This is especially true for narrow spectral bins. Furthermore, residuals are often uncorrelated, so a statistical cancellation of errors is observed when band model data is spectrally degraded.

Band Model Optical Depths: *The input for DISORT includes a vertical optical depth for each layer in the stratified atmosphere, but molecular absorption optical depth is not a well-defined band model quantity. One could envision setting these optical depths to the negative natural logarithm of the vertical molecular transmittance of each layer. However, that over-estimates the total vertical optical depth of the atmosphere, leading to under-prediction of scattered radiances. Instead, MODTRAN performs a vertical, ground to space, calculation of cumulative molecular absorption transmittances, $\langle t_i^{mol_abs}(\mu = +1) \rangle$; $\langle t_0(+1) \rangle$ is one and $\langle t_i(+1) \rangle$ is the vertical transmittance through layer i . The layer i band model vertical molecular absorption optical depth, $\langle \tau_{i-1 \rightarrow i}^{mol_abs}(+1) \rangle$, is then set to $-\ln [\langle t_i^{mol_abs}(+1) \rangle / \langle t_{i-1}^{mol_abs}(+1) \rangle]$, the natural logarithm of the incremental transmittance. This definition produces the correct total vertical optical depth, and insures that the surface emission attenuates properly with altitude. For modeling solar scatter, defining the layer optical depths from TOA nadir transmittances would be better, but MODTRAN does not provide that option. A problem arises when cumulative molecular absorption transmittances drop to zero, primarily an issue in the center of the strongest terrestrial molecular absorption band, the CO₂ asymmetric stretch fundamental at 4.3 μm . The natural logarithm cannot be evaluated, so the molecular absorption optical depth is estimated from the lower layer's value. This is seldom a problem because scattered radiances are negligible when absorption is this strong.*

MODTRAN4 introduced a statistical correlated- k (Ck) algorithm [Bernstein *et al.*, 1996; Chapter 4] to provide an option to rigorously model spectral band integrations. The MODTRAN Ck method recasts the molecular band model algorithm into a handful, n_k , of weighted monochromatic calculations, which can be summed to compute spectral band values. Consider, as an example, the calculation of in-band molecular absorption attenuated radiance. The MODTRAN Ck expression is

$$\left\langle t_\ell^{mol_abs} I_\ell(\Omega) \right\rangle = \sum_{i=1}^{n_k} \left[\int_{g(i-1)}^{g(i)} I_g(\Omega) \exp \left[-\tau_g^{mol_abs}(\ell) \right] dg \right], \quad (2.9)$$

where the integrand is obtained by exponentially interpolating the optical depth, $\tau_g^{mol_abs}(\ell)$, and radiant intensity, $I_g(\Omega)$, between bounding monochromatic calculated values

$$\tau_g^{mol_abs}(\ell) = \tau_{g(i)}^{mol_abs}(\ell) \left[\frac{\tau_{g(i-1)}^{mol_abs}(\ell)}{\tau_{g(i)}^{mol_abs}(\ell)} \right]^{\frac{g(i)-g}{g(i)-g(i-1)}} \quad \text{and} \quad I_g(\Omega) = I_{g(i)}(\Omega) \left[\frac{I_{g(i-1)}(\Omega)}{I_{g(i)}(\Omega)} \right]^{\frac{g(i)-g}{g(i)-g(i-1)}}. \quad (2.10)$$

The $g(i)$ are a fixed set of normalized cumulative k -distribution variables defined so that the fraction of molecular absorption coefficients (k 's) below $k(g)$ within a given band model spectral bin is g . By definition, $0 = g(0) < g(1) < \dots < g(n_k) = 1$. In MODTRAN, the number of monochromatic calculations per spectral bin, n_k , is generally 18 (the faster option), although one can opt for increased accuracy using 34 monochromatic calculations (the slower option). The MODTRAN Ck algorithm, in general, is more rigorous and accurate than the band model option, but these improvements come with a significant (order of magnitude) computation time penalty. It is recommended that users run sample calculations for their specific applications to determine if the differences in results warrant the additional computational costs.

To recap, the band model LOS radiance is equal to the monochromatic radiances integrated over the band model spectral bin, Eqs. (2.1) through (2.3). MODTRAN simplifies the surface reflected downwelling radiance term when viewing the ground, modeling it as proportional to the directional reflectivity, Eq. (2.6). The band model does not rigorously couple the spectral dependencies of multiplicative terms in the radiance equations, but the impact on the accuracy of the results is often small. A more computationally intensive statistical Ck algorithm is available for modeling the spectral correlations.

3. MODTRAN BAND MODEL MOLECULAR TRANSMITTANCE

The core MODTRAN radiative transfer algorithm is a statistical molecular band model (BM). In the BM approach, the exact location and strength of each molecular transition is not modeled. Instead, these distributions are statistically represented using a set of temperature and pressure dependent band model parameters. In this spirit, the formulation of the molecular transmittance calculation begins with the stipulation that the spectral bin molecular absorption from distinct species is randomly correlated, so that the combined species molecular transmittance $\langle t^{mol_abs} \rangle$ for a BM spectral bin can be computed as a product of individual BM species transmittances $\langle t^{m_abs} \rangle$:

$$\langle t^{mol_abs} \rangle = \prod_{m\ mol} \langle t^{m_abs} \rangle . \quad (3.1)$$

This approximation works well with the narrow (0.1, 1.0, 5.0 or 15.0 cm⁻¹) MODTRAN spectral bins for a number of reasons. As illustrated in Figure 3.1, in much of the visible through IR spectral region a single molecular species is the dominant absorption source. Where overlap between strong molecular bands of 2 species does occur, H₂O is generally one of the species; the H₂O transition frequency and line strength distributions appear quite randomly within narrow spectral bins since H₂O is an asymmetry top. Therefore, the correlation of H₂O absorption with the absorption from overlapping molecular species is well represented as random. Furthermore, when moderate spectral resolution results are sought, spectral averaging of the relatively fine resolution MODTRAN values tends to stochastically cancel statistical variations.

Within MODTRAN, the individual molecular species spectral bin transmittance is itself partitioned into three distinct components. These components are

- (1) *Line center transmittance*, $\langle t^{m_cen} \rangle$: the transmittance arising from molecular transitions centered within the spectral bin;
- (2) *Line tail transmittance*, $\langle t^{m_tail} \rangle$: the transmittance arising from molecular transitions centered outside of the spectral bin but not further than 25 cm⁻¹ from the spectral bin center; and
- (3) *Continuum transmittance*, $\langle t^{m_cont} \rangle$: the transmittance arising from molecular transitions centered more than 25 cm⁻¹ from a spectral bin.

Again, the assumption is made that these terms are randomly correlated:

$$\langle t^{m_abs} \rangle = \langle t^{m_cen} \rangle \langle t^{m_tail} \rangle \langle t^{m_cont} \rangle . \quad (3.2)$$

The line tail and continuum contributions vary relatively smoothly across the spectral interval, so the random correlation assumption is accurate. The modeling approach for each of these components is detailed in the following subsections.

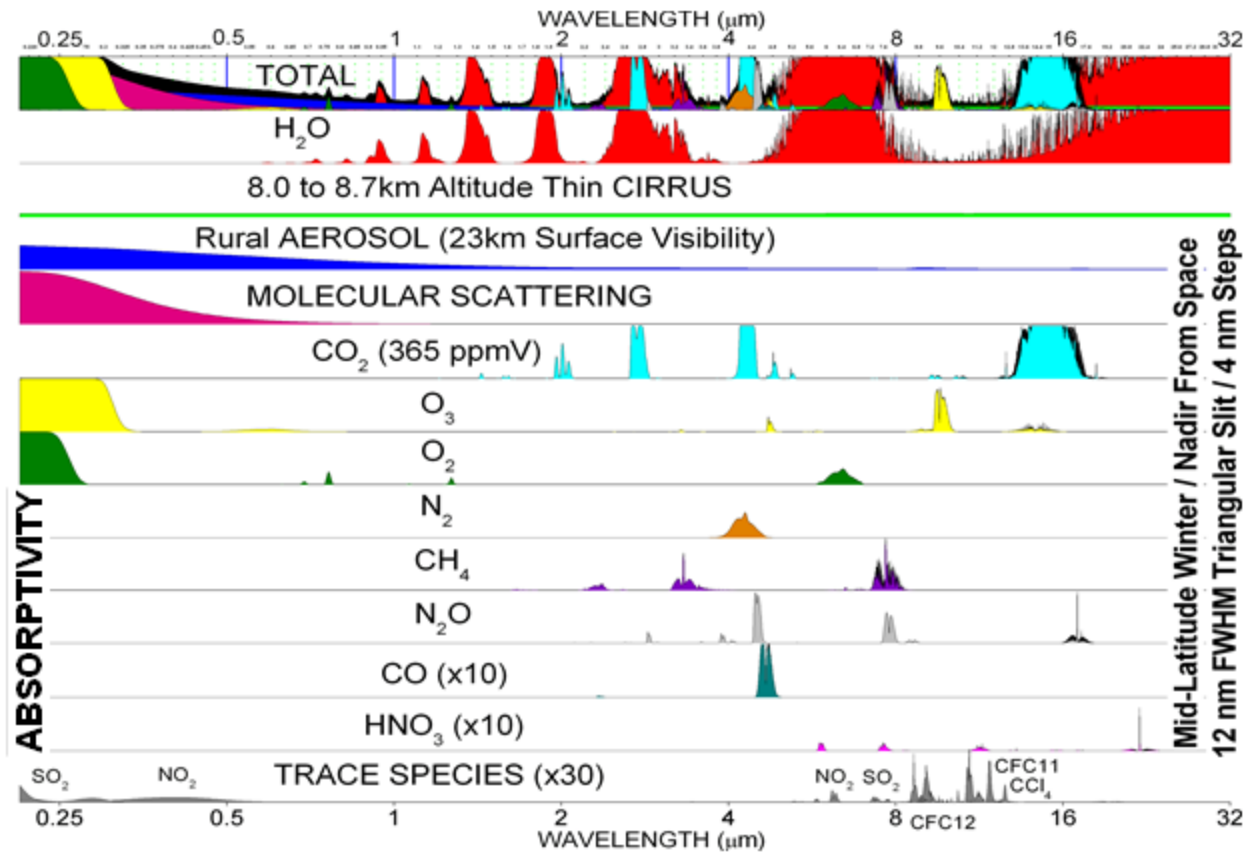


Figure 3.1. Spectral Absorptivity (1 – Transmittance) from the Dominant Sources of Atmospheric Extinction. These 12 nm Resolution Curves were Generated by MODTRAN for a Nadir Path from Space through its Mid-Latitude Winter Model Atmosphere

3.1 Line Center Transmittance

Explicit sampling of the molecular transitions centered within each spectral band on-the-fly, the MODTRAN line-by-line approach (Chapter 7), is computationally intensive, i.e., slow. To avoid this sampling, the MODTRAN BM determines line center spectral bin transmittances by defining a pseudo line-strength distribution representative of the true distribution. The chosen form of the pseudo distribution must enable rapid calculation of the in-band transmittance to be effective in speeding up processing; little time would be saved if the absorption from each line in the pseudo distribution had to be individually computed. Pre-computed temperature-dependent statistical data on the true distribution capture fundamental information on the collective effects of the molecular line strengths in each spectral interval, and the pseudo distributions are then constrained to replicate those statistics.

3.1.1 MODTRAN6 Line Strength Distributions

MODTRAN defines its pseudo line-strength distribution for a homogeneous path as consisting of

- n_s identical strong lines of strength S_s with transition frequencies randomly-distributed in the BM spectral bin, and
- n_w randomly-distributed identical weak lines of strength S_w .

This form for the distribution benefits from there being a simple expression, derived by Plass [1958], for the transmittance of n randomly distributed identical lines of strength S centered in a spectral bin of width $\Delta\nu$:

$$\left\langle t^{P_{\text{lass}}} \right\rangle = t_{sl}^n(Su, \gamma_c, \gamma_d; \Delta\nu, D) \quad . \quad (3.3)$$

Thus, the combined transmittance from the collection of n randomly-distributed identical lines equals the average spectral band transmittance of single-line, $t_{sl}(Su, \gamma_c, \gamma_d; \Delta\nu, D)$, raised to the n^{th} power. The single-line transmittance, t_{sl} , depends

- on the product of the line strength S [$\text{cm}^{-2}/\text{atm}$] and path column density u [atm cm],
- on the collision (Lorentz) γ_c and Doppler γ_d [cm^{-1}] half-widths,
- on the bin width $\Delta\nu$, and
- on the offset of the transition frequency from bin center, D .

In this document, γ_c is defined as the Lorentz half-width at half-maximum, but γ_d is defined as the Doppler half-width at $1/e$ of the maximum, a convention also used by Goody and Yung [1989].

Since the MODTRAN pseudo line-strength distribution contains n_s strong lines of strength S_s and n_w weak lines of strength S_w , the line-center transmittance is given by the product

$$\left\langle t^{m_{\text{-cen}}} \right\rangle = t_{sl}^{n_s}(S_s u, \gamma_c, \gamma_d; \Delta\nu, D) \times t_{sl}^{n_w}(S_w u, \gamma_c, \gamma_d; \Delta\nu, D) \quad . \quad (3.4)$$

For the MODTRAN BM, the single-line molecular absorption spectral dependence is modeled with a Voigt line-shape function, $f_v(\gamma_c, \gamma_d)$ [cm]:

$$t_{sl}(Su, \gamma_c, \gamma_d; \Delta\nu, D) \equiv \frac{1}{\Delta\nu} \int_{D-\Delta\nu/2}^{D+\Delta\nu/2} \exp[-Su f_v(\gamma_c, \gamma_d)] d\nu \quad ; \quad f_v(\gamma_c, \gamma_d) = \frac{\gamma_c}{\pi^{3/2}} \int_{-\infty}^{\infty} \frac{\exp(-x^2)}{\gamma_c^2 + (\nu - \gamma_d x)^2} dx \quad . \quad (3.5)$$

The Voigt spectral line-shape function at line center displacement frequency ν equals the convolution of Lorentz and Doppler line-shape functions. The BM curve-of-growth functional, $\text{CoG}[u] \equiv 1 - \langle t^{m_{\text{-cen}}}(u) \rangle$, defines the rate at which the spectral bin line center molecular absorption increases with column density for a given pressure and temperature. The true distribution of molecular lines in a specific spectral interval will exhibit its own unique CoG. The objective of BM theory, in general, is to find a transmittance form providing an accurate fit to the true CoG. Eq. (3.4) is the MODTRAN functional form.

The single-line finite-bin Voigt transmittance formula, Eq. (3.5), leaves ambiguous the specification of the line center displacement from bin center, D , i.e. the precise location of the molecular transition within the spectral bin. In the early days of MODTRAN development, the single-line transmittance was computed for a molecular line located in the center of the spectral band, a distance of $\Delta\nu/2$ from both edges ($D = 0$). That turned out to be problematic since an *average* single-line spectral band transmittance is desired; the central location creates a bias towards too much absorption. The correctly placed line is one whose transmittance equals the mean transmittance of a transition whose center is randomly distributed in the spectral bin. Determining that line center location is difficult for a Voigt molecular transition. However, under pure Lorentzian conditions [$\gamma_d = 0 \text{ cm}^{-1}$] in the weak line limit [u small; $\exp(-Su f_v) \approx 1 - Su f_v$], the exact analytic expression for the offset from bin center $D = D_{\Delta\nu}(r_c)$ [cm^{-1}] that produces a finite bin transmittance equal to the random distribution value can be derived as a function of the ratio of the Lorentz half-width to the bin width, $r_c = \gamma_c/\Delta\nu$:

$$\left(\frac{D_{\Delta\nu}(r_c)}{\Delta\nu}\right)^2 = \frac{1}{4} - (1+r_c^2) \frac{1 - r_c \tan\left[r_c \ln\left(1 + \frac{1}{r_c^2}\right)\right]}{2 + \left(\frac{1-r_c^2}{r_c}\right) \tan\left[r_c \ln\left(1 + \frac{1}{r_c^2}\right)\right]} . \quad (3.6)$$

As illustrated in Figure 3.2, the weak-line Lorentz line-shape displacement function $D_{\Delta\nu}(r_c)$ has a minimum value of 0.28335626 $\Delta\nu$ near $r_c = 0.743$. As the Lorentz half-width to bin width ratio approaches zero, the displacement function very slowly approaches $D_{\Delta\nu}(0) = \frac{1}{2} \Delta\nu$ (the edge of the spectral bin); for large r_c , $D_{\Delta\nu}(r_c \rightarrow \infty)$ approaches $\Delta\nu/\sqrt{12}$ ($= 0.288675 \Delta\nu$).

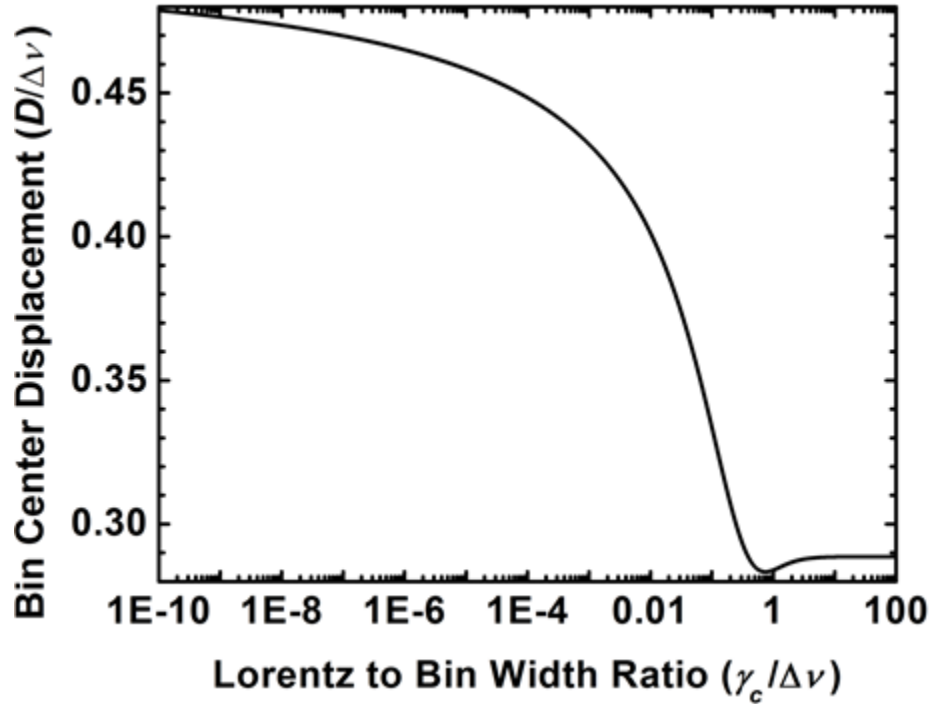


Figure 3.2. Plot of Bin Center Displacement at which the Transmittance of a Weak-Line Lorentz Transition equals the Mean Transmittance Value for a Line Centered Randomly in the Interval

The terrestrial surface Lorentz half-widths are typically of order 0.06 cm^{-1} . For the MODTRAN BM widths of 0.1, 1.0, 5.0 and 15.0 cm^{-1} , the 0.06 cm^{-1} Lorentz transition weak-line displacement values are $D_{0.1/\text{cm}}(0.06) \approx 0.0284 \text{ cm}^{-1}$, $D_{1.0/\text{cm}}(0.06) \approx 0.353 \text{ cm}^{-1}$, $D_{5.0/\text{cm}}(0.012) \approx 1.99 \text{ cm}^{-1}$, and $D_{15.0/\text{cm}}(0.004) \approx 6.25 \text{ cm}^{-1}$, respectively. These weak-line Lorentz transition offsets are only a guide for determining the MODTRAN values. The headers of the MODTRAN BM data files contain the band-width dependent displacements from bin edge ($\frac{1}{2}\Delta\nu - D_{\Delta\nu}$) actually used within MODTRAN:

- 0.025 $\text{cm}^{-1} = 0.05 \text{ cm}^{-1} - D_{0.1/\text{cm}}$ in the “DATA/p1_2013c.as4” BM data file;
- 0.20 $\text{cm}^{-1} = 0.50 \text{ cm}^{-1} - D_{1.0/\text{cm}}$ in the “DATA/01_2013c.as4” BM data file;
- 0.53 $\text{cm}^{-1} = 2.50 \text{ cm}^{-1} - D_{5.0/\text{cm}}$ in the “DATA/05_2013c.as4” BM data file; and
- 1.2 $\text{cm}^{-1} = 7.50 \text{ cm}^{-1} - D_{15.0/\text{cm}}$ in the “DATA/15_2013c.as4” BM data file;

These values are read in by MODTRAN routine openbm.f, a member of the mbmDta_mod.f90 module, as real variable EDGEN0.

3.1.2 MODTRAN6 Single-Line Finite-Bin Transmittance Calculation

Once the single-line offset from bin center D , the line strength S , the Lorentz γ_c and Doppler γ_d half-widths, and the bin width $\Delta\nu$ have all been specified, one can tackle the calculation of the single-line finite-bin transmittance, Eq. (3.5). This is not an easy problem. In the mathematical literature, the Voigt function is noted to be proportional to the real part of the Fadeeva function, a scaled complex complementary error function. Computing the single-line finite-bin transmittance requires exponentiating this transcendental function and integrating the result over a finite spectral region. Prior to the introduction of MODTRAN5 and its 0.1 cm^{-1} BM, the single line transmittance was estimated using a finite-bin equivalent width (EW) formulation:

$$t_{\Delta\nu}^{sl}(Su, \gamma_c, \gamma_d) = 1 - \frac{W_{D-\Delta\nu/2}^{D+\Delta\nu/2}(Su, \gamma_c, \gamma_d)}{\Delta\nu} ; \quad W_a^b(Su, \gamma_c, \gamma_d) \equiv \int_a^b \{1 - \exp[-Su f_v(\gamma_c, \gamma_d)]\} d\nu \quad (3.7)$$

This equation defines W_a^b as equal to the single line integrated absorptivity (one minus transmittance) between spectral frequencies $a < b$. MODTRAN computed W_a^b in two-steps. First, the total Voigt EW (W_∞^∞) was determined by interpolating between the pure Lorentz and Doppler limits using the Rodgers-Williams empirical formula [Rodgers and Williams, 1974]. Line tails were modeled using a far-wing Lorentzian form, spectral transmittances were integrated to determine their line tail EW's (W_∞^a and W_b^∞), and the result was subtracted from the total EW to obtain the required finite-bin EW:

$$W_a^b = W_\infty^\infty - (W_\infty^a + W_b^\infty) . \quad (3.8)$$

While the component calculations (Lorentz and Doppler total EW's, Rodgers-Williams interpolation, line tail EW's) were all accurate to better than 10%, differencing the two terms in Eq. (3.8) often produced significant errors when the two positive contributions were of similar magnitude. The errors became unacceptable large when the narrow 0.1 cm^{-1} spectral resolution BM was introduced into MODTRAN.

A truncated exact expansion for the single line transmittance [Berk, 2013] was incorporated into MODTRAN5, function `tvoigt` in module `tvoigt_mod.f90`. Deriving an exact expansion for t_{sl} , in of itself, is not difficult [e.g., expand the exponential in Eq. (3.5) in a power series]. The challenge is deriving an expansion which converges rapidly in both the weak (small u) and strong (large u) limits for the full range of Voigt line-shapes – pure Lorentz to pure Doppler. The MODTRAN expression is an expansion in terms of modified Bessel functions, $I_n(x)$, with a real (non-negative) argument:

$$\begin{aligned} t_{sl} = & \frac{Su}{\Delta\nu} \left[\frac{V_0^{D-\Delta\nu/2}}{2} I_0\left(\frac{Su}{2} f_{D-\Delta\nu/2}\right) + \sum_{n=1}^{\infty} V_n^{D-\Delta\nu/2} I_n\left(\frac{Su}{2} f_{D-\Delta\nu/2}\right) \right] \exp\left(-\frac{Su}{2} f_{D-\Delta\nu/2}\right) + \frac{D-\Delta\nu/2}{\Delta\nu} \exp(-Su f_{D-\Delta\nu/2}) \\ & + \frac{Su}{\Delta\nu} \left[\frac{V_0^{D+\Delta\nu/2}}{2} I_0\left(\frac{Su}{2} f_{D+\Delta\nu/2}\right) + \sum_{n=1}^{\infty} V_n^{D+\Delta\nu/2} I_n\left(\frac{Su}{2} f_{D+\Delta\nu/2}\right) \right] \exp\left(-\frac{Su}{2} f_{D+\Delta\nu/2}\right) + \frac{D+\Delta\nu/2}{\Delta\nu} \exp(-Su f_{D+\Delta\nu/2}) \quad (3.9) \\ & - \frac{2Su}{\Delta\nu} \left[\frac{V_0^0}{2} I_0\left(\frac{Su}{2} f_0\right) + \sum_{n=1}^{\infty} V_n^0 I_n\left(\frac{Su}{2} f_0\right) \right] \exp\left(-\frac{Su}{2} f_0\right) . \end{aligned}$$

The Voigt line-shape function, $f_\nu = f_\nu(\gamma_c, \gamma_d)$, must be evaluated at line center ($\nu = 0 \text{ cm}^{-1}$) and at the spectral bin edge displacement distances, $\nu = D \pm \frac{1}{2}\Delta\nu$. The dimensionless expansion coefficients V_n^ν ,

$$V_n^\nu \equiv V_n\left(\frac{\nu}{\gamma_d}, \frac{\gamma_c}{\gamma_d}\right) = \langle f_0(\nu) \rangle + n \sum_{k=1}^n \frac{(-4)^k (2k+1)_{n-k}}{(k+1)(n+k)(n-k)!} \{f_k(\nu)\} - \begin{cases} \frac{2\nu f_\nu}{n^2 - 1} & n \text{ even} \\ 0 & n \text{ odd} \end{cases} \quad (3.10)$$

$$\{f_k(\nu)\} \equiv \frac{2}{f_\nu^k} \int_\nu^\infty f_{\nu'}^{k+1} d\nu'$$

only depend on the ν/γ_d and γ_c/γ_d ratios, but Voigt line-shape moment integrals, $\{f_k(\nu)\}$, for non-negative integer k must be computed. The coefficients satisfy a strong-line limit sum rule,

$$\frac{V_0^\nu}{2} + \sum_{n=1}^{\infty} V_n^\nu = \sqrt{\pi \gamma_c f_\nu} \quad ; \quad V_{N+1}^\nu \approx \sqrt{\pi \gamma_c f_\nu} - \left(\frac{V_0^\nu}{2} + \sum_{n=1}^N V_n^\nu \right) \quad (3.11)$$

MODTRAN adds an $N+1$ term to the expansion of t_{sl} truncated after $n = N$ terms using the estimated V_{N+1}^ν coefficients; this significantly reduces the remaining residual.

Computing Line Center Voigt Line-Shape Moment Integrals, $\{f_k(0)\}$: *MODTRAN creates a table of the V_n^ν 's evaluated at the transition frequency ($\nu = 0 \text{ cm}^{-1}$). It would be preferred if these values could be efficiently calculated directly, not interpolated from a table. This would require calculation of transition frequency Voigt line-shape moment integrals, $\{f_k(0)\} \equiv (2 \int_{[0, \infty]} f_\nu^{k+1} d\nu) / f_0^k$ for $k = 0, 1, \dots$. The first of these integrals is simply the Voigt line-shape normalization, $\{f_0(0)\} = 1$. Reichel's paper [1969] entitled "The Integral of the n^{th} Power of the Voigt Function," provides an expression for second moment integral, but no derivation is given. In the notation of this manuscript, his expression can be written as follows:*

$$\{f_1(0)\} = \frac{f_0(\sqrt{2} \gamma_c, \gamma_d)}{\sqrt{2} f_0(\gamma_c, \gamma_d)} = \frac{\exp(\gamma_c^2 / \gamma_d^2) \operatorname{erfc}(\sqrt{2} \gamma_c / \gamma_d)}{\sqrt{2} \operatorname{erfc}(\gamma_c / \gamma_d)} .$$

The final equality is obtained by noting that the Voigt function f_ν evaluated at line center ($\nu = 0 \text{ cm}^{-1}$) is proportional to the scaled real complementary error function. Riestenberg and Berk have provided a derivation for the Reichel expression, and they generalized their method for application to higher order moments (see Appendix A). In the expression for the third ($k = 2$) moment, the complementary error function in the numerator of the expression for $\{f_0(0)\}$ is replaced by a related but somewhat more complicated integral:

$$\{f_2(0)\} = \frac{6 \exp(2 \gamma_c^2 / \gamma_d^2)}{\pi^2 \gamma_d^2 f_0^2(\gamma_c, \gamma_d) \sqrt{3}} \int_{\sqrt{3}}^{\infty} \frac{\exp\left(-2 t^2 \gamma_c^2 / 3 \gamma_d^2\right)}{1 + t^2} dt .$$

In principle, Riestenberg and Berk moment integral expressions could be integrated into MODTRAN to see if they improved accuracy and performance, but this effort was not completed before the release of MODTRAN6.

MODTRAN truncates the transmittance expansion, Eq. (3.9), when one of two convergence criteria is satisfied. Either the sum of the magnitude of consecutive transmittance terms must be less than 10^{-4} ,

$$|t_{sl}(N-1)| + |t_{sl}(N)| < 0.0001 \quad ; \quad N \geq 2 \quad ; \quad t_{sl}(n) \equiv \frac{Su}{\Delta\nu} \left[\begin{aligned} &V_n^{D-\Delta\nu/2} I_n \left(\frac{Su}{2} f_{D-\Delta\nu/2} \right) \exp \left(-\frac{Su}{2} f_{D-\Delta\nu/2} \right) \\ &+ V_n^{D+\Delta\nu/2} I_n \left(\frac{Su}{2} f_{D+\Delta\nu/2} \right) \exp \left(-\frac{Su}{2} f_{D+\Delta\nu/2} \right) \\ &- 2V_n^0 I_n \left(\frac{Su}{2} f_0 \right) \exp \left(-\frac{Su}{2} f_0 \right) \end{aligned} \right] , \quad (3.12)$$

or the sum must be less than 10^{-3} and the magnitude of the $t_{sl}(N+1)$ term, computed with the sum rule values for the V_{N+1}^ν coefficients [Eq. (3.11)], must be less than 10^{-7} . The consecutive terms criteria are used since a single $t_{sl}(n)$ term may be close to zero because the positive and negative contributions *coincidentally* cancel; this is much less likely to happen to consecutive terms.

The V_n^0 are only functions of the ratio γ_c/γ_d , and tabulated values were computed for n up to 20. Interpolations of the 'dat/vndata.h' data ($n \leq 7$) are performed in routine `get_vn`, a member of the `tvoigt_mod.f90` module. MODTRAN contains analytic expressions for V_n^ν up to $n = 7$, valid for $(\gamma_c^2 + \nu^2) / \gamma_d^2$ greater than ~ 2 [module `tvoigt_mod.f90` routines `v0smal`, `v1smal`, ..., `v7smal` if $\nu/\gamma_c < 3$; routines `v0larg`, `v1larg`, ..., `v7larg` if $\nu/\gamma_c \geq 3$]. Within MODTRAN6, spectral bin edge displacement frequency $\nu (= D \pm \Delta\nu/2)$ is never less than 0.025 cm^{-1} for the $\Delta\nu = 0.1 \text{ cm}^{-1}$ resolution BM, or less than 0.20 cm^{-1} with the $\Delta\nu = 1.0 \text{ cm}^{-1}$ resolution BM. The Lorentz and $1/e$ Doppler half-widths are defined by

$$\gamma_c = \gamma_c^0 \left(\frac{P(\text{atm})}{1 \text{ atm}} \right) \left(\frac{296 \text{ K}}{T(\text{K})} \right)^n \quad \text{and} \quad \gamma_d = 4.3 \times 10^{-7} \nu_0 \sqrt{T(\text{K})/m(\text{amu})} , \quad (3.13)$$

where γ_c^0 is the Lorentz half-width at 1 atm pressure and 296 K, P is the atmospheric pressure, n is the temperature dependent exponent, ν_0 is the calculation spectral frequency [cm^{-1}], $T(\text{K})$ is the atmospheric temperature and m is the molecular weight of the absorber gas. The $(\gamma_c^2 + \nu^2) / \gamma_d^2$ ratio is rarely if ever less than 2 for MODTRAN terrestrial calculations, so the V_n^ν analytic expressions are essentially always applicable (MODTRAN does include a rarely invoked and painstakingly slow option to compute on-the-fly the V_n^ν 's up to $n = 20$ using an adaptive numeric integration; this option is invoked by setting logical input `T_BEST` to `.true.` and was designed to be primarily used for code validation).

Occasionally, there is a spectral bin (BM or Ck), a molecule and a path combination for which the single-line finite-bin Voigt transmittance does not satisfy either of the convergence criteria for any value $n \leq 7$. Despite this, the last t_{sl} value is accepted and MODTRAN stores the largest value of $|t_{sl}(7)|$ [computed from Eq. (3.12)] for which this occurs. That value is written to standard output (*.tp6) to alert the user of a possible problem. The largest such residual arising from a MODTRAN6 test case occurs for a MODIS run (nadir from space, 0.1 cm^{-1} Ck run in the heart of the CO_2 $4.3 \text{ }\mu\text{m}$ absorption band), presumably in the Doppler regime where convergence is slowest [Berk, 2013]. The following message is generated:

```
VOIGT SINGLE LINE SPECTRAL BIN TRANSMITTANCES ARE COMPUTED FROM AN
INFINITE SERIES OF MODIFIED BESSEL FUNCTIONS OF THE FIRST KIND
TRUNCATED AFTER AT MOST 8 TERMS (I7). THE MAXIMUM MAGNITUDE OF THE
8th TERM WAS 0.12032 AND THIS VALUE OCCURRED AT SPECTRAL FREQUENCY
2361.45 CM-1.
```

This is an unusually large $|t_{sl}(7)|$ value; typically, the value is less than 0.025 and often by an order of magnitude or two.

The option to compute the V_n^ν 's up to $n = 20$ using an adaptive numeric integration (input `T_BEST = .true.`) was used to validate both the pre- and post-MODTRAN5 transmittance algorithms. A zenith

calculation from ground to space through the mid-latitude winter model atmosphere was run (the Figure 3.1 test case) using the 1.0 cm^{-1} spectral resolution BM between 0.2 and $3.0 \text{ }\mu\text{m}$ and the 0.1 cm^{-1} BM between 3.0 and $32.0 \text{ }\mu\text{m}$. The t_{sl} algorithm is called over 120,000 times. In Figure 3.3, the truncated expansion residuals are compared to t_{sl} residuals obtained using the Rodgers-Williams total Voigt EW approximation along with removal of the Lorentzian line-tails contributions. In general, the truncated Bessel function expansion reduces transmittance residuals by 2 orders of magnitude. Absolute residuals for the Bessel function expansion never exceed 0.0004 for the test problem (even though the maximum $|t_{sl}(7)|$ was 0.01494), and over 97% of the transmittances are accurate to better than 5 decimal places.

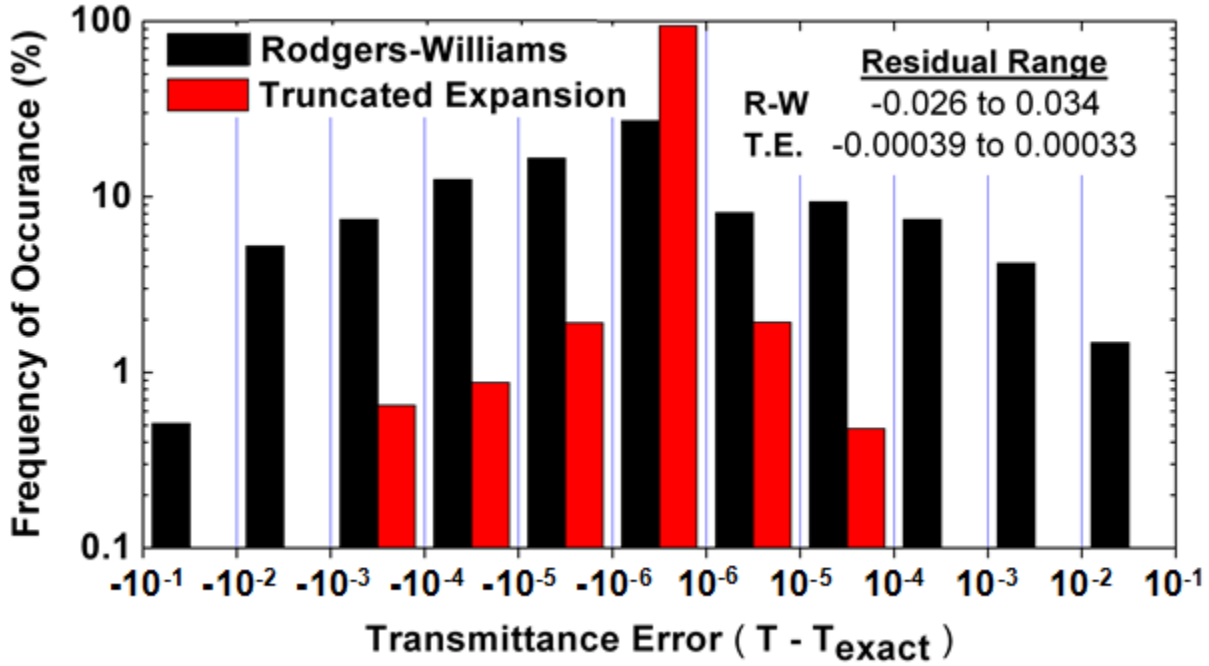


Figure 3.3. Comparison of finite-bin Voigt single-line transmittance residuals. Results are shown for both a Rodgers-Williams equivalent width algorithm (black) and the Bessel function expansion (red) truncated at $N+1$ equal 8 with the last term defined by the sum rule of Eq. (3.11)

3.1.3 MODTRAN6 Line Center Band Model Parameters

As noted in the beginning of Section 3.1.1, MODTRAN6 models line center transmittance as arising from n_s randomly distributed strong lines of strength S_s and n_w randomly distributed weak lines of strength S_w . MODTRAN pre-computes these 4 parameters on a grid of temperatures by requiring that the sum line of strengths raised to a linear progression of powers is the same for both the true and pseudo line strength distributions:

$$\begin{aligned}
 \Sigma_{1/3} &\equiv \sum_{i=1}^N s_i^{1/3} = n_s S_s^{1/3} + n_w S_w^{1/3} & \Sigma_{2/3} &\equiv \sum_{i=1}^N s_i^{2/3} = n_s S_s^{2/3} + n_w S_w^{2/3} \\
 \Sigma_1 &\equiv \sum_{i=1}^N s_i = n_s S_s + n_w S_w & \Sigma_{4/3} &\equiv \sum_{i=1}^N s_i^{4/3} = n_s S_s^{4/3} + n_w S_w^{4/3} .
 \end{aligned} \tag{3.14}$$

Here, N is the true number of lines centered in the spectral interval and s_i is the strength of line i . Traditional band models define a single absorption coefficient (i.e., an *average* line strength over the bin width) and a single line spacing parameter (inversely proportional to the *effective* number of lines. The Σ_1 conditions insures proper convergence to the weak line (column amount u small) limit; a $\Sigma_{1/2}$ condition insures proper convergence to the Lorentzian strong line (u large) limit given the conventional forms for

the BM transmittance [Goody and Yung, 1989]. The MODTRAN upgrade using 4 moment equations, originally introduced by Adler-Golden [circa 1997], improves modeling of the BM curve-of-growth for the troublesome 9.6 μm O_3 band; the chosen moments (1/3, 2/3, 1 and 4/3) extends the 2-parameter range $[\frac{1}{2}, 1]$ while also decreases the moment step size from $\frac{1}{2}$ to $\frac{1}{3}$.

MODTRAN BM data is generated at temperatures ranging from 180 to 330 K with a 5 K step-size and for all the HITRAN [Rothman *et al.*, 2013] molecular line species (except for those for whom line compilations are too sparse and integrated line strength is too low, namely ClONO_2 , SF_6 and CF_4). Nearly degenerate lines occur more often than a random distribution would predict. Furthermore, the absorption from nearly degenerate lines mimics that of a single line. To correct for these situations, all lines centered within each 0.001 cm^{-1} sub-bin are coalesced. In other words, their strengths are summed together and they are treated as a single line when $\Sigma_{1/3}$, $\Sigma_{2/3}$, Σ_1 and $\Sigma_{4/3}$ are computed.

Even though the set of 2N moment equations with exponents $\alpha, \alpha^2, \dots, \alpha^{2N}$ are non-linear, they only require solution of an N^{th} degree polynomial to be solved. The full derivation of the solution to Eq. (3.14) is provided in Appendix B (the notation of the appendix is somewhat different from that used below). The 4 BM parameters are defined by

$$\begin{aligned} S_s^{1/3} &= \left(\frac{\Delta_b}{2\Delta_a} + \sqrt{\frac{\Delta_b^2}{4\Delta_a^2} - \frac{\Delta_c}{\Delta_a}} \right) \Sigma_{1/3} \equiv \tilde{S}_s^{1/3} \Sigma_{1/3} \quad , \quad S_w^{1/3} = \left(\frac{\Delta_c/\Delta_a}{\tilde{S}_s^{1/3}} \right) \Sigma_{1/3} \equiv \tilde{S}_w^{1/3} \Sigma_{1/3} \quad , \\ n_s &= \frac{\tilde{\Sigma}_{2/3} - \tilde{S}_w^{1/3}}{\tilde{S}_s^{1/3} [\tilde{S}_s^{1/3} - \tilde{S}_w^{1/3}]} \quad \text{and} \quad n_w = \frac{\tilde{S}_s^{1/3} - \tilde{\Sigma}_{2/3}}{\tilde{S}_w^{1/3} [\tilde{S}_s^{1/3} - \tilde{S}_w^{1/3}]} \quad ; \quad \tilde{\Sigma}_{m/3} \equiv \frac{\Sigma_{m/3}}{\Sigma_{1/3}} \end{aligned} \quad (3.15)$$

where

$$\begin{aligned} \Delta_a &\equiv \tilde{\Sigma}_1 - \tilde{\Sigma}_{2/3}^2 = \sum_{i=1}^N \sum_{j=1}^{i-1} \tilde{S}_i^{1/3} \tilde{S}_j^{1/3} \left(\tilde{S}_i^{1/3} - \tilde{S}_j^{1/3} \right)^2 \\ \Delta_b &\equiv \tilde{\Sigma}_{4/3} - \tilde{\Sigma}_{2/3} \tilde{\Sigma}_1 = \sum_{i=1}^N \sum_{j=1}^{i-1} \tilde{S}_i^{1/3} \tilde{S}_j^{1/3} \left(\tilde{S}_i^{1/3} - \tilde{S}_j^{1/3} \right)^2 \left(\tilde{S}_i^{1/3} + \tilde{S}_j^{1/3} \right) \quad ; \quad \tilde{S}_i^{1/3} \equiv \frac{S_i}{\Sigma_{1/3}} \\ \Delta_c &\equiv \tilde{\Sigma}_{2/3} \tilde{\Sigma}_{4/3} - \Sigma_1^2 = \sum_{i=1}^N \sum_{j=1}^{i-1} \tilde{S}_i^{2/3} \tilde{S}_j^{2/3} \left(\tilde{S}_i^{1/3} - \tilde{S}_j^{1/3} \right)^2 \end{aligned} \quad (3.16)$$

The central expression in each line of Eq. (3.16) is a difference between terms that depend on the single summations $\Sigma_{1/3}$, $\Sigma_{2/3}$, Σ_1 and $\Sigma_{4/3}$. The right-most expressions contain double sums and are therefore computationally more intensive to evaluate; however, they eliminate numerical instabilities which arise from differencing positive terms of similar magnitude. The coalescing of lines within 0.001 cm^{-1} sub-bins helps to limit the size of N , so the MODTRAN data are determined using the right-most expressions. In these equations, the dimensionless terms denoted with a $\sim$ have been divided by $\Sigma_{1/3}$ raised to a power; this normalization helps avoid under- and over-flow problems.

The MODTRAN database files contain the n_s , n_w , S_s and S_w data expressed as traditional absorption coefficient (S/d) and line spacing ($1/d$) BM parameters. These are defined for both strong and weak lines as follows:

$$\left(\frac{S}{d} \right)_s = \frac{n_s S_s}{\Delta \nu} \quad , \quad \left(\frac{1}{d} \right)_s = \frac{n_s}{\Delta \nu} \quad , \quad \left(\frac{S}{d} \right)_w = \frac{n_w S_w}{\Delta \nu} \quad \text{and} \quad \left(\frac{1}{d} \right)_w = \frac{n_w}{\Delta \nu} \quad . \quad (3.17)$$

To improve the coarser resolution ($\Delta\nu = 1.0, 5.0$ and 15.0 cm^{-1}) MODTRAN5 band models, their line spacing BM parameters are tuned to provide better agreement with the 0.1 cm^{-1} BM predictions. In the first of a two-step process, the column amount U is determined for which the 0.1 cm^{-1} BM predicts a homogeneous path transmittance equal to $1/e$ when degraded to the resolution of the BM that is being tuned. In step 2, the coarse resolution line spacing parameters $(1/d)_s$ and $(1/d)_w$ are scaled until the coarse resolution BM also predicts a transmittance of $1/e$ for column density U .

For each of the 4 BM spectral bin sizes ($15.0, 5.0, 1.0$ and 0.1 cm^{-1}), the line center BM files ('15_2013c.as4', '05_2013c.as4', '01_2013c.as4' and 'p1_2013c.as4', respectively) contain the temperature grid of $(S/d)_s$, $(S/d)_w$, $(1/d)_s$ and $(1/d)_w$ for every spectral bin and the sub-set of the 12 default MODTRAN BM molecules (1=H₂O, 2=CO₂, 3=O₃, 4=N₂O, 5=CO, 6=CH₄, 7=O₂, 8=NO, 9=SO₂, 10=NO₂, 11=NH₃, 12=HNO₃) which have HITRAN lines centered in the given BM interval. Auxiliary HITRAN molecules, such as OH, have their BM data files stored individually in the DATA/HITRANtrace2013/ sub-directory. As an example, the '01_OH.cb4' file contains the 1.0 cm^{-1} resolution data for OH.

In Figure 3.4, the MODTRAN pseudo line-strength distribution is compared to the true distribution for CH₄ at 296 K. HITRAN contains 32 CH₄ lines centered in the 3000.0 to 3000.1 cm^{-1} spectral band, and their 296 K strengths vary between 10^{-9} and $0.0008 \text{ cm}^2/\text{atm}$. Based on solving Eq. (3-14), MODTRAN models the line center absorption in this bin as arising from 4.7 randomly distributed strong lines of strength $0.00051 \text{ cm}^2/\text{atm}$ and 17.1 randomly distributed weak lines of strength $2.46 \times 10^{-5} \text{ cm}^2/\text{atm}$. The line positions in the right in Figure 3.4 were randomly chosen for illustrative purposes; none of the MODTRAN modeled lines are assigned a specific transition frequency.

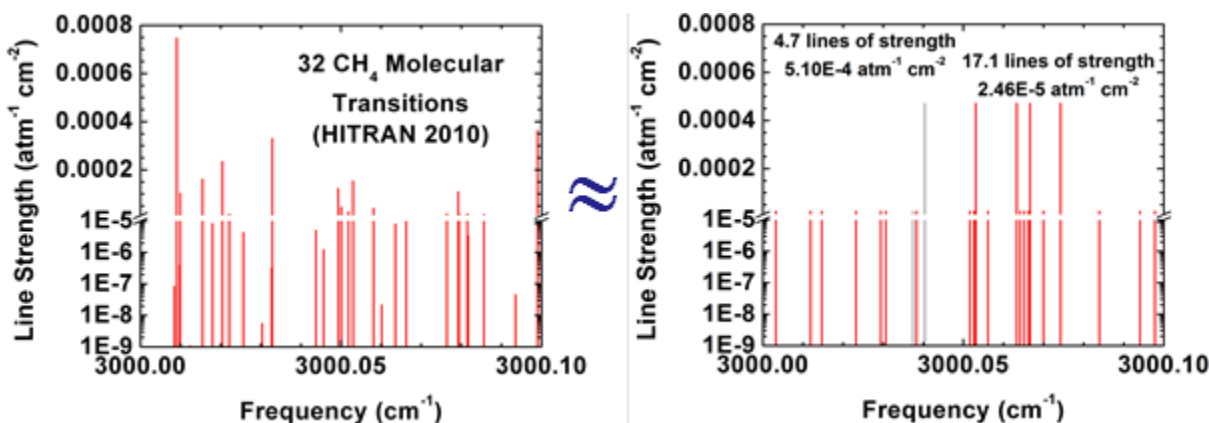


Figure 3.4. Comparison of the True Line Strength Distribution for CH₄ at 296 K to the MODTRAN Pseudo Distribution. The position of the 4.7 strong lines and 17.1 weak lines were randomly assigned in the plot on the right. Partial lines have been drawn in grey. Note that there is a break in the vertical scale at $10^{-5} \text{ cm}^2/\text{atm}$ with a switch in scale from logarithmic to linear.

The Curve-of-Growth (CoG) defines the dependence of absorptivity (one minus transmittance) integrated over a fixed spectral band on the molecular column density. The left-hand side of Figure 3.5 illustrates how it is calculated. For a series of column densities, line-by-line spectral absorptivity calculations are performed (lower left). For example, 10 atm-cm of CH₄ at 296K and 0.1 atm pressure yields the spectral curve above the magenta color. The spectral integration of this curve normalized by the 0.1 cm^{-1} bin width equals an integrated absorptivity, a value close to 0.2. The curve above the blue region is the 100 atm-cm of CH₄ spectral absorptivity; its integrated absorptivity is near 0.8. With the MODTRAN BM description, the CoG is given by

$$\text{CoG}[u] = 1 - t_{sl}^{n_s}(S_s u, \gamma_c, \gamma_d; \Delta\nu, D) \times t_{sl}^{n_w}(S_w u, \gamma_c, \gamma_d; \Delta\nu, D) \quad (3.18)$$

In this expression, all the arguments of t_{sl} other than u are held fixed when the CoG is generated.

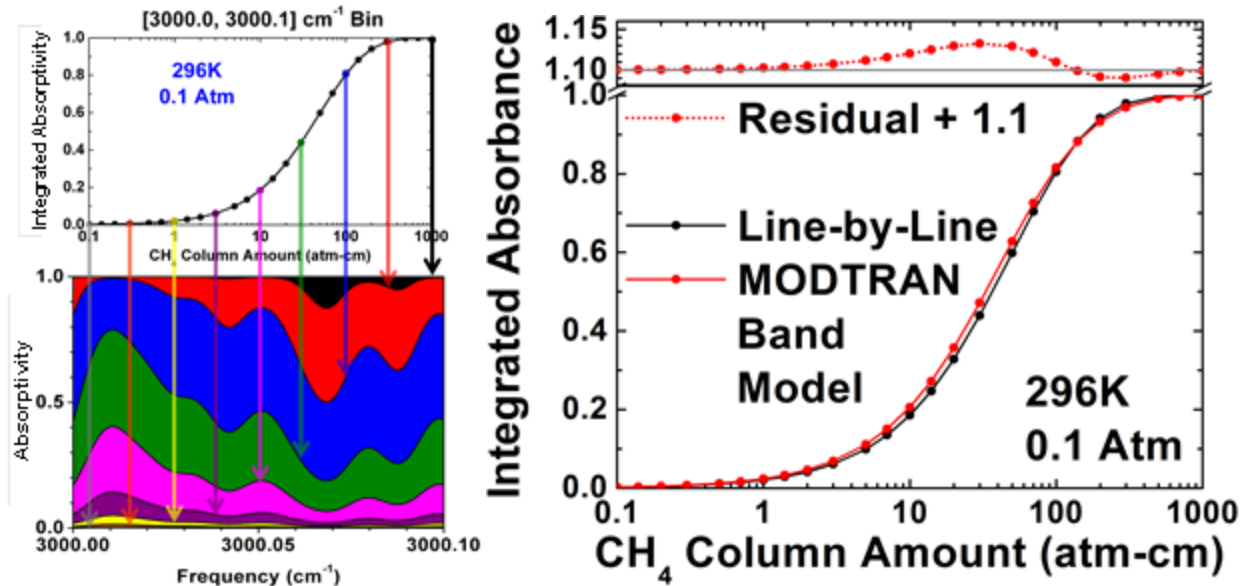


Figure 3.5. (left) LBL spectral absorptivities for a series of CH₄ column amounts at 296 K and 0.1 atm pressure are plotted in the lower left. The integrated absorptivities are plotted as a function of CH₄ column amount, the LBL CoG, in the upper left. (right) The LBL and MODTRAN BM CoG's are compared. Residuals, offset by +1.1 and plotted on an expanded scale, are plotted at the top.

The LBL and MODTRAN band model CoG's for the test problem are compared on the right in Figure 3.5. At first glance, the comparison looks quite good. However, a more careful examination reveals that the BM transmittance is too high by as much as 0.03 when the methane column amount is near 30 atm-cm. It is informative to investigate the source of the discrepancy. Figure 3.4 illustrates (on the left) that the 2nd strongest line in the interval is centered extremely close to the right edge of the spectral bin, somewhere between 3000.099 and 3000.100 cm⁻¹. Rarely would 5.1 randomly distributed strong lines have any single one positioned so close to the bin edge. The random distribution ansatz predicts more centrally located lines, and therefore increased absorption. This example illustrates the limitation of the statistical BM approach. Generally, the predictions are excellent, but occasional a statistical anomaly will produce a substantial residual.

3.1.4 Inhomogeneous Path MODTRAN Band Model Transmittance

The discussion on line center transmittance in Section 3.1 began with a description of the MODTRAN band model expression for the molecular transmittance of a *homogeneous* path. However, the molecular transmittances in the RTE for both the LBL [Eq. (1.14), e.g., $t_\ell^{abs}(\nu)$] and the BM [Eqs. (2.2) and (2.3), e.g., $\langle t_\ell^{abs} \rangle$] methods were defined for *inhomogeneous* paths. Eq. (1.3) defines the i^{th} species (molecular or particulate) density $\rho_i(\ell)$ and spectral absorption cross-section $\kappa_{\ell,i}^{abs}(\nu)$ as varying with path length ℓ along the line-of-sight. Similarly, the line center BM parameters are defined as functions of temperature, T_ℓ , which also varies with path length (the line tail parameters are also functions of pressure, P_ℓ). Methods for generalizing the BM transmittance calculation to inhomogeneous paths are required.

MODTRAN models the Earth's ambient atmosphere as stratified, with molecular and particulate densities, total pressure, and temperature defined as a function of altitude via vertical profiles³. In principle, the optical properties vary with altitude within each layer. MODTRAN assumes this variation is

³ There is an option in MODTRAN to alternatively define temperature and density profiles as a function of pressure with altitudes determined from the hydrostatic equation and a user-defined ground altitude.

minimal, and models absorption and scattering cross-sections as layer constants. For the BM, density-weighted effective temperatures and pressures are defined for every layer segment. Optimally, these density-weighted values should all be species dependent; for example, the O₃ BM data should be computed from O₃ density-weighted pressures and temperatures. For some segments, such as those used for solar illumination paths, species-dependent density weighted values are defined; for other segments, weighting of effective temperatures and pressures is based on air density.

In MODTRAN, the atmospheric layers are modeled as spherical shells above the local Earth radius. The MODTRAN spherical refractive geometry routines determine the LOS segment length inside a layer, mapping the path by adhering to the spherical medium form of Snell's Law. This dictates that the product of three terms,

- (1) the spectral index of refraction $n_\ell \equiv n(h_\ell; \nu)$ at altitude h_ℓ a distance ℓ along the LOS,
- (2) the Earth center distance, r_ℓ (the sum of the Earth radius and h_ℓ), and
- (3) the sine of the path zenith angle, $\sqrt{1-\mu_\ell^2}$,

is a path constant:

$$n_\ell r_\ell \sqrt{1-\mu_\ell^2} = C_{path} \quad . \quad (3.19)$$

MODTRAN defines a reference spectral frequency at which the spherical refractive geometry problem is solved. The user can define this value via input VRFRAC [cm⁻¹]; otherwise, the band pass central frequency is used. The MODTRAN spectral index of refraction is defined as a function of pressure, temperature, H₂O density and CO₂ density [Bodhaine, 1999]. The refracted path segment column densities and the density-weighted average temperatures and pressures are computed assuming refractivity (index of refraction minus one), densities and pressures all vary exponentially with altitudes, and temperature varies linearly with altitude. See Appendix C for a full description of the MODTRAN spherical refractive geometry and column density algorithms.

If Beer's Law were applicable within the BM approach, the LOS molecular line center transmittance could be computed by simply stringing together individual segment transmittances for each species. However, as detailed above, the negative natural logarithm of the in-band transmittance does not vary linearly with column density. An approximate solution to the BM inhomogeneous path problem was derived independently by van de Hulst [1945], Curtis [1952] and Godson [1953], although it is commonly referred to as the Curtis-Godson (CG) approximation. The approach involves approximating the inhomogeneous path for each molecule by a *quasi-equivalent* homogeneous path. To define the homogeneous path, the sum of Su products for each segment in the LOS is initially computed for both the strong and a weak line-strength band model parameters:

$$\overline{S_s u} \equiv \sum_{\sigma \text{ segments}} S_s(T_\sigma) u_\sigma \quad \text{and} \quad \overline{S_w u} \equiv \sum_{\sigma \text{ segments}} S_w(T_\sigma) u_\sigma \quad . \quad (3.20)$$

Next, CG path-averaged effective line numbers are defined weighted by the Su products

$$\overline{n_s} \equiv \frac{\sum_{\sigma \text{ segments}} n_s(T_\sigma) S_s(T_\sigma) u_\sigma}{\overline{S_s u}} \quad \text{and} \quad \overline{n_w} \equiv \frac{\sum_{\sigma \text{ segments}} n_w(T_\sigma) S_w(T_\sigma) u_\sigma}{\overline{S_w u}} \quad . \quad (3.21)$$

Finally, the CG Lorentz and Doppler line-width parameters are defined, but they are weighted by the product of Su and n :

$$\overline{\gamma_s} \equiv \frac{\sum_{\sigma \text{ segments}} \gamma_\sigma n_s(T_\sigma) S_s(T_\sigma) u_\sigma}{\overline{n_s} \overline{S_s u}} \quad \text{and} \quad \overline{\gamma_w} \equiv \frac{\sum_{\sigma \text{ segments}} \gamma_\sigma n_w(T_\sigma) S_w(T_\sigma) u_\sigma}{\overline{n_w} \overline{S_w u}} \quad (3.22)$$

The CG line center transmittance is computed by inserting the CG parameters, Eqs (3.20-3.22) into the MODTRAN BM line center transmittance formula, Eq. (3.4):

$$\left\langle t^{m_cen} \right\rangle_{CG} = t_{sl}^{\bar{n}_s}(\overline{S_s u}, \overline{\gamma_{cs}}, \overline{\gamma_{ds}}; \Delta\nu, D) \times t_{sl}^{\bar{n}_w}(\overline{S_w u}, \overline{\gamma_{cw}}, \overline{\gamma_{dw}}; \Delta\nu, D) . \quad (3.23)$$

The CG approximation works well for most terrestrial atmosphere ambient molecular species. The one important exception is ozone. The *absolute* O₃ concentration has a relatively flat profile from the Earth's surface up to the middle of the stratosphere, but the spectral shape of the absorption varies drastically over this altitude range. The homogeneous path approximation is poor. MODTRAN corrects for this problem by using a higher-order CG formulation derived by Goody [1964] and modified by Bernstein [1998].

Applying CG spectral path averaging for the solar illumination paths arising in the spherical Earth single scatter solar calculation, Figure 3.6, is computationally expensive. To simplify the solar illumination calculations, MODTRAN approximates the illumination path at each LOS altitude level intersection (and the tangent point) as a single homogeneous segment using cumulative column densities and molecular density weighted effective temperatures and pressures. An efficient and high-fidelity upgrade to the single layer approximation has been proposed and is anticipated to be included in a future release of MODTRAN. It is described in Appendix D.

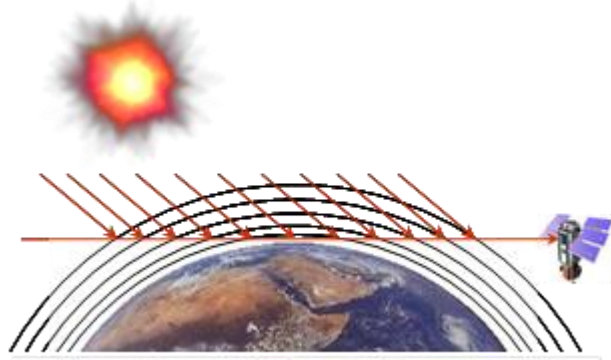


Figure 3.6. Illustration of LOS and Solar Illumination Paths

3.2 Line Tail Transmittance

The MODTRAN spectral bin *line tail* transmittance models the absorption from molecular transitions centered outside the current spectral bin, but no more than 25 cm⁻¹ from the bin center. Unlike the line center spectral attenuation, the line tail absorption varies smoothly across the bin with at most a single minimum. As a result, the line tail spectral absorption cross-section curve can be accurately fit to temperature and pressure dependent [2,2] Padé approximants. These functions are ratios of quadratic polynomials in δ_ν , the spectral frequency displacement from line center normalized to vary between -1 and +1. Three of the five Padé approximant parameters, σ_0 , σ_1 and σ_2 , have units of reciprocal atm-cm and other two, x_1 and x_2 , are dimensionless:

$$\frac{\kappa^{abs}(\delta_\nu)}{P/P_0} = \frac{\sigma_0 + \sigma_1 \delta_\nu + \sigma_2 \delta_\nu^2}{1 + x_1 \delta_\nu + x_2 \delta_\nu^2} ; \quad \delta_\nu \equiv \frac{\nu - \bar{\nu}}{\Delta\nu/2} ; \quad \left| \begin{array}{l} \sigma_0 + \sigma_1 \delta_\nu + \sigma_2 \delta_\nu^2 \geq 0 \\ 1 + x_1 \delta_\nu + x_2 \delta_\nu^2 > 0 \end{array} \right. . \quad (3.24)$$

In this parameterization, $\bar{\nu}$ is the spectral bin central frequency. Since the Lorentz line shape to leading order is proportional to pressure, the absorption cross-section is divided by pressure normalized to P_0 , i.e., 1 atm. One reason this form for the tail contributions was chosen is that it reduces to the exact line tail expression in the limit of a single Lorentz line. The five Padé parameters are determined from HITRAN line data by requiring that

- (1) the spectral curve passes through the summed Voigt line tail spectral absorption at the edge and center spectral frequencies [$\delta_\nu = -1, 0, 1$],
- (2) the data and curve fit spectral frequency derivatives at $\delta_\nu = 0$ are equal, and
- (3) the integrals of the spectral absorption over the spectral band are equal.

The derivation of the fitting parameters is included in Appendix E. The parameter fitting for CO₂ has one added factor; it also includes the CO₂ line shape sub-Lorentzian χ_v -factor.

Figure 3.7 contains the Padé spectral fit for a single 200.0 to 200.1 cm⁻¹ spectral bin. Hundreds of molecular transitions contribute to the 305K and 1 atm pressure H₂O line tail absorption spectrum, the thick red dashed curve. As is generally true, the spectral fit (green curve) lies directly on top of the high-resolution data (relative residuals less than 0.0001).

To achieve the level of agreement displayed in Figure 3.7, none of the line tail molecular transitions are centered too close to the spectral bin edge. Voigt line tails exhibit a strong Doppler dependence near line center, and this could impede Padé fitting. MODTRAN shifts inward the location of lines centered too near a bin edge to eliminate this potential problem. The relocation of line centers itself introduces inaccuracies into the band model; the MODTRAN band model tries to create a balance between curve fitting and line relocation errors.

Conserving Integrated Line Strength:

As just noted, lines centered too close to a spectral bin edge are relocated closer to bin center. The EW offset distance D defined in Section 3.1.1 is used for the bin edge distance test. Lines centered closer than 0.025, 0.20, 0.53 or 1.2 cm⁻¹ from a spectral bin edge for the 0.1, 1.0, 5.0 or 15.0 cm⁻¹ band model, respectively, are translated inward to that offset distance. Remember, this offset also defines the line center location used to calculate the average line center transmittance, t_{sl} . The subset of lines which are not relocated because they are centrally positioned in the bin actually have their integrated line strength under-predicted. This under-prediction, $\Delta S_p^{line\ shift}$, is explicitly computed for each centrally located line p ,

$$\Delta S_p^{line\ shift} = \left(\int_{(\nu_M + \frac{1}{2}\Delta\nu) - \nu_p}^{\infty} S f_\nu d\nu + \int_{\nu_p - (\nu_M - \frac{1}{2}\Delta\nu)}^{\infty} S f_\nu d\nu \right) - \left(\int_{\frac{1}{2}\Delta\nu - D}^{\infty} S f_\nu d\nu + \int_{\frac{1}{2}\Delta\nu + D}^{\infty} S f_\nu d\nu \right),$$

and added to the line tail Padé approximant, conserving integrated line-strength.

Re-Evaluation of Molecular Transition Frequency Shifting: Band model validation studies performed as part of the MODTRAN6 LBL algorithm development (Chapter 7) showed that the largest residuals arise when the line center position of a strong molecular transition is shifted away from a bin edge to facilitate the line tail fitting. As noted above, the presumption was made that this shifting was necessary because the functional form of the Padé fits were incompatible with the Doppler dominated line-shape characteristics near line center. In actuality, this shifting of line centers was introduced prior to evaluating the magnitude of these errors. Given the significance of the residuals created by the shifting, SSI plans to evaluate the extent to which the line shifting is truly necessary. It is anticipated that, at a minimum, the distance lines are displaced from bin edges will be reduced and possibly eliminated entirely if the Padé fits are found to remain highly accurate.

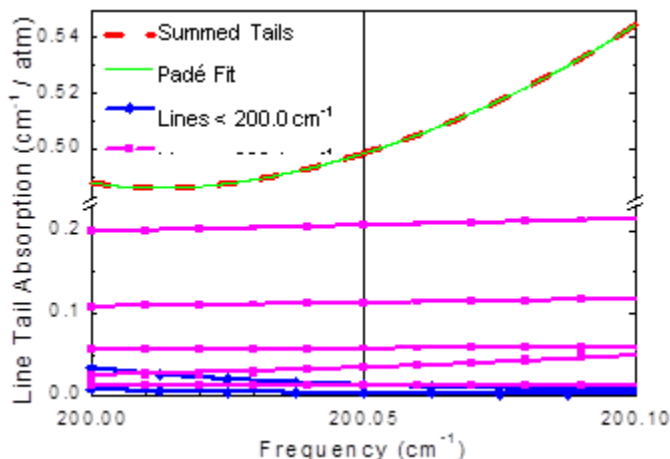


Figure 3.7. High Spectral Resolution Line Tail Absorption Curve (thick dash) and Padé Approximant Fit (thin line) for Lorentzian H₂O Lines at 305K and 1 Atm Pressure. The spectral curves from the largest contributors are also illustrated (lines with diamond for lower frequency transitions and lines with squares for higher frequency transitions). Note the break in the vertical scale near 0.24 cm⁻¹/atm.

The Padé parameter database is generated for pressures of 1.0 and 0.1 atm, for a series of temperatures (180 to 330 K in 5K increments), and for *all* the HITRAN line compilation molecules. The κ^{abs}/P interpolations are modeled as linear in temperature and linear in *pressure-squared*. The P^2 dependence arises because the Voigt line-shape over γ_c is a function pressure-squared. Analysis performed as part of the MODTRAN line-by-line (LBL) algorithm development (see Chapter 7) demonstrated that pressure interpolations of P/κ^{abs} , the reciprocal of the line tail absorption, Eq. 3-24, provides a much improved fit. Even better results, more accurate in the higher altitude Doppler regime, are obtained when a 4-point spline fit in pressure (not pressure-squared) is used to define P/κ^{abs} . These pressure interpolation upgrades have yet (as of early 2016) to be implemented in MODTRAN6.

This section has described the MODTRAN algorithm for modeling the spectral dependence of the line tail absorption. In the end, a method for computing the spectral bin line tail transmittance $\langle t^{m_tail} \rangle$, Eq. (3.2), is required. Since the line tail spectral contributions are smoothly varying, they can be coarsely sampled and interpolated. For each segment, the MODTRAN parameter NTLSUB (nominally set to 4) defines the number of sub-intervals into which each band model spectral bin is sub-divided. Monochromatic line tail absorption optical depth calculations are performed at the NTLSUB + 1 sub-bin edges. Full path line tail spectral transmittances are calculated assuming the optical depths vary exponentially between neighboring spectral frequency grid points.

3.3 Molecular Continua Transmittance

MODTRAN models lines tail contributions beyond 25 cm^{-1} from line center for two atmospheric absorbers, H_2O and CO_2 . The continuum algorithms employed in standard LBL models are used for both of these molecules.

The temperature dependent H_2O continuum spectral absorption cross-section $\kappa_v^{H_2O \text{ cont}}(T)$ is partitioned into a foreign and a temperature-dependent self-broadened component:

$$\begin{aligned} \kappa_v^{H_2O \text{ cont}}(T) = \nu \tanh\left(\frac{c_2 \nu}{2T}\right) \times \left\{ \left(\frac{\rho_{total} - \rho_{H_2O}}{\rho_{total}} \right) C_v^{H_2O \text{ foreign}}(296K) \right. \\ \left. + \left(\frac{\rho_{H_2O}}{\rho_{total}} \right) \left[\left(\frac{T - 260K}{36K} \right) C_v^{H_2O \text{ self}}(296K) + \left(\frac{296K - T}{36K} \right) C_v^{H_2O \text{ self}}(260K) \right] \right\} \end{aligned} \quad (3.25)$$

The spectral foreign ($C_v^{H_2O \text{ foreign}}$) and self ($C_v^{H_2O \text{ self}}$) broadened H_2O continuum parameters [$(\text{cm}^2/\text{mol}) / \text{cm}^{-1}$] are tabulated from 0 to $10,000 \text{ cm}^{-1}$ on a 5 cm^{-1} grid; the self-broadened term endpoint values are used for temperatures below 260K or above 296K rather than extrapolation. The MT_CKD_2.5 H_2O continuum data is used in MODTRAN6 [Clough *et al.*, 1989; Delamere, *et al.*, 2010; Payne, *et al.*, 2011; Mlawer, *et al.*, 2012]

The form of the CO_2 continuum spectral absorption cross-section is simpler than that for H_2O . CO_2 has a zero dipole moment, so partitioning into foreign and self broadened components is unnecessary:

$$\kappa_v^{CO_2 \text{ cont}}(T) = \nu \tanh\left(\frac{c_2 \nu}{2T}\right) C_v^{CO_2} \quad (3.26)$$

The CO_2 data is tabulated on a 10 cm^{-1} grid from 0 to $10,000 \text{ cm}^{-1}$. Since the CO_2 continuum cross-section values are themselves not functions of mole fraction, they are directly incorporated into the CO_2 line tail BM parameters, as mentioned above.

4. MODTRAN STATISTICAL CORRELATED-K ALGORITHM

Two documents have been generated describing MODTRAN's statistical correlated- k method. The first is based on a paper by Bernstein *et al.* [1996], which described the original approach. This paper has been updated to include all upgrades included in the initial release of MODTRAN6. Appendix F describes a proposed upgrade to the C_k treatment designed to improve the modeling of spectral correlations. A preliminary implementation of the upgraded C_k has been coded, but further testing is required before it can be released.

Approved for public release; distribution is unlimited; 1996 IRIS Targets & Backgrounds Meeting.

ADDITION OF A CORRELATED- k CAPABILITY TO MODTRAN

January 1996 (revised, April 2016)

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ABSTRACT

The development of a new Correlated- k (C_k) transmission model and its integration into MODTRAN are described. A Monte Carlo line-by-line spectral simulation is used to generate the C_k absorption coefficient distributions. A compact database representation of the C_k distributions is developed for real time use in MODTRAN calculations. Comparisons of transmittance spectra computed with the C_k and MODTRAN band model algorithms are presented.

4.1 Introduction

This paper describes the approach used to add a Correlated- k (C_k) capability to the MODTRAN atmospheric transmission/radiance computer code [Berk *et al.*, 1989]. The overall objective of the effort is to provide an accurate and fast means for evaluation of the effects of volcanic aerosols and clouds on atmospheric radiative heating/cooling and on retrievals of species concentration profiles. These radiative transfer computations require coupling the effects of gaseous molecular absorption due primarily to water vapor and carbon dioxide and particulate multiple scattering due to volcanic aerosols, ice crystals, and water droplets. The molecular absorption band model used in MODTRAN is not suitable for interfacing with standard multiple scattering algorithms. This is because the scattering models are based on a Beer's law description of the molecular absorption/transmission [i.e., $T = \exp(-k u)$] whereas molecular band models exhibit different and more complex dependencies on the absorption cross-section k [$\text{cm}^2/\text{molecule}$] and absorber column amount u [$\text{molecules}/\text{cm}^2$]. In order to adapt a band model approach for use in scattering calculations it is necessary to express the band model transmission function in terms of a weighted sum of Beer's law exponential terms. Thus, one of the fundamental problems addressed in this work is the development of a method for determining the weighing factors and monochromatic absorption coefficients corresponding to the MODTRAN band model.

4.2 Overview of C_k Method

For simplicity, consider the problem of determining the average transmittance, $T(u)$, for a homogeneous path over a finite spectral interval, $\Delta\omega \equiv \omega_2 - \omega_1$. The generalization to inhomogeneous paths is straightforward and is also described below. The path transmittance can be exactly determined through evaluation of

$$T(u) = \frac{1}{\Delta\omega} \int_{\omega_1}^{\omega_2} \exp[-k(\omega)u] d\omega \quad , \quad (4.1)$$

where ω is the spectral frequency [cm^{-1}] and $k(\omega)$ is the monochromatic absorption cross-section. The basis of the *Ck* approach is that evaluation of $T(u)$ by integration over frequency can be replaced by an equivalent integration over the probability density of absorption cross-section values $f(k)$ in the spectral interval

$$T(u) = \int_0^\infty f(k) \exp(-ku) dk \quad . \quad (4.2)$$

The difficulty in evaluation of $T(u)$ via Eq. (4.2) is the necessity of first determining $f(k)$; the advantage is that $k(\omega)$ is typically a highly repetitive function of ω (i.e., there are many nearly equivalent values of a given k in a spectral interval) and thus numerical evaluation of Eq. (4.2) requires far fewer grid points than Eq. (4.1). Note that $f(k)$ equals zero above the maximum $k_{\max}$ and below the minimum $k_{\min}$ absorption cross-sections in the spectral interval. Several methods are available for determining $f(k)$. It can be determined directly from $k(\omega)$ by binning the k 's into selected Δk sub-intervals. Another method takes advantage of the fact that Eq. (4.2) is in the form of a Laplace transform; thus, $f(k)$ can be determined from the inverse Laplace transform of $T(u)$

$$f(k) = L^{-1}[T(u)] \quad . \quad (4.3)$$

The utility of Eq. (4.3) is it requires that only $T(u)$ and not $k(\omega)$ be known *a priori* and therefore can be applied to any transmittance model, both line-by-line and band model. The distribution function $f(k)$ is not smooth or monotonic; it generally consists of a series of sharp spikes which reflects the sharp line structure of $k(\omega)$. It is computationally more convenient to work with the continuous and monotonic cumulative probability distribution function

$$g(k) = \int_0^k f(k') dk' \quad . \quad (4.4)$$

Physically, $g(k)$ simply equals the fraction of $k'(\omega)$'s within the finite spectral interval ω_1 to ω_2 that are less than k . By definition $g(k_{\min}) = 0$ and $g(k_{\max}) = 1$. The transmittance is related to g by

$$T(u) = \int_0^1 \exp[-k(g)u] dg \quad , \quad (4.5)$$

where $k(g)$ is given by the inverse of $g(k)$, $k(g) = g^{-1}(k)$. In practice, the evaluation of Eq. (4.5) proceeds via summation according to

$$T(u) = \sum_{i=1}^I \Delta g_i \exp(-\underline{k}_i u) \quad , \quad (4.6)$$

where the Δg_i 's and $\underline{k}_i$'s are the sub-interval weighing factors and effective absorption cross-sections we seek to define for the MODTRAN band model. The maximum number of intervals I and the specific selection of Δg_i interval boundaries (g_{i-1} , g_i) are chosen based on consideration of a number of factors including computational speed and accuracy, and the altitude and spectral ranges of interest. For multi-layer paths containing segments $j = 1$ to J , perfect spectral correlation is assumed among the Δg_i sub-intervals of each segment, resulting in a total path transmittance given by

$$T(u_1, \dots, u_J) = \sum_{i=1}^I \Delta g_i \exp\left(-\sum_{j=1}^J \underline{k}_{i,j} u_j\right) \quad . \quad (4.7)$$

4.3 MODTRAN k -Distributions

The MODTRAN band model line center transmittance for a single species (multiple species are discussed later) is based on six parameters: (a) strong and weak molecular transition integrated line strengths, S_s and S_w , in a spectral interval $\Delta\omega$ [$\Delta\omega = 0.1, 1.0, 5.0$ or 15 cm^{-1} in MODTRAN], (b) strong and weak molecular transition effective number of equivalent lines n_s and n_w [non-integer values of n_s and n_w are acceptable] in the interval, (c) the average pressure broadening Lorentz line width γ_L , and (4) the Doppler line width γ_D . These parameters are determined directly from the HITRAN line parameter compilation [Rothman *et al.*, 2013]. The line center molecular transmittance model is given by

$$T(u) = \left(1 - \frac{W(S_s^{sl} u, \gamma_L, \gamma_D)}{\Delta\omega} \right)^{n_s} \left(1 - \frac{W(S_w^{sl} u, \gamma_L, \gamma_D)}{\Delta\omega} \right)^{n_w}, \quad (4.8)$$

where $S_s^{sl} = S_s/n_s$ and $S_w^{sl} = S_w/n_w$ are the strong and weak strength for an effective single line, and W is the Voigt equivalent width of the line taking into consideration only the portion of the line shape which falls within the $\Delta\omega$ spectral interval. This functional form for the band model transmittance derives from the assumption that all the lines in the spectral interval are positionally uncorrelated and thus accounts for line overlap in a statistically average sense. The transmittance contributions due to the tails of lines which originate from outside the spectral interval are included by randomly distributing their sub-bin contributions among the Δg_i intervals. Detailed formulas for the equivalent width W are available elsewhere [Berk *et al.*, 2013]. As discussed earlier, the distribution of k values defined by the probability density function $f(k)$ for the band model could be determined directly from Eq. (4.3), the inverse Laplace transform of $T(u)$. Alternatively, and the method applied here, it can be determined from line-by-line (LBL) simulations which are physically constrained by the same assumptions used to derive Eq. (4.8). The use of real time LBL simulation or inverse Laplace transformation to derive $f(k)$ during a MODTRAN run is much too slow to be of practical value; however, the approach adopted here is to use LBL simulation off-line to pre-compute a table of k cumulative distributions which can be rapidly accessed and interpolated during a MODTRAN run.

4.3.1 Monte Carlo Line-by-Line Simulation of $f(k)$

For a given choice of n , γ_L , γ_D , and $S = 1 \text{ cm}^{-2} \text{ atm}^{-1}$ [$f(k)$ is directly proportional to S so any convenient reference value can be used] a series of LBL absorption coefficient spectra are generated for each of MODTRAN's 4 band model spectral widths, $\Delta\omega = 0.1, 1.0, 5.0$ or 15 cm^{-1} . The positions of the n lines for each spectrum are selected at random with uniform probability over the spectral interval. The $f(k)$ for each spectrum is determined and all the $f(k)$'s are averaged with equal weighting to determine the final band model equivalent k -distribution (here, the convention is to use the term k -distribution even though the $f(k)$'s are actually probability densities). The final $f(k)$ is then transformed into the more conveniently used $k(g)$ distribution as discussed earlier. The procedure is illustrated in Figures 4.1 and 4.2 for an example using $n = 5$ lines and a 1.0 cm^{-1} spectral bin. Three of the many LBL spectra used to determine $f(k)$ are shown in Figure 4.1. In determining the $f(k)$ distribution, each trial spectrum is interrogated over a fine grid of equally spaced frequency points $\delta\omega$ and the k value at each point is binned into a logarithmic sequence of k -bins

$$k_j = k_{min} 10^{j\alpha}, \quad (4.9)$$

where α controls the spacing of adjacent k -bins. In this study $\alpha = 0.005$ was used which results in spacing of $10^{0.005} = 1.012$. The Monte Carlo averaged $k(g)$ distribution is then calculated from

$$\Delta g_j = \frac{n_j \delta\omega}{n_{MC} \Delta\omega}, \quad (4.10)$$

where n_j is the number of binned k values between k_{j-1} and k_j , and n_{MC} is the number of Monte Carlo trial spectra. The function $g_j(k_j)$ is constructed from the Δg_j 's by

$$g_j = \sum_{i=1}^j \Delta g_i \quad , \quad (4.11)$$

which also defines the final desired distribution function $k_j(g_j)$.

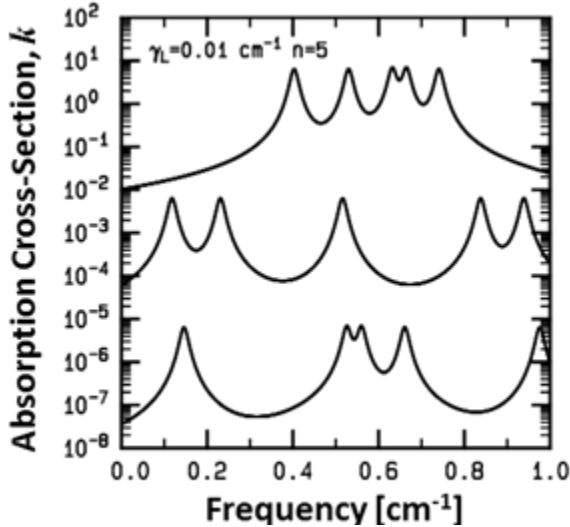


Figure 4.1. Sample Monte Carlo Trial Spectral Calculations. The bottom spectra are offset by factors of 10^{-3} and 10^{-6} from the top spectrum for display purposes

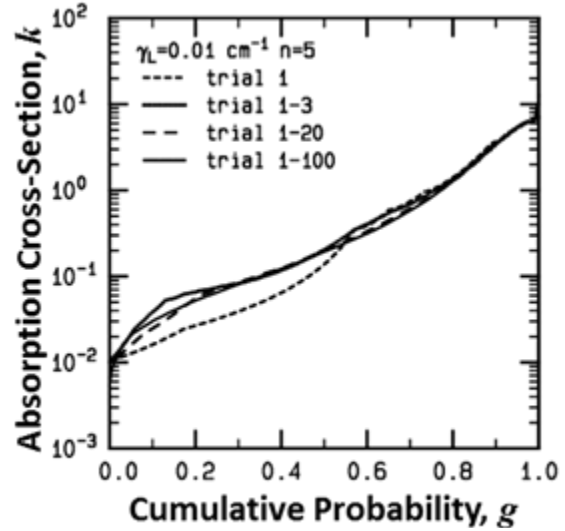


Figure 4.2. Convergence of the $k(g)$ Distribution as a Function of the Number of Monte-Carlo Trial Spectra

The convergence of the $k(g)$ distribution as a function of the number of Monte Carlo trial spectra is shown in Figure 4.2. Reasonable convergence is obtained after 20 trials, although some differences are noted between 20 and 100 trials. The number of trials needed to obtain a good distribution depends on the number of lines n ; as a general rule of thumb at least $n_{MC} = 100/n$ trials are required to obtain a useful result. It is difficult to see the convergence of $k(g)$ in the region of $g = 1$ in Figure 4.2; this region is shown in an expanded scale in Figure 4.3. It is seen that the peak value of k approximately doubles when 20 or more trials are considered. This reflects the probability that any two of the $n = 5$ lines will closely overlap for a given trial spectrum. If enough trials were considered, eventually overlaps of 3, 4, and 5 lines would be observed; however, the cumulative probability interval Δg associated with these cases would be insignificant and would have an inconsequential impact on radiative transfer calculations.

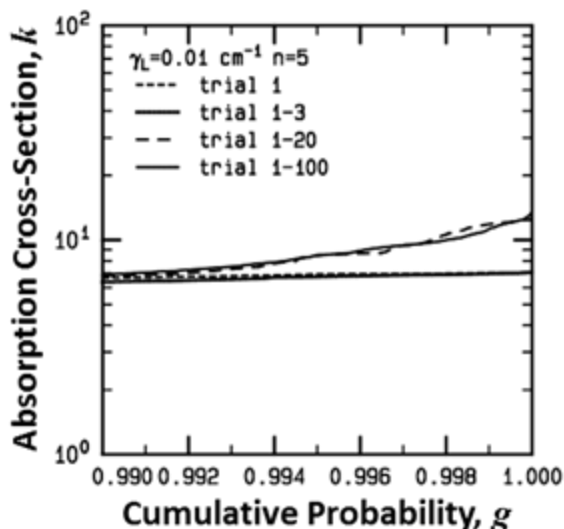


Figure 4.3. Same as Figure 4.2 but Showing the Convergence in the Region of $g = 1$ in More Detail

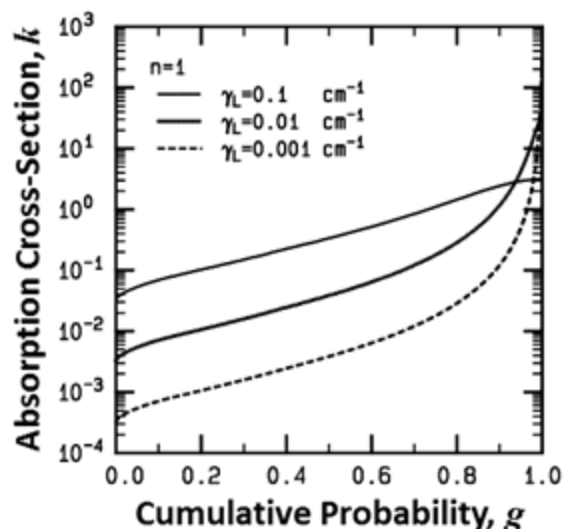


Figure 4.4. Dependence of $k(g)$ on the Lorentz Line Width γ_L for $\gamma_D = 0.0 \text{ cm}^{-1}$ and $n = 1$

Several examples of k -distributions and their variability with n , γ_L , and γ_D are shown in Figures 4.4 through 4.6. Figure 4.4 shows the variation with the Lorentz line width where it is seen that the line wing absorption coefficients (i.e., $g < 0.8$) are proportional to γ_L . Figure 4.5 shows that in the limit of a very large number of lines the absorption coefficients asymptote to a constant value for the entire spectral interval. Figure 4.6 illustrates the changes in the k -distribution when Doppler broadening is added to a pure Lorentz broadened line. Only the region near the peak absorption coefficients is shown (i.e., near $g = 1.0$) since the line tail absorption coefficient distribution is unaffected by the inclusion of Doppler broadening.

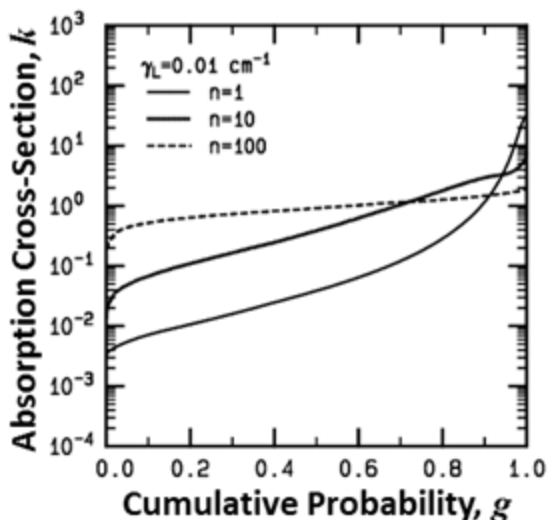


Figure 4.5. Dependence of $k(g)$ on the Number of Lines n for $\gamma_L = 0.01 \text{ cm}^{-1}$ and $\gamma_D = 0.0 \text{ cm}^{-1}$

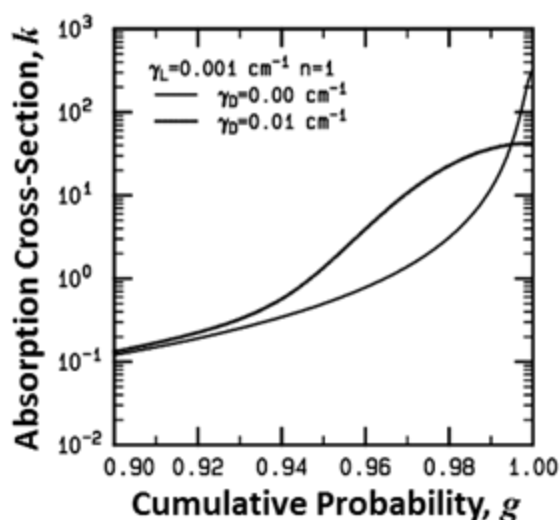


Figure 4.6. Effect on $k(g)$ for Inclusion of Doppler Broadening for $n = 1$, $\gamma_L = 0.01 \text{ cm}^{-1}$ and $\gamma_D = 0.001 \text{ cm}^{-1}$

A non-integer number of lines $n.x$ (x denotes a fractional line) is treated by consideration of a distribution consisting of n lines of strength S_1 and a single line of strength S_2 . The strengths are constrained to both preserve the total interval strength (i.e., $S = S_1 + S_2$) and the effective number of lines

$n.x$. These lines are used in the Monte Carlo simulation described above to determine $f(k)$ for a non-integer number of lines.

4.3.2 Determination of the MODTRAN $k(g)$ Table

Tables of $k(g)$ distributions for each of the four band model spectral bin resolutions (0.1, 1.0, 5.0 and 15.0 cm^{-1}) were constructed for real time use during MODTRAN runs. For each of the MODTRAN four spectral bin widths, a tabulated $k(g)$ distribution corresponds to a specific set of values for n , γ_L and γ_D , and a reference interval strength of $S = 1 \text{ cm}^{-2} \text{ atm}^{-1}$. The ranges of values considered for these parameters were chosen to be compatible with radiative transfer calculations covering the 0 – 70 km altitude range, the spectral range 10 – 10,000 cm^{-1} , and including all the MODTRAN atmospheric molecular band species. This translated into the ranges of $0.2 - 10^{-6} \text{ cm}^{-1}$ for γ_L , $0.02 - 10^{-5} \text{ cm}^{-1}$ for γ_D , and 1 – 90 for n . Each of these ranges was spanned by a logarithmic scale of values with a spacing factor of $2^{1/2}$. This resulted in approximately 10,000 tabulated distributions where each distribution was represented at 34 g values. The overall size of the $k(g)$ data base is 1.4 Mbytes. The g values were selected so that the $k(g)$ distribution with the largest dynamic range in k values would have each decade of k values covered by approximately 3 (k_i , g_i) points. It would be preferable from a computational speed point of view to span each decade with fewer points; however, as shown later, 3 points per decade is a reasonable tradeoff between computational speed and accuracy. The largest dynamic range occurs for $n = 1$, $\gamma_L = 10^{-6} \text{ cm}^{-1}$, and $\gamma_D = 10^{-5} \text{ cm}^{-1}$. Most of the g_i points were selected based on the logarithmic scale

$$g_i = 1 - 10^{-5} (1.46780)^i, \quad (4.12)$$

where i ranged from 1 to 30. Three additional points were added, $g = 1$ for the peak, and $g = 0.05$ and $g = 0.1$ to resolve structure near $g = 0.0$. The selection of g values [0.000000, 0.050000, 0.100000, 0.318707, 0.535841, 0.683772, 0.784556, 0.853220, 0.900000, 0.931871, 0.953584, 0.968377, 0.978456, 0.985322, 0.990000, 0.993187, 0.995358, 0.996838, 0.997846, 0.998532, 0.999000, 0.999319, 0.999536, 0.999684, 0.999785, 0.999853, 0.999900, 0.999932, 0.999954, 0.999968, 0.999978, 0.999985, 0.999990, 1.000000] for this stressing case is illustrated in Figure 4.7, which shows that the distribution spans 12 orders of magnitude. Figures 4.8 – 4.10 provide a more detailed focus on various portions of the distribution. Each (n , γ_L , γ_D) distribution is tabulated at the same g_i grid points as indicated in these figures. MODTRAN does provide an option to speed up the Ck calculations by reducing the number of g_i grid points from 34 down to 18; in principle, this is less accurate but in practice the predicted radiances do not differ significantly when the number of g_i grid points is reduced.

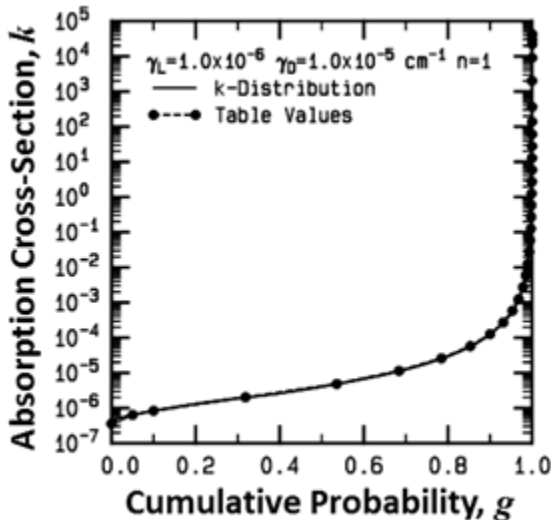


Figure 4.7. Selection of g_i Table Values for the $k(g)$ Distribution with the Largest Dynamic Range of k Values

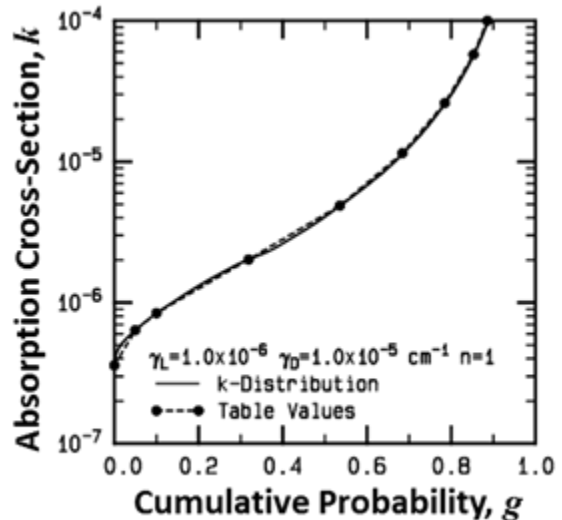


Figure 4.8. Same as Figure 7 but Showing an Expanded k Range Near k_{min}

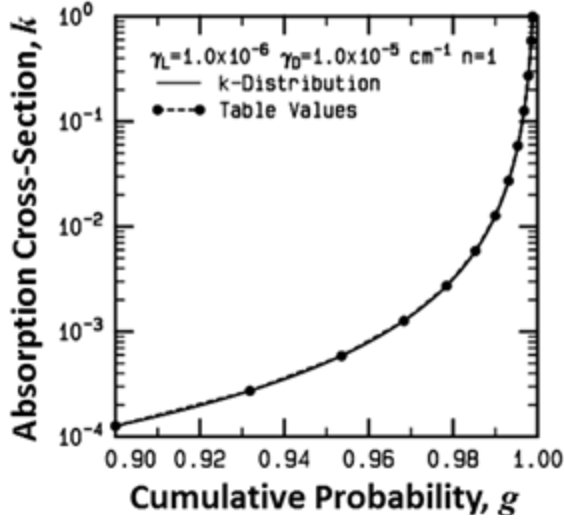


Figure 4.9. Same as Figure 7 but Showing Expanded k and g Scales

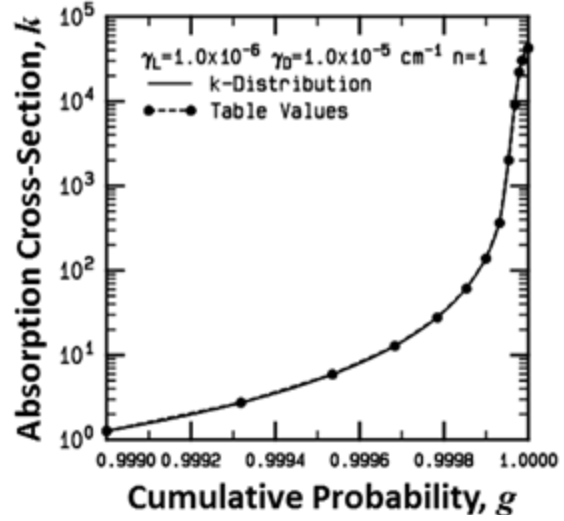


Figure 4.10. Same as Figure 7 but Showing Expanded k and g Scales

During a MODTRAN calculation, the (k_i, g_i) points are found by a three-dimensional linear interpolation over the grid of n, γ_L and γ_D values. For each atmospheric segment, values of γ_L and γ_D are determined as molecular optical depth weighted averages over the contributing band model species. The value of $\underline{n}$ is set so that the *pseudo gas* band model segment transmittance, computed with $\underline{Su}, \underline{n}, \gamma_L$ and γ_D , equals the product of the band model segment transmittances of the M molecules contributing to the pseudo gas:

$$\text{Find } \underline{n} \text{ such that } T_{BM}[\underline{Su}, \underline{n}, \gamma_L, \gamma_D] = \prod_{m=1}^M T_{BM}[(Su)_m, n_m, (\gamma_L)_m, (\gamma_D)_m] , \quad (4.13)$$

where

$$\underline{Su} \equiv \sum_{m=1}^M (Su)_m , \quad \gamma_L \equiv \frac{1}{\underline{Su}} \sum_{m=1}^M (Su)_m (\gamma_L)_m \quad \text{and} \quad \gamma_D \equiv \frac{1}{\underline{Su}} \sum_{m=1}^M (Su)_m (\gamma_D)_m \quad (4.14)$$

4.3.3 Determination of Effective Sub-Interval Absorption Cross-Sections

We adopt the approach of Lacis and Oinas for determining an effective absorption coefficient $\underline{k}_i$ for the sub-interval $\Delta g_i = g_i - g_{i-1}$ which preserves the average transmittance for each atmospheric layer. It is given by

$$\exp(-\underline{k}_i u) = \frac{1}{\Delta g_i} \int_{g_{i-1}}^{g_i} \exp[-k(g)u] dg . \quad (4.15)$$

Since we have tabulated the $k(g)$ distribution over a relatively coarse grid, an assumed functional form for the $k(g)$ variation over Δg_i is required in order to evaluate Eq. (4.15). An exponential variation is assumed

$$k(g) = k_{i-1} \left(\frac{k_i}{k_{i-1}} \right)^{\frac{g - g_{i-1}}{g_i - g_{i-1}}} . \quad (4.16)$$

This variation is a reasonable representation of $k(g)$ as can be seen in Figures 4.8 – 4.10 where the tabulated points are connected with straight lines [i.e., this is equivalent to Eq. (4.16)]. There are differences of several percent with the actual $k(g)$ in some places; however, this case is the most stressing one. When Eq. (4.16) is substituted into Eq. (4.15), the final expression for $\underline{k}_i$ becomes

$$\bar{k}_i = -\frac{1}{u} \ln \left(\frac{E_1(k_{i-1}u) - E_1(k_i u)}{\ln(k_i/k_{i-1})} \right), \quad (4.17)$$

where E_1 is a first order exponential integral. From Eq. (4.17) it is clear that the effective absorption cross-section is a function of the absorption column density u . While this approach preserves the average transmittance for each layer in a multi-layer calculation, it does not preserve the total path transmittance.

In general, the Ck segment transmittance T_{CK} and the band model segment transmittance T_{BM} for the same values of S , γ_L , γ_D , and n will not be identical. This is illustrated in Figure 4.11 for a 500 km horizontal path at 15 km altitude and in Figure 4.12 for a vertical path to space from 5 km altitude. Differences in the model transmittances as large as approximately $\pm 10\%$ are evident. Although both transmittance formulations are based on the same underlying physical assumptions (i.e., equal strength positionally uncorrelated lines), the implementation of each involves different numerical approaches and approximations. In order to maintain internal consistency in MODTRAN between the two approaches, the Ck line number n is adjusted to exactly preserve the band model transmittance for each segment.

In general, the Ck segment transmittance T_{CK} and the band model segment transmittance T_{BM} for the same values of S , γ_L , γ_D , and n will not be identical. This is illustrated in Figure 4.11 for a 500 km horizontal path at 15 km altitude; differences in the model transmittances as large as approximately $\pm 10\%$ are evident. Although both transmittance formulations are based on the same underlying physical assumptions (i.e., equal strength positionally uncorrelated lines), the implementation of each involves different numerical approaches and approximations. In order to maintain internal consistency in MODTRAN between the two approaches, the Ck absorption cross-sections are scaled up or down to exactly preserve the band model transmittance for each segment. This approach does not preserve the total path transmittance for multi-layer paths as is evident in Figure 4.12 for a vertical path from 5 km to space; the largest differences are approximately $\pm 7\%$.

In summary, a MODTRAN Ck approach was developed for and integrated into MODTRAN4. Users are encouraged to use Ck whenever multiple scattering is a dominant source of radiation. There is a significant time penalty with the Ck approach. If this is a problem for a specific application, users should perform comparisons between the Ck and BM calculations to quantify the magnitude of residuals for their application and determine whether the BM approach adequately meets their needs.

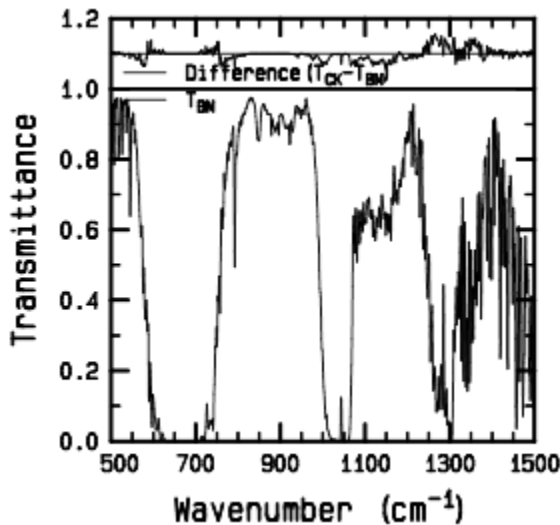


Figure 4.11. Comparison of Computed Transmittances for the Correlated- k (CK) and Band Model (BM) Approaches for a 500 km Horizontal Path at 15 km Altitude

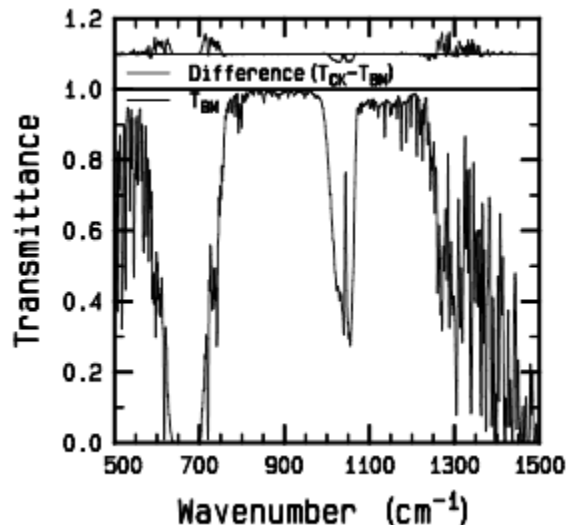


Figure 4.12. Same as Figure 4.11 Except for a Vertical Path from 5 km to Space

4.4 Acknowledgements

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5. MODTRAN DISORT MULTIPLE SCATTERING

MODTRAN computes line-of-sight (LOS) spectral radiant intensities by integrating the local attenuated spectral source function, with both scattered and Planck emissive components, along the LOS path. If the LOS terminates at a surface, e.g., the hard Earth, an emission and reflection boundary term is included in the radiant intensity. The path is defined by the MODTRAN spherical refractive geometry algorithm. This is the essence of Eq. (1.1). The multiple scattering component of the source function [Eq. (1.6)] and the downward flux component [Eq. (1.11)] of the boundary term both depend on LOS spectral radiant intensities themselves, coupling the radiant intensity, the source and the boundary terms. Not surprisingly, this complicates the path integration procedure. The equations are repeated here for quick reference, but the user is encouraged to review Chapter 1 for further details.

$$I_0(\Omega_0; \nu) = \exp[-\tau_\ell(\nu)] I_\ell(\Omega_\ell; \nu) + \int_0^{\tau_\ell(\nu)} \exp[-\tau_{\ell'}(\nu)] J_{\ell'}(\Omega_{\ell'}; \nu) d\tau_{\ell'}(\nu) \quad (1.1)$$

$$\begin{aligned} J_\ell(\Omega_\ell; \nu) &= J_\ell^{em}(\Omega_\ell; \nu) + J_\ell^{ss}(\Omega_\ell; \nu) + J_\ell^{ms}(\Omega_\ell; \nu) \\ &= \frac{b_\ell^{abs}(\nu)}{b_\ell(\nu)} B(T_\ell; \nu) \quad \text{local thermal emission} \\ &+ \frac{b_\ell^{sct}(\nu)}{b_\ell(\nu)} p_\ell(\Omega_\ell, \Omega_\ell^{sun}; \nu) \exp[-\tau_\ell^{sun}(\nu)] F^{sun}(\nu) \quad \text{single scatter solar} \\ &+ \frac{b_\ell^{sct}(\nu)}{b_\ell(\nu)} \int_{4\pi} p_\ell(\Omega_\ell, \Omega'_\ell; \nu) I_\nu(\Omega'_\ell; \nu) d\Omega'_\ell \quad \text{multiple scatter} \end{aligned} \quad (1.6)$$

$$\begin{aligned} I_s(\Omega_s; \nu) &= \varepsilon_s(\Omega_s; \nu) B(T_g, \nu) \quad \{ \text{Surface emission} \\ &+ \mu_s^{sun} f_s(\Omega_s, \Omega_s^{sun}; \nu) t_s^{sun}(\nu) F^{sun}(\nu) \quad \left\{ \begin{array}{l} \text{Directly - transmitted} \\ \text{reflected solar irradiance} \end{array} \right. \\ &+ \int_{2\pi} I_s(\Omega'_s; \nu) f_s(\Omega_s, \Omega'_s; \nu) \mu'_s d\Omega'_s \quad \{ \text{Surface reflected downward flux} \end{aligned} \quad (1.11)$$

MODTRAN resolves the self-consistency problem by using a plane-parallel multiple scattering (MS) algorithm to compute the LOS spectral MS contributions. Underlying this procedure is the assumption that multiple scattering in the terrestrial atmosphere is predominately a lower atmosphere effect and therefore well characterized with the locally flat Earth approximation. This *ansatz* can be problematic for low sun, twilight conditions.

The MODTRAN model provides three MS algorithm options. An analytic two-stream algorithm, dubbed the Isaacs approach [Isaacs *et al.*, 1986; Isaacs *et al.*, 1987; Meador and Weaver, 1980], approximates the MS source terms with minimal computations. These two-stream results are best suited for calculation of spatially integrated radiances (flat plate fluxes) and thermal scatter contributions. However, the accuracy of the 2-stream solar scatter radiant intensities is relatively poor, typically $\pm 20\%$, but factor of 2 errors are not uncommon in the shortwave end of visible spectral region (blue). For higher fidelity results, MODTRAN employs the DISORT *DIS*crete *OR*dina*Te* algorithm to compute MS radiances. Typically, DISORT requires considerably more processing time than Isaacs, although both models efficiently compute 2-stream thermal scattering. MODTRAN does provide a third MS option, in which Isaacs MS spectral radiances are scaled based on a handful of DISORT calculations at fixed pivot spectral points; this provides speeds comparable to Isaacs, and accuracy much closer to that of DISORT.

In the following section, the DISORT MS algorithm and the MODTRAN tailored implementation are described. References by Stamnes *et al.* [1988, 2000] provide greater detail on the DISORT discrete ordinate solution method. The latter of these two references, “DISORT, a General-Purpose Fortran Program for Discrete-Ordinate-Method Radiative Transfer in Scattering and Emitting Layered Media: Documentation of Methodology,” authored by K. Stamnes, S.-C. Tsay, W. Wiscombe and I. Laszlo, forms the basis for much of the DISORT description in this chapter. This reference will be referred to as [STWL] and equation numbers from STWL are provided in bold brackets (e.g. **[1]**) for direct reference.

5.1 The Discrete Ordinate Solution to the Plane-Parallel Atmosphere Radiative Transfer Equation (RTE) and its Integration into MODTRAN6

DISORT explicitly solves the *integral-differential* RTE for a horizontally-homogeneous plane-parallel atmosphere with (1) an emitting and reflecting ground surface, (2) Planck emission and scattering by molecules and particles within the atmosphere, and (3) parallel ray (solar/lunar) illumination from above. DISORT also provides an option to include a top of atmosphere (TOA) homogeneous radiation source, but MODTRAN sets this source to zero. Alternatively, this boundary condition could be used to model the 2.7 K cosmic microwave background or OH Meinel emissions. The RTE is defined by the following expressions:

$$\begin{aligned} \mu \frac{dI_{\tau}^{\downarrow}(\Omega; \nu)}{d\tau} &= I_{\tau}^{\downarrow}(\Omega; \nu) - J_{\tau}^{\downarrow}(\Omega; \nu) \quad [1] \\ \text{Source Function} \\ J_{\tau}^{\downarrow}(\Omega; \nu) &\equiv \left[1 - \omega_{\tau}^{\downarrow}(\nu) \right] B(T_{\tau}^{\downarrow}; \nu) + \omega_{\tau}^{\downarrow} \left[p_{\tau}^{\downarrow}(\Omega, \Omega^{sun}; \nu) e^{-\tau^{\downarrow}/\mu^{sun}} F^{sun}(\nu) \right. \\ &\quad \left. + \int_{4\pi} p_{\tau}^{\downarrow}(\Omega, \Omega'; \nu) I_{\tau}^{\downarrow}(\Omega'; \nu) d\Omega' \right] \quad [2a, 3a, 3d] \\ \omega_{\tau}^{\downarrow}(\nu) &\equiv b_{\tau}^{sct}(\nu) / b_{\tau}^{\downarrow}(\nu) \\ \text{Boundary Conditions} \\ I_{\tau_{gnd}}^{\downarrow}(|\mu|, \phi; \nu) &= \left\{ \begin{aligned} &\varepsilon(|\mu|, \phi) B(T_{gnd}; \nu) + \mu^{sun} f(|\mu|, \phi, \mu^{sun}, \phi^{sun}; \nu) e^{-\tau_{gnd}^{\downarrow}/\mu^{sun}} F^{sun} \\ &+ \int_0^{2\pi} \int_0^{\tau_{gnd}} I_{\tau}^{\downarrow}(-\mu', \phi'; \nu) f(|\mu|, \phi, -\mu', \phi'; \nu) \mu' d\phi' d\mu' \end{aligned} \right\} \quad [39] \\ I_0^{\downarrow}(-|\mu|, \phi; \nu) &= 0 \end{aligned} \quad (5.1)$$

The goal is to solve for the diffuse (scattered) radiant intensity $I_{\tau}^{\downarrow}(\Omega; \nu)$ at spectral frequency ν for a sensor path pointing in direction $\Omega \equiv (\mu, \phi)$ that begins at nadir optical depth $\tau^{\downarrow}$, measured from the TOA, and terminates at the ground for downward viewing paths or at the TOA for upward viewing paths. The path direction is defined by an azimuth angle ϕ and the cosine of the sensor path *nadir* angle. The solar direction is defined in terms of its azimuth angle, ϕ^{sun} , and its solar *zenith* angle, μ^{sun} ; the direction Ω^{sun} is defined as $(-\mu^{sun}, \phi^{sun})$. The top equation in Eq. (5.1) is derived by differentiating the plane-parallel atmosphere form of the integral RTE, Eq. (1.1), with respect to the path optical depth, $\tau_{\ell}(\nu)$. For the plane parallel atmosphere, the path length optical depth is equal to the path termination point to sensor nadir optical depth difference, $\tau_{\ell}^{\downarrow}(\nu) - \tau_0^{\downarrow}(\nu)$, over the cosine of the path nadir/zenith angle.

$$\tau_1(\nu) = \frac{\tau_1^\downarrow(\nu) - \tau_0^\downarrow(\nu)}{\mu} ; \quad \tau^\downarrow \equiv \tau_1^\downarrow(\nu) . \quad (5.2)$$

Since refraction and spherical Earth effects are excluded, the path direction $\Omega \equiv (\mu, \phi)$ at each point along the LOS is constant. There is a convention to define the path cosine, nadir or zenith, μ as always being positive. Rather than adopt this convention, the path nadir cosine μ is explicitly bracketed by absolute value vertical bars when only positive μ are being considered. When ‘ $\pm$ ’ or ‘ $\mp$ ’ signs are used, the upper sign will always correspond to a downward viewing path segment (positive cosine nadir), while the lower sign will always correspond to an upward viewing path segment (negative cosine nadir angle). The convention may seem backwards from the sensor’s perspective, but it is correct from the photon’s perspective; the positive sign corresponds to photons on an upward trajectory and the negative sign corresponds to photons on a downward trajectory.

The more general dependence on path length ℓ in Eq. (1.1) has been replaced in Eq. (5.1) with a dependence on nadir optical depth $\tau^\downarrow$; this is permissible for the plane-parallel atmosphere in which the radiant intensity varies spatially only with respect to the vertical coordinate. Furthermore, since the extended Earth surface is modeled as flat and horizontally homogeneous, the Bidirectional Reflectance Distribution Function (BRDF), $f(+\mu, \phi, -\mu', \phi')$, does not include a subscript labeling location as was the case in Eq. (1.11).

To facilitate solving the plane parallel atmosphere radiation transport problem, two additional model assumptions are made up front: (1) the scattering phase function is modeled as being only a function of the scattering angle Θ $\{ \cos \Theta = \mu \mu' + \cos(\phi - \phi') \sqrt{[(1 - \mu^2) (1 - \mu'^2)]} \}$ between the incident and outgoing beams, $p(\Omega, \Omega'; \tau^\downarrow) = p(\Theta; \tau^\downarrow)$; and (2) the surface BRDF is modeled as a function of the incoming and outgoing zenith angles and of the relative azimuth angle, but not of the incoming and outgoing azimuth angles independently, i.e., $f(\Omega, \Omega') = f(\mu, -\mu', \phi - \phi')$. These assumptions limit DISORT’s ability to precisely model physical phenomena such as preferential orientation of ice particles in cirrus clouds and plowed fields on the Earth’s surface. However, for most (not all) terrestrial atmospheric scenarios, it is quite reasonable to model suspended particles as randomly oriented with equal numbers of optical isomers, and the Earth surface as locally isotropic.

DISORT follows a multiple step approach to solve the integral-differential RTE. The concept of using a discrete ordinate method to solve the differential equations was originally laid out by Chandrasekhar [1960]. Many improvements to the original approach had to be instituted to solve the challenging problem of creating a *numerically stable* realization of the Chandrasekhar approach. Much of that work was performed Stamnes and his coworkers [1988, 2000].

For notational simplicity, the spectral frequency dependence has been dropped from all subsequent equations in this section, even though all quantities other than the temperatures and angles are spectrally dependent (due to refraction within a spherical atmosphere, LOS angles will vary spectrally, but MODTRAN does not model these dependencies).

Step 1: Factoring the Azimuth Dependence

The azimuth dependence of the integral-differential RTE is factored out by expanding the radiant intensity and bidirectional reflectivity in Fourier cosine series, and expanding the scattering phase function in Legendre polynomials, $P_\ell(\mu)$, and unit normalized associated Legendre polynomials, $\Lambda_\ell^m(\mu)$:

$$\begin{aligned}
I_{\tau^\downarrow}(\Omega) &= \sum_{m=0}^{2M-1} I_{\tau^\downarrow}^m(\mu) \cos m(\phi^{sun} - \phi) \quad [6] \quad ; \\
f(\mu, -\mu', \phi - \phi') &= \sum_{m=0}^{2M-1} f^m(\mu, -\mu') \cos m(\phi - \phi') \quad [40] \quad ; \quad \text{and} \\
p_{\tau^\downarrow}(\Theta) &= \sum_{\ell=0}^{2M-1} (2\ell+1) \frac{g_{\tau^\downarrow}^\ell}{4\pi} P_\ell(\cos \Theta) \quad [5a] \\
&= \sum_{\ell=0}^{2M-1} (2\ell+1) \frac{g_{\tau^\downarrow}^\ell}{4\pi} \left\{ P_\ell(\mu) P_\ell(\mu') + 2 \sum_{m=1}^{\ell} \Lambda_\ell^m(\mu) \Lambda_\ell^m(\mu') \cos(\phi - \phi') \right\} \quad [5c] \quad .
\end{aligned} \tag{5.3}$$

In these equations, the $I_{\tau^\downarrow}^m(\mu)$ are the m^{th} azimuth moment radiant intensities, the $f^m(\mu; -\mu')$ are the azimuth moment BRDFs, the $g_{\tau^\downarrow}^\ell$ are the scattering phase function Legendre moments, and $2M$ is the number of Legendre moments ($\ell = 0$ to $2M-1$). These expansions yield *independent/orthogonal* integral differential equations for each m component:

$$\mu \frac{d I_{\tau^\downarrow}^m(\mu)}{d \tau^\downarrow} = I_{\tau^\downarrow}^m(\mu) - J_{\tau^\downarrow}^m(\mu) \quad (m = 0, 1, \dots, 2M-1) \quad [7]$$

Source Function

$$J_{\tau^\downarrow}^m(\mu) = \frac{\omega_{\tau^\downarrow}}{2} \left\{ (2 - \delta_{m0}) \frac{e^{-\tau^\downarrow/\mu^{sun}} F^{sun}}{2\pi} \left[\sum_{\ell=m}^{2M-1} (-1)^{\ell+m} (2\ell+1) g_{\tau^\downarrow}^\ell \Lambda_\ell^m(\mu) \Lambda_\ell^m(\mu^{sun}) \right] \right. \\
\left. + \int_{-1}^1 I_{\tau^\downarrow}^m(\mu') \left[\sum_{\ell=m}^{2M-1} (2\ell+1) g_{\tau^\downarrow}^\ell \Lambda_\ell^m(\mu) \Lambda_\ell^m(\mu') \right] d\mu' \right\} + \delta_{m0} (1 - \omega_{\tau^\downarrow}) B(T_{\tau^\downarrow}) \quad [8a-d]$$

Boundary Conditions

$$I_{\tau_g}^m(+\mu) = \left\{ \delta_{m0} \varepsilon(\mu) B(T_g) + \mu^{sun} f^m(\mu, -\mu^{sun}) e^{-\tau_g^\downarrow/\mu^{sun}} F^{sun} \right. \\
\left. + (1 + \delta_{m0}) \int_0^1 I_{\tau_g}^m(-\mu') f^m(+\mu, -\mu') \mu' d\mu' \right\} ; \quad I_{\tau_g}^m(-\mu) = 0 \quad [42b, 43a] \tag{5.4}$$

These equations are similar to those of Eq. (5.1) except that there is no azimuth dependence. In the absence of solar radiation, only the $m=0$ problem needs to be solved. This is also true if there is azimuthal symmetry resulting from either a zenith sun or a vertical/nadir LOS.

Step 2: Introducing Discrete Ordinates

In the second step of the Chandrasekhar procedure, the $2M$ independent integral-differential equations (one for each value of m) are each converted into a system of $2N$ coupled ordinary differential equations (ODE's) in nadir optical depth, $\tau^\downarrow$. The continuous variable μ on $[-1,1]$, equal to the cosine of the path nadir angle, is replaced by $2N$ discrete ordinate Gaussian quadrature values, and the integrals over μ' in Eq. (5.4) are redefined as Gaussian quadrature sums:

$$\mu_i \frac{dI_{\tau}^m(\mu_i)}{d\tau} = I_{\tau}^m(\mu_i) - J_{\tau}^m(\mu_i) \quad (i = \pm 1, \pm 2, \dots, \pm N) \quad [10a]$$

Source Function

$$J_{\tau}^m(\mu_i) = e^{-\tau/\mu_i^{sun}} X_{\tau}^m(\mu_i) + \delta_{m0} (1 - \omega_{\tau}^{\downarrow}) B(T_{\tau}^{\downarrow}) + \sum_{j=-N, j \neq 0}^N w_j I_{\tau}^m(\mu_j) D_{\tau}^m(\mu_i, \mu_j) \quad [10b]$$

$$D_{\tau}^m(\mu_i, \mu_j) \equiv \frac{\omega_{\tau}^{\downarrow}}{2} \sum_{\ell=m}^{2M-1} (2\ell+1) g_{\tau}^{\ell} \Lambda_{\ell}^m(\mu_i) \Lambda_{\ell}^m(\mu_j) \quad [8b]$$

$$X_{\tau}^m(\mu_i) \equiv \frac{\omega_{\tau}^{\downarrow} F^{sun}}{4\pi} (2 - \delta_{m0}) \sum_{\ell=m}^{2M-1} (-1)^{\ell+m} (2\ell+1) \frac{g_{\tau}^{\ell}}{4\pi} \Lambda_{\ell}^m(\mu_i) \Lambda_{\ell}^m(\mu_i^{sun}) \quad [8d]$$

Boundary Conditions

$$I_{\tau_g}^m(+\mu_i) = \delta_{m0} \varepsilon(\mu_i) B(T_g) + \mu_i^{sun} f^m(\mu_i, -\mu_i^{sun}) e^{-\tau_g/\mu_i^{sun}} F^{sun} \quad ; \quad I^m(-\mu_i; 0) = 0$$

$$+ (1 + \delta_{m0}) \sum_{j=1}^N w_j \mu_j I_{\tau_g}^m(-\mu_j) f^m(+\mu_i, -\mu_j) \quad (i = +1, +2, \dots, +N) \quad [43d] \quad (5.5)$$

DISORT employs a double Gaussian quadrature, with paired cosine angles, $\mu_j = -\mu_{-j}$, symmetric in each hemisphere about $\mu = \pm 1/2$ and having a common Gaussian weight, $w_j = w_{-j}$.

Step 3: Simplifying Optical Depth Dependence via Atmospheric Layering

There is no closed-form analytic solution to Eq. (5.5) because the $D_{\tau}^m(\mu_i, \mu_j)$ coefficients of $I_{\tau}^m(\mu_j)$ are themselves functions of the nadir optical depth, $\tau^{\downarrow}$, the dependent variable. In this third step of the solution procedure, the atmospheric medium is partitioned into $l = 1$ to L adjacent homogeneous layers with $l = 0$ to L level boundaries. At each boundary altitude level from $z_0 = \text{TOA}$ to $z_L = \text{ground}$, nadir optical depths $\tau_0^{\downarrow} = 0$ to $\tau_L^{\downarrow} = \tau_g^{\downarrow}$ and temperatures T_0 to T_L are specified. The single scattering albedo and the scattering phase function are modeled as constants within each layer:

$$\omega_l \equiv \int_{z_{l-1}}^{z_l} b^{scat}(z) dz \bigg/ \int_{z_{l-1}}^{z_l} b(z) dz \quad ; \quad g_l^{\ell} \equiv \int_{z_{l-1}}^{z_l} g_{\tau}^{\ell}(z) b^{scat}(z) dz \bigg/ \int_{z_{l-1}}^{z_l} b^{scat}(z) dz \quad . \quad (5.6)$$

The Planck emission is characterized as varying linearly with nadir optical depth *through each layer*:

$$B_l(T_{\tau}^{\downarrow}) = B(T_{l-1}) + \frac{\tau^{\downarrow} - \tau_{l-1}^{\downarrow}}{\tau_l^{\downarrow} - \tau_{l-1}^{\downarrow}} [B(T_l) - B(T_{l-1})] \quad ; \quad \tau_{l-1}^{\downarrow} \leq \tau^{\downarrow} \leq \tau_l^{\downarrow} \quad . \quad [24c] \quad (5.7)$$

With these definitions, the $D_{\tau}^m(\mu_i, \mu_j)$ and $X_{\tau}^m(\mu_i)$ terms are no longer functions of nadir optical depth $\tau^{\downarrow}$ within each layer; instead, they are layer constants and are indexed by layer, $D_l^m(\mu_i, \mu_j)$ and $X_l^m(\mu_i)$.

The set of $2N$ coupled ordinary differential equations from Eq. (5.5) for each layer now have constant coefficients, and therefore the equations can be analytically solved. A general solution is derived for the m -dependent RTE within each atmospheric layer (for full details, see STWL):

$$I_l^m(\mu_i; \tau^\downarrow) = \sum_{j=1}^N \left\{ \hat{C}_{+jl}^m G_{+jl}^m(\mu_i) \exp \left[-k_{jl}^m(\tau^\downarrow - \tau_{l-1}^\downarrow) \right] + \hat{C}_{-jl}^m G_{-jl}^m(\mu_i) \exp \left[-k_{jl}^m(\tau_{l-1}^\downarrow - \tau^\downarrow) \right] \right\} \\ + \delta_{m0} \left[Y_{0l}(\mu_i) + Y_{1l}(\mu_i) \tau^\downarrow \right] + Z_l^m(\mu_i) \exp \left(-\tau^\downarrow / \mu^{\text{sun}} \right) ; \quad \tau_{l-1}^\downarrow \leq \tau^\downarrow \leq \tau_l^\downarrow \quad . \quad [27a-c, 48] \quad (5.8)$$

In this solution, the radiant intensity $I_l^m(\mu_i; \tau^\downarrow)$ dependence on nadir optical depth has been moved from a subscript to an argument for notational clarity. In Eq. (5.8), the $Y_{0l}(\mu_i)$ and $Y_{1l}(\mu_i)$ arise as a particular solution to the thermal source problem and are obtained by solving a system of $4N$ linear equations for each atmospheric layer:

$$\sum_{\substack{j=-N \\ j \neq 0}}^N \left[\delta_{ij} - w_j D_l^0(\mu_i, \mu_j) \right] Y_{1l}(\mu_j) = (1 - \omega_l) \frac{B(T_l) - B(T_{l-1})}{\tau_l^\downarrow - \tau_{l-1}^\downarrow} ; \quad (i = \pm 1, \pm 2, \dots, \pm N) \quad [24c, 26a] \\ (5.9) \\ \sum_{\substack{j=-N \\ j \neq 0}}^N \left[\delta_{ij} - w_j D_l^0(\mu_i, \mu_j) \right] Y_{0l}(\mu_j) = (1 - \omega_l) \left[B(T_{l-1}) - \tau_{l-1}^\downarrow \frac{B(T_l) - B(T_{l-1})}{\tau_l^\downarrow - \tau_{l-1}^\downarrow} \right] + \mu_i Y_{1l}(\mu_i) \quad [24c, 26b]$$

Similarly, the $Z_l^m(\mu_i)$ are a particular solution to the solar source problem, obtained by solving a system of $2N$ linear equations for each value of m and layer l :

$$\sum_{\substack{j=-N \\ j \neq 0}}^N \left[\left(1 + \frac{\mu_j}{\mu^{\text{sun}}} \right) \delta_{ij} - w_j D_l^m(\mu_i, \mu_j) \right] Z_l^m(\mu_j) = X_l^m(\mu_i) ; \quad (i = \pm 1, \pm 2, \dots, \pm N) \quad [24b] \quad (5.10)$$

The k_{+jl}^m , $k_{-jl}^m = -k_{+jl}^m$, and $G_{\pm jl}^m$ are the layer positive eigenvalues, negative eigenvalues and corresponding eigenvector components, respectively, that arise in solving the homogeneous layer l RTE problem for the m^{th} azimuth moment [$F^{\text{sun}} = 0$; $B_l = 0$]:

$$\left(\alpha_l^m - \beta_l^m \right) \left(\alpha_l^m + \beta_l^m \right) \left(\mathbf{G}_l^{m+} + \mathbf{G}_l^{m-} \right) = \left(k_l^m \right)^2 \left(\mathbf{G}_l^{m+} + \mathbf{G}_l^{m-} \right) \quad [23f] \\ \frac{d}{d\tau^\downarrow} \left[\mathbf{G}_l^{m+} \exp \left(-k_l^m \tau^\downarrow \right) - \mathbf{G}_l^{m-} \exp \left(-k_l^m \tau^\downarrow \right) \right] = -(\alpha + \beta) \left(\mathbf{G}_l^{m+} + \mathbf{G}_l^{m-} \right) \exp \left(-k_l^m \tau^\downarrow \right) \quad [23a, d] \quad (5.11a)$$

where

$$\alpha = \begin{pmatrix} \left[w_1 D_l^m(\mu_1, \mu_1) - 1 \right] / \mu_1 & w_2 D_l^m(\mu_1, \mu_2) / \mu_1 & \cdots & w_N D_l^m(\mu_1, \mu_N) / \mu_1 \\ w_1 D_l^m(\mu_2, \mu_1) / \mu_2 & \left[w_2 D_l^m(\mu_2, \mu_2) - 1 \right] / \mu_2 & \cdots & w_N D_l^m(\mu_2, \mu_N) / \mu_2 \\ \vdots & \vdots & \ddots & \vdots \\ w_1 D_l^m(\mu_N, \mu_1) / \mu_N & w_2 D_l^m(\mu_N, \mu_2) / \mu_N & \cdots & \left[w_N D_l^m(\mu_N, \mu_N) - 1 \right] / \mu_N \end{pmatrix} \quad [15b]$$

$$\beta = \begin{pmatrix} w_1 D_l^m(\mu_1, -\mu_1) / \mu_1 & w_2 D_l^m(\mu_1, -\mu_2) / \mu_1 & \cdots & w_N D_l^m(\mu_1, -\mu_N) / \mu_1 \\ w_1 D_l^m(\mu_2, -\mu_1) / \mu_2 & w_2 D_l^m(\mu_2, -\mu_2) / \mu_2 & \cdots & w_N D_l^m(\mu_2, -\mu_N) / \mu_2 \\ \vdots & \vdots & \ddots & \vdots \\ w_1 D_l^m(\mu_N, -\mu_1) / \mu_N & w_2 D_l^m(\mu_N, -\mu_2) / \mu_N & \cdots & w_N D_l^m(\mu_N, -\mu_N) / \mu_N \end{pmatrix} \quad [15b]$$

$$\mathbf{G}_l^{m+} = \begin{pmatrix} \mathbf{G}_{+1l}^m & \mathbf{G}_{+2l}^m & \cdots & \mathbf{G}_{+Nl}^m \end{pmatrix} = \begin{pmatrix} G_{1l}^m(\mu_1) & G_{2l}^m(\mu_1) & \cdots & G_{Nl}^m(\mu_1) \\ G_{1l}^m(\mu_2) & G_{2l}^m(\mu_2) & \cdots & G_{Nl}^m(\mu_2) \\ \vdots & \vdots & \ddots & \vdots \\ G_{1l}^m(\mu_N) & G_{2l}^m(\mu_N) & \cdots & G_{Nl}^m(\mu_N) \end{pmatrix}$$

$$\mathbf{G}_l^{m-} = \begin{pmatrix} \mathbf{G}_{-1l}^m & \mathbf{G}_{-2l}^m & \cdots & \mathbf{G}_{-Nl}^m \end{pmatrix} = \begin{pmatrix} G_{1l}^m(-\mu_1) & G_{2l}^m(-\mu_1) & \cdots & G_{Nl}^m(-\mu_1) \\ G_{1l}^m(-\mu_2) & G_{2l}^m(-\mu_2) & \cdots & G_{Nl}^m(-\mu_2) \\ \vdots & \vdots & \ddots & \vdots \\ G_{1l}^m(-\mu_N) & G_{2l}^m(-\mu_N) & \cdots & G_{Nl}^m(-\mu_N) \end{pmatrix} \quad (5.11b)$$

The coefficients $\hat{C}_{jl}^m$ in Eq. (5.8) are the constants of integration that are determined from boundary and layer continuity conditions.

Step 4: Applying Boundary and Continuity Equations

For each of the azimuth moments m , the L atmospheric layers each contain $2N$ constants of integration, $\hat{C}_{\pm 1, l}^m, \dots, \hat{C}_{\pm N, l}^m$. These values are determined from the $(2N) \times L$ linear boundary and continuity equations:

$$I_1^m(-|\mu_i|; \tau_0^\downarrow = 0) = 0 \quad (i = +1, +2, \dots, +N) \quad ; \quad [43a]$$

$$I_{l-1}^m(-|\mu_i|; \tau_{l-1}^\downarrow) = I_l^m(-|\mu_i|; \tau_{l-1}^\downarrow) \quad (l = 2, 3, \dots, L; i = \pm 1, \pm 2, \dots, \pm N) \quad ; \quad [43b]$$

$$I_L^m(+|\mu_i|; \tau_L^\downarrow = \tau_g^\downarrow) = \delta_{m0} \varepsilon(|\mu_i|) B(T_g) + \mu^{\text{sun}} f^m(|\mu_i|, -\mu^{\text{sun}}) e^{-\tau_g^\downarrow / \mu^{\text{sun}}} F^{\text{sun}} \\ + (1 + \delta_{m0}) \sum_{j=1}^N w_j |\mu_j| I^m(-|\mu_j|; \tau_g^\downarrow) f^m(+|\mu_i|, -|\mu_j|) \quad (i = +1, +2, \dots, +N) \quad . \quad [43c-d]$$

Once the constants of integration have been determined, the radiant intensity azimuth components at any atmospheric altitude level $\tau^\downarrow$ and for all the Gaussian quadrature nadir angle cosines can be evaluated via Eq. (5.8). The $m = 0$ azimuthal terms, in particular, can be used to compute total upward, $F_+^{\text{tot}}(\tau^\downarrow)$, and downward, $F_-^{\text{tot}}(\tau^\downarrow)$, level fluxes. Given a layer l nadir optical depth, i.e., $\tau_{l-1}^\downarrow \leq \tau^\downarrow \leq \tau_l^\downarrow$, level fluxes at $\tau^\downarrow$ are evaluated from

$$F_+^{tot}(\tau^\downarrow) \cong 2\pi \sum_{i=1}^N w_i \mu_i I_l^0(\mu_i; \tau^\downarrow) \quad ; \quad F_-^{tot}(\tau^\downarrow) \equiv F_-^{dir}(\tau^\downarrow) + F_-(\tau^\downarrow) \quad (5.13)$$

$$F_-^{dir}(\tau^\downarrow) \cong \mu^{sun} F^{sun} \exp\left(\frac{-\tau^\downarrow}{\mu^{sun}}\right) \quad \text{and} \quad F_-(\tau^\downarrow) \cong 2\pi \sum_{i=1}^N w_i \mu_i I_l^0(-\mu; \tau^\downarrow) \quad . \quad [9f]$$

The total downward flux is partitioned into its direct, $F_-^{dir}(\tau^\downarrow)$, and diffuse, $F_-(\tau^\downarrow)$, components.

Step 5: Computing Radiant Intensities at Arbitrary Angles

The expression for $I_l^m(\mu; \tau^\downarrow)$, Eq. (5.8), defines the m^{th} radiant intensity within any layer l of the 1D atmosphere at all the defined Gaussian quadrature cosine nadir angles on $[-1, +1]$. One could envision computing the radiant intensity in an arbitrary direction $\Omega \equiv (\mu, \phi)$ and from an arbitrary altitude (or, equivalently, nadir optical depth) by interpolating or extrapolating the $\{I_l^m(\mu_{\pm 1}; \tau^\downarrow), I_l^m(\mu_{\pm 2}; \tau^\downarrow), \dots, I_l^m(\mu_{\pm N}; \tau^\downarrow)\}$ values to μ and inserting the result in the Fourier expression for $I_{\tau^\downarrow}(\Omega)$, Eq. (5.3). However, accurately interpolating and/or extrapolating these values is difficult, particularly near $\mu = \mu^{sun}$, near $\mu = 0$ and near $\mu = \pm 1$.

Much more accurate radiant intensities at arbitrary nadir angles can be calculated by solving the integral form of the RTE, Eq. (1.1). For the plane parallel atmosphere, the m^{th} azimuth component is computed from the following expressions:

$$I_{\tau^\downarrow}^m(+|\mu|) = I_L^m(+|\mu|) \exp\left(-\frac{\tau_L^\downarrow - \tau^\downarrow}{|\mu|}\right) + \int_{\tau^\downarrow}^{\tau_L^\downarrow} J_{\tau^\downarrow}^m(+\mu) \exp\left(-\frac{t^\downarrow - \tau^\downarrow}{|\mu|}\right) \frac{dt^\downarrow}{\mu} \quad [28a]$$

$$I_{\tau^\downarrow}^m(-|\mu|) = I_0^m(-|\mu|) \exp\left(-\frac{\tau^\downarrow}{|\mu|}\right) + \int_0^{\tau^\downarrow} J_{\tau^\downarrow}^m(-\mu) \exp\left(-\frac{\tau^\downarrow - t^\downarrow}{|\mu|}\right) \frac{dt^\downarrow}{\mu} \quad . \quad [28b] \quad (5.14)$$

Here, a TOA boundary term $I_0^m(-|\mu|)$ is included to provide greatest generality even though MODTRAN does not provide an option to define an angularly varying TOA downward source.

After much manipulation, one can obtain the following equations for the downward viewing, $I_{\tau^\downarrow}^m(+|\mu|)$, and upward viewing $I_{\tau^\downarrow}^m(-|\mu|)$ radiant intensity azimuth components at $\tau^\downarrow$ located in layer p ($\tau_{p-1}^\downarrow \leq \tau^\downarrow < \tau_p^\downarrow$ for downward viewing and $\tau_{p-1}^\downarrow < \tau^\downarrow \leq \tau_p^\downarrow$ for upward viewing):

$$I_{\tau^\downarrow}^m(+|\mu|) = I_L^m(+|\mu|) \exp\left(-\frac{\tau_L^\downarrow - \tau^\downarrow}{|\mu|}\right) + \Delta I_p^m(+|\mu|; \tau^\downarrow \Rightarrow \tau_p^\downarrow)$$

$$+ \sum_{l=p+1}^L \Delta I_l^m(+|\mu|; \tau_{l-1}^\downarrow \Rightarrow \tau_l^\downarrow) \exp\left(-\frac{\tau_{l-1}^\downarrow - \tau^\downarrow}{|\mu|}\right) \quad [35a, 36a - c, 48] \quad (5.15a)$$

$$I_{\tau^\downarrow}^m(-|\mu|) = I_0^m(-|\mu|) \exp\left(-\frac{\tau^\downarrow}{|\mu|}\right) + \Delta I_p^m(-|\mu|; \tau^\downarrow \Rightarrow \tau_{p-1}^\downarrow)$$

$$+ \sum_{l=1}^{p-1} \Delta I_l^m(-|\mu|; \tau_l^\downarrow \Rightarrow \tau_{l-1}^\downarrow) \exp\left(-\frac{\tau^\downarrow - \tau_l^\downarrow}{|\mu|}\right) \quad [35b, 37a - c, 48] \quad ,$$

where the radiant intensity, $\Delta I_l^m(\mu; \tau_c^\downarrow \Rightarrow \tau_f^\downarrow)$, arising from the “ $\tau_c^\downarrow \Rightarrow \tau_f^\downarrow$ ” sub-segment of layer l ($\tau_c^\downarrow$ is the nadir optical depth of the segment endpoint closer to the sensor and $\tau_f^\downarrow$ is the nadir optical depth of the segment endpoint further from the sensor), is given by

$$\begin{aligned}
\Delta I_l^m(\mu; \tau_c^\downarrow \Rightarrow \tau_f^\downarrow) &= \left\{ I_{\tau_c^\downarrow}^m(\mu) - \exp\left[-\left(\tau_f^\downarrow - \tau_c^\downarrow\right)/\mu\right] I_{\tau_f^\downarrow}^m(\mu) \right\} \\
&= \sum_{j=1}^N \left\{ \hat{C}_{jl}^m \frac{G_{jl}^m(\mu)}{1+k_{jl}\mu} \left\{ \exp\left[-k_{jl}\left(\tau_c^\downarrow - \tau_{l-1}^\downarrow\right)\right] - \exp\left[-k_{jl}\left(\tau_f^\downarrow - \tau_{l-1}^\downarrow\right) - \left(\tau_f^\downarrow - \tau_c^\downarrow\right)/\mu\right] \right\} \right. \\
&\quad \left. + \hat{C}_{-jl}^m \frac{G_{-jl}^m(\mu)}{1-k_{jl}\mu} \left\{ \exp\left[-k_{jl}\left(\tau_l^\downarrow - \tau_c^\downarrow\right)\right] - \exp\left[-k_{jl}\left(\tau_l^\downarrow - \tau_f^\downarrow\right) - \left(\tau_f^\downarrow - \tau_c^\downarrow\right)/\mu\right] \right\} \right\} \\
&\quad + \frac{Z_l^m(\mu)}{1+\mu/\mu^{sun}} \left[\exp\left(-\tau_c^\downarrow/\mu^{sun}\right) - \exp\left(-\tau_f^\downarrow/\mu^{sun} - \frac{\tau_f^\downarrow - \tau_c^\downarrow}{\mu}\right) \right] \\
&\quad + \delta_{m0} \left\{ \left[V_{0l}^m(\mu) + \left(\tau_f^\downarrow + \mu\right) V_{1l}^m(\mu) \right] \left[1 - \exp\left(-\frac{\tau_f^\downarrow - \tau_c^\downarrow}{\mu}\right) \right] - \left(\tau_f^\downarrow - \tau_c^\downarrow\right) V_{1l}^m(\mu) \right\} \\
&\quad (0 < |\mu| \leq 1 \quad ; \quad \mu = \text{cosine of the off-nadir angle})
\end{aligned} \tag{5.15b}$$

The bracketed term on the first line of this equation emphasizes the fact that $\Delta I_l^m(\mu; \tau_c^\downarrow \Rightarrow \tau_f^\downarrow)$ is the segment incremental radiant intensity. The nadir angle dependent functions $G_{\pm jl}^m(\mu)$, $Z_l^m(\mu)$, $V_{0l}^m(\mu)$ and $V_{1l}^m(\mu)$ are defined by the following equations:

$$\begin{aligned}
G_{jl}^m(\mu) &\equiv \sum_{i=1}^N w_i \left[D_l^m(\mu, \mu_i) G_{jl}^m(\mu_i) + D_l^m(\mu, -\mu_i) G_{jl}^m(-\mu_i) \right] \quad [31b] \\
Z_l^m(\mu) &\equiv X_l^m(\mu) + \sum_{i=1}^N w_i \left[D_l^m(\mu, \mu_i) Z_l^m(\mu_i) + D_l^m(\mu, -\mu_i) Z_l^m(-\mu_i) \right] \quad [31a] \\
V_{0l}^m(\mu) &\equiv (1 - \omega_l) \left[B(T_{l-1}) - \tau_{l-1}^\downarrow \frac{B(T_l) - B(T_{l-1})}{\tau_l^\downarrow - \tau_{l-1}^\downarrow} \right] + \sum_{i=1}^N w_i \left[D_l^0(\mu, \mu_i) Y_{0l}^m(\mu_i) + D_l^0(\mu, -\mu_i) Y_{0l}^m(-\mu_i) \right] \quad [31c] \\
V_{1l}^m(\mu) &\equiv (1 - \omega_l) \frac{B(T_l) - B(T_{l-1})}{\tau_l^\downarrow - \tau_{l-1}^\downarrow} + \sum_{i=1}^N w_i \left[D_l^0(\mu, \mu_i) Y_{1l}^m(\mu_i) + D_l^0(\mu, -\mu_i) Y_{1l}^m(-\mu_i) \right] \quad [31c] .
\end{aligned} \tag{5.16}$$

Expressions for functions $D_l^m(\mu, \pm\mu_i)$ and $X_l^m(\mu)$ [Eq. (5.5)], $G_{jl}^m(\pm\mu_i)$ [Eqs. (5.11a-b)], $Z_l^m(\pm\mu_i)$ [Eq. (5.10)], and both $Y_{0l}^m(\pm\mu_i)$ and $Y_{1l}^m(\pm\mu_i)$ [Eq. (5.9)] have all been provided earlier.

Equations (5.15a) for the radiant intensity azimuth components are compared to Eqs. [35a] and [35b] from STWL. However, the form provided here explicitly illustrates the partitioning of each term into a layer (or boundary) radiant intensity and a foreground transmittance. Another difference between the expressions here and those in STWL is that the layer l radiant intensity, $\Delta I_l^m(\mu; \tau_c^\downarrow \Rightarrow \tau_f^\downarrow)$ of Eq. (5.15b) is defined for any “ $\tau_c^\downarrow \Rightarrow \tau_f^\downarrow$ ” sub-segment within the layer, where, as stated above, $\tau_c^\downarrow$ is the nadir optical depth at the segment end closer to the sensor and $\tau_f^\downarrow$ is the nadir optical depth at the segment end further from the sensor. As illustrated in Figure 5.1, the nadir optical depths satisfy the inequality $\tau_{l-1}^\downarrow \leq \tau_c^\downarrow < \tau_f^\downarrow \leq \tau_l^\downarrow$ for downward viewing sensor LOS segments, and the nadir optical depths satisfy the inequality $\tau_l^\downarrow \geq$

$\tau_c^\downarrow > \tau_f^\downarrow \geq \tau_{l-1}^\downarrow$ for upward viewing sensor LOS segments. In STWL, the ΔI_l^m expression is defined only for the case in which $\tau_f^\downarrow$ equals a layer boundary level, $\tau_{l-1}^\downarrow$ or $\tau_l^\downarrow$. Indeed, Eq. (5.15a) only contains terms of this form, $\Delta I_l^m(\mu; \tau^\downarrow \Rightarrow \tau_l^\downarrow)$ and $\Delta I_l^m(\mu; \tau^\downarrow \Rightarrow \tau_{l-1}^\downarrow)$. The more general expression for the segment radiant intensity, ΔI_l^m , will be required for modeling radiances in MODTRAN's spherical refraction geometry atmosphere.

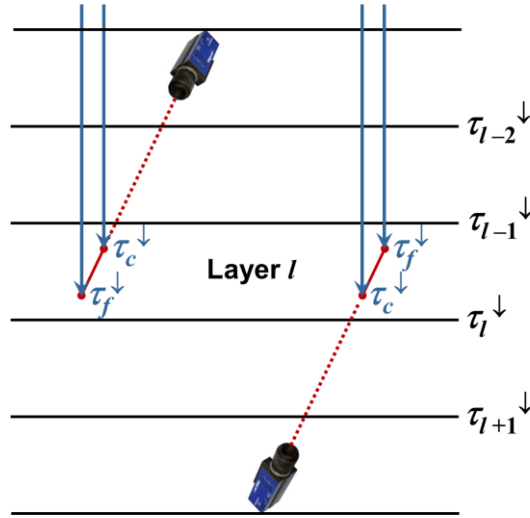


Figure 5.1. Definition of Nadir Optical Depths $\tau_{l-1}^\downarrow$, $\tau_l^\downarrow$, $\tau_c^\downarrow$ and $\tau_f^\downarrow$

Step 6: Integrating Plane-Parallel Atmosphere DISORT into Spherical Refractive Geometry Atmosphere MODTRAN

DISORT models the Earth surface as flat and the atmosphere as consisting of contiguous horizontal layers, each with homogeneous scattering properties. MODTRAN, on the other hand, treats the surface as a sphere with the Earth radius preferentially defined based on the local latitude and longitude values, models the atmosphere via contiguous spherical shells, and includes spherical refraction in its path geometry calculations, Figure 5.2 [left]. The dichotomy between the DISORT and MODTRAN atmosphere definitions greatly complicates the integration of DISORT radiant intensity data into the MODTRAN RT; this challenging integration is portrayed pictorially in Figure 5.2 [right].

An example of a scenario in which the DISORT-MODTRAN merge is problematic is the modeling of limb paths that pass through a tangent height, e.g. the green and blue paths in Figure 5.2 [left]. These paths begin with a downward trajectory, but terminate at the TOA. In DISORT, all paths beginning with a downward trajectory terminate at the ground. How can one use DISORT to determine the MODTRAN multiple scattering radiant intensity contributions for all the downward and upward limb path segments?

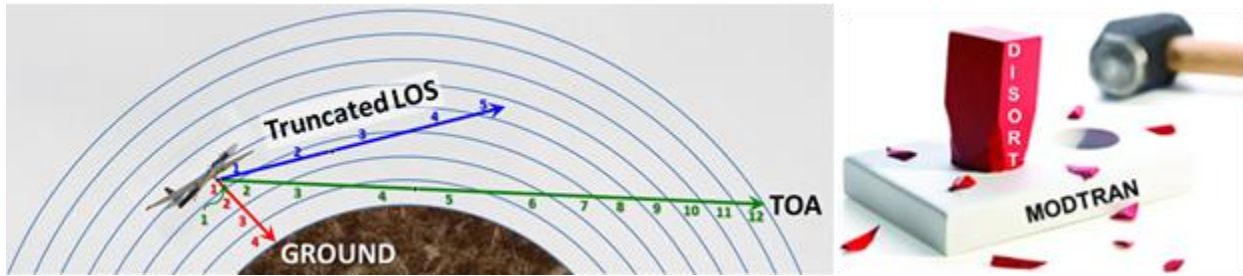


Figure 5.2. [left] MODTRAN Lines-of-Sight through a Locally Spherical Atmosphere and [right] Pictorial Representation of the DISORT-MODTRAN Integration Challenge

To answer this question, it is recalled that MODTRAN defines radiant intensity, Eq. (1.1) to be a function of the sensor path length spectral extinction optical depth, $\tau_\ell \equiv \tau_\ell(\nu)$. In contrast, DISORT models radiant intensity as a nadir extinction optical depth, $\tau^\downarrow$, dependent quantity. In fact, the nadir optical depth superscript $\downarrow$ has been retained throughout this chapter to emphasize this fact. Furthermore, MODTRAN lines-of-sight are defined via a sum over S path segments indexed by σ while DISORT LOS's are defined via a sum over layer contributions. It turns out that the DISORT radiant intensity, $I(\Omega; \tau_0) \equiv I_{\tau^\downarrow}(\Omega)$ [Eq. (5.3)], and azimuth moment radiant intensity, $I^m(\mu; \tau_0) \equiv I_{\tau^\downarrow}^m(\mu)$ [Eq. (15a)], expressions can be rewritten as a sum over path segments σ and a function of path length optical depths, τ_σ :

$$I(\Omega; \tau_0) = \sum_{m=0}^{2M-1} I^m(\mu; \tau_0) \cos m\Delta\phi^{sun} \quad \left(\Delta\phi^{sun} \equiv \phi^{sun} - \phi \quad ; \quad \tau_0 = 0 \right) \quad (5.17a)$$

$$I^m(\mu; \tau_0) = \exp(-\tau_S) I^m(\mu; \tau_S) + \sum_{\sigma=1}^S \exp(-\tau_{\sigma-1}) \Delta I_{l_\sigma}^m(\mu; \tau_{\sigma-1} \Rightarrow \tau_\sigma)$$

Here, τ_σ is defined as the spectral extinction optical depth from the sensor and through σ segments, $l_\sigma = l(\sigma)$ is the index for the layer containing segment σ , τ_S is the optical depth to the boundary (ground or TOA), and $I^m(\mu; \tau_\sigma)$ is the m^{th} azimuth component of the radiant intensity for a path beginning at an optical depth distance τ_σ from the sensor and proceeding in direction μ . Separate expressions are no longer required for down-look ($+\mu$) and up-look ($-\mu$) viewing. The m^{th} azimuth component *segment* radiant intensity, $\Delta I_{l_\sigma}^m(\mu; \tau_{\sigma-1} \Rightarrow \tau_\sigma)$, is given by

$$\begin{aligned} \Delta I_{l_\sigma}^m(\mu; \tau_{\sigma-1} \Rightarrow \tau_\sigma) &= \left\{ I^m(\mu; \tau_{\sigma-1}) - \Delta t_\sigma I^m(\mu; \tau_{\sigma-1}) \right\} \\ &= \sum_{j=1}^N \left\{ \hat{C}_{jl_\sigma}^m \frac{G_{jl_\sigma}^m(\mu)}{1 + k_{jl_\sigma} \mu} \left\{ \exp \left[-k_{jl_\sigma} \left(\tau_{\sigma-1}^\downarrow - \tau_{l_{\sigma-1}}^\downarrow \right) \right] - \Delta t_\sigma \exp \left[-k_{jl_\sigma} \left(\tau_\sigma^\downarrow - \tau_{l_{\sigma-1}}^\downarrow \right) \right] \right\} \right\} \\ &\quad + \left\{ \hat{C}_{-jl_\sigma}^m \frac{G_{-jl_\sigma}^m(\mu)}{1 - k_{jl_\sigma} \mu} \left\{ \exp \left[-k_{jl_\sigma} \left(\tau_{l_\sigma}^\downarrow - \tau_{\sigma-1}^\downarrow \right) \right] - \Delta t_\sigma \exp \left[-k_{jl_\sigma} \left(\tau_{l_\sigma}^\downarrow - \tau_\sigma^\downarrow \right) \right] \right\} \right\} \\ &\quad + \frac{Z_{l_\sigma}^m(\mu)}{1 + \mu/\mu} \left(t_{\sigma-1}^{sun} - t_\sigma^{sun} \Delta t_\sigma \right) + \delta_{m0} \left\{ \left[V_{0l_\sigma}^m(\mu) + \left(\tau_\sigma^\downarrow + \mu \right) V_{1l_\sigma}^m(\mu) \right] (1 - \Delta t_\sigma) - \mu (\tau_\sigma - \tau_{\sigma-1}) V_{1l_\sigma}^m(\mu) \right\} \\ &\quad \left\{ \Delta t_\sigma \equiv \exp \left[-(\tau_\sigma - \tau_{\sigma-1}) \right] = \exp \left(-\frac{\tau_\sigma^\downarrow - \tau_{\sigma-1}^\downarrow}{\mu} \right) \quad ; \quad 0 < |\mu| \leq 1 \quad ; \quad \mu = \text{cosine of the off-nadir angle} \right\} , \end{aligned} \quad (5.17b)$$

where Δt_σ is the transmittance of segment σ ; $\tau_{\sigma-1}^\downarrow$ and $\tau_\sigma^\downarrow$ are the nadir optical depths at the beginning and end of segment σ ; and $t_{\sigma-1}^{sun}$ and t_σ^{sun} are the solar path transmittances at the beginning and end of segment σ .

Eqs. (5.17a – b) exactly equal, term by term, Eqs. (5.15a – b) for the plane-parallel atmosphere. The only difference between these pairs of equations is that the latter set is defined in terms of LOS segments σ rather than atmospheric layers. Eqs. (5.17a – b) are preferred because they can be generalized for application to arbitrary MODTRAN spherical refractive paths. However, this does require that the definition of a few path quantities be generalized. First, the LOS termination point τ_S no longer has to occur at the ground or TOA (if no boundary term is defined at τ_S , $I^m(\mu; \tau_S)$ can be set to zero to compute the radiant intensity for the truncated path). Also, the cosine of the solar zenith angle, the cosine of the path nadir angle and the relative solar azimuth, $\Delta\phi^{sun} \equiv \phi^{sun} - \phi$, are not LOS constants within the spherical atmosphere. Segment averaged values are defined for each of these quantities: $\langle \mu^{sun} \rangle_\sigma$, $\langle \mu \rangle_\sigma$ and $\langle \Delta\phi^{sun} \rangle_\sigma$. The latter two segment averaged values are combined into one as $\langle \Omega \rangle_\sigma \equiv (\langle \mu \rangle_\sigma, \langle \Delta\phi^{sun} \rangle_\sigma)$. The sensor pointing direction Ω_0 is defined as $(\mu_0, \Delta\phi_0^{sun})$. In addition, the cosine of the path nadir angle and

the relative solar azimuth are required at the LOS termination point, μ_s and, $\Delta\phi_s^{sun}$ when a boundary term is to be included; these are not path averages. Since the relative solar azimuth angle is segment dependent, the sum over segments must be generalized to include this dependence, and the two expressions from Eq. (5.17a) must be combined into one:

$$\begin{aligned}
I(\Omega_0; \tau_0 \Rightarrow \tau_s) &= \sum_{m=0}^{2M-1} \left[\exp(-\tau_s) I^m(\mu_s; \tau_s) \cos(m\Delta\phi_s^{sun}) + \right. \\
&\quad \left. \sum_{\sigma=1}^S \exp(-\tau_{\sigma-1}) \Delta I_{l_\sigma}^m(\langle\mu\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) \cos(m\langle\Delta\phi^{sun}\rangle_\sigma) \right] \\
&= \exp(-\tau_s) \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos(m\Delta\phi_s^{sun}) + \sum_{\sigma=1}^S \exp(-\tau_{\sigma-1}) \Delta I_{l_\sigma}(\langle\Omega\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) \quad (5.18a) \\
\Delta I_{l_\sigma}(\langle\Omega\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) &\equiv \sum_{m=0}^{2M-1} \Delta I_{l_\sigma}^m(\langle\mu\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) \cos(m\langle\Delta\phi^{sun}\rangle_\sigma)
\end{aligned}$$

The extinction optical depth argument of the total radiant intensity, $I(\Omega_0; \tau_0 \Rightarrow \tau_s)$, is modified to explicitly list the LOS endpoint values, $\tau_0 \Rightarrow \tau_s$, reflecting the fact that the path need not end at the ground or TOA. The radiant intensity can still written as $I(\Omega_0; \tau_0)$ for paths terminating at the ground or TOA. Remember, by definition, τ_0 , the path length extinction optical depth at the sensor, is zero.

The m^{th} azimuth component of the segment radiant intensity is now a function of the segment-averaged μ and μ^{sun} values:

$$\begin{aligned}
\Delta I_{l_\sigma}^m(\langle\mu\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) &= \{I^m(\langle\mu\rangle_\sigma; \tau_{\sigma-1}) - \Delta t_\sigma I^m(\langle\mu\rangle_\sigma; \tau_{\sigma-1})\} \\
&= \sum_{j=1}^N \left\{ \hat{C}_{jl_\sigma}^m \frac{G_{jl_\sigma}^m(\langle\mu\rangle_\sigma)}{1 + k_{jl_\sigma} \langle\mu\rangle_\sigma} \left\{ \exp[-k_{jl_\sigma}(\tau_{\sigma-1}^\downarrow - \tau_{l_{\sigma-1}}^\downarrow)] - \Delta t_\sigma \exp[-k_{jl_\sigma}(\tau_\sigma^\downarrow - \tau_{l_{\sigma-1}}^\downarrow)] \right\} \right. \\
&\quad \left. + \hat{C}_{-jl_\sigma}^m \frac{G_{-jl_\sigma}^m(\langle\mu\rangle_\sigma)}{1 - k_{jl_\sigma} \langle\mu\rangle_\sigma} \left\{ \exp[-k_{jl_\sigma}(\tau_{l_\sigma}^\downarrow - \tau_{\sigma-1}^\downarrow)] - \Delta t_\sigma \exp[-k_{jl_\sigma}(\tau_{l_\sigma}^\downarrow - \tau_\sigma^\downarrow)] \right\} \right\} \quad (5.18b) \\
&\quad + \frac{Z_{l_\sigma}^m(\langle\mu\rangle_\sigma)}{1 + \langle\mu\rangle_\sigma / \langle\mu\rangle_\sigma^{sun}} (t_{\sigma-1}^{sun} - t_\sigma^{sun} \Delta t_\sigma) \\
&\quad + \delta_{m0} \{ [V_{0l_\sigma}^m(\langle\mu\rangle_\sigma) + (\tau_\sigma^\downarrow + \langle\mu\rangle_\sigma) V_{1l_\sigma}^m(\mu_\sigma)] (1 - \Delta t_\sigma) - \langle\mu\rangle_\sigma (\tau_\sigma - \tau_{\sigma-1}) V_{1l_\sigma}^m(\langle\mu\rangle_\sigma) \}
\end{aligned}$$

The version of DISORT in MODTRAN6 has been upgraded to ingest path segment view angle and solar geometry data so that the $I(\Omega_0; \tau_0 \Rightarrow \tau_s)$ radiant intensities can be computed. Segments endpoints are inserted at every vertical profile altitude level crossing between the sensor and the LOS termination point. If the LOS is a limb path, an additional segment endpoint is defined at the tangent point. The segment relative azimuth angles, $\langle\Delta\phi^{sun}\rangle_\sigma$, are defined as the average of the endpoint values, $(\Delta\phi_{\sigma+1}^{sun} + \Delta\phi_\sigma^{sun}) / 2$; the segment nadir cosines are defined as the ratio of the segment's nadir optical depth difference over the segment's path optical depth difference:

$$\langle\mu\rangle_\sigma \equiv \frac{\tau_{\sigma+1}^\downarrow - \tau_\sigma^\downarrow}{\tau_{\sigma+1} - \tau_\sigma} \cong -\frac{z_{\sigma+1} - z_\sigma}{d_\sigma} \quad (5.19)$$

The denominator, $\tau_{\sigma+1} - \tau_\sigma$ is always positive, but the numerator, $\tau_{\sigma+1}^\downarrow - \tau_\sigma^\downarrow$, is only positive for downward viewing path segments. The definition for $\langle\mu\rangle_\sigma$ is spectrally dependent. Currently, MODTRAN approximates the segment nadir cosine by the spectrally independent ratio of the segment endpoint

altitude, z , difference to the segment path length. The solar data has yet to be upgraded, so that $\langle \mu \rangle_{\sigma}^{sun}$ is simply set to a constant LOS μ^{sun} value (typically defined at the latitude and longitude of the LOS minimum altitude) and t_{σ}^{sun} is set to $\exp(-\tau_{\sigma}^{\downarrow}/\mu^{sun})$; this will be corrected in a later release of MODTRAN6.

It is important that readers acknowledge that Eqs. (5.18 a – b) are not an exact discrete ordinate solution to the 1D spherical shell atmosphere problem. Eqs. (5.8) provide an exact solution to the plane-parallel atmosphere problem. Usually, Eqs. (5.14) are used to integrate arbitrary lines-of-sight within the plane-parallel atmosphere. The approach here assumes that one can instead integrate arbitrary lines-of-sight within a spherical atmosphere, including the effects of refraction. For the thermal problem and for reasonable solar zenith angles (non-twilight conditions), this approach accurately predicts line-of-sight radiances.

Step 7: Defining the Delta-M Method

Atmospheric aerosol and cloud particulates typically possess highly forward peaked scattering phase functions. Accurately representing these functions using a Legendre expansion, Eq. (5.3), requires hundreds of terms. Computation times grow rapidly with the number of discrete ordinate streams, which DISORT couples to the number of azimuth terms. Accelerating convergence is critical for practical application of DISORT.

Wiscombe [1977] introduced the delta- M method to improve convergence of the discrete ordinate method. In his original derivation of the method, Wiscombe considered the case of a plane-parallel atmosphere with a parallel source impinging on the TOA; thermal source terms were not explicitly included. Here, Planck thermal emission is modeled as varying linearly with nadir optical depth between altitude levels, the standard approach. The method is generalized for application to spherical refractive lines-of-sight, modeled as described in Step 6. A modified formulation of the delta- M method is also presented at the end of Step 7 to provide some additional insight into the implications of the approach.

The basic premise of the delta- M approach is that the unit normalized scattering phase function, $p_{l(\sigma)}(\Theta)$, for segment σ in layer $l_{\sigma} = l(\sigma)$, can be represented as a weighted sum of a forward peaked Dirac-delta function $\delta(1 - \cos \Theta) / 2\pi$ and a less oscillatory, scaled phase function, $p'_{l(\sigma)}(\Theta)$, which itself is represented by a Legendre expansion:

$$p_{l_{\sigma}}(\Theta) = \frac{f_{l_{\sigma}}}{2\pi} \delta(1 - \cos \Theta) + (1 - f_{l_{\sigma}}) p'_{l_{\sigma}}(\Theta) ; \quad p'_{l_{\sigma}}(\Theta) \equiv \sum_{\ell=0}^{2M-1} (2\ell+1) \frac{g'_{l_{\sigma}}{}^{\ell}}{2\pi} P_{\ell}(\cos \Theta) ; \quad [54a] \quad (5.20)$$

$$g'_{l_{\sigma}}{}^{\ell} = \frac{g_{l_{\sigma}}{}^{\ell} - f_{l_{\sigma}}}{1 - f_{l_{\sigma}}} \quad (\ell = 0, \dots, 2M-1) ; \quad f_{l_{\sigma}} = g_{l_{\sigma}}^{2M} . \quad [54b,c]$$

For the spherical atmosphere, the scattering angle Θ varies (slowly) along the LOS segment, while the other variables listed here are modeled as constants within layer $l_{\sigma} = l(\sigma)$. The delta- M phase function expansion coefficients, $g'_{l(\sigma)}{}^{\ell}$, are defined by requiring preservation of the original phase function Legendre expansion up to order $2M-1$, and, as indicated here, an extra parameter $f_{l(\sigma)}$ is set by fitting one additional term from the expansion. The delta- M method changes the specification of higher order Legendre coefficients, $g_{l(\sigma)}{}^{\ell}$ for all $\ell \geq 2M$. For the unscaled normalized scattering phase function, $p_{l(\sigma)}(\Theta)$, these coefficients are all set to zero, Eq. (5.3). In the delta- M (scaled) phase function of Eq. (5.20), the higher order terms are instead set to the value of $f_{l(\sigma)} = g_{l_{\sigma}}^{2M}$, determined from the true scattering phase function. When Legendre expansion coefficients of typical terrestrial particulate phase functions are examined, one often finds that these coefficients fall off quite slowly. Setting $g_{l(\sigma)}{}^{\ell}$ for $\ell > 2M$ to $f_{l(\sigma)} = g_{l(\sigma)}^{2M}$ provides a better approximation than setting these terms to zero. Furthermore, the integrals used to compute the traditional phase function Legendre expansion coefficients are highly influenced by the forward peak, which produces ringing and a very poor representation for off-peak angles. Factoring out

the forward scattering component via Eq. (5.20) reduces the oscillatory ringing and allows the phase function in the $\varepsilon < \Theta < 180^\circ$ angular region (ε small) to be better characterized.

Spectral radiant intensities and fluxes can generally be computed with much fewer streams when the delta- M method is invoked. For MODTRAN built-in Mie (spherical) aerosol and cloud particles, 8 or 16 streams often produce truncation residuals commensurate with the accuracy of the MODTRAN band model.

The delta- M method is derived using the integral-differential equation for *total* radiant intensity, $I_{\tau\downarrow}^{tot}(\Omega)$, defined as the sum of the diffuse, $I_{\tau\downarrow}(\Omega)$, and direct, $I_{\tau\downarrow}^{dir}(\Omega)$, radiant intensities.

$$\mu \frac{dI_{\tau\downarrow}^{tot}(\Omega)}{d\tau} = I_{\tau\downarrow}^{tot}(\Omega) - (1 - \omega_{l_\sigma}) B(T_{\tau\downarrow}) - \omega_{l_\sigma} \int_{4\pi} p_{l_\sigma}(\Theta') I_{\tau\downarrow}^{tot}(\Omega') d\Omega' \quad (5.21)$$

Boundary Conditions

$$I_{\tau_{gnd}\downarrow}^{tot}(\mu, \phi; \nu) = \varepsilon(\mu, \phi) B(T_{gnd}; \nu) + \int_0^{1/2\pi} \int_0^{2\pi} I_{\tau_{gnd}\downarrow}^{tot}(-\mu', \phi'; \nu) f(\mu, -\mu', \phi - \phi'; \nu) \mu' d\phi' d\mu'$$

$$I_0^{tot}(-\mu, \phi; \nu) = \delta(\mu + \mu^{sun}) \delta(\phi - \phi^{sun}) F^{sun}/4 = \delta(\Omega - \Omega^{sun}) F^{sun}/4 \quad .$$

This set of equations differs from those in the diffuse radiant intensity RTE, Eq. (5.1), in that the solar source is defined as a TOA boundary condition, so the transmitted solar irradiance term is excluded. Note that the argument of the first delta function is a sum, $\mu + \mu^{sun}$, because μ is defined as the path nadir angle while μ^{sun} is defined as the solar zenith angle. The nadir optical depth, $\tau\downarrow$ is assumed to be within layer $l_\sigma = l(\sigma)$.

A delta- M or *scaled* RTE is obtained by inserting delta- M phase function, Eq. (5.20), into the first line of Eq. (5.21):

$$\mu \frac{dI_{\tau\downarrow}^{tot}(\Omega)}{d\tau} = I_{\tau\downarrow}^{tot}(\Omega) - (1 - \omega_{l_\sigma}) B(T_{\tau\downarrow}) - \omega_{l_\sigma} \left[f_{l_\sigma} I_{\tau\downarrow}^{tot}(\Omega) + \int_{4\pi} (1 - f_{l_\sigma}) p'_{l_\sigma}(\Theta) I_{\tau\downarrow}^{tot}(\Omega') d\Omega' \right] \quad (5.22)$$

$$= (1 - \omega_{l_\sigma} f_{l_\sigma}) I_{\tau\downarrow}^{tot}(\Omega) - (1 - \omega_{l_\sigma}) B(T_{\tau\downarrow}) - (1 - f_{l_\sigma}) \omega_{l_\sigma} \int_{4\pi} p'_{l_\sigma}(\Theta) I_{\tau\downarrow}^{tot}(\Omega') d\Omega' \quad .$$

The goal is to transform this scaled RTE into a form identical to that of the unscaled RTE, the first line of Eq. (5.21), so that the DISORT discrete ordinate solution approach can be directly applied. Begin by dividing Eq. (5.22) by the coefficient of the $I_{\tau\downarrow}^{tot}(\Omega)$ term, $1 - \omega_{l_\sigma} f_{l_\sigma}$:

$$\frac{\mu}{1 - \omega_{l_\sigma} f_{l_\sigma}} \frac{dI_{\tau\downarrow}^{tot}(\Omega)}{d\tau} = I_{\tau\downarrow}^{tot}(\Omega) - \frac{1 - \omega_{l_\sigma}}{1 - \omega_{l_\sigma} f_{l_\sigma}} B(T_{\tau\downarrow}) - \frac{1 - f_{l_\sigma}}{1 - \omega_{l_\sigma} f_{l_\sigma}} \omega_{l_\sigma} \int_{4\pi} p'_{l_\sigma}(\Theta) I_{\tau\downarrow}^{tot}(\Omega') d\Omega' \quad (5.23)$$

Next, define the delta- M nadir optical depth and single scattering albedo scaling relationships:

$$d\tau'\downarrow \equiv (1 - \omega_{l_\sigma} f_{l_\sigma}) d\tau\downarrow \quad ; \quad \omega'_{l_\sigma} \equiv \frac{1 - f_{l_\sigma}}{1 - \omega_{l_\sigma} f_{l_\sigma}} \omega_{l_\sigma} \quad . \quad [54d] \quad (5.24)$$

The scaled nadir optical depth, $\tau'\downarrow$, never exceeds the unscaled value, $\tau\downarrow$, while the scaled single scattering albedo, ω'_{l_σ} , is greater than or equal to the unscaled value. Their product, the scaled scattering nadir optical depth, like the scaled extinction nadir optical depth, is less than the unscaled value. With these definitions, the scaled RTE can be rewritten as follows:

$$\mu \frac{dI_{\tau^\downarrow}^{tot}(\Omega)}{d\tau^\downarrow} = I_{\tau^\downarrow}^{tot}(\Omega) - (1 - \omega'_{l_\sigma}) B(T_{\tau^\downarrow}) - \omega'_{l_\sigma} \int_{4\pi} p'_{l_\sigma}(\Theta) I_{\tau^\downarrow}^{tot}(\Omega') d\Omega' \quad (5.25)$$

Here, the total radiant intensity $I_{\tau^\downarrow}^{tot}(\Omega)$ and the Planck function $B(T_{\tau^\downarrow})$ are still written as functions of $\tau^\downarrow$ even though the derivative is respect to $\tau'^\downarrow$.

The expression for $\tau'^\downarrow$ as a function of $\tau^\downarrow$ is obtained by integrating the left-hand side of Eq. (5.24) along the nadir path from the TOA, noting that both $\tau^\downarrow$ and $\tau'^\downarrow$ are zero at the TOA:

$$\tau'^\downarrow = \tau'^\downarrow(\tau^\downarrow) = (1 - \omega_{l_\sigma} f_{l_\sigma}) (\tau^\downarrow - \tau_{l_\sigma-1}^\downarrow) + \tau_{l_\sigma-1}'^\downarrow \quad ; \quad \tau_{l_\sigma-1}'^\downarrow \equiv \sum_{i=1}^{l_\sigma-1} (1 - \omega_i f_i) (\tau_i^\downarrow - \tau_{i-1}^\downarrow) \quad (5.26)$$

This equation can be inverted to define $\tau^\downarrow$ as a function of $\tau'^\downarrow$:

$$\tau^\downarrow = \tau^\downarrow(\tau'^\downarrow) = \tau_{l_\sigma-1}^\downarrow + \frac{\tau'^\downarrow - \tau_{l_\sigma-1}'^\downarrow}{1 - \omega_{l_\sigma} f_{l_\sigma}} = \frac{\tau'^\downarrow}{1 - \omega_{l_\sigma} f_{l_\sigma}} + \left(\tau_{l_\sigma-1}^\downarrow - \frac{\tau_{l_\sigma-1}'^\downarrow}{1 - \omega_{l_\sigma} f_{l_\sigma}} \right) \quad (5.27)$$

One can now define the functions $I_{\tau^\downarrow}^{tot}(\Omega)$ and $B_{\tau^\downarrow} \equiv B(T_{\tau^\downarrow})$ in terms for $\tau'^\downarrow$:

$$I_{\tau'^\downarrow}^{tot}(\Omega) \equiv I_{\tau^\downarrow(\tau'^\downarrow)}^{tot}(\Omega) \quad \text{and} \quad B_{\tau'^\downarrow} \equiv B_{\tau^\downarrow(\tau'^\downarrow)} \quad (5.28)$$

Since $B_{\tau^\downarrow}$ is modeled as varying linearly with $\tau^\downarrow$ from B_{l-1} at $\tau_{l-1}^\downarrow$ to B_l at $\tau_l^\downarrow$, the function $B_{\tau'^\downarrow}$ will also vary linearly with $\tau'^\downarrow$ from B_{l-1} at $\tau_{l-1}'^\downarrow$ to B_l at $\tau_l'^\downarrow$. Inserting these functions into the scaled RTE of Eq. (5.25) gives

$$\mu \frac{dI_{\tau'^\downarrow}^{tot}(\Omega)}{d\tau'^\downarrow} = I_{\tau'^\downarrow}^{tot}(\Omega) - (1 - \omega'_{l_\sigma}) B_{\tau'^\downarrow}(T_{\tau'^\downarrow}) - \omega'_{l_\sigma} \int_{4\pi} p'_{l_\sigma}(\Theta) I_{\tau'^\downarrow}^{tot}(\Omega') d\Omega' \quad (5.29)$$

The diffuse radiant intensity RTE can now be written as

$$\mu \frac{dI_{\tau'^\downarrow}'(\Omega)}{d\tau'^\downarrow} = I_{\tau'^\downarrow}'(\Omega) - (1 - \omega'_{l_\sigma}) B_{\tau'^\downarrow}(T_{\tau'^\downarrow}) - \omega'_{l_\sigma} \left[p'_{l_\sigma}(\Theta^{sun}) \exp(-\tau'^\downarrow / \mu^{sun}) F^{sun} + \int_{4\pi} p'_{l_\sigma}(\Theta) I_{\tau'^\downarrow}'(\Omega') d\Omega' \right]$$

Boundary Conditions

$$I_{\tau_{gnd}'}^\downarrow(+|\mu|, \phi) = \left[\varepsilon(+|\mu|, \phi) B_{\tau_{gnd}'}(T_{\tau_{gnd}'}^\downarrow) + \mu^{sun} f(+|\mu|, \phi, \mu^{sun}, \phi^{sun}) e^{-\tau_{gnd}'^\downarrow / \mu^{sun}} F^{sun} + \int_0^1 \int_0^{2\pi} I_{\tau_{gnd}'}^\downarrow(-\mu', \phi') f(+|\mu|, \phi, -\mu', \phi') \mu' d\phi' d\mu' \right] \quad (5.30)$$

$$I_{\tau_0'}^\downarrow(-|\mu|, \phi) = 0 \quad .$$

The scaled total radiant intensity RTE, Eq. (5.29), has the same form as the unscaled RTE, Eq. (5.21). The DISORT solution approach can be applied directly to solve for the diffuse radiant intensity $I_{\tau^\downarrow}'(\Omega)$. For $\Omega \neq \Omega^{sun}$, $I_{\tau^\downarrow}^{dir}(\Omega)$ and $I_{\tau^\downarrow}^{dir}(\Omega)$ are zero, so $I_{\tau^\downarrow}'(\Omega)$ should equal $I_{\tau^\downarrow}(\Omega)$ given the definition on the left-hand side of Eq. (5.28). After defining the scaling relationships, Eqs. (5.24), up front, DISORT processing proceeds normally. This is the procedure currently implemented in MODTRAN6.

Since delta- M scales down the nadir optical depths, Eqs. (5.24) and (5.26), the delta- M direct solar flux exceeds the true value, $F_{\tau^\downarrow}'^{dir}(\tau^\downarrow) \geq F_{\tau^\downarrow}^{dir}(\tau^\downarrow)$. However, the sum of the scaled direct and diffuse solar downward fluxes, $F_{\tau^\downarrow}'^{tot}(\tau^\downarrow) = F_{\tau^\downarrow}'^{dir}(\tau^\downarrow) + F_{\tau^\downarrow}'(-\tau^\downarrow)$, equals the true total downward flux, $F_{\tau^\downarrow}^{tot}(\tau^\downarrow)$. Thus,

DISORT subtracts the unscaled direct solar flux, $F_-^{dir}(\tau^\downarrow)$, from this total flux to compute the unscaled diffuse downward flux, $F_-(\tau^\downarrow)$. This is somewhat disconcerting. Eq. (5.28) indicates that the unscaled and delta- M total radiant intensities are equal. Since the direct radiant intensity is zero for all Ω not equal to Ω^{sun} , one expects that $I'_{\tau^\downarrow}(\Omega)$ also equals $I_{\tau^\downarrow}(\Omega)$ for all Ω not equal to Ω^{sun} . How can it be that $I'_{\tau^\downarrow}(\Omega) = I_{\tau^\downarrow}(\Omega)$ for $\Omega \neq \Omega^{sun}$, but that the delta- M downward diffuse flux, defined by the integral of Eq. (5.13), is less than the unscaled diffuse flux?

The inconsistency dictates that the scaled radiant intensity $I'_{\tau^\downarrow}(\Omega)$ for upward viewing lines-of-sight does not equal but instead approaches $I_{\tau^\downarrow}(\Omega)$ as the number of streams increases. The delta- M upward viewing diffuse radiant intensity must include a delta-function contribution at a scattering angle Θ of zero, i.e. when $\Omega = \Omega^{sun}$. This contribution is not included in the calculation of the scaled diffuse downward flux. The upper hemisphere angular integral over the missing delta-function multiplied by the cosine of the solar zenith angle equals the delta- M *pseudo* direct flux minus the unscaled true direct flux. Thus, the upper hemisphere integral over $\mu I'_{\tau^\downarrow}(\Omega)$ for $\Omega \neq \Omega^{sun}$ falls short of the true value by this amount. The question arises as to whether one can and should correct/increase the delta- M upward viewing radiant intensities for $\Omega \neq \Omega^{sun}$ to produce the proper normalization? One could argue that only the $m = 0$ radiant intensity terms need to be corrected since they define the flux values. However, these terms are independent of azimuth angle, so such a correction could not account for the fact that the error arises from modeling near forward scatter as direct transmittance (except in the case of a zenith sun). In the spirit of Nakajima and Tanaka [1988], one could envision increasing layer radiant intensities proportional to the layer's scattering phase function. But then it would be difficult to argue that the correction should only be applied to upward viewing radiant intensities. If the downward viewing radiant intensities were perturbed, an upward flux imbalance would result. For now, the delta- M diffuse flux percent error will be reported as a gauge for the magnitude of the upward radiant intensity normalization problem.

In the previous paragraph, the statement that “the delta- M upward viewing diffuse radiant intensity must include a delta-function contribution at a scattering angle Θ of zero” was made without proof. It does make intuitive sense: the delta- M method models nearly forward diffuse scattered photons as directly transmitted photons. To prove this stipulation, a return is made to the unscaled diffuse radiant intensity RTE:

$$\mu \frac{d I_{\tau^\downarrow}(\Omega)}{d\tau} = I_{\tau^\downarrow}(\Omega) - (1 - \omega_{l(\sigma)}) B(T_{\tau^\downarrow}) - \omega_{l(\sigma)} \left[p_{l(\sigma)}(\Theta^{sun}) \exp\left(-\tau^\downarrow / \mu^{sun}\right) F^{sun} + \int_{4\pi} p_{l(\sigma)}(\Theta') I_{\tau^\downarrow}(\Omega') d\Omega' \right]$$

Boundary Conditions

$$I_{\tau_{gnd}^\downarrow} \left(+|\mu|, \phi \right) = \left[\varepsilon \left(+|\mu|, \phi \right) B(T_{\tau_{gnd}^\downarrow}) + \mu^{sun} f \left(+|\mu|, \phi, \mu^{sun}, \phi^{sun} \right) e^{-\tau_{gnd}^\downarrow / \mu^{sun}} F^{sun} + \int_0^1 \int_0^{2\pi} I_{\tau_{gnd}^\downarrow} \left(-\mu', \phi' \right) f \left(+|\mu|, \phi, -\mu', \phi' \right) \mu' d\phi' d\mu' \right] \quad (5.31)$$

$$I_{\tau_{gnd}^\downarrow} \left(-|\mu|, \phi \right) = 0 \quad .$$

As before, the delta- M phase function, Eq. (5.20), can be inserted into the first line of Eq. (5.31) to obtain a scaled RTE:

$$\begin{aligned}
\mu \frac{dI_{\tau, \downarrow}(\Omega)}{d\tau} &= I_{\tau, \downarrow}(\Omega) - (1 - \omega_{l_\sigma}) B(T_{\tau, \downarrow}) - \omega_{l_\sigma} \left\{ \left[\frac{f_{l_\sigma}}{2\pi} \delta(1 - \cos \Theta) + (1 - f_{l_\sigma}) p'_{l_\sigma}(\Theta^{sun}) \right] \exp(-\tau / \mu^{sun}) F^{sun} \right. \\
&\quad \left. + f_{l_\sigma} I_{\tau, \downarrow}(\Omega) + (1 - f_{l_\sigma}) \int_{4\pi} p'_{l_\sigma}(\Theta') I_{\tau, \downarrow}(\Omega') d\Omega' \right\} \\
&= (1 - \omega_{l_\sigma} f_{l_\sigma}) I_{\tau, \downarrow} - (1 - \omega_{l_\sigma}) B(T_{\tau, \downarrow}) - (1 - f_{l_\sigma}) \omega_{l_\sigma} \left[p'_{l_\sigma}(\Theta^{sun}) \exp(-\tau / \mu^{sun}) F^{sun} \right. \\
&\quad \left. + \int_{4\pi} p'_{l_\sigma}(\Theta') I_{\tau, \downarrow}(\Omega') d\Omega' \right] ; \quad (\Omega \neq \Omega^{sun})
\end{aligned} \tag{5.32}$$

The first line of Eq. (5.32) explicitly includes a delta-function solar source term, which must contribute to the unscaled downward diffuse flux calculation of Eq. (5.13). Dropping this term, leads to an equation for the diffuse radiant intensity for all values of Ω other than Ω^{sun} ; the DISORT solution without the delta-function solar source term will under-predict the downward diffuse flux. It is interesting to note this equation, the second line of Eq. (5.32), produces an alternative RTE for the radiant intensity. Applying the scaling relationships of Eq. (5.24), one obtains the following expression for the delta- M diffuse radiant intensity:

$$\mu \frac{dI'_{\tau, \downarrow}(\Omega)}{d\tau} = I'_{\tau, \downarrow}(\Omega) - (1 - \omega'_{l_\sigma}) B'_{\tau, \downarrow} - \omega'_{l_\sigma} \left[p'_{l_\sigma}(\Theta^{sun}) \exp(-\tau / \mu^{sun}) F^{sun} + \int_{4\pi} p'_{l_\sigma}(\Theta') I'_{\tau, \downarrow}(\Omega') d\Omega' \right] ; \quad (\Omega \neq \Omega^{sun})$$

Boundary Conditions

$$\begin{aligned}
I'_{\tau_{gnd}, \downarrow} (+|\mu|, \phi) &= \left[\varepsilon(+|\mu|, \phi) B(T_{\tau_{gnd}, \downarrow}) + \mu^{sun} f(+|\mu|, \phi, \mu^{sun}, \phi^{sun}) e^{-\tau_{gnd} / \mu^{sun}} F^{sun} \right. \\
&\quad \left. + \int_0^1 \int_0^{2\pi} I'_{\tau_{gnd}, \downarrow}(-\mu', \phi') f(+|\mu|, \phi, -\mu', \phi') \mu' d\phi' d\mu' \right] \\
I'_{\tau_{gnd}, \downarrow} (-|\mu|, \phi) &= 0
\end{aligned} \tag{5.33}$$

Inserting the expression for $\tau^\downarrow(\tau'^\downarrow)$, Eq. (5.27), into the solar attenuation terms, $\exp(-\tau^\downarrow / \mu^{sun})$, leads to an equation solely in terms of $\tau'^\downarrow$:

$$\begin{aligned}
\mu \frac{d I'_{\tau', \downarrow}(\Omega)}{d \tau'_{\downarrow}} &= I'_{\tau', \downarrow}(\Omega) - (1 - \omega'_{l_\sigma}) B'_{\tau'} \\
&- \omega'_{l_\sigma} \left\{ p'_{l_\sigma}(\Theta^{sun}) \exp \left[-\frac{\tau'_{\downarrow} / \mu^{sun}}{1 - \omega_{l_\sigma} f_{l_\sigma}} - \left(\frac{\tau'_{l_\sigma-1} / \mu^{sun}}{\mu^{sun}} - \frac{\tau'_{l_\sigma-1} / \mu^{sun}}{1 - \omega_{l_\sigma} f_{l_\sigma}} \right) \right] F^{sun} + \int_{4\pi} p'_{l_\sigma}(\Theta') I'_{\tau', \downarrow}(\Omega') d\Omega' \right\} \\
&= I'_{\tau', \downarrow}(\Omega) - (1 - \omega'_{l_\sigma}) B'_{\tau'} - \omega'_{l_\sigma} \left[p'_{l_\sigma}(\Theta^{sun}) \exp \left(-\tau'_{\downarrow} / \mu'_{l_\sigma}{}^{sun} \right) F'_{l_\sigma}{}^{sun} \right. \\
&\quad \left. + \int_{4\pi} p'_{l_\sigma}(\Theta') I'_{\tau', \downarrow}(\Omega') d\Omega' \right] \quad ; \quad \left(\Omega \neq \Omega^{sun} \right) ,
\end{aligned}$$

Boundary Conditions

$$\begin{aligned}
I'_{\tau', \downarrow}(|\mu|, \phi) &= \left\{ \begin{aligned} &\varepsilon(|\mu|, \phi) B \left(T_{\tau'_{\downarrow}} \right) + \int_0^{1/2} \int_0^{2\pi} I'_{\tau', \downarrow}(-\mu', \phi') f(|\mu|, \phi, -\mu', \phi') \mu' d\phi' d\mu' \\ &+ \mu^{sun} f(|\mu|, \phi, \mu^{sun}, \phi^{sun}) \exp \left[-\frac{\tau'_{\downarrow} / \mu^{sun}}{1 - \omega_{l_{\downarrow}} f_{l_{\downarrow}}} - \left(\frac{\tau'_{l_{\downarrow}-1} / \mu^{sun}}{\mu^{sun}} - \frac{\tau'_{l_{\downarrow}-1} / \mu^{sun}}{1 - \omega_{l_{\downarrow}} f_{l_{\downarrow}}} \right) \right] F^{sun} \end{aligned} \right\} \\
&= \left\{ \begin{aligned} &\varepsilon(|\mu|, \phi) B \left(T_{\tau'_{\downarrow}} \right) + \int_0^{1/2} \int_0^{2\pi} I'_{\tau', \downarrow}(-\mu', \phi') f(|\mu|, \phi, -\mu', \phi') \mu' d\phi' d\mu' \\ &+ \mu^{sun} f(|\mu|, \phi, \mu^{sun}, \phi^{sun}) \exp \left(\tau'_{\downarrow} / \mu'_{l_{\downarrow}}{}^{sun} \right) F'_{l_{\downarrow}}{}^{sun} \end{aligned} \right\} \quad (5.34) \\
I'_{0, \downarrow}(-|\mu|, \phi) &= 0 .
\end{aligned}$$

where

$$\mu'_{l_\sigma}{}^{sun} \equiv (1 - \omega_{l_\sigma} f_{l_\sigma}) \mu^{sun} \quad \text{and} \quad F'_{l_\sigma}{}^{sun} \equiv \exp \left[-\left(\tau'_{l_\sigma-1} / \mu^{sun} - \tau'_{l_\sigma-1} / \mu'_{l_\sigma}{}^{sun} \right) \right] F^{sun} \quad (5.35)$$

The final scaled RTE, the last line of Eq. (5.34), has the same form as the unscaled RTE, Eq. (5.21). The only difference is that the scaled cosine solar zenith $\mu'_{l_\sigma}{}^{sun}$ and TOA solar irradiance $F'_{l_\sigma}{}^{sun}$ terms in Eq. (5.35) depend on the layer index l_σ . The DISORT approach can be applied directly to solve for the diffuse radiant intensity $I'_{\tau', \downarrow}(\Omega) = I_{\tau', \downarrow}(\Omega)$. After defining the scaling relationships, Eqs. (5.24) and (5.35), up front, DISORT processing could proceed normally. Would this value for $I'_{\tau', \downarrow}(\Omega)$ equal the value obtained from Eq. (5.30)? That is hard to fathom given that the source terms differ. Is this solution for the diffuse radiant intensity better than the standard, Eq. (5.30)? For now, that is left as an open question.

Step 8: Extracting Multiple Scatter Solar and Thermal Scatter Components

One obvious drawback of the delta- M method is that the transformed radiant intensities are somewhat inaccurate for very near forward scattering scenarios, such as occurs in the study of the solar aureole. The finite width of the forward peak is not properly characterized. To improve near forward scattering accuracy, DISORT provides an option to invoke a series of radiant intensity corrections developed by Nakajima and Tanaka [1988]. In these corrections, the plane-parallel atmosphere single scatter solar radiant intensity is explicitly subtracted from the diffuse solar radiant intensity and then added back in using a high order Legendre expansion for the scattering phase function. Additional, less dramatic Nakajima and Tanaka corrections improve higher order solar scattering intensity terms.

For DISORT runs from within MODTRAN, the solar and Planck emission calculations are performed separately. It is important that the Nakajima and Tanaka corrections *not* be invoked for the DISORT solar calculations! MODTRAN itself subtracts the DISORT single scatter solar contribution from the DISORT *at-source* solar radiant intensity, $\Delta I_{l\sigma}^{sun}$, for each segment σ :

$$\Delta I_{l\sigma}^{mss} \left(\langle \mu \rangle_{\sigma}, \langle \Delta \phi^{sun} \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) = \Delta I_{l\sigma}^{sun} \left(\langle \mu \rangle_{\sigma}, \langle \Delta \phi^{sun} \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) - \frac{F^{sun} \omega_{l\sigma} p'_{l\sigma}(\Theta) \left(t_{\sigma-1}^{sun} - \Delta t_{\sigma} t_{\sigma}^{sun} \right)}{1 + \langle \mu \rangle_{\sigma} \langle \mu^{sun} \rangle_{\sigma}} . \quad (5.36)$$

In this equation, $\Delta I_{l\sigma}^{mss}$ is the σ segment *multiple* (no single) scatter solar radiant intensity; $\omega_{l\sigma}$ and $p'_{l\sigma}(\Theta)$ are the unscaled single scattering albedo and the scaled (delta- M) scattering phase function within layer l_{σ} . MODTRAN subsequently adds in its calculation of the *spherical refractive geometry* single scatter solar radiant intensity. MODTRAN computes the scattering phase function for each scattering source either from an analytic form (Rayleigh and Henyey-Greenstein) or by interpolating into a scattering angle dependent table. For user-defined scattering phase functions, both Legendre expansion coefficients and the tabulated values are required inputs. The single scattering calculations use the tabulated values, while DISORT relies on the Legendre expansion of the scattering phase function.

MODTRAN also subtracts out the *linear-in-tau* path segment thermal emission component, $\Delta I_{l\sigma}^{0\ thm_em}$, from the DISORT segment thermal radiance, $\Delta I_{l\sigma}^{0\ thm}$, leaving the segment thermal scatter, $\Delta I_{l\sigma}^{0\ thm_sct}$. Only the $m = 0$ thermal radiant intensity component is non-zero. The segment thermal scatter radiant intensity, $\Delta I_{l\sigma}^{0\ thm_sct}$, is obtained from

$$\begin{aligned} \Delta I_{l\sigma}^{0\ thm_sct} \left(\langle \mu \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) &= \Delta I_{l\sigma}^{0\ thm} \left(\langle \mu \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) - \Delta I_{l\sigma}^{0\ thm_em} \left(\langle \mu \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) \\ \Delta I_{l\sigma}^{0\ thm_em} \left(\langle \mu \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) &\equiv (1 - \omega_{l\sigma}) \left\{ (1 - \Delta t_{\sigma}) B(T_{\sigma-1}) + \left(\frac{1 - \Delta t_{\sigma}}{\tau_{\sigma} - \tau_{\sigma-1}} - \Delta t_{\sigma} \right) [B(T_{\sigma}) - B(T_{\sigma-1})] \right\} \end{aligned} \quad (5.37)$$

The MODTRAN value for the path segment thermal emission is subsequently added to DISORT segment thermal scatter. This subtraction is performed because DISORT does not treat optically very thin layers, $\Delta \tau_l^{\downarrow} \equiv \tau_{l\sigma}^{\downarrow} - \tau_{l\sigma-1}^{\downarrow} \leq 0.0001$, as a special case. Therefore, these layers can produce numerical instabilities within the DISORT processing when layer differences are computed. To avoid these problems, MODTRAN merges optically thin layers to insure that each $\Delta \tau_l^{\downarrow}$ exceeds 0.0001 for every DISORT layer l . This actually provides the added benefit of speeding up the DISORT calculations, because computation time is proportional to the number of computation layers. The problem with this procedure is that temperature local extrema in a profile can be bypassed. When MODTRAN computes the path thermal emission, the original layering is preserved and the true temperature profile is captured. The DISORT thermal scatter values may be inaccurate for some optically thin layers if temperature extrema are bypassed, but these scattered thermal contributions to the total radiance are insignificant under these optically thin conditions.

Step 9: Delta- M Segment Radiances

The primary output of the MODTRAN modified version of DISORT is the array of spherical atmosphere line-of-sight segment radiances,

$$\Delta I_{\sigma} \equiv \Delta I_{l\sigma} \left(\langle \Omega \rangle_{\sigma}; \tau_{\sigma-1} \Rightarrow \tau_{\sigma} \right) , \quad (5.38)$$

defined by Eqs. (5.18a-b) as part of Step 6, *Integrating Plane-Parallel Atmosphere DISORT into Spherical Refractive Geometry Atmosphere MODTRAN*. These values are used in Step 8 to compute multiple scatter solar and thermal scatter path radiance components. The question arises, “do the unscaled segment radiances of Eq. (5.38) equal the delta- M (scaled) segment radiance, $\Delta I'_{\sigma} \equiv \Delta I'_{l\sigma}(\langle \Omega \rangle_{\sigma}; \tau'_{\sigma-1} \Rightarrow$

τ'_σ)?" The answer turns out to be no! Since DISORT invokes the delta- M method, the scaled $\Delta I'_\sigma$ values are calculated but the unscaled ΔI_σ values are required. In this section, a method for computing the residual between $\Delta I'_\sigma$ and ΔI_σ is laid out so the true segment radiant intensity values can be generated.

To begin, recall that delta- M radiant intensity, $I'_{\tau'\downarrow}(\Omega)$, measured by a sensor located at scaled nadir optical depth $\tau'^\downarrow$ and facing direction Ω along a path that terminates at the ground or TOA, in principle, equals the unscaled radiant intensity, $I_{\tau\downarrow}(\Omega)$, measured by a sensor located at the true nadir optical depth $\tau^\downarrow$ and facing the same direction, Eq. (5.28):

$$I'_{\tau'\downarrow}(\Omega) \equiv I_{\tau\downarrow}(\Omega) \quad \text{with} \quad \tau'^\downarrow = \tau'^\downarrow \left(\tau^\downarrow \right) \quad \text{and} \quad \Omega \neq \Omega^{\text{sun}}. \quad (5.39)$$

The spherical atmosphere DISORT expression for the unscaled radiant intensity, $I_{\tau\downarrow}(\Omega)$, is given by Eq. (5.18a):

$$I_{\tau\downarrow}(\Omega_0) = I(\Omega_0; \tau_0 \Rightarrow \tau_s) = \exp(-\tau_s) \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos\left(m\Delta\phi_s^{\text{sun}}\right) + \sum_{\sigma=1}^S \exp(-\tau_{\sigma-1}) \Delta I_{l_\sigma}(\langle\Omega\rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma) \quad (5.40)$$

where $\tau^\downarrow$ is in layer l_1 and τ_s is the line-of-sight optical depth to the ground or TOA. One can define the radiant intensities, I_σ , from arbitrary τ_σ points along the line-of-sight to the boundary at τ_s as follows:

$$I_\sigma \equiv \exp[-(\tau_s - \tau_\sigma)] \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos\left(m\Delta\phi_s^{\text{sun}}\right) + \sum_{s=\sigma+1}^S \exp[-(\tau_{s-1} - \tau_\sigma)] \Delta I_s; \quad \Delta I_s \equiv \Delta I_{l_s}(\langle\Omega\rangle_s; \tau_{s-1} \Rightarrow \tau_s) \quad (5.41)$$

One immediately observes that

$$I_0 = I_{\tau\downarrow}(\Omega_0) = I(\Omega_0; \tau_0 \Rightarrow \tau_s) \quad \text{and} \quad I_s = \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos\left(m\Delta\phi_s^{\text{sun}}\right) \quad (5.42)$$

One also expects that

$$I_\sigma = \Delta I_{\sigma+1} + I_{\sigma+1} \exp(-\Delta\tau_{\sigma+1}) \quad ; \quad 0 \leq \sigma < S \quad ; \quad \Delta\tau_{\sigma+1} \equiv \tau_{\sigma+1} - \tau_\sigma \quad (5.43a)$$

This simply says that the radiant intensity, I_σ , at τ_σ equals the radiant intensity, $I_{\sigma+1}$, at $\tau_{\sigma+1}$ attenuated by the segment transmittance, $\exp(-\Delta\tau_{\sigma+1})$, plus the intervening segment's radiant intensity, $\Delta I_{\sigma+1}$. Below, this relationship is derived explicitly by first separating out the $s = \sigma + 1$ term from the summation in Eq. (5.41), and then factoring out the common exponential term from the remaining terms:

$$\begin{aligned} I_\sigma &= \Delta I_{\sigma+1} + \exp[-(\tau_s - \tau_\sigma)] \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos\left(m\Delta\phi_s^{\text{sun}}\right) + \sum_{s=\sigma+2}^S \exp[-(\tau_{s-1} - \tau_\sigma)] \Delta I_s \\ &= \Delta I_{\sigma+1} + \left\{ \exp[-(\tau_s - \tau_{\sigma+1})] \sum_{m=0}^{2M-1} I^m(\mu_s; \tau_s) \cos\left(m\Delta\phi_s^{\text{sun}}\right) + \sum_{s=\sigma+2}^S \exp[-(\tau_{s-1} - \tau_{\sigma+1})] \Delta I_s \right\} \exp[-(\tau_{\sigma+1} - \tau_\sigma)] \quad (5.43b) \\ &= \Delta I_{\sigma+1} + I_{\sigma+1} \exp(-\Delta\tau_{\sigma+1}) \end{aligned}$$

The relationship between the unscaled segment radiant intensity, $\Delta I_{\sigma+1}$, and the delta- M segment radiant intensity, $\Delta I'_{\sigma+1}$, can now be determined by first solving Eq. (5.43a) for $\Delta I_{\sigma+1}$:

$$\Delta I_{\sigma+1} = I_\sigma - I_{\sigma+1} \exp(-\Delta\tau_{\sigma+1}) \quad ; \quad 0 \leq \sigma < S \quad (5.44)$$

Since the unscaled radiant intensities, I_σ and $I_{\sigma+1}$, are both defined for paths terminating at the ground or TOA, they equal the delta- M radiant intensities, I'_σ and $I'_{\sigma+1}$, respectively. Thus,

$$\begin{aligned} \Delta I_{\sigma+1} &= I'_\sigma - I'_{\sigma+1} \exp(-\Delta\tau_{\sigma+1}) = [I'_\sigma - I'_{\sigma+1} \exp(-\Delta\tau'_{\sigma+1})] + [\exp(-\Delta\tau'_{\sigma+1}) - \exp(-\Delta\tau_{\sigma+1})] I'_{\sigma+1} \\ &= \Delta I'_{\sigma+1} + [\exp(-\Delta\tau'_{\sigma+1}) - \exp(-\Delta\tau_{\sigma+1})] I'_{\sigma+1} = \Delta I'_{\sigma+1} + [\exp(-\Delta\tau'_{\sigma+1}) - \exp(-\Delta\tau_{\sigma+1})] I_{\sigma+1} \end{aligned} \quad (5.45a)$$

where

$$\Delta I'_{\sigma+1} \equiv I'_\sigma - I'_{\sigma+1} \exp(-\Delta \tau'_{\sigma+1}) \Rightarrow I'_\sigma = \Delta I'_{\sigma+1} + I'_{\sigma+1} \exp(-\Delta \tau'_{\sigma+1}) = \Delta I'_{\sigma+1} + I_{\sigma+1} \exp(-\Delta \tau'_{\sigma+1}) = I_\sigma \quad (5.45b)$$

The delta- M method, with its representation of the scattering phase function as the weighted sum of a delta function in the forward direction and a remainder phase function, yields an increased segment transmittance and a decreased segment radiant intensity. In Eq. (5.45a), this loss in radiant intensity is compensated for by adding an attenuated background term. Integrating from the boundary S forward towards the sensor, Eqs. (5.45a-b) combine to produce the following set of relationships:

$$\begin{aligned} \Delta I_S &= \Delta I'_S + [\exp(-\Delta \tau'_S) - \exp(-\Delta \tau_S)] I_S \quad ; \quad I_{S-1} = \Delta I'_S + I_S \exp(-\Delta \tau'_S) \\ \Delta I_{S-1} &= \Delta I'_{S-1} + [\exp(-\Delta \tau'_{S-1}) - \exp(-\Delta \tau_{S-1})] I_{S-1} \quad ; \quad I_{S-2} = \Delta I'_{S-1} + I_{S-1} \exp(-\Delta \tau'_{S-1}) \\ &\dots \\ \Delta I_1 &= \Delta I'_1 + [\exp(-\Delta \tau'_1) - \exp(-\Delta \tau_1)] I_1 \quad ; \quad I_0 = \Delta I'_1 + I_1 \exp(-\Delta \tau'_1) \end{aligned} \quad (5.46)$$

As Eqs. (5.46) suggests, the MODTRAN modified version of DISORT sums over the sensor line-of-sight segments from back to front, first computing unscaled segment radiant intensities, ΔI_σ and then the unscaled path radiant intensities, $I_{\sigma-1}$. The delta- M segment radiances equal

$$\begin{aligned} \Delta I'_{\sigma+1} &\equiv \sum_{m=0}^{2M-1} \Delta I'^m_{l_{\sigma+1}}(\langle \Omega \rangle_{\sigma+1}; \tau'_\sigma \Rightarrow \tau'_{\sigma+1}) \\ \Delta I'^m_{l_{\sigma+1}}(\langle \Omega \rangle_{\sigma+1}; \tau'_\sigma \Rightarrow \tau'_{\sigma+1}) &\equiv \Delta I'^m_{l_{\sigma+1}}(\langle \mu \rangle_{\sigma+1}; \tau'_\sigma \Rightarrow \tau'_{\sigma+1}) \cos \left(m \left\langle \Delta \phi^{sun} \right\rangle_{\sigma+1} \right) \end{aligned} \quad (5.47)$$

where the $\Delta I'^m_{l_{\sigma+1}}(\langle \mu \rangle_\sigma; \tau'_\sigma \Rightarrow \tau'_{\sigma+1})$ terms are determined from the delta- M version of Eq. (5.18b). In actuality, the sum over path segments is internal to the sum of azimuth moments. For that reason, an m -dependent version of Eq. (5.46) is applied:

$$\begin{aligned} \Delta I^m_S &= \Delta I'^m_S + [\exp(-\Delta \tau'_S) - \exp(-\Delta \tau_S)] I^m_S \quad ; \quad I^m_{S-1} = \Delta I'^m_S + I^m_S \exp(-\Delta \tau'_S) \\ \Delta I^m_{S-1} &= \Delta I'^m_{S-1} + [\exp(-\Delta \tau'_{S-1}) - \exp(-\Delta \tau_{S-1})] I^m_{S-1} \quad ; \quad I^m_{S-2} = \Delta I'^m_{S-1} + I^m_{S-1} \exp(-\Delta \tau'_{S-1}) \\ &\dots \\ \Delta I^m_1 &= \Delta I'^m_1 + [\exp(-\Delta \tau'_1) - \exp(-\Delta \tau_1)] I^m_1 \quad ; \quad I^m_0 = \Delta I'^m_1 + I^m_1 \exp(-\Delta \tau'_1) \end{aligned} \quad (5.48)$$

The segment radiant intensities are then determined from the running sum over the m -dependent contributions:

$$\Delta I_{\sigma+1} \equiv \sum_{m=0}^{2M-1} \Delta I^m_{\sigma+1} \quad (5.49)$$

An interesting question arises with the application of Eq. (5.46), or equivalently, Eq. (5.48). MODTRAN includes an option to model the adjacency effect. This refers to the phenomenon in which light scattered from surfaces adjacent to a directly imaged pixel interferes with the observed signal. MODTRAN models adjacency by assigning a different surface reflectance to the imaged pixel, used to calculate surface radiant intensity contributions directly transmitted to the sensor, from the surface reflectance read into DISORT to model the surface radiation indirectly scattered into the line-of-sight. Thus, MODTRAN essentially models the imaged pixel as having infinitesimal area. When Eq. (5.46) is used to model lines-of-sight that intersect the Earth, the term containing I_S models a highly forward scattered radiance contribution. As such, one could argue that this surface contribution should be defined using the reflectance properties of the imaged pixel. In effect, this provides a finite width to the imaged pixel. In MODTRAN, this assignment of a reflectance spectrum for I_S in Eq. (5.46) [or for I_S^m in Eq.

(5.48)] has been made a user option. If input “WIDERP” (for wider pixel) is true, forward scattered I_s term is modeled with the imaged pixel reflectance; otherwise, I_s is modeled with the area-averaged pixel reflectance. Note that as the number of DISORT streams is increased, the difference between the unscaled and delta- M segment transmittances diminishes, so the width of the forward scattered pixel effectively diminishes.

A final question that must be answered is “how should MODTRAN model the scattered radiant intensity for lines-of-sight that terminate within the atmosphere?” For example, suppose one wishes to model a 5 km LOS from 2 to 4 km altitude, or from 4 to 2 km altitude. One could argue that the truncated path (tp) radiance intensity, ΔI_{tp} , should be calculated as the difference between the full path radiant intensity, I_0 , and background path radiant intensity, I_s :

$$\Delta I_{tp} = I_0 - I_s \exp(-\Delta \tau_{tp}) \quad (5.50)$$

Here, τ_{tp} is the optical depth of the truncated path and S is not the ground or TOA. However, this assumes that the path was truncated rather arbitrarily. Another option, which MODTRAN has adopted, is to assume that the truncated path was defined because an object is to be modeled at the path endpoint. Since we do not know the emission and reflectance from that object, I_s is simply set to zero. In the future, an option could be introduced into MODTRAN to allow the user to define the boundary radiant intensity, I_s , even when the path does not terminate at the ground.

5.2 Summary of the DISORT Integration into MODTRAN6

The integration of DISORT into MODTRAN has evolved considerably since its initial introduction in the late 1980's. As described in this chapter, the current processing is quite involved. A summary is warranted.

MODTRAN models a spherical shell atmosphere including the effects of refraction, but it also defines a plane-parallel atmosphere for use by DISORT. DISORT provides a discrete ordinate solution to the plane-parallel atmosphere RTE, defining all terms in the Gaussian angle expression for the LOS radiant intensity, $I_l^m(\mu_i; \tau^\downarrow)$ [Eq. (5.8)]. These quantities are defined for every atmospheric layer l and for $2M - 1$ azimuth components. The standard DISORT distribution uses this information to compute plane-parallel atmosphere LOS radiant intensities for arbitrary view angles and at arbitrary nadir optical depths, $\tau^\downarrow$. For MODTRAN, a customized version of DISORT has been developed that instead defines segment radiant intensities, $\Delta I_{l(\sigma)}^m(\langle \mu \rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma)$ [Eq. (5.18b)], which are defined for refracted paths through the MODTRAN spherical atmosphere. The MODTRAN LOS radiant intensities, $I(\Omega_0; \tau_0 \Rightarrow \tau_5)$, are then computed via Eq. (5.18a).

DISORT accelerates convergence by implementing the delta- M method. Unfortunately, the delta- M segment radiances, $\Delta I_{\sigma}^{\prime m} \equiv \Delta I_{l(\sigma)}^{\prime m}(\langle \mu \rangle_\sigma; \tau'_{\sigma-1} \Rightarrow \tau'_\sigma)$, do not equal the unscaled segment radiant intensities, $\Delta I_{\sigma}^m \equiv \Delta I_{l(\sigma)}^m(\langle \mu \rangle_\sigma; \tau_{\sigma-1} \Rightarrow \tau_\sigma)$ required by MODTRAN. The recursive algorithm of Eq. (5.48) defines the method for calculating the ΔI_{σ}^m segment radiant intensities from the $\Delta I_{\sigma}^{\prime m}$ delta- M values.

MODTRAN performs separate solar and thermal source DISORT calculations. MODTRAN subtracts the discrete ordinate single scatter solar segment radiant intensity from the DISORT total solar segment radiant intensity, replacing it with the MODTRAN single scatter solar calculated value. Similarly, MODTRAN removes the thermal emission component from the DISORT total thermal segment radiant intensity, replacing it with the MODTRAN value.

A few additional DISORT outputs are used by MODTRAN. These include the level fluxes of Eq. (5.13). For lines-of-sight that terminate at the ground, DISORT can also generate LOS and solar path diffuse spectral transmittances and the atmosphere spectral spherical albedos measured from the ground. If requested, MODTRAN writes this raw (unconvolved) atmospheric correction data to its *.acd output file. This chapter does not describe the calculation of these latter quantities because this calculation follows the standard DISORT method detailed in the STWL document.

6. LOCALIZED GAS CLOUDS

6.1 Scattered Radiance

Let ΔI_s be the scattered source radiance arising from a single path segment:

$$\Delta I_s = \int_0^{\tau_s} S_s(t_s) \exp(-t_s - t_a) dt_s \quad , \quad (6.1)$$

where τ_s is the path segment scattering optical depth, S_s is the scattered source function at a scattering optical depth t_s along the path segment, and $t_a \equiv t_a(t_s)$ is the absorption optical depth at t_s . To approximate this integral, two assumptions are made:

- The scattering source function does not vary significantly along the path segment, so it can be replaced by a path segment average $\langle S_s \rangle$, and
- The absorption optical depth is proportional to the scattering optical depth, i.e., $t_a / t_s \approx \tau_a / \tau_s$, where τ_a is the path segment absorption optical depth

Invoking these approximations, the path segment scattered source radiance can be expressed as follows

$$\Delta I_s \approx \langle S_s \rangle \int_0^{\tau_s} \exp \left[\left(-\frac{\tau_s + \tau_a}{\tau_s} \right) t_s \right] dt_s = \langle S_s \rangle \int_0^{\tau_s} e^{-t_s (\tau / \tau_s)} dt_s = \langle S_s \rangle \int_0^{\tau_s} e^{-t_s / \omega} dt_s = \omega (1 - e^{-\tau}) \langle S_s \rangle \quad (6.2)$$

where the segment extinction optical depth, τ , is defined as the sum of the segment absorption and scattering optical depths [$\tau \equiv \tau_a + \tau_s$], and the segment single scattering albedo, ω , is defined as the ratio of the segment scattering optical depth τ_s to the segment extinction optical depth [$\omega \equiv \tau_s / \tau$].

Now consider the effect of introducing a localized absorbing gas. This will increase the path segment extinction optical depth from τ to $\tau' [\equiv \tau + \Delta \tau_a]$, and decrease the path segment single scattering albedo from ω to ω' [Here, the prime (') is being used to denote a perturbed quantity and it has nothing to do with the use of the same notation for the delta- M method of Chapter 5]. Using the above approximate form for the path segment scattered source radiance, the following scaling relationship is obtained for the perturbed scattered source, $\Delta I'_s$:

$$\Delta I'_s \approx \frac{\omega' (1 - e^{-\tau'})}{\omega (1 - e^{-\tau})} \Delta I_s = \frac{\tau (1 - e^{-\tau'})}{\tau' (1 - e^{-\tau})} \Delta I_s = \frac{\tau (1 - e^{-\tau - \Delta \tau_a})}{(\tau + \Delta \tau_a) (1 - e^{-\tau})} \Delta I_s \quad (6.3)$$

This equation can be used to derive an expression for the change in the scattered radiance contribution:

$$\begin{aligned}
\Delta I'_s - \Delta I_s &\approx \left(\frac{\tau (1 - e^{-\tau - \Delta \tau_a})}{(\tau + \Delta \tau_a) (1 - e^{-\tau})} - 1 \right) \Delta I_s \\
&= \left(\frac{\tau (1 - e^{-\tau - \Delta \tau_a}) - (\tau + \Delta \tau_a) (1 - e^{-\tau})}{(\tau + \Delta \tau_a) (1 - e^{-\tau})} \right) \Delta I_s \\
&= - \left(\frac{\Delta \tau_a - e^{-\tau} [(\tau + \Delta \tau_a) - \tau e^{-\Delta \tau_a}]}{(\tau + \Delta \tau_a) (1 - e^{-\tau})} \right) \Delta I_s \\
&= \frac{-\Delta \tau_a}{(\tau + \Delta \tau_a) (1 - e^{-\tau})} \left\{ 1 - e^{-\tau} \left[1 + \tau (1 - e^{-\Delta \tau_a}) / \Delta \tau_a \right] \right\} \Delta I_s \\
&= \frac{-\Delta \tau_a}{\tau + \Delta \tau_a} \left(1 - e^{-\tau} \frac{\tau (1 - e^{-\Delta \tau_a})}{\Delta \tau_a (1 - e^{-\tau})} \right) \Delta I_s \\
&= \left(e^{-\tau} \frac{\tau (1 - e^{-\Delta \tau_a})}{1 - e^{-\tau}} - \Delta \tau_a \right) \frac{\Delta I_s}{\tau + \Delta \tau_a}
\end{aligned} \tag{6.4}$$

6.2 Emitted Radiance

Let ΔI_t be the emitted source radiance arising from a single path segment of extinction optical depth τ and single scattering albedo ω . With the linear-in-tau approximation, ΔI_t has the following form:

$$\begin{aligned}
\Delta I_t(\tau, \omega) &= (1 - \omega) \left[\left(1 - e^{-\tau} \right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau} \right) \Delta B \right] \\
\Delta B &= B_b - B_f \quad ; \quad B_b = B(T_b) \quad \text{and} \quad B_f = B(T_f) \quad ,
\end{aligned} \tag{6.5}$$

where B_f and B_b are the Planck function at the front and back of the path segment, respectively. The spectral index is suppressed to simplify the notation, and T_f and T_b are the temperature at the front and back of the path segment. If a chemical release layer at constant temperature T_f with absorption optical depth τ_g were placed *in front of* this segment, the thermal radiance would be given by

$$\begin{aligned}
\Delta I_t(\tau, \omega; \tau_g) &= \left(1 - e^{-\tau_g} \right) B_f + (1 - \omega) e^{-\tau_g} \left[\left(1 - e^{-\tau} \right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau} \right) \Delta B \right] \\
&= \left(1 - e^{-\tau - \tau_g} \right) B_f - e^{-\tau_g} \left[\left(1 - e^{-\tau} \right) \omega B_f - (1 - \omega) \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau} \right) \Delta B \right]
\end{aligned} \tag{6.6}$$

If instead, the chemical release is embedded in the original layer, the thermal emission is given by

$$\Delta I_t(\tau + \tau_g, \omega') = (1 - \omega') \left[\left(1 - e^{-\tau - \tau_g} \right) B_f + \left(\frac{1 - e^{-\tau - \tau_g}}{\tau + \tau_g} - e^{-\tau - \tau_g} \right) \Delta B \right] \quad , \tag{6.7}$$

As before, we can determine the relative change in radiance resulting from adding the chemical release via the two different methods. Consider first the placing of the chemical release in front of the layer:

$$\begin{aligned}\frac{\Delta I_t(\tau, \omega; \tau_g)}{\Delta I_t(\tau, \omega)} &= \frac{\left(1 - e^{-\tau_g}\right) B_f}{(1 - \omega) \left[\left(1 - e^{-\tau}\right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau}\right) \Delta B \right]} + e^{-\tau_g} \\ &= \frac{\left(1 - e^{-\tau_g}\right) B_f}{\left(1 - \omega' \frac{\tau + \tau_g}{\tau}\right) \left[\left(1 - e^{-\tau}\right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau}\right) \Delta B \right]} + e^{-\tau_g}\end{aligned}\tag{6.8}$$

In the second line, the following relationship involving the single scattering albedos is utilize:

$$\frac{\omega}{\omega'} = \frac{\tau_s / \tau}{\tau_s / (\tau + \tau_g)} = \frac{\tau + \tau_g}{\tau}\tag{6.9}$$

When the chemical release is inserted into the layer, a quite different result is obtained:

$$\begin{aligned}\frac{\Delta I_t(\tau + \tau_g, \omega')}{\Delta I_t(\tau, \omega)} &= \left(\frac{1 - \omega'}{1 - \omega}\right) \frac{\left(1 - e^{-\tau - \tau_g}\right) B_f + \left(\frac{1 - e^{-\tau - \tau_g}}{\tau + \tau_g} - e^{-\tau - \tau_g}\right) \Delta B}{\left(1 - e^{-\tau}\right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau}\right) \Delta B} \\ &= \left(\frac{1 - \omega'}{1 - \omega' \frac{\tau + \tau_g}{\tau}}\right) \frac{\left(1 - e^{-\tau - \tau_g}\right) B_f + \left(\frac{1 - e^{-\tau - \tau_g}}{\tau + \tau_g} - e^{-\tau - \tau_g}\right) \Delta B}{\left(1 - e^{-\tau}\right) B_f + \left(\frac{1 - e^{-\tau}}{\tau} - e^{-\tau}\right) \Delta B}\end{aligned}\tag{6.10}$$

For simplicity, consider the constant temperature case:

$$\begin{aligned}\left. \frac{\Delta I_t(\tau, \omega; \tau_g)}{\Delta I_t(\tau, \omega)} \right|_{\Delta B=0} &= e^{-\tau_g} + \frac{\tau (1 - e^{-\tau_g})}{[\tau - \omega'(\tau + \tau_g)](1 - e^{-\tau})} \\ &= e^{-\tau_g} + \frac{\tau (1 - e^{-\tau_g})}{\tau_a (1 - e^{-\tau})}\end{aligned}\tag{6.11}$$

and

$$\begin{aligned}
\frac{\Delta I_t(\tau + \tau_g, \omega')}{\Delta I_t(\tau, \omega)} \Big|_{\Delta B=0} &= \frac{\tau(1-\omega')(1-e^{-\tau-\tau_g})}{[\tau - \omega'(\tau + \tau_g)](1-e^{-\tau})} \\
&= e^{-\tau_g} + \frac{\tau(1-e^{-\tau_g})}{\tau_a(1-e^{-\tau})} - \frac{\tau - \tau_a}{\tau_a(\tau + \tau_g)} \left[\tau \left(\frac{1-e^{-\tau_g}}{1-e^{-\tau}} \right) - \tau_g e^{-\tau_g} \right] \\
&= \frac{\Delta I_t(\tau, \omega; \tau_g)}{\Delta I_t(\tau, \omega)} \Big|_{\Delta B=0} - \frac{\tau - \tau_a}{\tau_a(\tau + \tau_g)} \left[\tau \left(\frac{1-e^{-\tau_g}}{1-e^{-\tau}} \right) - \tau_g e^{-\tau_g} \right] \\
&< \frac{\Delta I_t(\tau, \omega; \tau_g)}{\Delta I_t(\tau, \omega)} \Big|_{\Delta B=0} \quad \text{if } \tau_g > 0 \quad \text{and} \quad \tau > \tau_a
\end{aligned} \tag{6.12}$$

since

$$\frac{\tau - \tau_a}{\tau_a(\tau + \tau_g)} \left[\tau \left(\frac{1-e^{-\tau_g}}{1-e^{-\tau}} \right) - \tau_g e^{-\tau_g} \right] > 0 \quad \text{if } \tau_g > 0 \quad \text{and} \quad \tau > \tau_a \tag{6.13}$$

7. THE MODTRAN LINE-BY-LINE ALGORITHM

A Line-By-Line (LBL) option has been developed for MODTRAN6. The motivation for this development was two-fold. Firstly, when MODTRAN is validated against an independent LBL model, it is often difficult to isolate the source of discrepancies. One must verify consistency between pressure, temperature and density profiles, between column density calculations, between continuum and particulate data, between spectral convolution methods, and more. Introducing a LBL option directly within MODTRAN will insure common elements for all calculations other than those used to compute molecular transmittances. Thus, the new option will facilitate testing and subsequent refinement of the MODTRAN band model.

The second motivation for the LBL upgrade is that it will enable users to compute high spectral resolution transmittances and radiances not only for the full range of current MODTRAN applications, but also for radiative transfer problems outside of current purview of MODTRAN. In particular, MODTRAN current applications include methods for solving the multiple scattering problem. Introducing the LBL feature into MODTRAN will enable first-principle calculations of scattered radiances, an option that is often not readily available with LBL models.

MODTRAN6 computes LBL transmittances within 0.1 cm^{-1} spectral bins, marching through the full requested band pass. The LBL algorithm uses the highly accurate, pressure- and temperature-dependent MODTRAN Padé approximant fits to the contribution from line tails to define the absorption from all molecular transitions centered more than 0.05 cm^{-1} from each 0.1 cm^{-1} spectral bin. The beauty of this approach is that the on-the-fly computations for each 0.1 cm^{-1} bin will only require explicit LBL summing of transitions centered within a 0.2 cm^{-1} spectral region. That is, the contribution from the more distant lines will be pre-computed via the Padé approximants.

7.1 The Traditional Line-By-Line Method

LBL methods compute monochromatic spectral transmittances and radiances on an arbitrarily fine grid of spectral points by explicitly solving the radiative transfer (RT) equation. Most LBL codes model the temperature- and pressure-dependent molecular absorption by summing line-shape contributions from every molecular transition centered within 25 cm^{-1} of each spectral frequency grid point. Distant lines, i.e., those centered more than 25 cm^{-1} from the calculational frequency, are modeled via temperature-dependent continua databases. Given a typical spectral resolution of 0.001 cm^{-1} , an 8.0 to $12.5\text{ }\mu\text{m}$ band would contain 450,000 spectral points ($800.0005, 800.0015, \dots, 1249.995\text{ cm}^{-1}$). At each one of these points, the absorption sum must be computed for each segment along a line-of-sight (LOS). The 2012 HITRAN molecular transition database [Rothman *et al.*, 2013] contains 27,651 lines centered within 25 cm^{-1} of 1000.05 cm^{-1} ($\sim 10\text{ }\mu\text{m}$) for the default MODTRAN band model species, the first 12 molecules in the HITRAN compilation. Clearly, the computation of high-resolution molecular transmittances involves significant computational effort.

The LBLRTM radiative transfer software [Clough *et al.*, 1992; 2005] distributed by the Atmospheric and Environmental Research (AER) company generates spectral transmittances and thermal emitted radiances at a 0.001 cm^{-1} increment over the 8.0 to $12.0\text{ }\mu\text{m}$ band in about 20 seconds for a 60° off-nadir view through the 1976 U.S. Standard (USS) model atmosphere. By comparison, the MODTRAN6 band model run with its statistical correlated- k algorithm at 0.1 cm^{-1} spectral resolution performs the same calculation on the same machine, a 32 core Advanced Micro Devices, Inc. (AMD) Opteron with 512 Gbytes of RAM, more than 35 times faster.

7.2 The MODTRAN Band Model RT Approach

Band model RT algorithms use statistical data on the distribution of line strengths in contiguous, regularly spaced spectral intervals to compute bin-averaged transmittances and radiances, bypassing explicit sampling of individual spectral points. The MODTRAN band model (BM) transmittance formulation partitions molecular absorption into 3 components. Absorption from distant lines is modeled

via temperature-dependent continua databases, identical to the method used in LBL models. However, MODTRAN treats the molecular absorption within each BM spectral bin, whether it be of width 0.1, 1.0, 5.0 or 15.0 cm^{-1} , as arising from two additional sources: the contribution from lines centered within the spectral bin, and the remaining line-tail absorption from molecular transitions centered outside of the bin but within 25 cm^{-1} . Figure 7.1 (left) illustrates the absorption optical depth from the set of HITRAN 2012 CH_4 lines centered within the single 0.1 cm^{-1} bin between 3000.0 and 3000.1 cm^{-1} (30 atm-cm CH_4 at $T=296\text{ K}$ and $P=0.1\text{ atm}$). As the figure illustrates, one obtains significant spectral structure from the line center contributions, but the line-tail absorption is a smooth function that falls off precipitously with distance from the spectral bin at 0.1 atm pressure. Upon summing the line-tail contributions of all CH_4 transitions centered outside of a spectral bin (but within 25 cm^{-1}), the blue curves on the right side of Figure 7.1, one obtains a smooth curve with at most one minimum, the black curve in Figure 7.1 (right).

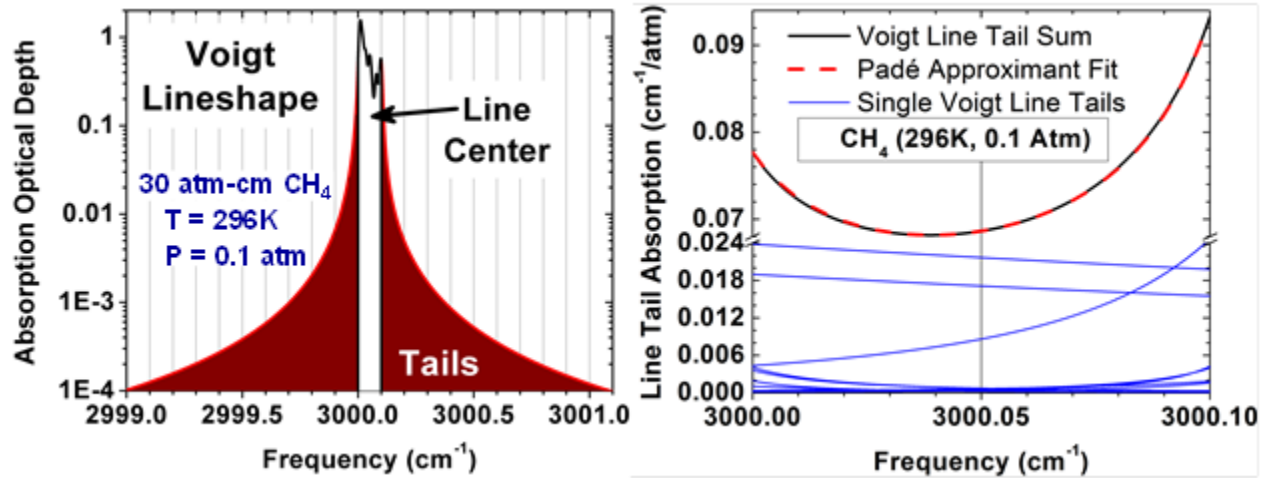


Figure 7.1. (left) Plot of the Absorption Optical Depth Arising from the set of HITRAN CH_4 Lines Centered in a 0.1 cm^{-1} Spectral Bin between 3000.0 and 3000.1 cm^{-1} , and (right) Plot of Line Tail Absorption from Molecular Transitions Centered in Neighboring Spectral Bins Out to 25 cm^{-1}

For MODTRAN, the line-tail absorption cross-section spectral curves (divided by pressure, P), $a_p(\delta_\nu)$, within each band model bin of width $\Delta\nu$, are fit to a [2, 2] Padé approximant, i.e., the ratio of quadratic polynomials in δ_ν ; δ_ν is defined as the spectral frequency, ν , displacement from bin center, ν_{cen} , normalized to range from -1 to $+1$:

$$a_p(\delta_\nu) = \frac{1}{P} \left(\frac{\sigma_0 + \sigma_1 \delta_\nu + \sigma_2 \delta_\nu^2}{1 + d_1 \delta_\nu + d_2 \delta_\nu^2} \right) ; \quad \delta_\nu \equiv \frac{\nu - \nu_{cen}}{\Delta\nu/2} ; \quad \begin{cases} \sigma_0 + \sigma_1 \delta_\nu + \sigma_2 \delta_\nu^2 \geq 0 \\ 1 + d_1 \delta_\nu + d_2 \delta_\nu^2 > 0 \end{cases} . \quad (7.1)$$

The curves are fit for 2 values of pressure (1.0 and 0.1 atm); at a series of temperatures (180 to 330 K in 5K increments); and for *all* the HITRAN line compilation molecules. The 5 parameters [σ_0 , σ_1 and σ_2 with cross-section units ($\text{cm}^{-1}/\text{atm}$), and dimensionless d_1 and d_2] that define each Padé approximant are determined from summed absorption cross-sections at the bin center and bin edges; from the integral of the absorption cross-section curve over the interval; and from the spectral derivative of the absorption cross-section at bin center. As the red dashed curve on the right in Figure 7.1 illustrates for a single sample case, the fits are extremely accurate; generally, relative residuals are of order 10^{-5} . To help insure excellent fits, MODTRAN translates inward all line center positions that are too close to a bin edge. This helps avoid the zero slope at line center and the relative sharp Gaussian (Doppler) drop off that can occur at the lower 0.1 atm pressure. The stipulation is that the residuals introduced by shifting line center positions a fraction of a spectral bin (no more than 0.25 $\Delta\nu$) will be small compared to the inherent errors

of the statistical band model. In sub-section 7.3.2, this hypothesis is tested by comparing the MODTRAN6 BM and LBL 0.1 cm⁻¹ spectral bin transmittances.

7.3 The Initial MODTRAN6 Line-By-Line RT Approach

The initial MODTRAN6 LBL approach simply provided an option to replace the statistical band model *line center* absorption calculation with a first-principles LBL integration of the absorption line-shape of all molecular transitions centered in each 0.1 cm⁻¹ interval. A new MODTRAN6 input, *N_RES*, defines the number of spectral grid points in each 0.1 cm⁻¹ bin. For example, if *N_RES* is set to 100, the default value, then 0.001 cm⁻¹ spectral resolution calculations are performed; the 100 spectral frequencies for the 3000.0 to 3000.1 cm⁻¹ bin are 3000.0005, 3000.0015, ..., 3000.9995 cm⁻¹. At each spectral grid point, ν_i , the explicit LBL calculation of line center absorption is added to the sum of the Padé fit of the line-tails and the spectrally interpolated continuum at ν_i .

7.3.1 Features of the Initial MODTRAN6 Line-By-Line RT Approach

Three features of the MODTRAN6 LBL approach are noted. First, the number of calculations is reduced dramatically from the traditional LBL approach. As enumerated earlier, 27,651 HITRAN lines for the MODTRAN default BM molecules fall within the 50 cm⁻¹ domain centered at 1000.05 cm⁻¹; the MODTRAN6 LBL algorithm only performs on-the-fly line-shape absorption cross-section calculations for molecular transitions centered within the 0.1 cm⁻¹ bin containing 1000.05 cm⁻¹, 55 lines.

A second feature of the MODTRAN6 LBL approach is that it provides an option to directly measure of the accuracy of the statistical band model line-center transmittance calculation, since the computation of extinction from all other sources is essentially the same for the two methods.

The third feature of the MODTRAN6 LBL approach is that it enables specification of segment and spectral dependent optical depth (OD) cut-offs for use in speeding up processing. Within an initial loop over all candidate lines for a given spectral bin, a pre-computed 296 K line center absorption cross-section is multiplied by the molecule's largest column density from the current MODTRAN geometry calculations, a value determined prior to entering the frequency loop. If the resulting OD exceeds the spectrally-independent cut-off value, *ODcut*, currently set at 10⁻⁸, then the molecular transition is added to a list of active lines. Prior to beginning the explicit LBL calculations for each path segment, the particulate extinction, the Rayleigh scattering and the molecular continuum and line-tail optical depths are summed at each grid point within the current spectral bin. A path segment and spectral bin *dependent* cut-off value, *ODmin*, is set to the minimum spectral grid point extinction OD multiplied by 10⁻⁴, but not be less than *ODcut*. In the loop over active lines, spectral Voigt ODs are computed outward from line center. Once a line's absorption OD drops below *ODmin*, the computationally expensive calculation is terminated. For baseline 8.0 to 12.0 μm test cases, downward viewing from 20 km, the OD tolerance technique reduces processing time by ~50%.

7.3.2 Validation of MODTRAN's Band Model Line-center Transmittances

Spectral transmittances, generated using the initial MODTRAN6 LBL approach, were convolved with the MODTRAN 0.1 cm⁻¹ non-overlapping, contiguous rectangle slit functions. The resulting data was compared to MODTRAN6's highest fidelity spectral bin transmittances, generated using the MODTRAN 0.1 cm⁻¹ BM and its statistical Correlated-*k* (CK) algorithm. The results of an 8.0 to 12.0 μm comparison are illustrated on the left in Figure 7.2 for a nadir path from 20 km altitude through the model mid-latitude summer (MLS) atmosphere. The CK (black) and LBL (red) spectral transmittances are plotted in the lower half of the figure; above, LBL minus CK residuals, offset by 1, are plotted on an expanded scale.

The 9.6 μm O₃ band has always been the most difficult molecular band for MODTRAN to model; the relative flat O₃ vertical density profile below 35 km (number densities between 1 and 6 × 10¹² / cm³) and overlap with low altitude H₂O absorption create problems for both the BM and CK methods. This is clearly evident in Figure 7.2, where 9.6 μm band transmittance residuals range between -0.05 to +0.13.

Outside of this region, most residuals have a magnitude is less than 0.02, and there appears to be little bias towards too much or too little absorption. This latter fact suggests that degrading the data to coarser resolution will significantly lower these residuals; it is generally recommended that MODTRAN spectral output be degraded with a slit function whose full-width at half-maximum (FWHM) is at least twice the BM resolution (0.2 cm^{-1} in this case).

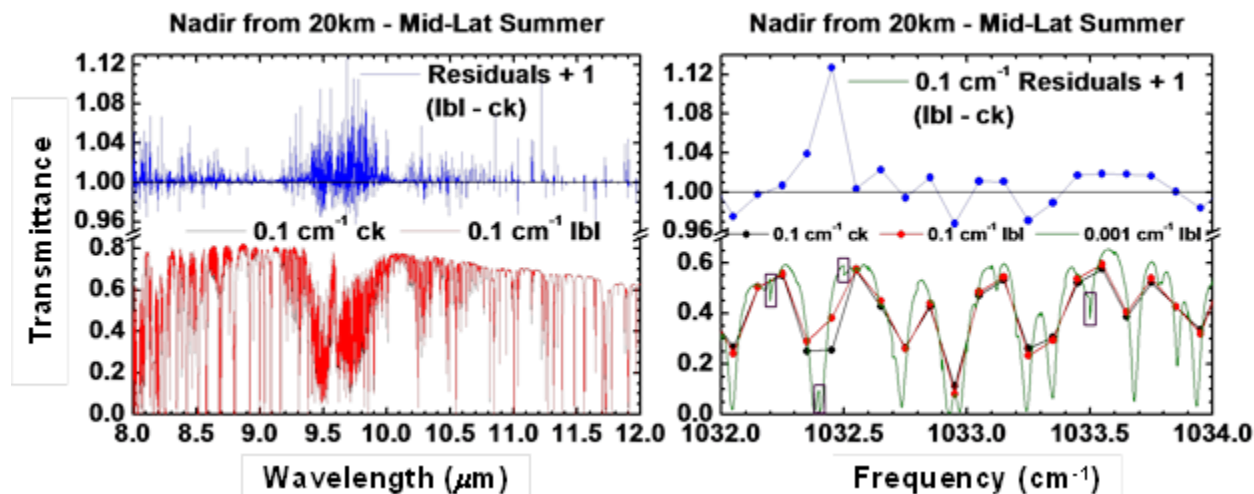


Figure 7.2. (left) Validation of the MODTRAN6 Correlated- k Transmittances (black) against the initial MODTRAN6 LBL Calculations (red) at 0.1 cm^{-1} spectral Resolution over the 8.0 to $12.0 \text{ }\mu\text{m}$ Band-Pass, and (right) a 2 cm^{-1} Sub-Region within the $9.6 \text{ }\mu\text{m}$ O_3 Band Containing the Largest Transmittance Residual Located at 1032.45 cm^{-1}

The LBL to CK comparison was performed at the MODTRAN band model finest spectral resolution to facilitate studying the source of the residuals. To this end, a 2.0 cm^{-1} region containing the largest residual, $t_{\text{lbl}} - t_{\text{ck}} = 0.13$, is shown on the right in Figure 7.2. This figure also contains the fine (0.001 cm^{-1}) spectral resolution LBL transmittance curve, shown in green. Focusing on the 0.1 cm^{-1} spectral bin between 1032.4 and 1032.5 cm^{-1} , the LBL spectrum seems to illustrate that there is a strong absorption doublet near the left edge. In actuality, there are 3 strong O_3 lines near the bin's left edge, with 1 atm pressure transition frequencies of 1032.3799 , 1032.4007 and $1032.4057 \text{ cm}^{-1}$; the 296 K strength of these lines are 0.9514 , 0.2754 and $0.6018 \text{ cm}^{-2}/\text{atm}$, respectively. The MODTRAN band model assumes absorption lines are randomly located within each spectral bin. Having two strong lines centered so close to an edge and to each other is a statistical anomaly. MODTRAN translates the line centers 0.025 cm^{-1} away from the edge. The shifting is performed to avoid possible problems fitting the line-tail contributions to a Padé approximant in the neighboring spectral bin, as was described above. For the 1032.4 to 1032.5 cm^{-1} bin, these small shifts lead to a large over-prediction of the absorption by the band model. A smaller over-prediction occurs for the neighboring spectral bin to the left, because it has an even stronger line located 0.0201 cm^{-1} from the edge is also shifted inward. The LBL 0.1 cm^{-1} spectral bin transmittance at 1032.45 cm^{-1} is 0.39 , while the CK transmittance is 0.26 . One observes from this analysis the current line center shifting is introducing errors in absorption that are unacceptably large, which strongly suggests that the band model methodology needs to be refined to reduce these residuals.

7.3.3 Spectral Discontinuities

A second problem is evident on the right in Figure 7.2. Purple boxes centered at 1032.2 , 1032.4 , 1032.5 and 1033.5 highlight large spectral bin edge discontinuities in the LBL spectral transmittances. This is unacceptable for a LBL model, which should provide the highest possible accuracy. The need to eliminate these discontinuities led to the reformulation of the MODTRAN LBL algorithm described in Section 7.4.

7.4 The Finalized MODTRAN6 Line-By-Line RT Approach

The challenge of the MODTRAN6 LBL algorithm is to eliminate discontinuities at spectral bin edges. The problem arises because MODTRAN6 solves the LBL problem within 0.1 cm^{-1} bins, and then concatenates these distinct solutions. A line centered in one bin is modeled via an explicit LBL calculation in that bin, but its contributions to the neighboring bins are determined from temperature- and pressure-interpolated pre-computed line-tail data. With the initial MODTRAN6 LBL model, the discontinuity error is largely due to the fact that the line-tail cross-sections are generated for lines whose centers are shifted if centered too close to an edge; no lines are shifted due to their proximity to a bin edge for the LBL line-center calculations. Simply revising the approach to eliminate the need for line shifting reduces the magnitude of the discontinuities, but they persist. As is shown below, additional upgrades to the formalism were required to produce smooth transitions across all the spectral bin edges.

A number of other problems had to be dealt with to validate the MODTRAN6 LBL algorithm against LBLRTM. MODTRAN has never concerned itself with modeling Lorentz self-broadening or modeling air pressure (or density) induced line shifts. These details cannot be ignored at LBL spectral resolutions. Another problem is that MODTRAN uses the Voigt line-shape function, which does not exhibit correct behavior in the microwave spectral region.

All of the above problems are addressed by the revised MODTRAN6 LBL algorithm. The following sub-sections describe the algorithm's updates to the line shape function, its use of a 2-bin line-center approach, the treatment of air-pressure (or density) induced line shifts and the modeling of Lorentz self-broadening.

7.4.1 Updating the Line-shape Function

MODTRAN6 models the absorption cross-section, $k_v(T; \gamma_c, \gamma_d)$, at spectral frequency ν for a molecular transition of strength $S_{if}(T)$ with a Voigt line-shape function, f_v :

$$k_v(T; \gamma_c, \gamma_d) = S_{if}(T) [f_\Delta(\gamma_c, \gamma_d) + f_\Sigma(\gamma_c, \gamma_d)]; \quad \Delta = \nu - \nu_{if}; \quad \Sigma = \nu + \nu_{if}$$

$$S_{if}(T) \propto \frac{\nu_{if}}{Q(T)} \left(1 - e^{-c_2 \nu_{if}/T}\right) e^{-c_2 E_i/T}; \quad f_\Delta(\gamma_c, \gamma_d) = \int_{-\infty}^{\infty} \frac{\gamma_c \exp(-t^2)}{\gamma_c^2 + (\Delta - \gamma_d t)^2} \frac{dt}{\pi^{3/2}}. \quad (7.2a)$$

Here, γ_c is the Lorentz (collision) half-width at half maximum (HWHM), γ_d is the Doppler half-width at $1/e$ of maximum, ν_{if} is the transition frequency from the initial (i) lower state to the final (f) upper state, $Q(T)$ is the total partition function at temperature T , c_2 is the second radiation constant, and E_i is the lower state energy. The Voigt line-shape function does not have the correct form at low spectral frequencies; in particular, the absorption cross-section should approach zero as the frequency approaches zero, but it does not. To remedy this problem, LBLRTM uses the Van Vleck and Huber (VVH) [1977] line-shape function, $\tilde{f}_v$, a form that transitions to the Voigt line-shape whenever ν/ν_{if} is near unity:

$$\tilde{k}_v(T; \gamma_c, \gamma_d) = \tilde{S}_{if}(T) [\tilde{f}_\Delta(\gamma_c, \gamma_d) + \tilde{f}_\Sigma(\gamma_c, \gamma_d)]$$

$$\tilde{S}_{if}(T) \equiv \frac{S_{if}(T)}{\nu_{if} \tanh(\frac{1}{2} c_2 \nu_{if}/T)}; \quad \tilde{f}_\Delta(\gamma_c, \gamma_d) \equiv \nu \tanh(\frac{1}{2} c_2 \nu/T) f_\Delta(\gamma_c, \gamma_d). \quad (7.2b)$$

Throughout most of the infrared, ν/ν_{if} is indeed near unity for frequencies within 25 cm^{-1} of ν_{if} ; so there is essentially no difference between the Voigt and VVH treatments in these regions.

In the finalized MODTRAN6 Line-By-Line RT implementation, the MODTRAN LBL calculations have been upgraded to use the VVH line-shape function when appropriate. In principle, the MODTRAN

line-tail database should also be generated using the VVH function. This upgrade has yet to be implemented. As noted in Section 7.2, one of the quantities used to determine the Padé approximant parameters is the integral of the line-shape curve over the spectral bin. Forcing the integral conditions insures preservation of the total integrated strength. In the program that generates these parameters, the Voigt line-shape function is computed using the original expansion due to Humlíček [1979]. The advantage of the Humlíček expansion over more modern algorithms is that its terms can be analytically integrated spectrally. Multiplying the Voigt line-shape function by the $(\nu/\nu_{if}) \tanh(\frac{1}{2} c_2 \nu/T) / \tanh(\frac{1}{2} c_2 \nu_{if}/T)$ factor complicates the computation of these integrals. To solve this problem, the $\tanh(\frac{1}{2} c_2 \nu/T)$ term can be expanded in a Taylor series about the central frequency for each spectral bin, ν_{cen} , yielding an analytically integrable expression; the variation of the hyperbolic tangent over the 0.1 cm^{-1} spectral bin should be small and the convergence of the Taylor series quick. This upgrade has yet to be implemented.

There is one additional upgrade that will have to be made to MODTRAN6 to provide a consistent implementation of the VVH line-shape function. As noted earlier, line-tail absorption contributions more than 25 cm^{-1} from line center are modeled via continua databases. This suggests that the continuum absorption cross-section contribution from a single Voigt molecular transition has the *No Center* form shown on the left in Figure 7.3. Such contributions would produce spectral continua with multiple discontinuities. To create smoothly varying continua, the continuum data for a single Voigt line is defined to include a flat plateau within $\pm 25 \text{ cm}^{-1}$ of line center, Figure 7.3 (center). When molecular absorption within this 50 cm^{-1} central region is modeled, the absorption cross-section value at $\pm 25 \text{ cm}^{-1}$ must be subtracted. The VVH line-shape function is not symmetric about the transition frequency. The question arises as to what value must be subtracted, or, equivalently, how is the continuum contribution defined within the central region? The answer is that the flat plateau is multiplied by the same factor as that used to convert the Voigt to a VVH line-shape, namely, $(\nu/\nu_{if}) \tanh(\frac{1}{2} c_2 \nu/T) / \tanh(\frac{1}{2} c_2 \nu_{if}/T)$. On the right in Figure 7.3, the effect of this factor is illustrated at standard temperature (273.15 K) for a molecular transition centered at 200 cm^{-1} . Although the plateau appears to simply be tilted, the slope is not constant due to the slight variations in the hyperbolic tangent term.

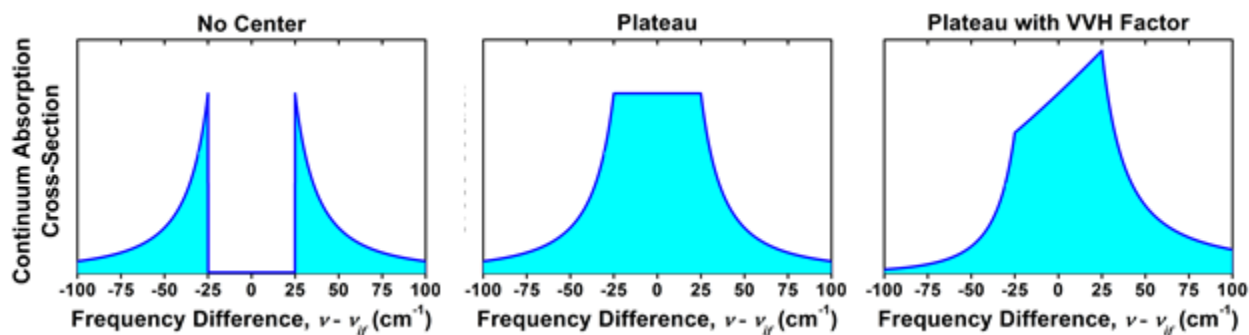


Figure 7.3. Continuum Absorption Cross-Section Contributions from a Single Line

7.4.2 A Two-bin line Center Approach Incorporating Air-density Induced Line Shifts

In the MODTRAN band model, molecular lines that are located too close to a band model spectral bin edge are translated inward towards the center of the bin. This is done because Padé approximant spectral fits of line-tails may be inaccurate for molecular transitions located just beyond a bin edge. The spectral slope for such lines approach zero and Doppler (Gaussian) contributions can be strong at low pressures. The stipulation is that the form of the Padé fits does not provide the flexibility to properly capture this spectral shape.

To avoid the need to shift line center positions within the MODTRAN6 LBL implementation, the Voigt line-shape of each molecular transition is explicitly modeled in two 0.1 cm^{-1} bins *instead of one*, Figure 7.4. Line-tails are still fit to a Padé approximant, but the extra bin provides a spectral buffer between the line center and its line-tail contributions. As Figure 7.4 illustrates, each spectral bin has a set

of line center transitions that it shares with its left neighbor (Figure 7.4 left) and a set that it shares with its right neighbor (Figure 7.4 right). In principle, each line is assigned the bin containing the line and its nearest neighbor. With such a selection, each line would be centered half a bin width or more from its line-tail region. This is illustrated in Figure 7.4 where the fine spectral structure for both the red and blue curves is separated from the modeled line-tails by $\sim 1/2$ of a spectral bin, i.e., 0.05 cm^{-1} .

The spectral bin assignment procedure is more complicated due to air pressure- or density-induced line shifts. According to HITRAN documentation [Rothman, *et al.*, 2013], transition frequencies vary with pressure according to the equation

$$\text{HITRAN : } \nu_{if}^H(P) \equiv \nu_{if}^{\text{vacuum}} + \nu_{if}^{\text{shift}} \left(\frac{P}{1 \text{ atm}} \right) . \quad (7.3a)$$

Within LBLRTM, temperature dependence is added, by defining an air density induced line shift:

$$\text{LBLRTM : } \nu_{if}^L(P, T) = \nu_{if}^L(P/T) \equiv \nu_{if}^{\text{vacuum}} + \nu_{if}^{\text{shift}} \left(\frac{P}{1 \text{ atm}} \right) \frac{296 \text{ K}}{T} . \quad (7.3b)$$

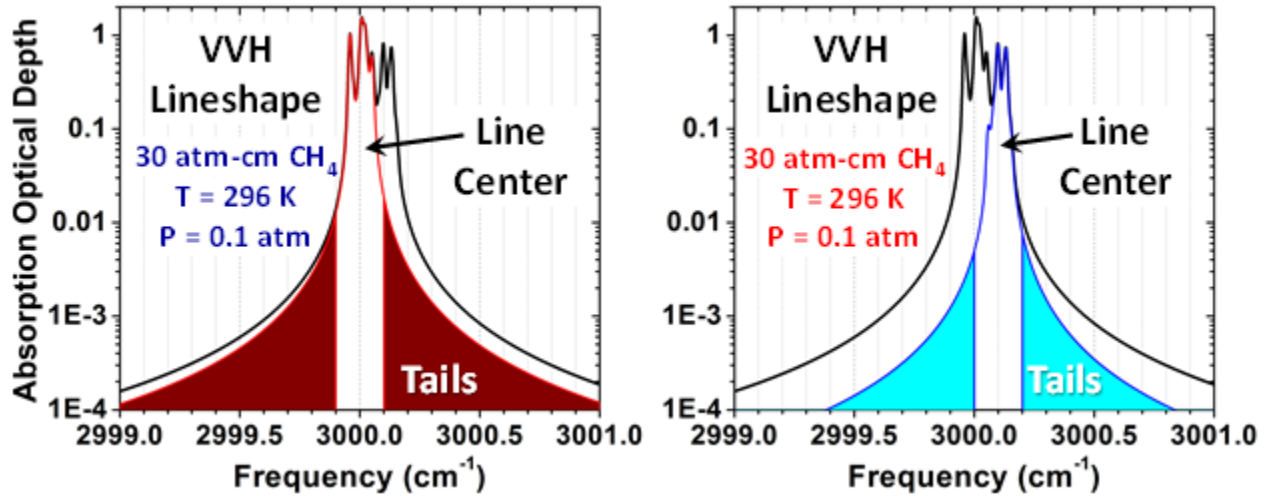


Figure 7.4. Total Optical Depth from all lines Contributing to the 0.1 cm^{-1} Spectral Bin between 3000.0 and 3000.1 cm^{-1} for a Homogeneous Path Containing 30 atm-cm of CH_4 at 296 K and 0.1 atm Pressure (black)

The MODTRAN6 LBL option is being validated against LBLRTM, so adopting the air density-induced line shift is the natural choice. However, there is a concern that Eq. (7.3b) may be inaccurate or untested for very low temperatures. The minimum MODTRAN band model temperature for terrestrial calculations is set to 180 K . For the LBL algorithm, this same temperature is used to provide a cut-off on the temperature dependence of the induced line shift:

$$\text{MODTRAN6 : } \nu_{if}^M(P, T) = \nu_{if}^{\text{vacuum}} + \nu_{if}^{\text{shift}} \left(\frac{P}{1 \text{ atm}} \right) \frac{296 \text{ K}}{\max(T, 180 \text{ K})} . \quad (7.3c)$$

Returning to the two-bin line center approach, the air density induced line shift can alter which spectral bin is closest to the transition frequency as pressures and temperatures vary along a LOS. If the molecular transitions contributing to line-tail data for a given spectral bin were allowed to change depending on the pair of (P, T) values, then interpolating line-tail data over pressure and temperature

would itself introduce spectral discontinuities. To avoid such problems, each HITRAN line is pre-assigned its two line center bins based on the range of line center positions between the vacuum frequency $\nu^{vac} \equiv \nu_{if}^{vacuum}$ and the frequency at 1 atm pressure and 180 K, $\nu_{180} \equiv \nu_{if}^M(1 \text{ atm}, 180 \text{ K})$. The intermediate transition frequency value, ν_{296} , defined as $\nu_{if}^M(1 \text{ atm}, 296 \text{ K})$, is also used in the assignment process. The spectral bin containing ν_{296} is bin N , where $N \times 0.1 \text{ cm}^{-1} < \nu_{296} \leq (N + 1) \times 0.1 \text{ cm}^{-1}$. The following hierarchical procedure is used to assign the two-line center bins:

1. If ν^{vac} and ν_{180} are both in spectral bin N , then
 - a. If $\min(\nu^{vac}, \nu_{180})$ is closer to bin $N - 1$ then $\max(\nu_0, \nu_1)$ is to bin $N + 1$, **select bins $N - 1$ and N**
 - b. Else, **select bins N and $N + 1$**
2. Else if $\min(\nu^{vac}, \nu_{180})$ is in bin $N - 1$ and $\max(\nu^{vac}, \nu_{180})$ is in bin N , **select bins $N - 1$ and N**
3. Else if $\min(\nu^{vac}, \nu_{180})$ is in bin N and $\max(\nu^{vac}, \nu_{180})$ is in bin $N + 1$, **select bins N and $N + 1$**
4. Else if ν^{vac} (and ν_{296}) are in spectral bin N , then
 - a. If ν_{180} is below bin N , **select bins $N - 1$ and N**
 - b. Else, **select bins N and $N + 1$**
5. Else if ν^{vac} is below bin N , **select bins $N - 1$ and N**
6. Else, **select bins N and $N + 1$.**

For all HITRAN 2012 lines, either criterion 1a, 1b, 2 or 3 is satisfied. If the other criteria were needed, it would imply that the range of line centers spans more than 2 spectral bins. If that had been the case for one of the HITRAN molecular transitions, it would have required that more than 2 bins be assigned as LBL bins for that line.⁴

After each HITRAN line is assigned its set of two line-center bins, the HITRAN database is reordered so that all lines assigned to bins 0 and 1 are listed first, followed by all lines assigned to bins 1 and 2, and so on. The first pair, bins 0 and 1, has the most lines, a total of 2,883. The N^{th} record of a binary, direct-access file named HITRAN2013.map contains a single integer equal to the number of the last line that contributed to bins $N - 1$ and N (if no lines contributed to bins $N - 1$ and N , the N^{th} record contains the number of the last line that did contribute to an earlier bin pair). Record #1 of HITRAN2012.map has the integer value 2,883 and Record #473,647, the last record, has the integer value 4,425,596 equal to the total number of HITRAN2012 lines.

The MODTRAN6 LBL algorithm reads in structured data for 4,096 HITRAN lines at a time. Two sets of the structured arrays are stored. Whenever new lines are needed for a given spectral bin pair, they are read into the older of the two structured arrays. The older array is guaranteed to no longer be required since no bin pair contains more than 2,883 ($< 4,096$) lines.

7.4.3 Modeling Lorentz Self-broadening

The Lorentz collisional HWHM is a function of air temperature T and pressure P as well as the partial pressure of the absorbing molecule. These dependencies are generally modeled via the equations

⁴ Computational speed could be increased by assigning a single line-center bin to the subset of lines which have transition frequencies that remain in the central portion of a single bin; that option complicates programming, so it has not been implemented.

$$\gamma_c \equiv \gamma_c(P_{self}, P, T) = \gamma_c^{air}(T)(P - P_{self}) + \gamma_c^{self}(T) P_{self}$$

$$\gamma_c^{air}(T) \equiv \gamma_c^{air} \left(\frac{296K}{T} \right)^n \quad \text{and} \quad \gamma_c^{self}(T) \equiv \gamma_c^{self} \left(\frac{296K}{T} \right)^n . \quad (7.4a)$$

Here, γ_c^{air} and γ_c^{self} are the air- and pressure-broadened half-widths [$\text{cm}^{-1}/\text{atm}$] at 1 atm pressure and 296 K, respectively; $\gamma_c^{air}(T)$ and $\gamma_c^{self}(T)$ are the corresponding temperature-dependent half-widths at 1 atm pressure; and n is temperature dependence exponent. MODTRAN has, to date, modeled the self-broadened half-width as being equal to the air-broadened value, but such an assumption is inappropriate for high spectral resolution LBL modeling. For H_2O , the ratio of the self-broadened half-width to its air-broadened complement, $\gamma_c^{self}/\gamma_c^{air}$, is typically of order 5. Since the partial pressure of H_2O at the ground for a moist atmosphere can exceed 2%, the self-broadened correction can lead to a 10% increase in the Lorentz half-width.

The Voigt line-shape is defined as the spectral convolution of the Doppler and Lorentz forms. The challenge associated with implementation of Eq. (7.4a) is that pre-calculated data, such as the line-tail absorption cross-sections, are defined before atmospheric constituent partial pressures are known and the Voigt line-shape depends on these quantities. The tables of Padé approximant line-tail parameters are defined as functions of air pressure and temperature; these tables would be unwieldy if they were to also depend on partial pressures. One avoids this complication by noting that the Lorentz line-shape varies with pressure only through its dependence on γ_c . Since the Doppler line-shape is independent of pressure, the Voigt line-shape must also only depend on pressure through γ_c .⁵ For a given line, one can define a Lorentzian pressure, P_L , by rearranging Eq. (7.4a):

$$\gamma_c = \gamma_c^{air}(T) \left[P + \left(\frac{\gamma_c^{self}}{\gamma_c^{air}} - 1 \right) P_{self} \right] \equiv \gamma_c^{air}(T) P_L . \quad (7.4b)$$

It follows that the Voigt line-shape for this molecular transition satisfies the relationship

$$f_v(\gamma_c, \gamma_d) = f_v[\gamma_c^{air}(T) P_L, \gamma_d(T)] . \quad (7.5)$$

This equation implies that one can pre-compute a line-tail database based solely on the Doppler half-width and Lorentz air-broadened half-width, and interpolate the pressure-dependence based on the Lorentzian pressure, P_L , to model self-broadening.

The MODTRAN6 line-tail Padé approximant fits of spectral absorption cross-sections over pressure, Eq. (7.1), are stored at two pressures, 1.0 and 0.1 atm. Often, these line-tails can be approximated by their Lorentzian form:

$$a_p(\delta_\nu) \approx \frac{1}{P} \sum_j \frac{(\gamma_c)_j / \pi}{\Delta_j^2 + (\gamma_c)_j^2} = \sum_j \frac{[\gamma_c^{air}(T)]_j / \pi}{\Delta_j^2 + [\gamma_c^{air}(T)]_j^2 P^2} ; \quad \Delta_j \equiv \nu - (\nu_{if})_j . \quad (7.6)$$

The summation is over all molecular lines j contributing line-tail absorption to a given band model spectral bin (see Figures 7.1 and 7.4). This approximation illustrates why the absorption cross-section curves are divided by pressure; since $\gamma_c = \gamma_c^{air}(T) P$, factoring out the pressure dependence in the numerator removes the dominant pressure dependence. The remaining term varies much more slowly

⁵ This is not completely accurate. The line-shape also depends on the pressure-induced line shift; however, the difference between the nominal line-shift and that due to the Lorentzian pressure of Eq. (4b) is inconsequential.

with pressure, which facilitates accurate interpolation and enables coarse pressure gridding. For the MODTRAN band model, the line-tail values are linearly interpolated in pressure-squared using the atmospheric pressure, and the resulting value is multiplied by atmospheric pressure to obtain band model spectral cross-sections, $\sigma_{bm}(\delta_\nu)$:

$$\sigma_{bm}(\delta_\nu) = \left(a_{1.0}(\delta_\nu) + [a_{0.1}(\delta_\nu) - a_{1.0}(\delta_\nu)] \frac{1.00 - P^2}{1.00 - 0.01} \right) P \quad (7.7)$$

To incorporate the effects of self-broadening, one needs to replace the use of atmospheric pressure, P , for interpolation with the Lorentzian pressure, P_L . There is a problem, however, because the partial pressure coefficient in Eq. (7.4b), $\gamma_c^{self}/\gamma_c^{air} - 1$, can differ for every molecular line. In actuality, the value is generally relatively constant or modeled as such if measurements or calculations are lacking. For the MODTRAN6 LBL algorithm, we introduce a new temperature-dependent line-tail band model parameter equal to an absorption cross-section weighted-average value of $\gamma_c^{self}/\gamma_c^{air} - 1$:

$$\left\langle \frac{\gamma_c^{self}}{\gamma_c^{air}} - 1 \right\rangle \equiv \frac{\sum_j \left(\frac{(\gamma_c^{self})_j}{(\gamma_c^{air})_j} - 1 \right) k_{v_cen}[T; (\gamma_c)_j, (\gamma_d)_j]}{\sum_j k_{v_cen}[T; (\gamma_c)_j, (\gamma_d)_j]} \quad (7.8)$$

The sums are computed over all lines j contributing to the line-tail parameters for a given spectral bin, and v_cen is the bin's central frequency. The parameter $\langle \gamma_c^{self}/\gamma_c^{air} - 1 \rangle$ is computed for each band model pressure and temperature grid point, but there is, understandably, no appreciable pressure dependence, so only the values at 1 atm pressure are stored.

For each spectral bin, the temperature-dependent Lorentzian pressure, P_L , is defined by the equation

$$P_L = P + \left\langle \frac{\gamma_c^{self}}{\gamma_c^{air}} - 1 \right\rangle P_{self} \quad (7.9)$$

Using P_L for interpolations immediately introduces a new problem. The Lorentzian pressure at the ground often exceeds 1 atm since γ_c^{self} usually exceeds γ_c^{air} . As noted earlier, the original line-tail data only had two pressure grid points, 1.0 and 0.1 atm. Extrapolating below 0.1 atm poses no serious problems since line-tail contributions are very weak at lower pressures. On the other hand, extrapolating above 1.0 atm is suspect. Since interpolations are performed on the pressure-squared and since uniform gridding is advantageous, two pressure-square grid points were added. The first was chosen half-way between 1.000 atm² and 0.010 atm², at 0.505 atm²; the second value was chosen to exceed 1.000 atm² at 1.495 atm². Correspondingly, the new set of pressure values is 0.1000, 0.7196, 1.0000 and 1.2227 atm. No Lorentzian pressure above this upper limit is expected for terrestrial calculations.

The Lorentzian pressure was inserted into Eq. (7.7) to determine the line-tail cross-sections for the MODTRAN6 LBL algorithm:

$$\sigma_{lbt}(\delta_\nu) = \left(a_i(\delta_\nu) + [a_{i+1}(\delta_\nu) - a_i(\delta_\nu)] \frac{P_i^2 - P_L^2}{P_i^2 - P_{i+1}^2} \right) P_L ; \left\{ \begin{array}{l} P_i \geq P_L \geq P_{i+1} \\ a_i(\delta_\nu) \equiv a_{p_i}(\delta_\nu) \end{array} \right. \left(\begin{array}{l} \text{1st interpolation} \\ \text{approach:} \\ \text{Unsuccessful} \end{array} \right) \quad (7.10a)$$

The MODTRAN6 LBL algorithm was run and compared to LBLRTM at high spectral resolution (0.001 cm⁻¹). Unfortunately, the residual curves continued to exhibit small discontinuities at bin edges

when the interpolation scheme of Eq. (7.10a) was implemented (this is illustrated below in Figure 7.8). Exponential interpolation in pressure-squared,

$$\sigma_{lbl}(\delta_\nu) = a_i(\delta_\nu) \left(\frac{a_{i+1}(\delta_\nu)}{a_i(\delta_\nu)} \right)^{\frac{P_i^2 - P_L^2}{P_i^2 - P_{i+1}^2}} P_L ; \quad P_i \geq P_L \geq P_{i+1} \quad \left(\begin{array}{c} 2^{\text{nd}} \text{ interpolation} \\ \text{approach:} \\ \text{Unsuccessful} \end{array} \right) , \quad (7.10b)$$

fared no better. An examination of the test case exhibiting the largest discontinuity showed that it was occurring at a spectral bin edge dominated by the contribution from a single strong line. On one side of the edge, the line was modeled explicitly using a direct LBL calculation; on the other side, the line-tail data was interpolated in pressure-squared. Eq. (7.6) was revisited to consider the case of a single strong dominant line; in that case,

$$a_p(\delta_\nu) \approx \sum_j \frac{(\gamma_c^{air})_j / \pi}{\Delta_j^2 + (\gamma_c^{air})_j^2 P^2} \approx \frac{(\gamma_c^{air})_J / \pi}{\Delta_J^2 + (\gamma_c^{air})_J^2 P^2} ; \quad (\text{Single strong line}) , \quad (7.6')$$

where $j = J$ is the dominant line. In this case, the reciprocal of $a_p(\delta_\nu)$ truly is a linear function of pressure-squared. This dependence suggests that the *reciprocal* of the line-tail absorption cross-section (divided by P) should be linearly interpolated in pressure-squared:

$$\sigma_{lbl}(\delta_\nu) = \frac{P_L}{\frac{1}{a_i(\delta_\nu)} + \left[\frac{1}{a_{i+1}(\delta_\nu)} - \frac{1}{a_i(\delta_\nu)} \right] \frac{P_i^2 - P_L^2}{P_i^2 - P_{i+1}^2}} ; \quad \left\{ \begin{array}{l} P_i \geq P_L \geq P_{i+1} \\ \frac{1}{a_i(\delta_\nu)} = P_i \left(\frac{1 + d_1 \delta_\nu + d_2 \delta_\nu^2}{\sigma_0 + \sigma_1 \delta_\nu + \sigma_2 \delta_\nu^2} \right) \end{array} \right. \quad (7.10c)$$

(3rd interpolation approach: *Successful*)

When this method of interpolation was implemented, all significant observed transmittance discontinuities were eliminated. One discontinuity is observed in residual plots between MODTRAN LBL and LBLRTM, but its transmittance magnitude is only 0.001 (see description of Figure 7.6 below).

7.4.4 Modeling Doppler Line-Tail Contributions

At lower pressures, Doppler contributions become important. Even though the line tails are less important at lower pressures, their contributions are relevant. One should not expect that the expression for $P_L / \sigma_{lbl}(\delta_\nu)$, based on linear interpolation in pressure-squared between pressure grid points, would remain accurate for higher altitude (lower pressure) path segments. Since the line tails are computed at 4 pressure points, uniformly spaced in P^2 , a set of four 4-pt interpolation/extrapolation formulas were examined to see which approach minimized the spectral discontinuities. Each of the four proposed methods included a constant and pressure-squared term so that the dominant dependence of Eq. (7.10c) could be retained:

a) 4-Point Spline Interpolation in P-squared

$$g(P_L) \equiv \frac{P_L}{\sigma_{lbl}(\delta_\nu)} = A + B P_L^2 + C P_L^4 + D P_L^6 \quad (7.11a)$$

b) 4-Point Asymptotic Series in P-squared

$$g(P_L) \equiv \frac{P_L}{\sigma_{ibl}(\delta_v)} = \frac{D}{P_L^4} + \frac{C}{P_L^2} + B + A P_L^2 \quad (7.11b)$$

c) 4-Point Padé Approximate in P-squared

$$g(P_L) \equiv \frac{P_L}{\sigma_{ibl}(\delta_v)} = A + B P_L^2 + \frac{C}{D + P_L^2} \quad (7.11c)$$

d) 4-Point Spline Interpolation in P [with $P_i^2 = P_4^2 + (4 - i) \Delta^2$; $g_i \equiv g(P_i)$; $i = 1, 2, 3, 4$]

$$\begin{aligned} g(P_L) &\equiv \frac{P_L}{\sigma_{ibl}(\delta_v)} = A + B P_L + C P_L^2 + D P_L^3 \\ &= \frac{(P_L - P_2)(P_L - P_3)(P_L - P_4)}{(P_1 - P_2)(P_1 - P_3)(P_1 - P_4)} g_1 + \frac{(P_L - P_1)(P_L - P_3)(P_L - P_4)}{(P_2 - P_1)(P_2 - P_3)(P_2 - P_4)} g_2 \\ &\quad + \frac{(P_L - P_1)(P_L - P_2)(P_L - P_4)}{(P_3 - P_1)(P_3 - P_2)(P_3 - P_4)} g_3 + \frac{(P_L - P_1)(P_L - P_2)(P_L - P_3)}{(P_2 - P_4)(P_2 - P_3)(P_2 - P_1)} g_4 \end{aligned} \quad (7.11d)$$

The 4-point spline fit in pressure-squared, method (a), is a series expansion generalization of the quadratic fit. Since the $g(P_L)$ term is the denominator of the line tail absorption coefficient, $\sigma_{ibl}(\delta_v)$, the asymptotic expansion, method (b), introduces a P_L^4 factor in the numerator when P_L is small; this expansion insures that the line tail rapidly drops towards zero as the pressure decreases. Padé approximates often exhibit better convergence than polynomial fits, so method (c) was introduced; the drawback of the Padé approach is that one must protect against the denominator approaching zero.

These first 3 methods all model $g(P_L) \equiv P_L / \sigma_{ibl}(\delta_v)$ as a function of Lorentzian pressure-squared. Indeed, the form of Eq. (7.2a) verifies that P^2 dependence for positive pressures since the Lorentz half-width is proportional to pressure and the Doppler half-width is independent of pressure. However, the Doppler limit is independent of pressure. This suggests that as pressure approaches zero, many terms may be needed to make $P_L / g(P_L) \equiv P_L / \hat{g}(P_L^2)$ converge to a pressure-independent value. The 4-point spline interpolation in P_L was introduced to help improve convergence of $g(P_L)$ to a value proportional to pressure.

All four methods were tested by focusing on an observed discontinuity in optical depth at 6704.6 cm^{-1} arising from a path containing 44.43 atm-cm of H_2O at 240.1 K and a Lorentzian pressure of 0.3095 atm . Short of 6704.6 cm^{-1} , the LBL calculations account for more than 90% of the optical depth; long of 6704.6 cm^{-1} that contribution is modeled via the pre-computed line tails interpolated over pressure and temperature. As illustrated in the inset on the left in Figure 7.5, all the pressure-interpolation techniques produce curves that pass through the gridded (magenta diamonds) values. For pressures above $\sim 0.7 \text{ atm}$ ($P_L^2 = 0.49 \text{ atm}^2$), they all yield similar interpolated values. However, as Doppler contributions become increasingly important, the methods diverge. The asymptotic interpolation, method (b), completely fails, producing negative line tails between 0.01 and 0.83 atm^2 ; based on these results, this form was eliminated from further consideration.

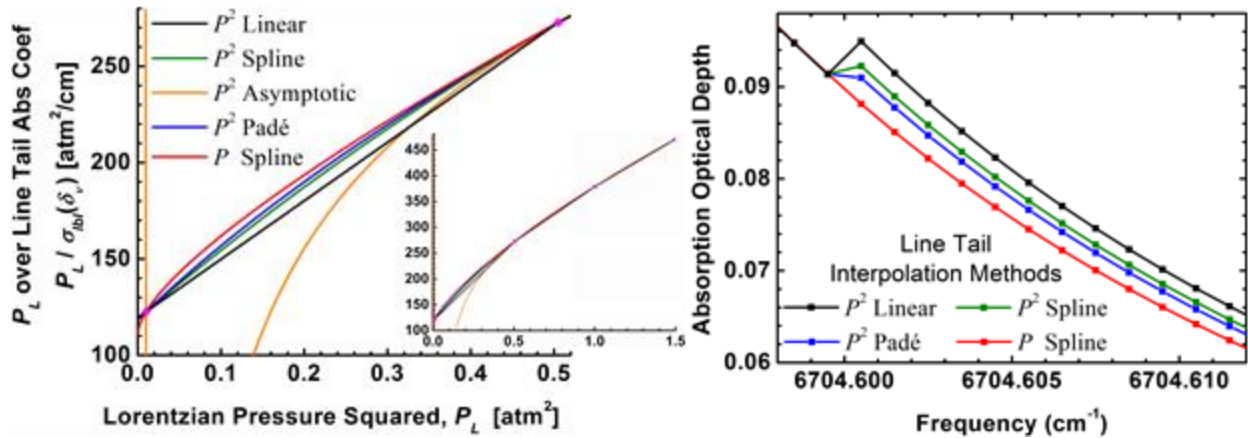


Figure 7.5. (left) Candidate Interpolation/Extrapolation Curves through 4 Values of the Pressure to line Tail Absorption Coefficient Ratio, and (right) Plots of Spectral Absorption Optical Depth Versus Frequency (right) Plots of spectral absorption optical depth versus frequency. Below 6704.6 cm⁻¹, first-principle line-by-line calculations were performed; above 6704.6 cm⁻¹, the pressure interpolated values at P_L^2 equal to 0.09579 atm² are plotted.

The other 3 methods produce physically reasonable curves (Figure 7.5 left), with $g(P_L)$ values exceeding the linear fit for P_L^2 values between 0.010 and 0.505 atm²; this translates to smaller optical depths in this region. These three 4-point methods and the linear interpolation all extrapolate to a positive constant value as P_L approaches zero atm with $g^{\text{linear}}(0) = 119.486$, $g^{\text{P2spline}}(0) = 118.825$, $g^{\text{P2pade}}(0) = 118.399$, and $g^{\text{Pspline}}(0) = 110.434$. This would suggest that the line tail optical depth drops to zero in the Doppler limit. While the Doppler line tail is not exactly zero, it does become inappreciably small.

To select between the 4 interpolation techniques (P_L^2 linear, P_L^2 spline, P_L^2 Padé and P_L spline), spectral absorption optical depth curves near 6704.6 cm⁻¹ are plotted in Figure 7.5 (right) for the selected test case scenario (44.43 atm-cm of H₂O at 240.1 K and a Lorentzian pressure of 0.3095 atm). As the figure illustrates, the P_L^2 linear interpolated optical depth exceeds the exact value by more than 0.007 (8% too high) at 6704.6005 cm⁻¹. The P_L^2 spline and P_L^2 Padé methods do better, but they still predict too much absorption optical depth by 0.004 and 0.003, respectively. For this one case, at least, the P_L spline interpolation clearly yields the best result, a smooth transition from LBL to line-tail pressure interpolation. Further testing confirms that the P_L spline interpolation method is best.

7.5 Validations of the MODTRAN6 Line-By-Line and Band Model RT Approach

Initial 8.0 to 12.0 μm band calculations have been run to validate the new MODTRAN6 LBL algorithm against LBLRTM and, in turn, to validate the MODTRAN6's band model against its new LBL algorithm. The band model validations are based on the scenario of Figure 7.2, a nadir view from 20 km altitude through the MLS model atmosphere. The scenario for validation against LBLRTM is similar, but the USS model atmosphere was chosen and the view angle was changed to 60° off-nadir to increase opacity and the influence of upper altitudes. The results are described in the following two sub-sections.

7.5.1 Validation of MODTRAN6 LBL against LBLRTM

Figure 7.6 contains the 8.0 to 12.0 μm comparison of MODTRAN6 LBL to LBLRTM at 0.001 cm⁻¹ spectral resolution. For frequencies below ~1180 cm⁻¹ (> 8.47 μm), the vast majority of the transmittance residuals (shown offset by 1 as the blue curve in Figure 7.6) have a magnitude less 0.01; the residuals ramp up to ~0.03 near 1240 cm⁻¹ (8.06 μm). Individual spectral points exhibit large differences between the two LBL models. Questions that need to be answered include:

- Why do low spectral resolution base-lines differ?
- What is the source of the *residual hash* in the 9.6 μm O_3 band spectral region?
- Why are there large residuals at sporadic and isolated spectral points?

The narrow spectral regions highlighted with green boxes are examined in more detail to shed light on these issues.

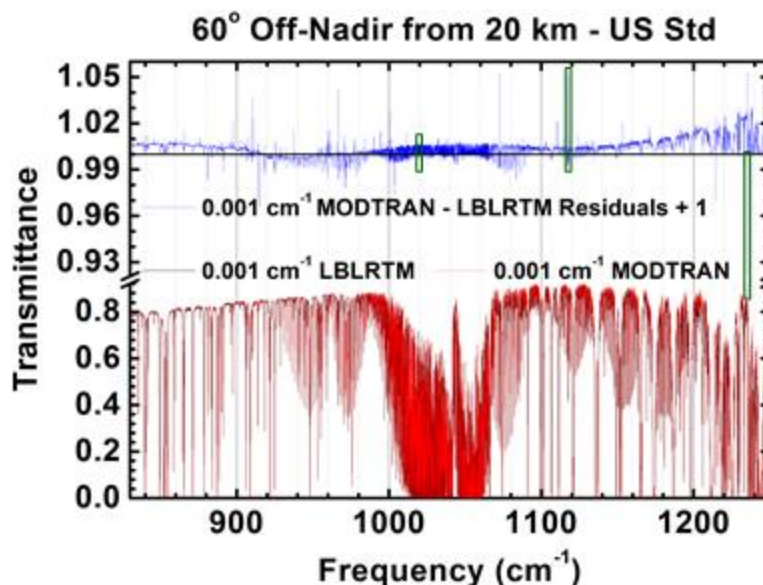


Figure 7.6. Validation of the MODTRAN6 LBL Algorithm (red) Against LBLRTM (black) on a 0.001 cm^{-1} Grid for a 60° Off-nadir View from 20 km through the USS Model Atmosphere

Figure 7.7 contains the plot of the *residual hash* between 1020.0 and 1020.9 cm^{-1} . One immediately observes that the MODTRAN (red) and LBLRTM (black) curves are very close; this good agreement is difficult to decipher from overcrowded spectra of Figure 7.6. The residual blue curve has an upper baseline value near 1.006; the upper baseline value occurs for transmittances between ~ 0.25 and ~ 0.65 . MODTRAN transmittances are too high, absorptivity too low. This translates to an error in absorption optical depth of ~ 0.013 [$\approx \ln(0.456 / 0.450)$]. At low transmittances (< 0.1), the MODTRAN and LBLRTM values are in good agreement. For a transmittance of 0.1, an optical depth error of 0.013 translates to a small transmittance error of 0.0013 $\{ \approx 0.1 - \exp[\ln(0.1) - 0.013] \}$. This seems to suggest that the *residual hash* arises from a relatively constant absorption optical depth difference. The only low-resolution extinction source that is modeled differently by MODTRAN6 and the current release of LBLRTM is the H_2O continuum. In these calculations, MODTRAN relied on the Clough-Kneizys H_2O continuum data extracted from LBLRTM in circa 2000. LBLRTM uses the state-of-the-art MT_CKD H_2O continuum [Delamere *et al.*, 2010; Payne *et al.*, 2011; Mlawer *et al.*, 2012]. The H_2O continuum in MODTRAN6 has since been updated to MT_CKD. This upgrade eliminated the low-resolution optical depth differences between the two models, the source of the *residual hash* in the 9.6 μm O_3 band spectral region.

It should be noted that the largest remaining bin edge discontinuity that has been observed is in Figure 7.7. There is no evidence of the discontinuity in the MODTRAN transmittance plot itself. However, the residual plot, on its expanded vertical scale, does exhibit a jump of magnitude 0.001 at 1020.6 cm^{-1} .

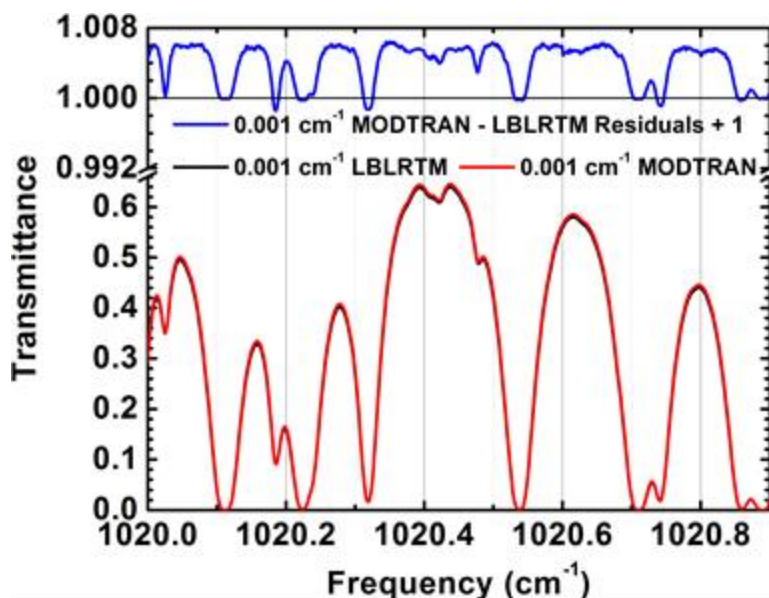


Figure 7.7. A 0.9 cm^{-1} Wide Sub-region of the $9.6 \text{ } \mu\text{m}$ O_3 Band Exhibiting the ‘Residual Hash’ of Figure 7.6

The residual of greatest magnitude from Figure 7.6 occurs near 1235.2 cm^{-1} ($8.096 \text{ } \mu\text{m}$). The MODTRAN6 LBL to LBLRTM comparison is re-plotted for a 0.2 cm^{-1} spectral region centered on the residual in Figure 7.8. The calculations exhibit a 0.025 baseline transmittance difference, suggesting that the MODTRAN6 H_2O continuum optical depth for this calculation is ~ 0.03 [$\approx \ln(0.850/0.825)$] less than the MT_CKD value (this baseline was eliminated with the introduction of MT_CKD into MODTRAN6). The absorption at 1235.2 cm^{-1} results from a doublet of H_2O lines. The two LBL calculations differ because they use different line compilation data. The MODTRAN6 LBL algorithm relies on HITRAN2012 data directly. AER, the distributors of LBLRTM, tailor the data for individual HITRAN lines based on their validations against field measurements. For the lines in question, the total HITRAN strength, $3.2 \times 10^{-25} \text{ cm/molec}$, is about 10% higher than the LBLRTM value, $2.9 \times 10^{-25} \text{ cm/molec}$. The half-widths and lower state energies are the same in the two line files, but AER assigns slightly higher frequencies to the line centers. These changes produce the spectral transmittance difference illustrated in Figure 7.8.

A final residual from Figure 7.6 is highlighted in Figure 7.9. This residual, located at 1176.2 cm^{-1} ($8.502 \text{ } \mu\text{m}$), corresponds to the largest MODTRAN minus LBLRTM transmittance difference. In this case, AER has introduced very small changes to the H_2O line’s transition frequency and line strength, but the air-broadened half-width has been decreased by 20%. This produces a significant increase in the line center absorption, as illustrated.

Figure 7.9 also contains small snippets of the residual curve (in green) that was obtained when the Padé line-tail data was interpolated linearly in pressure-squared, Eq. (7.10a). A significant (0.003) discontinuity is present at the 1117.7 cm^{-1} bin edge. Upgrading to a four-point spline fit in pressure of the reciprocal of the Padé line-tail data, Eq. (7.11d), completely eliminates both this continuity and the smaller one two spectral bins away, at 1117.5 cm^{-1} , in the blue curve.

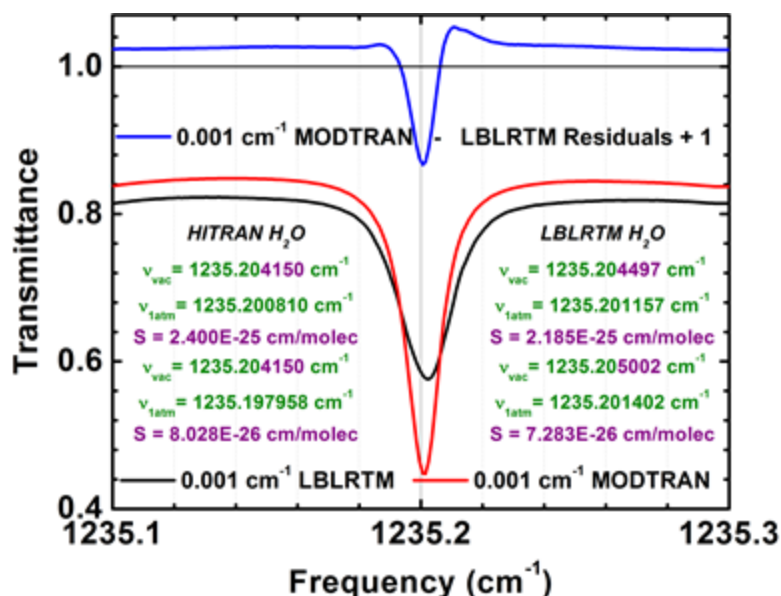


Figure 7.8. A Narrow Spectral Region Containing the Largest Magnitude Residual from Figure 7.6, and listing line positions and strengths for MODTRAN (HITRAN) on the left and for LBLRTM on the right

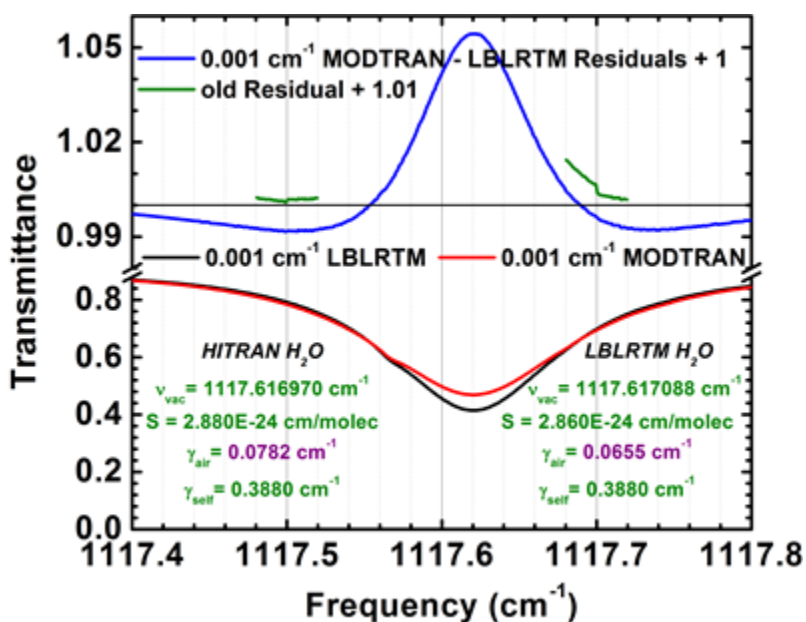


Figure 7.9. A Narrow Spectral Region Containing a Figure 7.6 Residual that Arises from Different Air-broadened Half-widths

7.5.2 Validation of the MODTRAN6 Band Model against MODTRAN6 LBL

In Sub-section 7.3.2 and Figure 7.2, the MODTRAN band model line-center transmittances were validated against the initial MODTRAN6 LBL results. In some respects, these were unfair comparisons. The LBL line center calculations were performed for the true molecular transition frequencies, but the band model line center data included shifted line centers. The calculations used the same set of line-tail data. For the correct validation, one must compare the 0.1 cm⁻¹ MODTRAN6 band model molecular transmittances against the MODTRAN6 LBL molecular transmittances convolved over the same 0.1 cm⁻¹ bins, rather than just comparing the line center transmittance contributions. Those comparisons are presented in this section.

The nadir calculations from 20 km through the MLS model atmosphere are plotted in Figure 7.10. In contrast to Figure 7.2, the data are plotted as a function of frequency not wavelength. This was done to facilitate selection of sub-regions for further investigation. The goal here is to understand the source of the MODTRAN band model errors so that refinements to the statistical approach can be proposed, developed and validated. Three of the largest residuals will be examined. They are highlighted by orange boxes in Figure 7.10.

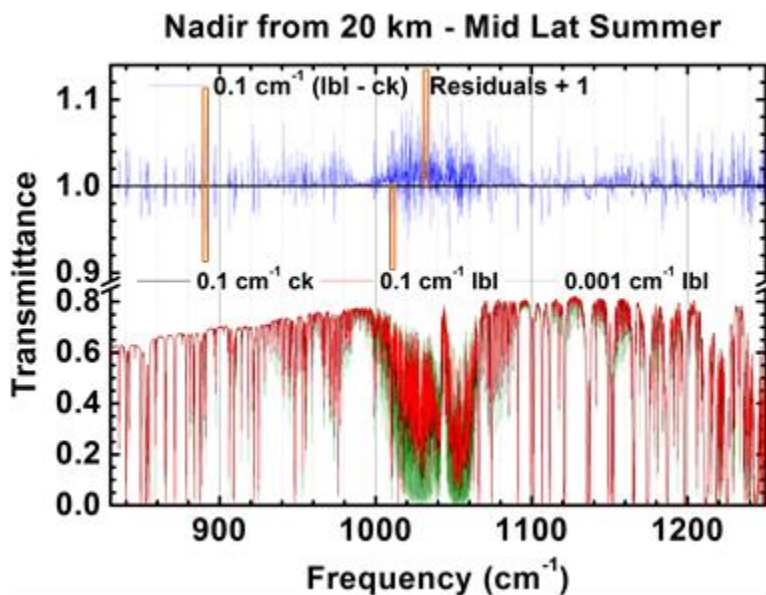


Figure 7.10. Validation of the MODTRAN6 Band Model against the MODTRAN6 LBL Algorithm for a Nadir Calculation from 20 km through the MLS Model Atmosphere

Figure 7.11 contains the updated version of Figure 7.2 (right) using the final MODTRAN6 LBL formalism. All of the spectral bin edge discontinuities are gone (it may appear as if some discontinuities are still present, but examination of the high resolution data verifies that all changes in slope actually occur before reaching any of the edges). The large transmittance residual at 1032.45 cm^{-1} has the same magnitude (0.13) as before. The shifting of the three H_2O lines located near the 1032.5 cm^{-1} by the band model approach still produces an error (see Sub-section 7.3.2 for the detailed description). Correcting the LBL approach did not have any appreciable effect of the magnitude of the residual.

A pair of positive and negative transmittance residuals of substantial magnitude (~ 0.1) are observed for the two spectral bins sharing the 891.3 cm^{-1} edge, Figure 7.12. The source of this absorption is an isolated H_2O line with vacuum and 1 atm pressure transition frequencies of 891.30071 and 891.29522 cm^{-1} , respectively, at 296 K. The air pressure-induced line-shift pushes the line center into the 891.25 cm^{-1} centered spectral bin. This is a classic case of a line straddling a bin edge. The band model shifts the molecular transition inward to 891.275 cm^{-1} . This produces too much absorption for the 891.25 cm^{-1} centered bin and too little for the 891.35 cm^{-1} centered bin. This is once again an example where the band model line shifting is generating unacceptably large errors.

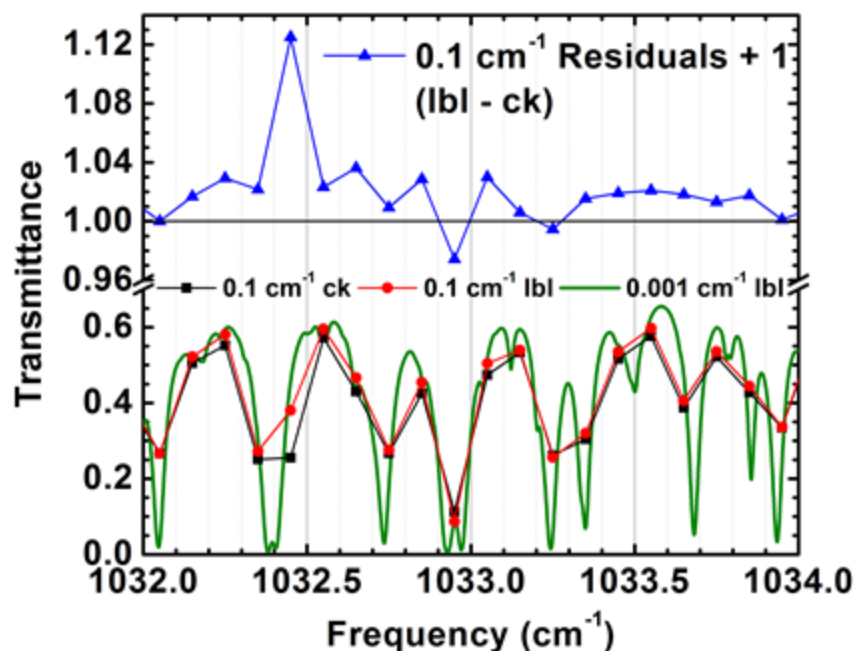


Figure 7.11. Same Conditions as Figure 7.10, but Highlighting Residuals at 1032.45 cm⁻¹

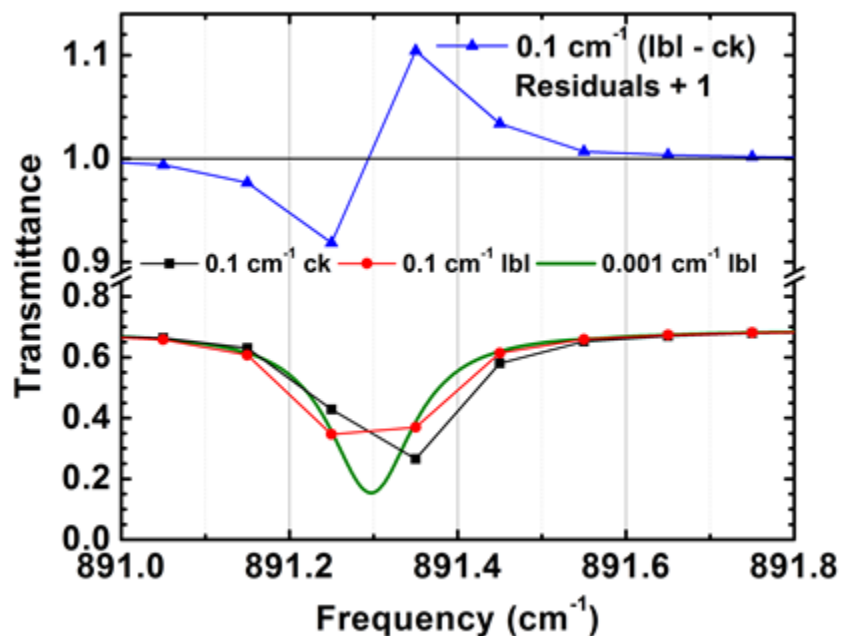


Figure 7.12. Same Conditions as Figure 7.10, but Highlighting Residuals at 891.3 cm⁻¹

Possibly, the most interesting 0.1 cm⁻¹ spectral bin transmittance residual case is the one illustrated in Figure 7.13. As before, there is a H₂O line straddling a bin edge with a 1 atm, 296 K air pressure-induced line shifted transition frequency of 1010.79863 cm⁻¹. The band model shifting of this line center towards the center of the 1010.75 cm⁻¹ bin produces too much BM absorption in that bin, and too little in the higher frequency neighboring bin.

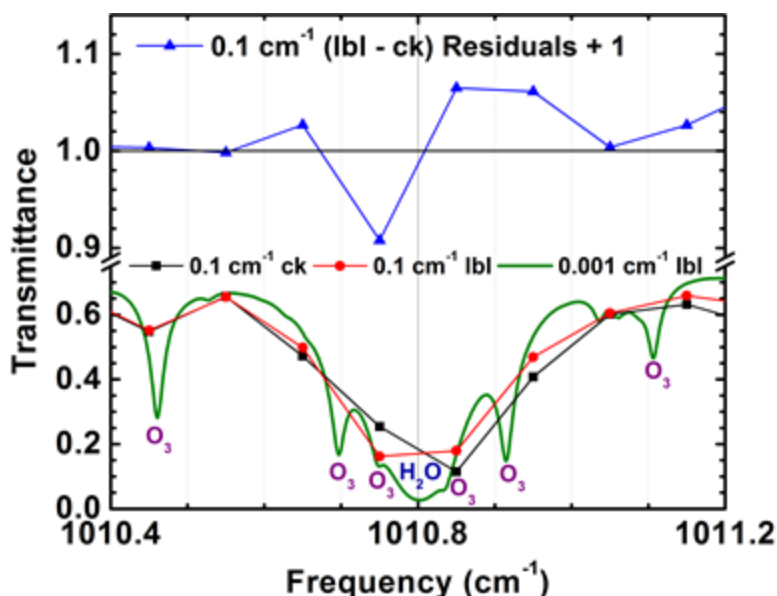


Figure 7.13. Same Conditions as Figure 7.10, but Highlighting Residuals at 1010.8 cm⁻¹

There are also multiple strong O₃ absorption lines in the region, the strongest being a line centered near 1010.835 cm⁻¹. The 1 atm, 296 K air-broadened Lorentz half-width for this line, γ_c , is 0.0723 cm⁻¹, substantially larger than the 0.0425 cm⁻¹ air-broadened Lorentz half-width of the H₂O line. Even so, the absorption from O₃ lines is much *less* broad than that from the H₂O line. The reason is fairly obvious. The geometry of the observation is nadir from 20 km. A significant fraction of the ozone absorption occurs at the higher altitudes due to the relatively constant O₃ profile over the altitude range of the simulation. The H₂O, on the other hand, primarily resides in the boundary and lower tropospheric layers. Since the Lorentzian width varies proportional to pressure, the O₃ absorption features are much narrower than the H₂O absorption feature.

The overlap between H₂O and O₃ absorption within the two spectral bins sharing the 1010.8 cm⁻¹ edge introduces another concern, and source of error when compared to the LBL calculations. The MODTRAN band model calculations assume that the absorption from all molecular species is randomly correlated. In practice, that means the product of their path transmittances equals the combined species transmittance. That is a reasonable approximation for the spectrum of Figure 7.13. However, the band model calculation was run using MODTRAN's statistical Correlated-*k* algorithm. With CK, the working assumption regarding line overlap changes. CK models absorption as being spectrally correlated among path segments. In other words, the location within a spectral bin of strongest absorption source for path segments near the ground needs to be the same as its location for upper altitude path segments. Clearly, the H₂O and O₃ absorption features are not spectrally correlated for the two spectral bins under consideration. One must conclude that the lack of spectral correlation is a second source of error in the comparisons against the LBL data.

7.6 Summary and Future Work

A novel approach has been developed and validated for performing LBL spectral transmittance calculations using MODTRAN6. The algorithm is unique in that high spectral resolution calculations are performed independently within 0.1 cm⁻¹ spectral bins, and these pockets of spectral data are subsequently pieced together. The processing is efficient because on-the-fly spectral line-shape calculations are only performed for lines centered within a 0.2 cm⁻¹ range for each spectral grid point. Line-tail absorption cross-sections are pre-computed on a fine temperature and coarse pressure grid for each molecule. A Lorentzian pressure band model parameter is introduced for accurately modeling self-broadening.

Spectral discontinuities at bin edges are essentially eliminated by a 4-term spline fit in pressure for the reciprocal of line-tail cross-sections over pressure.

Validations of the MODTRAN6 LBL algorithm against LBLRTM found two sources of discrepancies. Large residuals for individual lines arose because of differences in the line data files. Future comparisons will be made by utilizing common lines files for two LBL models. Low resolution residuals arose due to different molecular continua. MODTRAN6 has since been upgraded with the current version of the MT_CKD H₂O continuum. Remaining continua utilized by LBLRTM should also be integrated into MODTRAN6.

The MODTRAN6 LBL algorithm development and the MODTRAN6 band model validation efforts both led to recommendations for the band model. The development effort illustrated how the band model can be upgraded to include the effects of Lorentz self-broadening, to model air density-induced line shifts and to improve the interpolation of line-tail data. The validation test cases highlighted problems associated with shifting line centers to avoid bin edges, and with assuming spectral correlation; both of these problems will require additional study to find effective solutions.

APPENDIX A: INTEGRATING THE VOIGT FUNCTION RAISED TO A POWER

J. Maxwell Riestenberg and A. Berk

The MODTRAN expression for single-line finite-bin molecular transmittance requires calculation of spectral integrals over the Voigt lineshape function raised to integer powers. Analytic expansions are defined for these integrals when the domain of integration is restricted to line tail regions. In this appendix, expressions are derived for the spectral integration from line center out to infinity. The first step involves determining an expression for the Voigt lineshape that facilitates evaluation of the moment integrals. This expression is not new, but its derivation is provided in Section A.1 for completeness. The basic notation and problem are laid out in Section A.2, and the equation for the 2nd moment integral derived. Although this expression appears in the open literature [Reichel, 1969], attempts to find its derivation were unsuccessful. In Section A.3, the 3rd moment integral is solved, and Section 3.2.4 provides the beginning for a derivation of the 4th moment integral.

A.1 Deriving the Voigt Cosine Expression

“Voigt Cosine Expression”

Max Riestenberg

January 2, 2014

The (normalized) Lorentzian distribution is defined to be

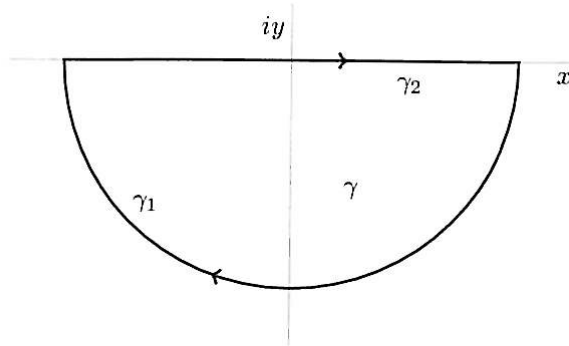
$$L(\nu) = \frac{\alpha/\pi}{\nu^2 + \alpha^2} \quad (1)$$

Its Fourier Transform is then

$$FL(t) = \frac{\alpha}{\pi} \int_{-\infty}^{\infty} \frac{e^{-it\nu}}{\nu^2 + \alpha^2} d\nu \quad (2)$$

This integral can be evaluated by the Residue Theorem. Consider the contour, defined for $R > \alpha, t > 0$

$$\begin{aligned} \gamma_1(s) &= Re^{-is}, & 0 \leq s \leq \pi \\ \gamma_2(s) &= s, & -R \leq s \leq R \end{aligned}$$



$$\int_{\gamma} \frac{e^{-itz}}{z^2 + \alpha^2} dz = 2\pi i \cdot \text{Int}_{\gamma}(-i\alpha) \text{Res}_f(-i\alpha) = 2\pi i(-1) \frac{e^{-it(-i\alpha)}}{-2i\alpha} = \frac{\pi}{\alpha} e^{-t\alpha}$$

On the other hand,

$$\int_{\gamma_1} \frac{e^{-itz}}{z^2 + \alpha^2} dz \leq \text{length}(\gamma_1) \cdot \max_{\pi \leq t \leq 2\pi} |f \circ \gamma(t)| \leq \pi R \cdot \frac{1}{R^2 - \alpha^2} \sim \frac{\pi}{R} \rightarrow 0$$

$$\int_{\gamma_2} \frac{e^{-itz}}{z^2 + \alpha^2} dz = \int_{-R}^R \frac{e^{-its}}{s^2 + \alpha^2} ds \rightarrow \int_{-\infty}^{\infty} \frac{e^{-its}}{t^2 + \alpha^2} ds$$

The $k < 0$ case works out similarly, and we obtain:

$$FL(t) = \frac{\alpha}{\pi} \int_{-\infty}^{\infty} \frac{e^{-its}}{s^2 + \alpha^2} ds = \frac{\alpha}{\pi} \int_{\gamma} \frac{e^{-itz}}{z^2 + \alpha^2} dz = e^{-|t|\alpha}$$

The (normalized) Gaussian distribution is defined as

$$G(\nu) = \frac{\beta}{\sqrt{\pi}} e^{-\beta^2 \nu^2} \quad (3)$$

and its Fourier Transform is

$$FG(t) = \frac{\beta}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\beta^2 \nu^2 + it\nu} d\nu \quad (4)$$

which may be computed by completing the square:

$$-\beta^2 \nu^2 + it\nu = -(\beta\nu + it/2\beta)^2 - t^2/4\beta^2$$

$$FG(t) = \frac{\beta}{\sqrt{\pi}} e^{-t^2/4\beta^2} \int_{-\infty}^{\infty} e^{-(\beta\nu + it/2\beta)^2} d\nu = e^{-t^2/4\beta^2}$$

Now we put the facts together. By evenness of L and G , the Voigt function may be given by:

$$V = L \star G = F \circ F(L \star G) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-t^2/4\beta^2 - |t|\alpha} \cos(\nu t) dt \quad (5)$$

I have to include the prefactor of $1/2\pi$ because I used a non-unitary transformation above. Consider the change of variables $s = t/\beta$

$$V = \frac{1}{\pi} \int_0^{\infty} e^{-t^2/4\beta^2 - t\alpha} \cos(\nu t) dt = \frac{\beta}{\pi} \int_0^{\infty} e^{-s^2/4 - s\alpha\beta} \cos(\nu\beta s) ds$$

Now the scaling $x = \nu\beta$ and $y = \alpha\beta$ (since $\beta = 1/\gamma_d$). We know the real part of the Fadeeva function is related to the Voigt function, so:

$$U = \frac{\sqrt{\pi}}{\beta} \cdot V = \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-t^2/4 - yt} \cos(xt) dt \quad (6)$$

is the final result.

A.2 Integration of the Square of the Voigt Line-Shape Function

Voigt Power Integrals at $\Delta = \infty$ Quick Write-Up

Our interest began with integrals of the form

$$\int_0^\Delta U(x, y)^n dx, \quad U(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-t^2/4-yt} \cos(xt) dt$$

$\Delta = \infty$ presents a somewhat friendlier integral. For example:

$$\begin{aligned} \int_0^\infty U(x, y) dx &= \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-t^2/4-yt} \int_0^\infty \cos(xt) dx dt \\ &= \sqrt{\pi} \int_0^\infty e^{-t^2/4-yt} \delta(t) dt \\ &= \sqrt{\pi} e^0 / 2 \end{aligned}$$

This follows from the fact that the Fourier Cosine Transform of $\sqrt{2\pi}\delta(t)$ is 1, which in turn has Fourier Cosine Transform $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \cos(xt) dx$. By evenness, these are equal. Note that the domain of integration “splits” the delta function, so we must divide by 2.

We can use the Werner formula:

$$\begin{aligned} \int_0^\infty \cos(xs) \cos(xt) dx &= \frac{1}{2} \int_0^\infty \cos x(s-t) + \cos x(s+t) dx \\ &= \frac{\pi}{2} [\delta(s-t) + \delta(s+t)] \end{aligned}$$

to obtain:

$$\begin{aligned} \int_0^\infty U(x, y)^2 dx &= \frac{1}{\pi} \iiint e^{-s^2/4-yt} e^{-t^2/4-yt} \cos(xs) \cos(xt) ds dt dx \\ &= \frac{1}{2} \int_0^\infty e^{-t^2/4-yt} e^{-t^2/4-yt} dt \\ &= \sqrt{\frac{\pi}{8}} U(0, \sqrt{2}y) \end{aligned}$$

Note that the domain of integration no longer “splits” the delta function, since $s = t = 0$ is now “only” the corner of our 2-dimensional domain (ie measure zero).

Now, it gets a bit tricky. We attempt to compute the $n = 3$ case.

A.3 Integration of the Voigt Line-Shape Function Raised to the 3rd Power

To simplify integration of the Voigt line-shape function raised to the 3rd power, the Voigt cosine expression from Section A.1 is inserted into the integrand:

$$\int_0^\infty U(x, y)^3 dx = \frac{1}{\pi^{3/2}} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2}{4} - y(s+t+u) \right] \cos(xs) \cos(xt) \cos(xu) ds dt du dx \quad . \quad (\text{A.3.1})$$

The product of cosines is re-written as a sum of cosines, and the order of integration can be switched so that the x integration is performed first:

$$\int_0^\infty U(x, y)^3 dx = \frac{1}{4\pi^{3/2}} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2}{4} - y(s+t+u) \right] \int_0^\infty \left\{ \begin{aligned} &\cos[x(s+t+u)] + \cos[x(s+t-u)] \\ &+ \cos[x(s-t+u)] + \cos[x(s-t-u)] \end{aligned} \right\} dx ds dt du \quad . \quad (\text{A.3.2})$$

Applying the Werner formula, the integral over x is evaluated:

$$\int_0^\infty U(x, y)^3 dx = \frac{1}{4\sqrt{\pi}} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2}{4} - y(s+t+u) \right] \left[\begin{aligned} &\delta(s+t+u) + \delta(s+t-u) \\ &+ \delta(s-t+u) + \delta(s-t-u) \end{aligned} \right] ds dt du \quad . \quad (\text{A.3.3})$$

The integral over s is evaluated next. The $\delta(s+t+u)$ term yields a value only when s, t and u are all zero; the subsequent integration over t (or u) equals zero. The s integral over $\delta(s-t-u)$ sets s equal to $t+u$. By symmetry, the integrals over $\delta(s+t-u)$ and $\delta(s-t+u)$ are equal. The s integral over $\delta(s-t-u)$ sets s equal to $t-u \geq 0$. The inequality arises because the domain of s . The implication is that the integral over t must begin at u . Thus, the s integration equals

$$\begin{aligned} \int_0^\infty U(x, y)^3 dx &= \frac{1}{4\sqrt{\pi}} \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u)^2 + t^2 + u^2}{4} - 2y(t+u) \right] dt du \\ &+ \frac{1}{2\sqrt{\pi}} \int_0^\infty \int_u^\infty \exp \left[-\frac{(t-u)^2 + t^2 + u^2}{4} - 2yt \right] dt du \quad . \end{aligned} \quad (\text{A.3.4})$$

Replacing the integration variable t with $t' = t - u$, the integral on the second line becomes equal to the one on the first line:

$$\begin{aligned} \int_0^\infty U(x, y)^3 dx &= \frac{3}{4\sqrt{\pi}} \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u)^2 + t^2 + u^2}{4} - 2y(t+u) \right] dt du \\ &= \frac{3}{4\sqrt{\pi}} \int_0^\infty \int_0^\infty \exp \left[-\frac{t^2 + tu + u^2}{2} - 2y(t+u) \right] dt du \quad . \end{aligned} \quad (\text{A.3.5})$$

Next, integration over t is replaced by integration over β where $-\beta$ is defined equal to the argument of the exponent:

$$\beta = \frac{t^2 + tu + u^2}{2} + 2y(t+u) \Rightarrow t^2 + (u+4y)t + u(u+4y) - 2\beta = 0 \quad . \quad (\text{A.3.6})$$

Solving the quadratic equation, t is defined as a function of u and β :

$$t = -\frac{u+4y}{2} + \sqrt{\left(\frac{u+4y}{2}\right)^2 - u(u+4y) + 2\beta}$$

$$dt = \frac{d\beta}{\sqrt{2\beta + 4y^2 - 2yu - 3u^2/4}}$$
(A.3.7)

Since the integration of t extends from 0 to ∞ , the integration of β extends from $u(u+4y)/2$ to ∞ :

$$\int_0^\infty U(x, y)^3 dx = \frac{3}{4\sqrt{\pi}} \int_0^\infty \int_{u(u+4y)/2}^\infty \frac{\exp(-\beta)}{\sqrt{2\beta + 4y^2 - 2yu - 3u^2/4}} d\chi du \quad .$$
(A.3.8)

Switching order of integration, noting that

$$\beta = u(u+4y)/2 \Rightarrow u^2 + 4yu - 2\beta = 0 \Rightarrow u = -2y + \sqrt{4y^2 + 2\beta} \quad ,$$
(A.3.9)

One obtains

$$\begin{aligned} \int_0^\infty U(x, y)^3 dx &= \frac{3}{4\sqrt{\pi}} \int_0^\infty \exp(-\beta) \int_0^{\sqrt{2\beta+4y^2}-2y} \frac{du}{\sqrt{2\beta + 4y^2 - 2yu - 3u^2/4}} d\chi \\ &= \sqrt{\frac{3}{4\pi}} \int_0^\infty \exp(-\beta) \int_0^{\sqrt{2\beta+4y^2}-2y} \frac{du}{\sqrt{8\beta/3 + 16y^2/3 - 8yu/3 - u^2}} d\chi \quad . \end{aligned}$$
(A.3.10)

Introduce $v = u + 4y/3$ to eliminate the linear term in u from the square root:

$$\int_0^\infty U(x, y)^3 dx = \sqrt{\frac{3}{4\pi}} \int_0^\infty \exp(-\beta) \int_{4y/3}^{\sqrt{2\beta+4y^2}-2y/3} \frac{dv}{\sqrt{8\beta/3 + 64y^2/9 - v^2}} d\chi \quad .$$
(A.3.11)

Trigonometric substitution,

$$v = \frac{2}{3} \sqrt{6\beta + 16y^2} \sin \theta \quad ; \quad dv = \frac{2}{3} \sqrt{6\beta + 16y^2} \cos \theta \quad ; \quad \theta = \arcsin \left(\frac{3v/2}{\sqrt{6\beta + 16y^2}} \right) \quad ,$$
(A.3.12)

enables the inner integral to be solved:

$$\begin{aligned} \int_0^\infty U(x, y)^3 dx &= \sqrt{\frac{3}{4\pi}} \int_0^\infty \exp(-\beta) \int_{\arcsin\left(\frac{2y}{\sqrt{6\beta+16y^2}}\right)}^{\arcsin\left(\frac{3\sqrt{\beta/2+y^2}-y}{\sqrt{6\beta+16y^2}}\right)} d\theta d\chi \\ &= \sqrt{\frac{3}{4\pi}} \int_0^\infty e^{-\beta} \left[\arcsin \left(\frac{3\sqrt{\beta/2+y^2}-y}{\sqrt{6\beta+16y^2}} \right) - \arcsin \left(\frac{2y}{\sqrt{6\beta+16y^2}} \right) \right] d\beta \quad . \end{aligned}$$
(A.3.13)

The two arcsin terms can be combined using the relationship

$$\begin{aligned}\arcsin(x) - \arcsin(y) &= \arcsin\{\sin[\arcsin(x) - \arcsin(y)]\} \\ &= \arcsin\{x \cos[\arcsin(y)] - y \cos[\arcsin(x)]\} .\end{aligned}\tag{A.3.14}$$

In preparation, the cosine of the arc-sine terms are evaluated.

$$\begin{aligned}\cos\left[\arcsin\left(\frac{3\sqrt{\beta/2 + y^2} - y}{\sqrt{6\beta + 16y^2}}\right)\right] &= \sqrt{1 - \frac{9\beta/2 + 10y^2 - 6y\sqrt{\beta/2 + y^2}}{6\beta + 16y^2}} \\ &= \sqrt{\frac{3\beta/2 + 6y^2 + 6y\sqrt{\beta/2 + y^2}}{6\beta + 16y^2}} = \frac{\sqrt{3}\left(y + \sqrt{\beta/2 + y^2}\right)}{\sqrt{6\beta + 16y^2}} .\end{aligned}\tag{A.3.15}$$

$$\cos\left[\arcsin\left(\frac{2y}{\sqrt{6\beta + 16y^2}}\right)\right] = \sqrt{1 - \frac{2y^2}{3\beta + 8y^2}} = \sqrt{\frac{3\beta + 6y^2}{3\beta + 8y^2}} = \frac{\sqrt{3}\sqrt{\beta + 2y^2}}{\sqrt{3\beta + 8y^2}} .\tag{A.3.16}$$

Inserting into Eq. (A.3.13) gives

$$\begin{aligned}\int_0^\infty U(x, y)^3 dx &= \sqrt{\frac{3}{4\pi}} \int_0^\infty e^{-\beta} \arcsin \left[\frac{\frac{\sqrt{3}\sqrt{\beta + 2y^2}}{\sqrt{3\beta + 8y^2}} \left(\frac{3\sqrt{\beta/2 + y^2} - y}{\sqrt{6\beta + 16y^2}} \right) - \frac{\sqrt{3}\left(y + \sqrt{\beta/2 + y^2}\right)}{\sqrt{6\beta + 16y^2}} \left(\frac{2y}{\sqrt{6\beta + 16y^2}} \right)} \right] d\beta \\ &= \sqrt{\frac{3}{4\pi}} \int_0^\infty e^{-\beta} \arcsin \left[\frac{3\left(\beta + 2y^2\right) - y\sqrt{2}\sqrt{\beta + 2y^2}}{2\left(3\beta + 8y^2\right)/\sqrt{3}} - \frac{2y^2 + y\sqrt{2}\sqrt{\beta + 2y^2}}{2\left(3\beta + 8y^2\right)/\sqrt{3}} \right] d\beta \\ &= \sqrt{\frac{3}{4\pi}} \int_0^\infty e^{-\beta} \arcsin \left[\frac{3\beta + 4y^2 - y\sqrt{2}\sqrt{\beta + 2y^2}}{2\left(3\beta + 8y^2\right)/\sqrt{3}} \right] d\beta \\ &= \sqrt{\frac{3}{4\pi}} \int_0^\infty e^{-\beta} \arcsin \left[\frac{\left(\sqrt{\beta + 2y^2} - y\sqrt{2}\right)\left(3\sqrt{\beta + 2y^2} + y\sqrt{2}\right)}{2\left(3\beta + 8y^2\right)/\sqrt{3}} \right] d\beta .\end{aligned}\tag{A.3.17}$$

Given an integral, I , of the form

$$I = \int_a^b \exp(-x) g(x) dx\tag{A.3.18}$$

where $g(x)$ is an invertible function such that $x(g)$ can be defined on $[a, b]$, then one can integrate by parts as follows:

$$u = g \quad , \quad dv = \exp[-x(g)]dx \quad ; \quad du = dg \quad , \quad v = -\exp[-x(g)]\tag{A.3.19}$$

giving

$$\begin{aligned}
 I &= -g \exp[-x(g)] \Big|_{g=g(a)}^{g=g(b)} + \int_{g(a)}^{g(b)} \exp[-x(g)] dg \\
 &= g(a) \exp(-a) - g(b) \exp(-b) + \int_{g(a)}^{g(b)} \exp[-x(g)] dg
 \end{aligned} \tag{A.3.20}$$

This relationship was used previously by Berk [2013] to derive the expansion for t_{sl} , Eq. (5.9).

To apply Eq. (A.3.20) to the integral of Eq. (A.3.17), the arc-sine in the integrand is set equal to $\phi(\beta)$ and an expression for $\beta(\phi)$ is derived.

$$\begin{aligned}
 \phi \equiv \phi(\beta) &= \arcsin \left[\frac{\left(\sqrt{\beta + 2y^2} - y\sqrt{2} \right) \left(3\sqrt{\beta + 2y^2} + y\sqrt{2} \right)}{2 \left(3\beta + 8y^2 \right) / \sqrt{3}} \right] \\
 \Rightarrow \sin \phi &= \frac{\left(\sqrt{\beta + 2y^2} - y\sqrt{2} \right) \left(3\sqrt{\beta + 2y^2} + y\sqrt{2} \right)}{\frac{2}{\sqrt{3}} \left[3 \left(\beta + 2y^2 \right) + 2y^2 \right]} \\
 \Rightarrow 0 &= 3 \left(\beta + 2y^2 \right) - y 2\sqrt{2} \sqrt{\beta + 2y^2} - 2y^2 - \sin \phi \left[2\sqrt{3} \left(\beta + 2y^2 \right) + \frac{4}{\sqrt{3}} y^2 \right] \\
 \Rightarrow 0 &= \left(3 - 2\sqrt{3} \sin \phi \right) \left(\beta + 2y^2 \right) - y 2\sqrt{2} \sqrt{\beta + 2y^2} - \left(2 + \frac{4}{\sqrt{3}} \sin \phi \right) y^2
 \end{aligned} \tag{A.3.21}$$

The quadratic equation in $\sqrt{\beta + 2y^2}$ is solved [the positive root insures that $\sqrt{\beta + 2y^2}$ will be positive since the denominator is non-negative ($\sin \phi$ does not exceed $\frac{1}{2}\sqrt{3}$ for any β)]:

$$\begin{aligned}
 \sqrt{\beta + 2y^2} &= \frac{y\sqrt{2} + \sqrt{2y^2 + \left(\sqrt{3} - 2\sin \phi \right) \left(2\sqrt{3} + 4\sin \phi \right) y^2}}{3 - 2\sqrt{3} \sin \phi} = \frac{y\sqrt{2}}{2\sqrt{3}} \left(\frac{1 + \sqrt{4 - 4\sin^2 \phi}}{\sqrt{3}/2 - \sin \phi} \right) \\
 &= \frac{y}{\sqrt{3}/2} \left(\frac{1/2 + \cos \phi}{\sqrt{3}/2 - \sin \phi} \right) = \frac{y}{\sqrt{3}/2} \left(\frac{\cos(\pi/3) + \cos \phi}{\sin(\pi/3) - \sin \phi} \right) = \frac{y}{\sqrt{3}/2} \left(\frac{\cos(\pi/6 + \phi/2) \cos(\pi/6 - \phi/2)}{\cos(\pi/6 + \phi/2) \sin(\pi/6 - \phi/2)} \right) \\
 &= \frac{y}{\sqrt{3}/2} \cot \left(\frac{\pi}{6} - \frac{\phi}{2} \right)
 \end{aligned} \tag{A.3.22}$$

It follows that $\beta(\phi)$ equals

$$\beta(\phi) = \frac{2y^2}{3} \cot^2 \left(\frac{\pi}{6} - \frac{\phi}{2} \right) - 2y^2 \tag{A.3.23}$$

From the first line of Eq. (A.3.21), one observes that $\phi(0) = 0$ and $\lim_{\beta \rightarrow \infty} \phi(\beta) = \arcsin(\frac{1}{2}\sqrt{3}) = \pi/3$. Therefore, the boundary terms from Eq. (A.3.20) are zero:

$$\phi(0) \exp(0) - \lim_{\beta \rightarrow \infty} [\phi(\beta) \exp(-\beta)] = 0 \tag{A.3.24}$$

Inserting Eq. (A.3.23) into Eq. (A.3.20) produces a much simplified expression for the Voigt line-shape function 3rd power integral:

$$\int_0^{\infty} U(x, y)^3 dx = \sqrt{\frac{3}{4\pi}} \exp\left(2y^2\right) \int_0^{\pi/3} \exp\left[-\frac{2y^2}{3} \cot^2\left(\frac{\pi}{6} - \frac{\phi}{2}\right)\right] d\phi . \quad (\text{A.3.25})$$

Setting ϕ equal to $\pi/6 - \phi/2$ gives

$$\int_0^{\infty} U(x, y)^3 dx = \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_0^{\pi/6} \exp\left[-\frac{2y^2}{3} \cot^2(\phi)\right] d\phi . \quad (\text{A.3.26})$$

Finally, t is set equal to $\cot(\phi)$

$$\int_0^{\infty} U(x, y)^3 dx = \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_{\sqrt{3}}^{\infty} \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt . \quad (\text{A.3.27})$$

Optimally, this integral could be expressed in terms of special functions. Two related integral can be expressed as such [Abramowitz and Stegun, Eqs. 7.4.11 and 7.4.12]:

$$\begin{aligned} \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_0^{\infty} \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt &= \frac{\sqrt{3}\pi}{2} \exp\left(\frac{8y^2}{3}\right) \operatorname{erfc}\left(\frac{y}{\sqrt{3/2}}\right) \quad \text{and} \\ \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_0^1 \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \left[1 - \operatorname{erf}^2\left(\frac{y}{\sqrt{3/2}}\right)\right] . \end{aligned} \quad (\text{A.3.28})$$

These can be combine to determine the integral from 1 to ∞ :

$$\begin{aligned} \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_1^{\infty} \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \operatorname{erfc}\left(\frac{y}{\sqrt{3/2}}\right) \left\{2 - \left[1 + \operatorname{erf}\left(\frac{y}{\sqrt{3/2}}\right)\right]\right\} \\ &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \operatorname{erfc}^2\left(\frac{y}{\sqrt{3/2}}\right) . \end{aligned} \quad (\text{A.3.29})$$

This expression is used to re-write Eq. (A.3.27) in terms of a finite integral

$$\begin{aligned} \int_0^{\infty} U(x, y)^3 dx &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \operatorname{erfc}^2\left(\frac{y}{\sqrt{3/2}}\right) - \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_1^{\sqrt{3}} \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt \\ &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \left\{ \operatorname{erfc}^2\left(\frac{y}{\sqrt{3/2}}\right) - \frac{4}{\pi} \int_1^{\sqrt{3}} \frac{\exp\left[-\frac{2}{3}y^2(1+t^2)\right]}{1+t^2} dt \right\} \\ &= \frac{\sqrt{3}\pi}{4} \exp\left(\frac{8y^2}{3}\right) \left[\operatorname{erfc}^2\left(\frac{y}{\sqrt{3/2}}\right) - \frac{4}{\pi} \int_{\pi/4}^{\pi/3} \exp\left(-\frac{2y^2 \sec^2 \eta}{3}\right) d\eta \right] . \end{aligned} \quad (\text{A.3.30})$$

The final expression is obtained by setting $t = \tan \eta$.

A final comment is warranted. Considerable effort was expended transforming the integral of Eq. (A.3.13) into Eq. (A.3.27). Many unanticipated simplifications arose during the manipulations. This suggests that a much more direct path to the final answer may exist. If so, the authors have failed to find it.

A.4 Integration of the Voigt Line-Shape Function Raised to the 4th Power

The integration of the Voigt line-shape function raised to the 4th power can also be initiated by inserting the Voigt cosine expression from Section A.1 into the integrand:

$$\int_0^\infty U(x, y)^4 dx = \frac{1}{\pi^2} \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2 + v^2}{4} \right] \cos(xs) \cos(xt) \cos(xu) \cos(xv) ds dt du dv dx \quad (\text{A.4.1})$$

As before, the product of cosines is re-written as a sum of cosines, and the order of integration can be switched so that the x integration is performed first:

$$\begin{aligned} & \int_0^\infty U(x, y)^4 dx \\ &= \frac{1}{8\pi^2} \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2 + v^2}{4} \right] \left[\begin{aligned} & \cos[x(s+t+u+v)] + \cos[x(s+t+u-v)] \\ & + \cos[x(s+t-u+v)] + \cos[x(s+t-u-v)] \\ & + \cos[x(s-t+u+v)] + \cos[x(s-t+u-v)] \\ & + \cos[x(s-t-u+v)] + \cos[x(s-t-u-v)] \end{aligned} \right] dx ds dt du dv \quad (\text{A.4.2}) \end{aligned}$$

Applying the Werner formula, the integral over x yields the sum of delta functions:

$$\int_0^\infty U(x, y)^4 dx = \frac{1}{8\pi^2} \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2 + v^2}{4} \right] \left[\begin{aligned} & \delta(s+t+u+v) + \delta(s+t+u-v) \\ & + \delta(s+t-u+v) + \delta(s+t-u-v) \\ & + \delta(s-t+u+v) + \delta(s-t+u-v) \\ & + \delta(s-t-u+v) + \delta(s-t-u-v) \end{aligned} \right] ds dt du dv \quad (\text{A.4.3})$$

The integral over s is evaluated next. The $\delta(s+t+u+v)$ term yields a value only when s, t, u and v are all zero; the subsequent integration over t (or u or v) equals zero. By symmetry, the integrals over $\delta(s+t+u-v)$, $\delta(s+t-u+v)$ and $\delta(s-t+u+v)$ are equal. Similarly, symmetry requires that the integrals over $\delta(s+t-u-v)$, $\delta(s-t+u-v)$ and $\delta(s-t-u+v)$ be equal.

$$\int_0^\infty U(x, y)^4 dx = \frac{1}{8\pi^2} \int_0^\infty \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{s^2 + t^2 + u^2 + v^2}{4} \right] \left[\begin{aligned} & 3\delta(s-t+u+v) + 3\delta(s-t+u-v) \\ & + \delta(s-t-u-v) \end{aligned} \right] ds dt du dv \quad (\text{A.4.4})$$

The s integral over $\delta(s-t-u-v)$ sets s equal to $t+u+v$. The s integral over $\delta(s-t+u+v)$ sets s equal to $t-u-v \geq 0$. The inequality arises because the domain of s . The implication is that the integral over t must begin at $u+v$. The s integral over $\delta(s-t+u-v)$ sets s equal to $t-u+v \geq 0$. The implication is that the integral over t must begin at $u-v$, and the integral over u must begin at v . Thus, the s integration equals

$$\begin{aligned}
\int_0^\infty U(x, y)^4 dx &= \frac{3}{8\pi} \int_0^\infty \int_0^\infty \int_{u+v}^\infty \exp \left[-\frac{(t-u-v)^2 + t^2 + u^2 + v^2}{4} - 2yt \right] dt du dv \\
&+ \frac{3}{8\pi} \int_0^\infty \int_v^\infty \int_{u-v}^\infty \exp \left[-\frac{(t-u+v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+v) \right] dt du dv \\
&+ \frac{1}{8\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u+v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+u+v) \right] dt du dv .
\end{aligned} \tag{A.4.5}$$

Replace the integration variable t with $t' = t - u - v$ in the first line and t with $t' = t - u + v$ in the second line (and then drop the prime):

$$\begin{aligned}
\int_0^\infty U(x, y)^4 dx &= \frac{3}{8\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u+v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+u+v) \right] dt du dv \\
&+ \frac{3}{8\pi} \int_0^\infty \int_v^\infty \int_0^\infty \exp \left[-\frac{(t+u-v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+u) \right] dt du dv \\
&+ \frac{1}{8\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u+v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+u+v) \right] dt du dv .
\end{aligned} \tag{A.4.6}$$

Finally, replace the integration variable u with $u' = u - v$ in the second line, and combine first and third lines:

$$\begin{aligned}
\int_0^\infty U(x, y)^4 dx &= \frac{1}{2\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u+v)^2 + t^2 + u^2 + v^2}{4} - 2y(t+u+v) \right] dt du dv \\
&+ \frac{3}{8\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{(t+u)^2 + t^2 + (u+v)^2 + v^2}{4} - 2y(t+u+v) \right] dt du dv \\
&= \frac{1}{2\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{t^2 + u^2 + v^2 + tu + uv + vt}{2} - 2y(t+u+v) \right] dt du dv \\
&+ \frac{3}{8\pi} \int_0^\infty \int_0^\infty \int_0^\infty \exp \left[-\frac{t^2 + u^2 + v^2 + tu + uv}{2} - 2y(t+u+v) \right] dt du dv .
\end{aligned} \tag{A.4.7}$$

For the 4th power integral, two integrals must be evaluated. The previous analysis would suggest setting the argument of each exponential to an integration variable, β , that replaces t , and then switching order of integration so that the outer integration is over β . This analysis has not been yet been pursued.

$$\begin{aligned}
\int_0^\infty U(x, y)^3 dx &= \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{1+t^2} dt \\
&= \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{t^2\left(1+1/t^2\right)} dt \\
&= \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{t^2} \left(1 - \frac{1}{t^2} + \frac{1}{t^4} - \dots\right) dt \\
&= \sqrt{\frac{3}{\pi}} \exp\left(2y^2\right) \sum_{j=0}^\infty \int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{-\left(-t^2\right)^{j+1}} dt \quad .
\end{aligned}
\tag{A.3.27}$$

$$u = \exp\left(-2y^2 t^2/3\right) \quad , \quad dv = \frac{dt}{t^2} \quad , \quad du = \frac{-4y^2 t}{3} \exp\left(-2y^2 t^2/3\right) dt \quad , \quad v = \frac{-1}{t}$$

$$\int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{t^2} dt = -\frac{\exp\left(-2y^2 t^2/3\right)}{t} \Bigg|_{t=\sqrt{3}}^{t=\infty} - \frac{4y^2}{3} \int_{\sqrt{3}}^\infty \exp\left(-2y^2 t^2/3\right) dt$$

$$= \frac{\exp\left(-2y^2\right)}{\sqrt{3}} - \sqrt{\frac{2\pi}{3}} y \operatorname{erfc}\left(\sqrt{2} y\right)$$

$$u = \exp\left(-2y^2 t^2/3\right) \quad , \quad dv = \frac{-dt}{t^4} \quad , \quad du = \frac{-4y^2 t}{3} \exp\left(-2y^2 t^2/3\right) dt \quad , \quad v = \frac{1}{3t^3}$$

$$-\int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{t^4} dt = \frac{\exp\left(-2y^2 t^2/3\right)}{3t^3} \Bigg|_{t=\sqrt{3}}^{t=\infty} + \frac{4y^2}{9} \int_{\sqrt{3}}^\infty \frac{\exp\left(-2y^2 t^2/3\right)}{t^2} dt$$

$$= \frac{-\exp\left(-2y^2\right)}{9\sqrt{3}} + \frac{4y^2}{9} \left[\frac{\exp\left(-2y^2\right)}{\sqrt{3}} - \sqrt{\frac{2\pi}{3}} y \operatorname{erfc}\left(\sqrt{2} y\right) \right]$$

$$= \frac{\exp\left(-2y^2\right)}{9\sqrt{3}} \left(-1 + 4y^2\right) - \frac{4}{9} \sqrt{\frac{2\pi}{3}} y^3 \operatorname{erfc}\left(\sqrt{2} y\right)$$

$$\int_{\sqrt{3}}^\infty \left(\frac{1}{t^2} - \frac{1}{t^4}\right) \exp\left(-2y^2 t^2/3\right) dt = \frac{4\left(2 + y^2\right) \exp\left(-2y^2\right)}{9\sqrt{3}} - \sqrt{\frac{2\pi}{3}} y \left(1 + \frac{4y^2}{9}\right) \operatorname{erfc}\left(\sqrt{2} y\right)$$

$$\begin{aligned}
\exp\left(-8y^2/3\right)\int_0^\infty U(x,y)^3 dx &= \sqrt{\frac{3}{\pi}} \int_{\sqrt{3}}^\infty \frac{\exp\left[-\frac{2y^2}{3}\left(1+t^2\right)\right]}{1+t^2} dt \quad . \\
\frac{d}{dy}\left[\exp\left(-8y^2/3\right)\int_0^\infty U(x,y)^3 dx\right] &= \sqrt{\frac{3}{\pi}} \frac{d}{dy} \int_{\sqrt{3}}^\infty \frac{\exp\left[-\frac{2y^2}{3}\left(1+t^2\right)\right]}{1+t^2} dt \\
&= -\frac{4y}{\sqrt{3\pi}} \int_{\sqrt{3}}^\infty \exp\left[-\frac{2y^2}{3}\left(1+t^2\right)\right] dt \\
&= -\frac{4y}{\sqrt{3\pi}} \exp\left(-2y^2/3\right) \int_{\sqrt{3}}^\infty \left(-\frac{2y^2 t^2}{3}\right) dt \\
&= -2\sqrt{2/\pi} \exp\left(-2y^2/3\right) \int_{y\sqrt{2}}^\infty \left(-t^2\right) dt \\
&= -\sqrt{2} \exp\left(-2y^2/3\right) \operatorname{erfc}(y\sqrt{2}) \\
&= -\sqrt{2} \exp\left(-2y^2/3\right) \left[1 - \sqrt{8/\pi} \sum_{n=0}^\infty \frac{(-2)^n y^{2n+1}}{n!(2n+1)}\right] \quad \text{or} \\
&= -\sqrt{2} \exp\left(-2y^2/3\right) \left[1 - \sqrt{8/\pi} \exp\left(-2y^2\right) \sum_{n=0}^\infty \frac{(4)^n y^{2n+1}}{1 \cdot 3 \dots (2n+1)}\right] \quad .
\end{aligned}$$

$$\begin{aligned}
\exp\left(-8y^2/3\right)\int_0^\infty U(x,y)^3 dx &= -\sqrt{2}\int_y^\infty \exp\left(-2\tilde{y}^2/3\right)\left[1-\sqrt{8/\pi}\sum_{n=0}^\infty \frac{(-2)^n \tilde{y}^{2n+1}}{n!(2n+1)}\right] d\tilde{y} \\
&= -\sqrt{3}\int_{y\sqrt{2/3}}^\infty \exp\left(-\tilde{y}^2\right)\left[1-2\sqrt{3/\pi}\sum_{n=0}^\infty \frac{(-3)^n \tilde{y}^{2n+1}}{n!(2n+1)}\right] d\tilde{y} \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{6}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{(-3)^n}{n!(2n+1)}\int_{y\sqrt{2/3}}^\infty \tilde{y}^{2n+1}\exp\left(-\tilde{y}^2\right)d\tilde{y} \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{3}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{(-3)^n}{n!(2n+1)}\int_{2y^2/3}^\infty x^n \exp(-x)dx \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{n=0}^\infty \frac{(-3)^n}{n!(2n+1)}\sum_{k=0}^n (-n)_k \left(-\frac{2y^2}{3}\right)^{n-k} \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty \sum_{n=k}^\infty \frac{(-3)^n}{n!(2n+1)}(-n)_k \left(-\frac{2y^2}{3}\right)^{n-k} \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty \sum_{n=k}^\infty \frac{3^n}{(n-k)!(2n+1)}\left(\frac{2y^2}{3}\right)^{n-k} \\
&= \frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty 3^k \sum_{n=k}^\infty \frac{\left(2y^2\right)^{n-k}}{(n-k)!(2n+1)} - \frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty \frac{3^k}{y^{2k+1}}\sum_{n=0}^\infty \frac{2^n y^{2n+2k+1}}{n!(2n+2k+1)} - \frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty \frac{3^k}{y^{2k+1}}\sum_{n=0}^\infty \frac{2^n}{n!}\int_0^y \tilde{y}^{2n+2k} d\tilde{y} - \frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{3}{\sqrt{\pi}}\exp\left(-2y^2/3\right)\sum_{k=0}^\infty \frac{3^k}{y^{2k+1}}\int_0^y \tilde{y}^{2k} \exp\left(2y^2\right)d\tilde{y} - \frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{3}{y\sqrt{\pi}}\exp\left(-2y^2/3\right)\int_0^y \frac{\exp\left(2y^2\right)}{1-3\tilde{y}^2/y^2} d\tilde{y} - \frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) .
\end{aligned}$$

$$\begin{aligned}
\exp\left(-8y^2/3\right)\int_0^\infty U(x,y)^3 dx &= -\sqrt{2}\int_y^\infty \exp\left(-2\tilde{y}^2/3\right)\left[1-\sqrt{8/\pi}\exp\left(-2\tilde{y}^2\right)\sum_{n=0}^\infty \frac{(4)^n \tilde{y}^{2n+1}}{1\cdot 3\cdots(2n+1)}\right]d\tilde{y} \\
&= -\sqrt{3}\int_{y\sqrt{2/3}}^\infty \exp\left(-\tilde{y}^2\right)\left[1-2\sqrt{3/\pi}\exp\left(-3\tilde{y}^2\right)\sum_{n=0}^\infty \frac{(6)^n \tilde{y}^{2n+1}}{1\cdot 3\cdots(2n+1)}\right]d\tilde{y} \\
&= -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right)+\frac{6}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{6^n}{1\cdot 3\cdots(2n+1)}\int_{y\sqrt{2/3}}^\infty \tilde{y}^{2n+1}\exp\left(-3\tilde{y}^2\right)d\tilde{y} \\
&= \frac{1}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{2^{n+1}}{1\cdot 3\cdots(2n+1)}\int_{y\sqrt{2}}^\infty \tilde{y}^{2n+1}\exp\left(-\tilde{y}^2\right)d\tilde{y}-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{1}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{2^n}{1\cdot 3\cdots(2n+1)}\int_{2y^2}^\infty x^n \exp(-x)dx-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= -\frac{1}{\sqrt{\pi}}\sum_{n=0}^\infty \frac{2^n \exp\left(-2y^2\right)}{1\cdot 3\cdots(2n+1)}\sum_{k=0}^n (-1)^{k+1}(-n)_k \left(2y^2\right)^{n-k}-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{1}{\sqrt{\pi}}\sum_{k=0}^\infty (-1)^k \sum_{n=k}^\infty (-n)_k \left(2y^2\right)^{n-k} \frac{\exp\left(-2y^2\right)}{(3/2)_n}-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{\exp\left(-2y^2\right)}{\sqrt{\pi}}\sum_{k=0}^\infty (-1)^k \sum_{n=0}^\infty \frac{(-n-k)_k}{(3/2)_{n+k}} \left(2y^2\right)^n -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{\exp\left(-2y^2\right)}{\sqrt{\pi}}\sum_{k=0}^\infty \sum_{n=0}^\infty \frac{(n+1)_k}{(3/2)_{n+k}} \left(2y^2\right)^n -\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{\exp\left(-2y^2\right)}{\sqrt{\pi}}\sum_{k=0}^\infty \frac{1}{(3/2)_k} \sum_{n=0}^\infty \frac{(n+k)!}{n!} \frac{\left(2y^2\right)^n}{(3/2+k)_n}-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{\exp\left(-2y^2\right)}{\sqrt{\pi}}\sum_{k=0}^\infty \frac{k!}{(3/2)_k} \sum_{n=0}^\infty \frac{(k+1)_n}{n!} \frac{\left(2y^2\right)^n}{(3/2+k)_n}-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) \\
&= \frac{1}{\sqrt{\pi}}\sum_{k=0}^\infty \frac{k!}{(3/2)_k} {}_1F_1\left(1/2;3/2+k;-2y^2\right)-\frac{\sqrt{3\pi}}{2}\operatorname{erfc}\left(y\sqrt{2/3}\right) .
\end{aligned}$$

APPENDIX B: SOLVING FOR THE 4 LINE CENTER BAND MODEL PARAMETERS

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$$n_1 S_1^\alpha + n_2 S_2^\alpha = \Sigma_1 \equiv \sum_i S_i^\alpha \quad (\text{B.1a})$$

$$n_1 S_1^{2\alpha} + n_2 S_2^{2\alpha} = \Sigma_2 \equiv \sum_i S_i^{2\alpha} \quad (\text{B.1b})$$

$$n_1 S_1^{3\alpha} + n_2 S_2^{3\alpha} = \Sigma_3 \equiv \sum_i S_i^{3\alpha} \quad (\text{B.1c})$$

$$n_1 S_1^{4\alpha} + n_2 S_2^{4\alpha} = \Sigma_4 \equiv \sum_i S_i^{4\alpha} \quad (\text{B.1d})$$

Solve for $n_2 S_2^\alpha$ using Eq. (B.1a):

$$n_2 S_2^\alpha = \Sigma_1 - n_1 S_1^\alpha \quad (\text{B.2})$$

Eliminating $n_2 S_2^\alpha$:

$$n_1 S_1^{2\alpha} + (\Sigma_1 - n_1 S_1^\alpha) S_2^\alpha = \Sigma_2 \Rightarrow n_1 S_1^\alpha (S_1^\alpha - S_2^\alpha) = \Sigma_2 - \Sigma_1 S_2^\alpha \quad (\text{B.3b})$$

$$n_1 S_1^{3\alpha} + (\Sigma_1 - n_1 S_1^\alpha) S_2^{2\alpha} = \Sigma_3 \Rightarrow n_1 S_1^\alpha (S_1^{2\alpha} - S_2^{2\alpha}) = \Sigma_3 - \Sigma_1 S_2^{2\alpha} \quad (\text{B.3c})$$

$$n_1 S_1^{4\alpha} + (\Sigma_1 - n_1 S_1^\alpha) S_2^{3\alpha} = \Sigma_4 \Rightarrow n_1 S_1^\alpha (S_1^{3\alpha} - S_2^{3\alpha}) = \Sigma_4 - \Sigma_1 S_2^{3\alpha} \quad (\text{B.3d})$$

Eliminate $n_1 S_1^\alpha (S_1^\alpha - S_2^\alpha)$ using Eq. (B.3b) and substituting into Eqs. (B.3c) and (B.3d):

$$(\Sigma_2 - \Sigma_1 S_2^\alpha)(S_1^\alpha + S_2^\alpha) = \Sigma_3 - \Sigma_1 S_2^{2\alpha} \Rightarrow (\Sigma_2 - \Sigma_1 S_2^\alpha) S_1^\alpha = \Sigma_3 - \Sigma_2 S_2^\alpha \quad (\text{B.4c})$$

$$(\Sigma_2 - \Sigma_1 S_2^\alpha)(S_1^{2\alpha} + S_1^\alpha S_2^\alpha + S_2^{2\alpha}) = \Sigma_4 - \Sigma_1 S_2^{3\alpha} \Rightarrow (\Sigma_2 - \Sigma_1 S_2^\alpha) S_1^\alpha (S_1^\alpha + S_2^\alpha) = \Sigma_4 - \Sigma_2 S_2^{2\alpha} \quad (\text{B.4d})$$

Substitute the expression for $(\Sigma_2 - \Sigma_1 S_2^\alpha) S_1^\alpha$ from Eq. (B.4c) into Eq. (B.4d):

$$(\Sigma_3 - \Sigma_2 S_2^\alpha)(S_1^\alpha + S_2^\alpha) = \Sigma_4 - \Sigma_2 S_2^{2\alpha} \Rightarrow (\Sigma_3 - \Sigma_2 S_2^\alpha) S_1^\alpha = \Sigma_4 - \Sigma_3 S_2^\alpha \quad (\text{B.5})$$

Ratio Eqs. (B.4c) and (B.5), and solve for S_2^α :

$$\begin{aligned} \frac{\Sigma_2 - \Sigma_1 S_2^\alpha}{\Sigma_3 - \Sigma_2 S_2^\alpha} &= \frac{\Sigma_3 - \Sigma_2 S_2^\alpha}{\Sigma_4 - \Sigma_3 S_2^\alpha} \Rightarrow (\Sigma_2 - \Sigma_1 S_2^\alpha)(\Sigma_4 - \Sigma_3 S_2^\alpha) = (\Sigma_3 - \Sigma_2 S_2^\alpha)^2 \\ &\Rightarrow (\Sigma_1 \Sigma_3 - \Sigma_2^2) S_2^{2\alpha} - (\Sigma_1 \Sigma_4 - \Sigma_2 \Sigma_3) S_2^\alpha + (\Sigma_2 \Sigma_4 - \Sigma_3^2) = 0 \end{aligned} \quad (\text{B.6})$$

Define coefficients Δ_a , Δ_b and Δ_c :

$$\Delta_a \equiv \Sigma_1 \Sigma_3 - \Sigma_2^2 \quad ; \quad \Delta_b \equiv \Sigma_1 \Sigma_4 - \Sigma_2 \Sigma_3 \quad ; \quad \text{and} \quad \Delta_c \equiv \Sigma_2 \Sigma_4 - \Sigma_3^2 \quad (\text{B.7})$$

Solve the quadratic:

$$S_2^\alpha = \frac{\Delta_b}{2\Delta_a} \pm \sqrt{\frac{\Delta_b^2}{4\Delta_a^2} - \frac{\Delta_c}{\Delta_a}} = \frac{\Delta_b \pm \sqrt{\Delta_b^2 - 4\Delta_a \Delta_c}}{2\Delta_a} \quad (\text{B.8})$$

By convention, S_1^α is greater than or equal to S_2^α . Therefore,

$$S_1^\alpha = \frac{\Delta_b}{2\Delta_a} + \sqrt{\frac{\Delta_b^2}{4\Delta_a^2} - \frac{\Delta_c}{\Delta_a}} \quad \text{and} \quad S_2^\alpha = \frac{\Delta_b}{2\Delta_a} - \sqrt{\frac{\Delta_b^2}{4\Delta_a^2} - \frac{\Delta_c}{\Delta_a}} = \frac{\Delta_c/\Delta_a}{S_1^\alpha} \quad (\text{B.9})$$

Now consider the case of $N = 2$ lines, s_1 and s_2 :

$$\begin{aligned} \Delta_a &\equiv \Sigma_1 \Sigma_3 - \Sigma_2^2 = (s_1^\alpha + s_2^\alpha)(s_1^{3\alpha} + s_2^{3\alpha}) - (s_1^{2\alpha} + s_2^{2\alpha})^2 \\ &= s_1^\alpha s_2^\alpha (s_1^{2\alpha} + s_2^{2\alpha}) - 2s_1^{2\alpha} s_2^{2\alpha} = s_1^\alpha s_2^\alpha (s_1^\alpha - s_2^\alpha)^2 \geq 0 \\ \Delta_b &\equiv \Sigma_1 \Sigma_4 - \Sigma_2 \Sigma_3 = (s_1^\alpha + s_2^\alpha)(s_1^{4\alpha} + s_2^{4\alpha}) - (s_1^{2\alpha} + s_2^{2\alpha})(s_1^{3\alpha} + s_2^{3\alpha}) \\ &= s_1^\alpha s_2^\alpha (s_1^{3\alpha} - s_1^{2\alpha} s_2^\alpha - s_1^\alpha s_2^{2\alpha} + s_2^{3\alpha}) = s_1^\alpha s_2^\alpha (s_1^\alpha + s_2^\alpha)(s_1^\alpha - s_2^\alpha)^2 \geq 0 \\ \Delta_c &\equiv \Sigma_2 \Sigma_4 - \Sigma_3^2 = (s_1^{2\alpha} + s_2^{2\alpha})(s_1^{4\alpha} + s_2^{4\alpha}) - (s_1^{3\alpha} + s_2^{3\alpha})^2 \\ &= s_1^{2\alpha} s_2^{2\alpha} (s_1^{2\alpha} + s_2^{2\alpha}) - 2s_1^{3\alpha} s_2^{3\alpha} = s_1^{2\alpha} s_2^{2\alpha} (s_1^\alpha - s_2^\alpha)^2 \geq 0 \end{aligned} \quad (\text{B.10})$$

If you have $N \geq 2$ lines,

$$\begin{aligned} \Delta_a &= \sum_{i=1}^N \sum_{j=1}^{i-1} (s_i^\alpha s_j^{3\alpha} + s_i^{3\alpha} s_j^\alpha - 2s_i^{2\alpha} s_j^{2\alpha}) = \sum_{i=1}^N \sum_{j=1}^{i-1} s_i^\alpha s_j^\alpha (s_i^\alpha - s_j^\alpha)^2 \geq 0 \\ \Delta_b &= \sum_{i=1}^N \sum_{j=1}^{i-1} (s_i^\alpha s_j^{4\alpha} + s_i^{4\alpha} s_j^\alpha - s_i^{2\alpha} s_j^{3\alpha} - s_i^{3\alpha} s_j^{2\alpha}) = \sum_{i=1}^N \sum_{j=1}^{i-1} s_i^\alpha s_j^\alpha (s_i^\alpha + s_j^\alpha)(s_i^\alpha - s_j^\alpha)^2 \geq 0 \\ \Delta_c &= \sum_{i=1}^N \sum_{j=1}^{i-1} (s_i^{2\alpha} s_j^{4\alpha} + s_i^{4\alpha} s_j^{2\alpha} - 2s_i^{3\alpha} s_j^{3\alpha}) = \sum_{i=1}^N \sum_{j=1}^{i-1} s_i^{2\alpha} s_j^{2\alpha} (s_i^\alpha - s_j^\alpha)^2 \geq 0 \end{aligned} \quad (\text{B.11})$$

It is clear from these expressions that the coefficients Δ_a , Δ_b and Δ_c have been defined such that they are always non-negative and only zero when all the line strengths are equal.

Although Eq. (B.11) exchanges the single sums of Eq. (B.7) with double sums, the double sum expressions are numerically more stable. Consider the two-line case where $s_1^\alpha \gg s_2^\alpha$. Direct calculation of Δ_a from Eq. (B.7) involves differencing two terms which both equal $s_1^{4\alpha}$ to leading order. In the double sum expressions, the leading order terms analytically cancel. The same argument holds true for both Δ_b and Δ_c except the leading order terms are $s_1^{5\alpha}$ and $s_1^{6\alpha}$, respectively. Unfortunately, computing the double sums increases computational time and memory requirements. As a result, for large N , there may be no choice but to rely on the single sum approach.

One might expect that both n_1 and n_2 will never be less than 1 if there are at least 2 lines. This turns out not to be true. Consider the case of three lines. For notational simplicity, define x, y and z as follows:

$$x \equiv s_1^\alpha \quad ; \quad y \equiv s_2^\alpha \quad ; \quad z \equiv s_3^\alpha \quad ; \quad x \geq y \geq z \quad (\text{B.12})$$

The double sums of Eq. (B.11) have the following form:

$$\begin{aligned}
\Delta_a &= x y (y-x)^2 + y z (z-y)^2 + z x (x-z)^2 \quad ; \\
\Delta_b &= x y (y+x)(y-x)^2 + y z (z+y)(z-y)^2 + z x (x+z)(x-z)^2 \quad ; \quad \text{and} \quad \text{(B.13)} \\
\Delta_c &= x^2 y^2 (y-x)^2 + y^2 z^2 (z-y)^2 + z^2 x^2 (x-z)^2 \quad .
\end{aligned}$$

Determining the discriminant requires some manipulations:

$$\begin{aligned}
\Delta_b^2 - 4\Delta_a\Delta_c &= x^2 y^2 (y+x)^2 (y-x)^4 - 4x^3 y^3 (y-x)^4 + 2x y z^2 (z+y)(z-y)^2 (x+z)(x-z)^2 \\
&\quad - 4y x^2 z^3 (x-z)^2 (z-y)^2 - 4x y^2 z^3 (z-y)^2 (x-z)^2 \\
&\quad \quad \quad + (xyz \rightarrow yzx) + (xyz \rightarrow zxy) \\
&= x^2 y^2 (y-x)^4 [(y+x)^2 - 4xy] \\
&\quad + 2x y z^2 (z-y)^2 (x-z)^2 [(z+y)(x+z) - 2(x+y)z] \\
&\quad \quad \quad + (xyz \rightarrow yzx) + (xyz \rightarrow zxy) \quad \text{(B.14)} \\
&= x^2 y^2 (y-x)^6 - 2x y z^2 (z-y)^3 (x-z)^3 + (xyz \rightarrow yzx) + (xyz \rightarrow zxy) \\
&= x^2 y^2 (y-x)^6 + y^2 z^2 (z-y)^6 + z^2 x^2 (x-z)^6 \\
&\quad - 2x y z^2 (z-y)^3 (x-z)^3 - 2y z x^2 (x-z)^3 (y-x)^3 - 2z x y^2 (y-x)^3 (z-y)^3 \quad .
\end{aligned}$$

Clearly, the first 3 of the terms in the final expression for the discriminant are non-negative. With the convention that $x \geq y \geq z$, the next two terms are also non-negative. The only negative term is $-2 z x y^2 (y-x)^3 (z-y)^3$. This term can be combined with the first two terms to obtain square $[x y (y-x)^3 - y z (z-y)^3]^2$, which, of course, is non-negative. Summarizing,

$$\begin{aligned}
\Delta_b^2 - 4\Delta_a\Delta_c &= [x y (y-x)^3 - y z (z-y)^3]^2 + z^2 x^2 (x-z)^6 \\
&\quad - 2x y z^2 (z-y)^3 (x-z)^3 - 2y z x^2 (x-z)^3 (y-x)^3 \geq 0 \quad . \quad \text{(B.15)}
\end{aligned}$$

The discriminant will be positive unless $x = y = z$.

Now consider the case where z is small compared to y (and x since $x \geq y \geq z$). In this case, Eqs. (B.13) and (B.14) can be re-written as follows:

$$\left. \begin{aligned}
\Delta_a &= x y (y-x)^2 + (y^3 + x^3)z + O(z^2) \\
\Delta_b &= x y (y+x)(y-x)^2 + (y^4 + x^4)z + O(z^2) \\
\Delta_b^2 - 4\Delta_a\Delta_c &= x^2 y^2 (y-x)^6 + 2x y (y-x)^3 (y^4 - x^4)z + O(z^2)
\end{aligned} \right\} \quad z \ll y \leq x \quad . \quad \text{(B.16)}$$

The second condition, that $x - y \gg z$, must be enforced to obtain a small z expression for the square root of the discriminant:

$$\left. \begin{aligned} \mp \left| \sqrt{\Delta_b^2 - 4\Delta_a\Delta_c} \right| &= \pm x y (y-x)^3 \sqrt{1 + 2z \frac{y^4 - x^4}{x y (y-x)^3}} + O(z^2) \\ &= \pm x y (y-x)^3 \left(1 + z \frac{y^4 - x^4}{x y (y-x)^3} \right) + O(z^2) \\ &= \pm \left[x y (y-x)^3 + (y^4 - x^4) z \right] + O(z^2) \end{aligned} \right\} ; \quad z \ll \min(y, x-y) \quad . \quad (\text{B.17})$$

Remembering that the coefficient of the linear term in Eq. (B.6) is preceded by a negative sign, $+\Delta_b$ from Eq. (B.16) is added to the root of the discriminant:

$$\begin{aligned} \Delta_b \mp \left| \sqrt{\Delta_b^2 - 4\Delta_a\Delta_c} \right| &= \left[x y (y+x)(y-x)^2 + (y^4 + x^4) z \right] \pm \left[x y (y-x)^3 + (y^4 - x^4) z \right] + O(z^2) \\ &= \begin{cases} 2y \left[x y (y-x)^2 \right] + 2y^4 z + O(z^2) \\ 2x \left[x y (y-x)^2 \right] + 2x^4 z + O(z^2) \end{cases} ; \quad z \ll \min(y, x-y) \quad . \end{aligned} \quad (\text{B.18})$$

Divide by $2\Delta_a$, Eq. (B.16), to obtain the small z expression for S_-^α :

$$\left. \begin{aligned} S_-^\alpha &= \frac{\Delta_b - \left| \sqrt{\Delta_b^2 - 4\Delta_a\Delta_c} \right|}{2\Delta_a} = \frac{y \left[x y (y-x)^2 \right] + y^4 z}{x y (y-x)^2 + (y^3 + x^3) z} + O(z^2) \\ &= \frac{y + \frac{y^4 z}{x y (y-x)^2}}{1 + \frac{y^3 + x^3}{x y (y-x)^2} z} + O(z^2) \end{aligned} \right\} ; \quad z \ll \min(y, x-y) \quad . \quad (\text{B.19a})$$

$$\left. \begin{aligned} S_-^\alpha &= \left(y + \frac{y^4 z}{x y (y-x)^2} \right) \left(1 - \frac{y^3 + x^3}{x y (y-x)^2} z \right) + O(z^2) \\ &= y - \frac{x^2 z}{(y-x)^2} + O(z^2) \end{aligned} \right\} ; \quad z \ll \min(y, x-y) \quad . \quad (\text{B.19b})$$

Symmetry dictates that the small z expression for S_+^α is

$$S_+^\alpha = x - \frac{y^2 z}{(y-x)^2} + O(z^2) ; \quad z \ll \min(y, x-y) \quad . \quad (\text{B.20})$$

The number of lines associated with n_- is determined from Eq. (B.3b):

$$\begin{aligned}
n_- &= \frac{\Sigma_2 - \Sigma_1 S_+^\alpha}{S_-^\alpha (S_-^\alpha - S_+^\alpha)} = \frac{z^2 + y^2 + x^2 - (z + y + x) \left[x - \frac{y^2 z}{(y-x)^2} + O(z^2) \right]}{\left(y - \frac{x^2 z}{(y-x)^2} \right) \left[\left(y - \frac{x^2 z}{(y-x)^2} \right) - \left(x - \frac{y^2 z}{(y-x)^2} \right) \right] + O(z^2)} \\
&= \frac{y^2 + x^2 - x(y+x) - \left[x - \frac{y^2(y+x)}{(y-x)^2} \right] z}{\left(y - \frac{x^2 z}{(y-x)^2} \right) \left[y - x + \frac{y^2 - x^2}{(y-x)^2} z \right]} + O(z^2) \\
&= \frac{y(y-x) - \left[\frac{x^3 - 2x^2 y - y^3}{(y-x)^2} \right] z}{y(y-x) + \left[\frac{y^2 - x^2}{(y-x)^2} y - \frac{x^2}{y-x} \right] z} + O(z^2) = \frac{1 - \left(\frac{x^3 - 2x^2 y - y^3}{y(y-x)^3} \right) z}{1 + \left(\frac{y^2 + xy - x^2}{y(y-x)^2} \right) z} + O(z^2) \\
&= \left[1 + \left(\frac{y^3 + 2x^2 y - x^3}{y(y-x)^3} \right) z \right] \left[1 - \left(\frac{y^2 + xy - x^2}{y(y-x)^2} \right) z \right] + O(z^2) \\
&= 1 + \left(\frac{y^3 + 2x^2 y - x^3 - (y-x)(y^2 + xy - x^2)}{y(y-x)^3} \right) z + O(z^2) \\
n_- &= 1 + \frac{4x^2 y - 2x^3}{y(y-x)^3} z + O(z^2) = 1 + \frac{2x^2(2y-x)}{y(y-x)^3} z + O(z^2) \quad ; \quad z \ll \min(y, x-y) \quad .
\end{aligned} \tag{B.21}$$

By symmetry, the expression for n_+ is:

$$n_+ = 1 + \frac{2y^2(y-2x)}{x(y-x)^3} z + O(z^2) \quad ; \quad z \ll \min(y, x-y) \quad . \tag{B.22}$$

Since $x \geq y$, n_+ will not be less than one; however, if x exceeds y but is less than $2y$, then n_- will be less than one. Physically, what is happening is that the line of strength $y^{1/\alpha}$ is being split with some of its strength placed in a strong line with strength between $x^{1/\alpha}$ and $y^{1/\alpha}$, and the remainder is placed in a weak line with strength between $y^{1/\alpha}$ and $z^{1/\alpha}$. Since z is so much smaller than y , only a fraction of the weak line contributes.

A problem can arise if the weak lines lead to excessive underflows. Calculation of Δ_b^2 , for example, involves terms of order $s_i^{10\alpha}$. Therefore, while s_i itself may exceed the underflow limit, $s_i^{10\alpha}$ may not. To alleviate this problem, terms are normalized by Σ_1 raised to a power. The following dimensionless variables are defined:

$$\begin{aligned}
r_2 &\equiv \Sigma_2 / \Sigma_1^2 \quad ; \quad r_3 \equiv \Sigma_3 / \Sigma_1^3 \quad ; \quad r_4 \equiv \Sigma_4 / \Sigma_1^4 \\
r_a &\equiv \Delta_a / \Sigma_1^4 = r_3 - r_2^2 \quad ; \quad r_b \equiv \Delta_b / \Sigma_1^5 = r_4 - r_2 r_3 \quad ; \quad r_c \equiv \Delta_c / \Sigma_1^6 = r_2 r_4 - r_3^2
\end{aligned}
\tag{B.23}$$

S_1^α then has the form (plus sign is used to insure that $S_1^\alpha > S_2^\alpha$):

$$\begin{aligned}
\frac{S_1^\alpha}{\Sigma_1} &= \frac{r_b}{2r_a} + \sqrt{\frac{r_b^2}{4r_a^2} - \frac{r_c}{r_a}} \quad and \quad \frac{S_2^\alpha}{\Sigma_1} = \frac{r_c/r_a}{S_1^\alpha/\Sigma_1} \\
\left[S_1 &= \left(\Sigma_1 \frac{S_1^\alpha}{\Sigma_1} \right)^{1/\alpha} \quad and \quad S_2 = \left(\Sigma_1 \frac{S_2^\alpha}{\Sigma_1} \right)^{1/\alpha} \right]
\end{aligned}
\tag{B.24}$$

$$n_1 = \frac{r_2 - S_2^\alpha / \Sigma_1}{(S_1^\alpha / \Sigma_1 - S_2^\alpha / \Sigma_1) S_1^\alpha / \Sigma_1} \quad and \quad n_2 = \frac{1 - n_1 S_1^\alpha / \Sigma_1}{S_2^\alpha / \Sigma_1}
\tag{B.25}$$

As a final comment, it should be noted that these equations cannot be used if there is only one line in an interval because Δ_a , Δ_b and Δ_c are all zero. In this case, $n_1 = 1$, $S_1 = s_1$, $n_2 = 0$ and $S_2 = 0$.

For MODTRAN database generation, α is set to $1/3$. Furthermore, the data is stored using the traditional line spacing ($1/d$) and absorption coefficient (S/d) band model parameters:

$$\left(\frac{1}{d} \right)_1 = \frac{n_1}{\Delta \nu} \quad ; \quad \left(\frac{S}{d} \right)_1 = \frac{n_1 S_1}{\Delta \nu} \quad ; \quad \left(\frac{1}{d} \right)_2 = \frac{n_2}{\Delta \nu} \quad ; \quad \left(\frac{S}{d} \right)_2 = \frac{n_2 S_2}{\Delta \nu}
\tag{B.26}$$

APPENDIX C: ANALYTIC, HIGH-FIDELITY SPHERICAL REFRACTIVE GEOMETRY BASED ON CIRCULAR ARC PATH SEGMENTS

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Abstract

A new approach has been developed for computing the refractive path through the terrestrial atmosphere based on circular arc path segments. The approach eliminates singularity problems associated with solving the refractive path differential equation near path tangent points. Simple analytic expressions are obtained for the path length, the bend angle and the Earth center angle. The circular arcs are the exact solution to the refractive geometry problem when the reciprocal of the index of refraction varies linearly with the square of the Earth center distance between altitude levels, a result known to Maxwell and the basis for the “fish-eye” lens. The path segments typically encountered in modeling atmospheric visible and infrared radiation transport would essentially follow a circular arc if the refractivity varied linearly with altitude. Since the true refractivity profile variation is closer to exponential, an altitude bisection method is used to provide better than 1 mm accuracy for >100 km half-tangent path segments. Column densities expressions are derived assuming constituent concentrations vary exponentially with altitude. The new and validated refractive geometry package has been integrated into the MODTRAN band model radiative transfer model.

Key Words

Spherical Refraction

Maxwell’s fish-eye

Radiative Transfer

MODTRAN

Atmospheric Refraction

Anomalous Refraction

1. Introduction

Terrestrial stratified atmospheric radiative transfer models of highest fidelity incorporate the effects of refraction to compute path lengths, bend angles, Earth center angles and column densities. Typically, the altitude above sea-level in the region of interest is defined by a constant Earth radius – the local spherical Earth approximation. Pressure, temperature and constituent densities are modeled as varying only with altitude and defined via vertical profiles. Optical lines-of-sight are governed by Snell’s Law for a spherically symmetric medium, the formula of Bouguer [Born and Wolf, 1999]. This requires that the product of the Earth centered distance r , the sine of the path zenith angle θ_r at r , and the spectral index of refraction n_r at r are a path constant for any line-of-sight (LOS):

$$n_r r \sin \theta_r = (1 + N_r) r \sin \theta_r = C_{path} \quad . \quad (\text{C.1})$$

Here, the index of refraction has been expanded as one plus the spectral refractivity N_r and C_{path} is the path constant. The spectral refractivity is predominately a linear function of atmospheric

density, but more rigorous treatments [Bodhaine, *et al.*, 1999], such as the one in the MODTRAN band model radiative transfer model [Berk *et al.*, 2006; Berk *et al.*, 2008], also include weaker dependencies on the relative humidity and CO₂ concentration.

As illustrated in Figure C.1, the differential increase in path length at any point along a path equals the differential increase in altitude over the cosine of the path zenith angle:

$$ds = \frac{dr}{\cos \theta_r} \quad . \quad (C.2)$$

This differential equation becomes problematic at the path tangent point, i.e., where the path zenith angle is $\pi/2$, because of the singularity. Thus, if Eq. (C.2) is discretized, it is difficult to accurately determine Δs for a given Δr when the path approaches the singularity. To avoid this problem, Kneizys *et al.* [1983] introduced the independent variable $x \equiv r \cos \theta$, where x equals the straight line distance to the tangent point as shown in Figure C.1. Replacing r with x as the independent variable in the path length differential equation, one obtains the following differential equation:

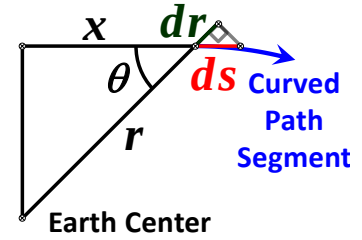


Figure C.1. Diagram of Differential Increase in Path Length

$$ds = \left[1 + \left(\frac{C_{path}^2}{n_r^3 r} \right) \frac{dn_r}{dr} \right]^{-1} dx \quad ; \quad x \equiv r \cos \theta_r \quad . \quad (C.3)$$

The formula of Bouguer, Eq. (C.1), was used to eliminate the path zenith angle from this equation. Although refractivity generally decreases with increasing altitude (i.e., as the air become thinner), so that dn_r/dr is negative, the term in square bracket seldom deviates significantly from unity for the ambient terrestrial atmosphere, and Eq.(C.3) contains no singularity.

Most radiative transfer models compute a refractivity vertical profile at a single reference, centrally located wavelength, and use that refractivity profile for the full spectral range of the calculations. Path lengths are determined by discretizing Eq. (C.3):

$$\Delta s_j \equiv s_j - s_{j-1} = \left(\left. \frac{ds}{dx} \right|_{j-1} + \left. \frac{ds}{dx} \right|_j \right) \frac{\Delta x_j}{2} \quad . \quad (C.4)$$

Here, j is both the index of a path segment (as in Δs_j and Δx_j) and the index of the segment's final endpoint (as in s_j and $ds/dx|_j$); $j-1$ is the index of the segment's initial end point. Generally, the path length integrations are performed upward from the minimum path altitude with the initial path zenith determined from Snell's Law. If the path passes through a tangent point, then the full path is partitioned into 2 sub-paths which meet at the tangent point. Within each atmospheric layer (or, more precisely, spherical shell), Δx_j values are defined to not exceed a pre-defined maximum step size, Δx^{max} .

This procedure for computing path lengths generally works well, and it was used for many years in industry standard atmospheric radiative transfer models, such as the original MODTRAN [Berk *et al.*, 1989] band model and the LBLRTM [Clough *et al.*, 2005] monochromatic algorithms. However, on rare occasions, problems arise. One issue involves the calculation of x .

Using the formula of Bouguer, Eq. (C.1), to define the cosine of the path zenith, x has the following form:

$$x = \sqrt{r^2 - C_{path}^2/n_r^2} = \sqrt{r^2 - r_t^2 (n_t/n_r)^2} \quad . \quad (C.5)$$

On the far right, the path constant C_{path} has been replaced by the product of the Earth centered distance, r_t , and the index of refraction, n_t , extrapolated to the tangent point t , where $\sin\theta$ equals one. A very small change in Earth centered distance near the tangent point (say, 1 mm) results in a large displacement of x (> 100 m). In fact, dx/dr approaches infinity as r approaches r_t^+ . Minimally, the calculation of x requires double precision. Another issue that arises with application of Eq. (C.4) is the selection of the maximum step size, Δx^{max} . In both MODTRAN4 and LBLRTM, Δx^{max} is hardwired to insure that the incremental path length, Δs_j , will not exceed ~ 5 km. The introduction of large temperature and/or humidity gradients within the atmosphere can cause significant bending for paths shorter than the 5 km distance.

In response to the difficulties with the Kneizys *et al.* refraction algorithm in MODTRAN, a new approach has been developed for calculating path length, Earth centered angle, the bending of curved path segments through atmospheric layers, and column densities. For many applications, the simple analytic expressions for these path parameters provide sufficient accuracy without requiring any partitioning of the layer path segments into sub-segments. When greater accuracy is required, a simple altitude-based bisection method can be used to meet these requirements.

The simplicity of the spherical geometry approach presented here also provides an opportunity to explicitly model the spectral dependence of the refractivity. Currently, essentially all infrared, visible and ultraviolet spectral region radiative transfer models pre-compute column densities at a fixed wavelength. These fixed path amounts are then used in subsequent spectral transmittance, radiance and flux calculations. This approach ignores the fact that the path itself has a slight dependence on wavelength due to the spectral dependence of the refractivity. The simplicity of the proposed algorithm adds little overhead when compared to that of the spectral radiative transfer calculations themselves, enabling the spectral dependence of the line-of-sight to be explicitly modeled.

2. An Analytic Approach to Spherical Refractivity

If refraction is negligible, atmospheric path lengths can be computed using straight-line paths with spherical geometry. Refraction causes the paths to bend. The ansatz made here is that path segments between neighboring altitude levels can be approximated by a circular arc. The Bouguer formula fixes the zenith angle at the second altitude if the value at the first altitude is specified. As will be shown, this approach introduces an alternative ansatz for defining the refractivity variation between altitudes. A question that must be answered is “how similar is the implied refractivity altitude dependence to truth, truth essentially being an exponential variation of refractivity with altitude?” Before answering that question, the approach for defining the circular arcs is presented. The following problem is posed:

Problem: *Can a circular arc be defined from a fixed point P_1 , with Earth centered distance r_1 and path zenith angle θ ($= \theta_1$), to a final point P_2 , with Earth centered distance $r_2 \geq r_1$ and reverse path zenith angle ϕ ($= \pi - \theta_2$)? If such an arc does exist, what is its length?*

The geometry of the problem is illustrated in Figure C.2. At first, it may seem that a conic section more general than a circle, such as an ellipse, is required to fit the 4 defined inputs: two Earth center distances and two zenith angles. The family of circular arcs that pass through two fixed points P_1 and P_2 does not provide sufficient flexibility to allow independent specification of both zenith angles. However, the point P_2 is not fixed, but instead it can be located at any point on the Earth centered circle of radius r_2 . The Cartesian distance between points P_1 and P_2 is unspecified and to be determined from the radial and angular inputs. The derivation of this circular arc begins with determination of the radius of curvature, R .

3. The Radius of Curvature

The blue circular arc in Figure C.2 connects point P_1 to point P_2 . In this diagram, P_E is the Earth center, R is the radius of curvature of the arc, and P_0 is the center of the circle. A Cartesian coordinate system has been defined with P_0 located at the origin [$P_0 = (0, 0)$], with the circular arc endpoint P_1 located on the positive y-axis [$P_1 = (0, R)$] and with the Earth center located in the negative y half-plane [$P_E = (-r_1 \cos \theta_1, R - r_1 \sin \theta_1)$].

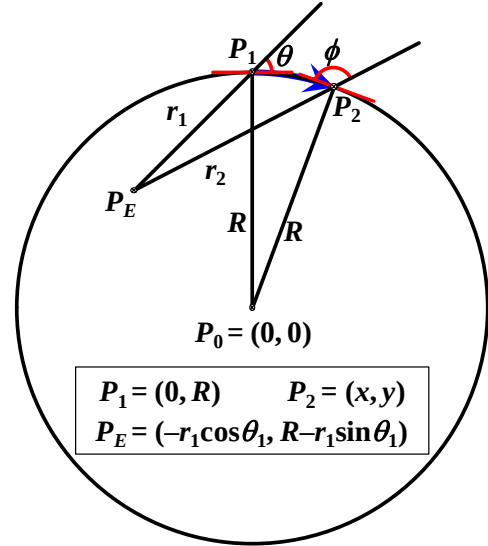


Figure C.2. Diagram of the Circular Arc from P_1 to P_2

There are 3 unknowns in Figure C.2: the radius of curvature R and the x and y coordinates of point P_2 . Therefore, 3 independent equations involving R , x and y are needed. Since P_2 is on the circumference of the circle centered at P_0 , its distance to the origin is R :

$$x^2 + y^2 = R^2 \quad . \quad (\text{C.6})$$

In addition, the radial distance r_2 is defined as the distance from P_2 to the Earth center, P_E :

$$(x + r_1 \cos \theta)^2 + (y + r_1 \sin \theta - R)^2 = r_2^2 \quad . \quad (\text{C.7})$$

A third equation is obtained by noting that cosine of the backward path zenith angle at P_2 , $\cos \phi$, equals the dot product of unit vector from P_E to P_2 with the unit vector at P_2 tangent to the circle in the reverse path direction:

$$\cos \phi = \left(\frac{x + r_1 \cos \theta}{r_2} \hat{\mathbf{i}} + \frac{y + r_1 \sin \theta - R}{r_2} \hat{\mathbf{j}} \right) \cdot \left(\frac{-y}{R} \hat{\mathbf{i}} + \frac{x}{R} \hat{\mathbf{j}} \right) \quad . \quad (\text{C.8})$$

The Cartesian coordinate system origin was defined as the circle center to simplify the equation of the tangent unit vector at P_2 .

The algebra required to solve these 3 equations is substantial. Appendix α is provided for those interested in the details. The final expression for the radius of curvature, Eq. (C.9), is repeated here:

$$R = \frac{(r_2^2 - r_1^2)/2}{r_2 \sin \phi - r_1 \sin \theta} \quad (\text{C.9})$$

4. The Cartesian Distance between P_1 and P_2

The problem posed in the introduction states that an expression for arc length s from P_1 to P_2 , Figure C.3, is desired. In the previous section, an expression for the radius of curvature R was determined in terms of the Earth center distances, r_1 and r_2 , and the Earth forward and reverse path zenith angles, θ and ϕ , respectively. The arc length simply equals ψR , where ψ is the angle subtended by the circular arc. The subtended angle is also of interest because, as illustrated in Figure C.3, it equals the path bending angle, i.e., the angle between tangent lines at the circular arc end points P_1 and P_2 .

The Cartesian distance d between the path segment endpoints must be computed to determine the ψ angle:

$$d^2 = x^2 + (R - y)^2 = 2R(R - y) \quad (\text{C.10})$$

Here, the quantity $x^2 + y^2$ has been replaced with R^2 , Eq. (C.6). The derivation of the expression for d as a function of r_1 , r_2 , θ and ϕ is provided in Appendix β . The final result is

$$d^2 = \frac{(r_2^2 - r_1^2)^2}{r_2^2 - 2r_2r_1 \cos(\phi - \theta) + r_1^2} \quad (\text{C.11})$$

This expression is correct, as it must be, in both the vertical path and half-limb straight (no refraction) path limits:

$$d_{\text{vertical}}^2(\theta = 0; \phi = \pi) = \frac{(r_2^2 - r_1^2)^2}{r_2^2 + 2r_2r_1 + r_1^2} = (r_2 - r_1)^2 \quad ; \quad (\text{C.12})$$

$$d_{\text{no refraction half-limb}}^2\left(\theta = \frac{\pi}{2}; r_2 \sin \phi = r_1\right) = \frac{(r_2^2 - r_1^2)^2}{r_2^2 - 2r_2r_1 \sin \phi + r_1^2} = \frac{(r_2^2 - r_1^2)^2}{r_2^2 - r_1^2} = r_2^2 - r_1^2 \quad .$$

5. The Path Bending Angle

The bending ψ associated with a given path segment, Figure C.3, is a measure of the severity of the refraction. The bending angle has the nice property that if multiple circular arcs are strung together with matching tangents at end points, then the total path bending is simply the sum of the contributions from the collection of sub-segments.

Since, as noted above, the bend angle for a circular arc equals the circular arc subtended angle, the bend angle can be expressed as a function of the radius of curvature R and the Cartesian distance d between circular arc end points:

$$\sin \psi = 2 \sin \frac{\psi}{2} \cos \frac{\psi}{2} = \frac{d}{R} \sqrt{1 - \frac{d^2}{4R^2}} \quad (\text{C.13})$$

The derivation of the final expression for the path bending angle is presented in Appendix γ :

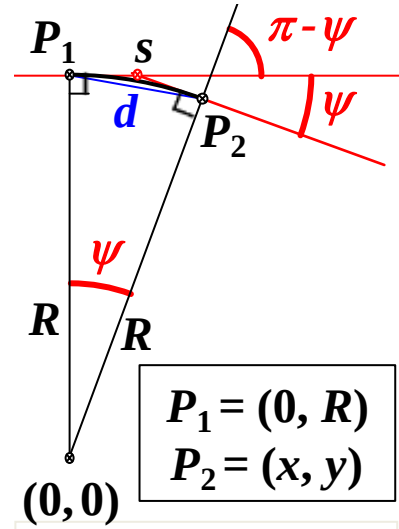


Figure C.3. Illustration of Subtended Angle ψ (equal to the path bending angle), Cartesian Distance d , and Path Length s for the Circular Arc from P_1 to P_2

$$\begin{aligned}\psi &= \arcsin \left(\frac{2(r_2 \sin \phi - r_1 \sin \theta)(r_1 \cos \theta - r_2 \cos \phi)}{(r_2 \sin \phi - r_1 \sin \theta)^2 + (r_1 \cos \theta - r_2 \cos \phi)^2} \right) \\ &= \arcsin \left(\frac{2(r_2 \sin \phi - r_1 \sin \theta)(r_1 \cos \theta - r_2 \cos \phi)}{r_2^2 - 2r_2 r_1 \cos(\phi - \theta) + r_1^2} \right) .\end{aligned}\tag{C.14}$$

Although the second line of Eq. (C.14) contains the simpler expression for the denominator, implementing the first line is computationally advantageous since the denominator contains terms already in the numerator.

6. The Arc Length

The arc length is calculated as the product of the circular arc subtended angle and the radius of curvature:

$$s = \psi R .\tag{C.15}$$

Inserting Eqs. (C.54) and (C.59), the arc length equals

$$s = \frac{r_2^2 - r_1^2}{2(r_2 \sin \phi - r_1 \sin \theta)} \arcsin \left(\frac{2(r_2 \sin \phi - r_1 \sin \theta)(r_1 \cos \theta - r_2 \cos \phi)}{(r_2 \sin \phi - r_1 \sin \theta)^2 + (r_1 \cos \theta - r_2 \cos \phi)^2} \right) .\tag{C.16}$$

When the argument of the arcsin is small ($< 10^{-4}$), the two term expansion provides double precision accuracy:

$$s = \frac{(r_2^2 - r_1^2)(r_1 \cos \theta - r_2 \cos \phi)}{r_2^2 - 2r_2 r_1 \cos(\phi - \theta) + r_1^2} \left(1 + \frac{2(r_2 \sin \phi - r_1 \sin \theta)^2 (r_1 \cos \theta - r_2 \cos \phi)^2}{3[r_2^2 - 2r_2 r_1 \cos(\phi - \theta) + r_1^2]^2} + \dots \right) .\tag{C.17}$$

Here, the more compact representation for the denominator has been used for notational simplicity.

In the straight line (no refraction) limit, $r_2 \sin \phi$ equals $r_1 \sin \theta$. This follows from Eq. (C.1) with a constant index of refraction. In this limit, all higher order terms in Eq. (C.17) are zero. Given the following no refraction limit identities

$$\begin{aligned}r_2^2 - r_1^2 &= (r_2 \cos \phi - r_1 \cos \theta)(r_2 \cos \phi + r_1 \cos \theta) + (r_2 \sin \phi - r_1 \sin \theta)(r_2 \sin \phi + r_1 \sin \theta) \\ &= (r_2 \cos \phi - r_1 \cos \theta)(r_2 \cos \phi + r_1 \cos \theta) \quad [no\ refraction] \quad and\end{aligned}\tag{C.18}$$

$$\begin{aligned}r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2 &= (r_2 \cos \phi - r_1 \cos \theta)^2 + (r_2 \sin \phi - r_1 \sin \theta)^2 \\ &= (r_2 \cos \phi - r_1 \cos \theta)^2 \quad [no\ refraction] ,\end{aligned}\tag{C.19}$$

Eq. (C.17) reduces to the correct straight line path segment distance:

$$s_{no\ refraction} = r_2 \cos \phi + r_1 \cos \theta .\tag{C.20}$$

7. The Earth Centered Angle

It is convenient to determine the path Earth center angle, β , to help track the location of a path segment's end point. The final location above the Earth surface, i.e. the end point latitude and longitude, can be computed from the

- the total Earth center angle,
- initial latitude and longitude, and
- initial azimuthal direction [deg E of N].

As illustrated in Figure C.4, the Earth center angle is a simple function of the path bending, ψ , and zenith angles, θ and ϕ :

$$\beta = \psi + \phi + \theta - \pi = \psi + \theta_1 - \theta_2 \quad . \quad (C.21)$$

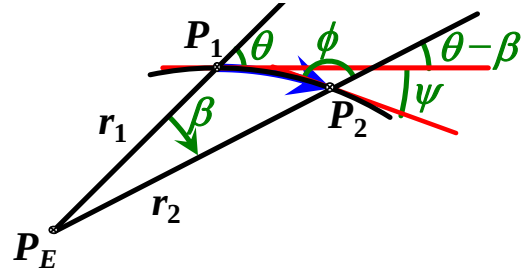


Figure C.4. Illustration of Earth Center Angle, β , and its Relationship to the Path Bending ψ and Path Zenith Angles, θ and ϕ . The blue ray is the circular arc and the red lines are the tangents at P_1 and P_2

8. Refractivity Profile

The ansatz that the path segment between altitude levels for a given initial view angle θ can be approximated as a circular arc is an exact solution to the spherical refractive geometry problem if the variation of refractivity with altitude is modeled in accordance with the circular arc constraint. In this section, that refractivity altitude dependence for a given view angle is derived.

By definition, the radius of curvature is constant along the circular arc. As a result, Eq. (C.9) must be applicable for any point (r, θ_r) along the arc:

$$R = \frac{(r^2 - r_1^2)/2}{r \sin \theta_r - r_1 \sin \theta_1} \quad ; \quad r_1 < r \leq r_2 \quad . \quad (C.22)$$

Here, θ_1 is the zenith angle at r_1 and θ_r is the zenith angle at r . The forward and reverse path zenith angles at a fixed point are complementary, i.e., $\theta + \phi = \pi$, so the sine of their zenith angles are equal. Solving for the product of the Earth center distance, r , and sine of the zenith angle at r , one obtains

$$r \sin \theta_r = r_1 \sin \theta_1 + \frac{r^2 - r_1^2}{2R} \quad . \quad (C.23)$$

Remembering that $n r \sin \theta$ is a path constant, Eq. (C.1), $n_r r \sin \theta_r = n_1 r_1 \sin \theta_1$ can be divided by Eq. (C.23) to determine the dependence of index of refraction on altitude:

$$n_r = \frac{2 n_1 r_1 R \sin \theta_1}{2 r_1 R \sin \theta_1 + r^2 - r_1^2} \quad . \quad (C.24)$$

Replacing n_r with $1 + N_r$, the refractivity profile is

$$N_r = \frac{2 N_1 r_1 R \sin \theta_1 - (r^2 - r_1^2)}{2 r_1 R \sin \theta_1 + r^2 - r_1^2} \quad . \quad (C.25)$$

This result is true for all r in the $(r_1, r_2]$ interval. In particular,

$$N_2 = \frac{2N_1 r_1 R \sin \theta_1 - (r_2^2 - r_1^2)}{2r_1 R \sin \theta_1 + r_2^2 - r_1^2} \quad (\text{C.26})$$

Solving for $2 r_1 R \sin \theta_1$,

$$2r_1 R \sin \theta_1 = \frac{(1 + N_2)(r_2^2 - r_1^2)}{N_1 - N_2} , \quad (\text{C.27})$$

and substituting back into Eq. (C.25), produces a rather surprising result:

$$N_r = \frac{N_1(1 + N_2)(r_2^2 - r_1^2) - (N_1 - N_2)(r^2 - r_1^2)}{(1 + N_2)(r_2^2 - r_1^2) + (N_1 - N_2)(r^2 - r_1^2)} ; \quad (\text{C.28a})$$

$$n_r = \frac{n_1 n_2 (r_2^2 - r_1^2)}{n_2 (r_2^2 - r_1^2) + (n_1 - n_2)(r^2 - r_1^2)} . \quad (\text{C.28b})$$

The refractivity and index of refraction profiles depend on the altitude and on the refractivity and index of refraction, respectively, at the bounding altitudes, but not on the view angle! The refractivity profile between r_1 and r_2 does not change depending upon whether the circular arc path segment is vertical, half-tangent or defined with an intermediate view angle. The vertical profile is inherent to the circular path construct. More specifically, Eq. (C.28b) indicates that *if the reciprocal of the index of refraction varies linearly with the square of the Earth center distance within bounding altitudes, then the exact solution to the refractive path geometry problem in that region is a circular arc*. As it turns out, this result was known to Maxwell and is the basis for the “fish-eye” lens [Born and Wolf, 1999]. What is new is the application of Maxwell’s solution to the computation of atmospheric refractive paths.

To illustrate the variation of refractivity implied by Eq. (C.28a) for the terrestrial atmosphere, it is useful to define h as the altitude above the lower boundary, $h \equiv r - r_1$ and h_2 as the thickness of the layer or spherical shell, $h_2 \equiv r_2 - r_1$. With these variables defined, the refractivity profile can be rewritten as

$$N_{r_1+h} = \left(N_1 - \frac{(N_1 - N_2)(2r_1 + h)h}{(1 + N_2)(2r_1 + h_2)h_2} \right) / \left(1 + \frac{(N_1 - N_2)(2r_1 + h)d}{(1 + N_2)(2r_1 + h_2)h_2} \right) . \quad (\text{C.29})$$

Typically, layer spacing for atmospheric profiles is of order 1 km, much less than the mean Earth center distance of 6371.23 km ($h \ll r_1$). In addition, the maximum ambient atmospheric refractivity generally occurs near the ground, where the value is of order 0.0003 in the center of the visible spectral region, much less than 1 ($N_2 < N_1 \ll 1$). Invoking these conditions, the refractivity profile simplifies to

$$N_{r_1+h} \approx N_1 - (N_1 - N_2)h . \quad (\text{C.30})$$

Thus, if the refractivity were to vary linearly with altitude, the atmospheric refractive path would essentially be a circular arc.

While the terrestrial spectral refractivity depends on relative humidity and CO₂ mixing ratio, it is predominantly a linear function of atmospheric density [Bodhaine *et al.*, 1999]. This is illustrated

in Figure C.5, which contains refractivity vertical profiles in the first one kilometer above ground level for the Mid-Latitude Winter (MLW) model atmosphere. The MLW model was chosen for this figure because its relative change in refractivity over this altitude range exceeds that of the other model atmospheres defined within MODTRAN. Four profiles are plotted in Figure C.5. The grey curve contains values generated directly from application of the Bodhaine formula. The red dots, which overlay the Bodhaine curve, were obtained by exponentially interpolated in altitude the 0 and 1 km refractivity values. Eq. (C.29) was used to generate the aqua curve, and the overlaying green dots are linearly interpolated values. The vertical profile in the upper right-hand corner of Figure C.5 shows the refractivity residuals that arise from using the circular arc refractivities of Eq. (C.29); for this case, the residuals never exceeds a quarter of a percent.

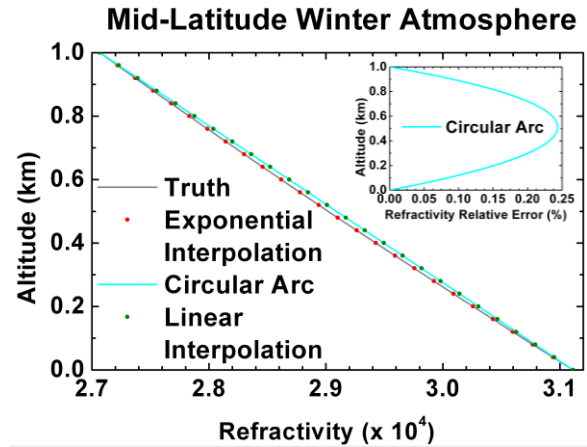


Figure C.5. Mid-Latitude Winter Model Atmosphere Refractivity Vertical Profiles between 0 and 1 km at a Spectral Wavelength of 550nm with Data Shown for MODTRAN Calculated Refractivity (Grey Line), Exponential Interpolation (Red Dots) between 0 and 1 km, the Circular Arc ansatz (Aqua Line), and Linear Interpolation (Green Dots); the Insert in the Upper Right Contains Circular Arc Residuals Against MODTRAN

9. Application

In the introduction, the claim is made that the newly derived analytic expressions for the bending angle ψ , the length s and the Earth center angle β of curved path segments provide sufficient accuracy for many applications, with no need to partition layer path segments into sub-segments. In this section, this claim is quantified and a method described for sub-dividing these segments when higher fidelity is required.

HALF-LIMB PATH FROM GROUND TO 1KM ALTITUDE <i>Mid-Latitude Winter Atmosphere with 6371.25 km Earth Radius</i> <i>550 nm Refractivities: 3.1112914E-04 at Ground; 2.7066910E-04 at 1 km Altitude</i> (percent residuals listed in parentheses)			
	Converged Result	Single Circular Arc	No Refraction
Bend Angle (Deg)	0.313410565	0.3036 (3.1%)	0.0 (100%)
Earth Centered Angle (Deg)	1.328482702	1.3187 (0.74%)	1.0151 (24%)
Path Length (km)	132.1115225	131.02 (0.82%)	112.9 (14%)

Table C.I. Accuracy of Single Circular Arc Approach for the Most Stressful MODTRAN Standard Model Atmosphere Single Segment Case.

The atmospheric layer defined in Figure C.5, the first kilometer of the MLW model atmosphere, is the focus of this analysis. The refractivity in the center of the visible spectral region (550 nm) drops 13% across this layer, more than that of any of the other MODTRAN ambient model atmospheric layer. The numbers are provided in the header of Table C.I. Figure C.6 contains plots of bend angle (left) and path length (right) as a function of the path zenith angle at the ground. Fully converged results (thin line) are plotted along with the single circular arc (thick

dash) calculations. Not surprisingly, both the bend angle and path length increase dramatically in these log plots as the path zenith approaches horizontal. The figures also contain inserts with percent error log plots. The relative errors also increase dramatically as the path zenith angle approaches 90°.

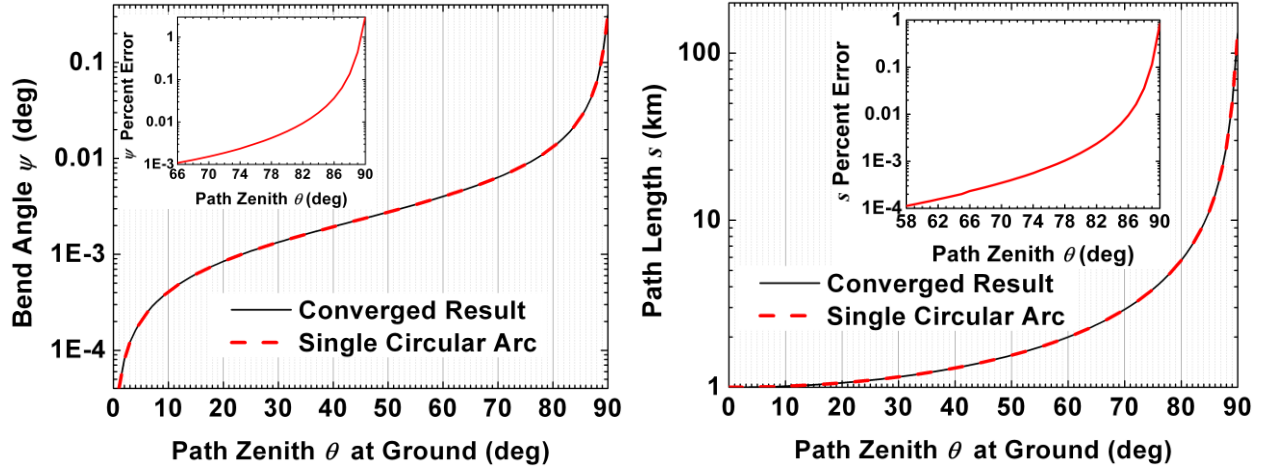


Figure C.6. Bend Angle and Path Length Curves as a Function of Path Zenith at the Ground for Spherical Refractive Paths through the First Kilometer of the Mid-Latitude Winter Model Atmosphere, with Inserts Containing the Percent Error between the Converged and Single Arc Calculations

Since residuals are largest for the half-limb (90° zenith angle) path, this geometry is examined further and results presented in Table I. As the table illustrates, the accuracy of the single circular arc approach for this particularly stressful selection of a path segment is 3.1% for the bend angle, 0.74% for the Earth centered angle and 0.82% for the path length. For comparison, the length of the straight-line, no refraction path is 14% too short. A pertinent question for radiative transport applications is “How significant is a 0.82% error in path length?” To answer this question, one can compute the maximum Beer’s Law transmittance, T , absolute error given a 0.82% path length residual and assuming, for the purpose of demonstration, an altitude independent extinction coefficient, κ :

$$\text{Find } \max_{\kappa \geq 0} \Delta T; \quad \Delta T \equiv \exp(-s_{\text{arc}} \kappa) - \exp(-s_{\text{exact}} \kappa) \quad (\text{C.31})$$

Setting the derivative of ΔT with respect to κ to zero and solving, the maximum transmittance residual is ~ 0.003 . As the path elevation angle increases, this residual drops dramatically; on the other hand, an increased refractivity vertical gradient would produce an even larger residual. This calculation is designed to provide a rough estimate for the magnitude of the maximum error. Transmittance residuals of less than, say 0.005, are perfectly acceptable for many applications. It is of concern, however, that the residuals create a bias; the circular arc path length will always be less than the true path length if the refractivity varies exponentially with altitude. For radiative transfer problems requiring highest fidelity, layer path segments should be subdivided with intermediate endpoint refractivities defined using altitude dependent exponential interpolation. To optimize convergence, one could partition each segment under consideration at the altitude corresponding to the maximum residual between the linearly and exponentially interpolated refractivity. The insert of Figure C.6 illustrates that for a $\eta \equiv N_2 / N_1$ refractivity ratio of 0.87, the maximum residual occurs very near the mid-point altitude. In general, the altitude

interpolation fraction α at which the maximum difference between linear and exponential interpolations occurs is an analytic function of the refractivity ratio:

$$\alpha \equiv \alpha(\eta) = \ln \left(\frac{\eta - 1}{\ln \eta} \right) / \ln \eta = \frac{1}{2} + \frac{\eta - 1}{24} - \frac{(\eta - 1)^2}{48} + O[(\eta - 1)^3] \quad (\text{C.32})$$

A plot of this function, Figure C.7, illustrates that for a very large range of refractivity ratios the maximum difference interpolation fraction is near $\frac{1}{2}$. This suggests that an altitude bisection procedure should work well, and its application is considerably simpler than one based on selecting the exact altitude at which the refractivity residual is maximum. Implementation of the bisection method demonstrates that the path length and bending angle residuals decrease exponentially with the number of iterations (linearly with the number of path sub-segments). In Figure C.8, this is illustrated for the half-tangent MLW path from 0 to 1 km altitude.

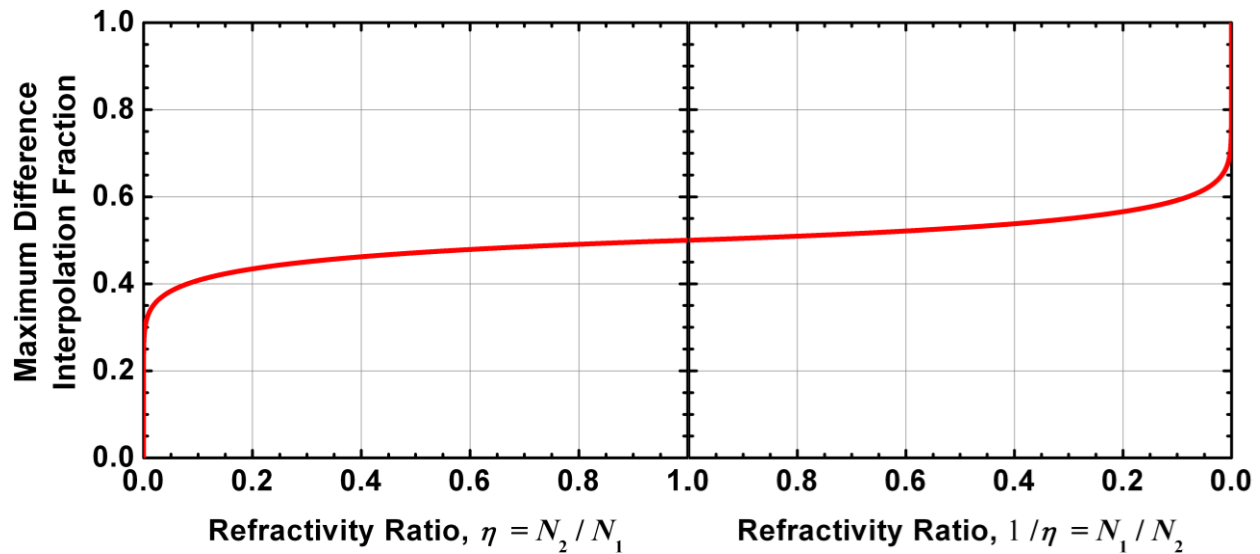


Figure C.7. Altitude Interpolation Fraction at which the Difference between Linearly and Exponentially Interpolated Refractivities is Maximum. Curve is Plotted as a Function of the Ratio of the Positive Refractivities

The bisection method has been integrated into the MODTRAN atmospheric radiative transfer model, with convergence defined to occur when the path length residual difference between successive iterations is less than 1 mm. This corresponds to 8 place accuracy for a 100 km path segment. For shorter paths, of order < 10 m, the single circular arc provides much finer than 1 mm accuracy under typical terrestrial atmospheric conditions.

10. Circular Arc Column Density

A method has been established for computing the length, bend angle and earth center angle of circular arc refractive path segments. To complete the refractive geometry package, a method is also required for computing the column densities along these circular arcs. This methodology is described in this section.

The column density of an atmospheric species whose density profile varies exponentially with altitude from $\rho_1 > 0$ at r_1 to $\rho_2 > 0$ at r_2 , with $r_2 > r_1$, is computed as a path integral over density:

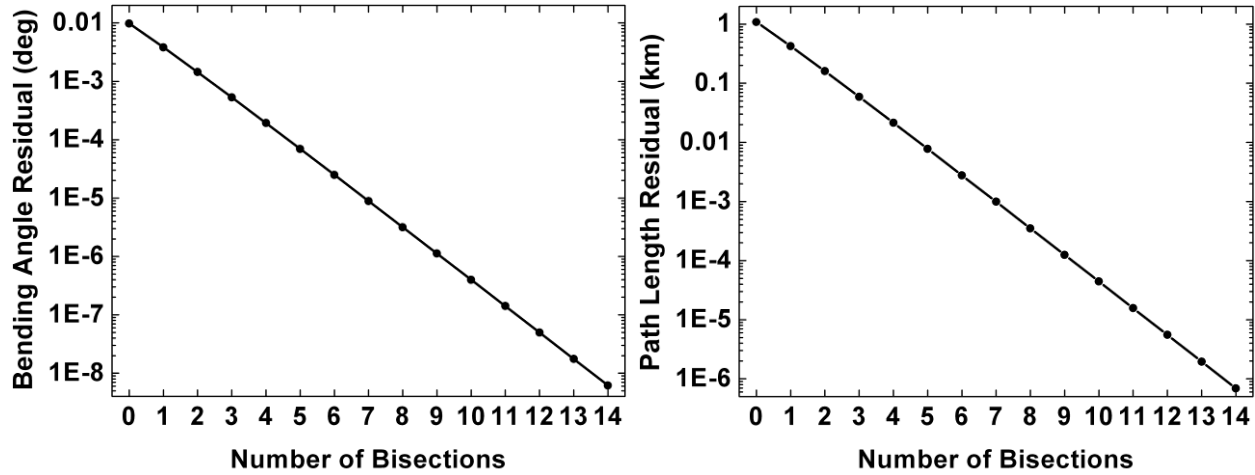


Figure C.8. Convergence of Bend Angle and Path Length as a Function of the Number of Segment Bisections for Spherical Refractive Half-Tangent Path through the First Kilometer of the MLW Model Atmosphere.

$$\begin{aligned}
 u &= \rho_1 \int_0^s (\rho_2/\rho_1)^{X(\sigma)} d\sigma \quad ; \quad X(\sigma) \equiv \frac{r(\sigma) - r_1}{r_2 - r_1} \\
 &= \rho_1 \int_0^{as} (\rho_2/\rho_1)^{X(\sigma)} d\sigma + \rho_2 \int_0^{bs} (\rho_1/\rho_2)^{Y(\sigma)} d\sigma \quad ; \quad a+b=1 \quad ; \quad Y(\sigma) \equiv \frac{r_2 - r(s-\sigma)}{r_2 - r_1} .
 \end{aligned}
 \tag{C.33}$$

As illustrated in the center drawing of Figure C.9, s is the arc length between points P_1 and P_2 ; $r(\sigma)$ is the Earth centered distance a path length σ from P_1 and $s - \sigma$ from P_2 along the circular arc; at σ equal 0, $r(\sigma)$ equals r_1 , and at σ equal s , $r(\sigma)$ equals r_2 ; and the interpolation fractions $X(\sigma)$ and $Y(s - \sigma)$ sum to one. In the second line of Eq. (C.33), the path integral from 0 to s is partitioned into 2 symmetric components.

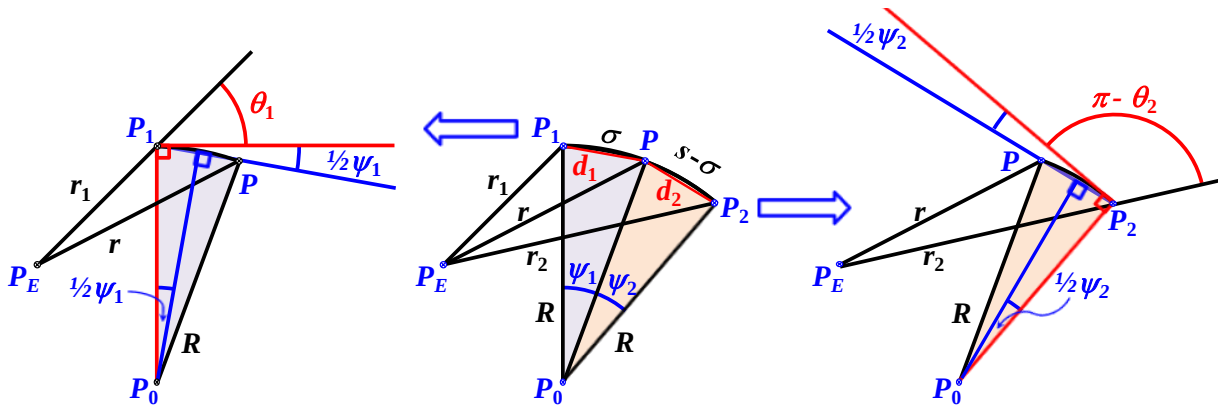


Figure C.9. Drawings for Deriving an Expansion for the Earth Centered Distance $r(\sigma)$.

Since the path integrals of Eq. (C.33) cannot be evaluated analytically, $X(\sigma)$ and $Y(\sigma)$ are each expanded about σ equals 0. This requires small σ expressions for the Earth centered distances $r(\sigma)$ and $r(s - \sigma)$. As the central drawing in Figure C.9 illustrates, the circular arc path lengths equal the product of the radius of curvature, R , and the circular arc angles, ψ_1 and ψ_2 :

$$\sigma = R\psi_1 \quad ; \quad s - \sigma = R\psi_2 \quad .
 \tag{C.34}$$

As illustrated on the left and right side drawings of Figure C.9, the Cartesian distances, d_1 and d_2 , between the circular arc endpoints can be expressed as twice the product of the radius of curvature and the sine of half the circular arc angles:

$$d_1 = 2 R \sin\left(\frac{\psi_1}{2}\right) = 2 R \sin\left(\frac{\sigma}{2 R}\right) \quad ; \quad d_2 = 2 R \sin\left(\frac{\psi_2}{2}\right) = 2 R \sin\left(\frac{s-\sigma}{2 R}\right) . \quad (\text{C.35})$$

The Figure C.9 left-hand side drawing illustrates that the Earth centered distance $r(\sigma)$ can be computed from the cosine law using the (P, P_1, P_E) triangle; the triangle's internal angle at P_1 equals $\pi - (\theta_1 + \frac{1}{2} \psi_1)$:

Expansion About P_1

$$\begin{aligned} r^2(\sigma) &= r_1^2 + 2 r_1 d_1 \cos\left(\theta_1 + \frac{\psi_1}{2}\right) + d_1^2 \\ &= r_1^2 + 4 r_1 R \sin\left(\frac{\sigma}{2R}\right) \cos\left(\theta_1 + \frac{\sigma}{2R}\right) + 4R^2 \sin^2\left(\frac{\sigma}{2R}\right) \\ &= r_1^2 + 2 r_1 \sigma \left\{ \cos \theta_1 - \frac{\sigma}{2R} \sin \theta_1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} + \sigma^2 \left\{ 1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} . \end{aligned} \quad (\text{C.36})$$

Similarly, the Earth centered distance $r(s - \sigma)$ can be computed from the cosine law using the triangle defined by points (P, P_2, P_E) ; as illustrated on the right in Figure C.9, this triangle's internal angle at P_2 equals $\theta_2 - \frac{1}{2} \psi_2$:

Expansion About P_2

$$\begin{aligned} r^2(s - \sigma) &= r_2^2 - 2 r_2 d_2 \cos\left(\theta_2 - \frac{\psi_2}{2}\right) + d_2^2 \\ &= r_2^2 - 4 r_2 R \sin\left(\frac{\sigma}{2R}\right) \cos\left(\theta_2 - \frac{\sigma}{2R}\right) + 4R^2 \sin^2\left(\frac{\sigma}{2R}\right) \\ &= r_2^2 - 2 r_2 \sigma \left\{ \cos \theta_2 + \frac{\sigma}{2R} \sin \theta_2 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} + \sigma^2 \left\{ 1 - O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} . \end{aligned} \quad (\text{C.37})$$

In Eqs. (C.36) and (C.37), the circular arc angles, ψ_1 and ψ_2 , and the Cartesian distances, d_1 and d_2 , are eliminated by substituting in the expressions from Eqs. (C.34) and (C.35). The notation $O[\varepsilon^n]$ is used to represent terms of order ε raised to the n^{th} power and higher. The last line of Eqs. (C.36) and (C.37) provide quadratic approximations for $r(\sigma)$ assuming arc lengths are small compared to the radius of curvature. For the ambient atmosphere in the infrared through ultraviolet spectral regions, the radius of curvature is generally more than an order of magnitude larger than the Earth radius, very much larger than the segment arc lengths. Thus, these approximations are extremely accurate.

The expressions for the Earth centered distance are inserted into the equations for the interpolation fractions $X(\sigma)$ and $Y(\sigma)$ [Eq. (C.33)] to obtain their linear and quadratic dependence on path length:

$$\begin{aligned}
X(\sigma) &= \frac{r_1}{r_2 - r_1} \left[\sqrt{1 + \frac{4R}{r_1} \sin\left(\frac{\sigma}{2R}\right) \cos\left(\theta_1 + \frac{\sigma}{2R}\right) + \frac{4R^2}{r_1^2} \sin^2\left(\frac{\sigma}{2R}\right)} - 1 \right] \\
&= \frac{r_1}{r_2 - r_1} \left[\sqrt{1 + \frac{2\sigma}{r_1} \left\{ \cos\theta_1 - \frac{\sigma}{2R} \sin\theta_1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} + \frac{\sigma^2}{r_1^2} \left\{ 1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\}} - 1 \right] \\
&= \frac{r_1}{r_2 - r_1} \left[\frac{\sigma}{r_1} \left\{ \cos\theta_1 - \frac{\sigma}{2R} \sin\theta_1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} + \frac{\sigma^2}{2r_1^2} \left\{ 1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} \right. \\
&\quad \left. - \frac{\sigma^2}{2r_1^2} \left\{ \cos\theta_1 - \frac{\sigma}{2R} \sin\theta_1 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\}^2 + \frac{\sigma}{r_1} O\left[\left(\frac{\sigma}{r_1}\right)^2\right] \right] \\
&= \frac{\sigma}{r_2 - r_1} \left\{ \cos\theta_1 + \frac{\sigma \sin\theta_1}{2r_1} \left(\sin\theta_1 - \frac{r_1}{R} \right) + O\left[\left(\frac{\sigma}{\min(r_1, 2R)}\right)^2\right] \right\} ; \\
Y(\sigma) &= \frac{r_2}{r_2 - r_1} \left[1 - \sqrt{1 - \frac{4R}{r_2} \sin\left(\frac{\sigma}{2R}\right) \cos\left(\theta_2 - \frac{\sigma}{2R}\right) + \frac{4R^2}{r_2^2} \sin^2\left(\frac{\sigma}{2R}\right)} \right] \\
&= \frac{r_2}{r_2 - r_1} \left[1 - \sqrt{1 - \frac{2\sigma}{r_2} \left\{ \cos\theta_2 + \frac{\sigma}{2R} \sin\theta_2 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} + \frac{\sigma^2}{r_2^2} \left\{ 1 - O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\}} \right] \\
&= \frac{r_2}{r_2 - r_1} \left[\frac{\sigma}{r_2} \left\{ \cos\theta_2 + \frac{\sigma}{2R} \sin\theta_2 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} - \frac{\sigma^2}{2r_2^2} \left\{ 1 - O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\} \right. \\
&\quad \left. + \frac{\sigma^2}{2r_2^2} \left\{ \cos\theta_2 + \frac{\sigma}{2R} \sin\theta_2 + O\left[\left(\frac{\sigma}{2R}\right)^2\right] \right\}^2 + \frac{\sigma}{r_1} O\left[\left(\frac{\sigma}{r_1}\right)^2\right] \right] \\
&= \frac{\sigma}{r_2 - r_1} \left\{ \cos\theta_2 - \frac{\sigma \sin\theta_2}{2r_2} \left(\sin\theta_2 - \frac{r_2}{R} \right) + O\left[\left(\frac{\sigma}{\min(r_2, 2R)}\right)^2\right] \right\} .
\end{aligned} \tag{C.38}$$

In these equations, the square roots of the form $\sqrt{1 + 2x}$ have been expanded as $1 + x - x^2/2 + O[x^3]$, exploiting the assumption that the segment lengths are small compared to the Earth center distances.

It is convenient to define parameters

$$D_1 \equiv \frac{\sin \theta_1}{r_1} \left(\sin \theta_1 - \frac{r_1}{R} \right) \quad \text{and} \quad D_2 \equiv \frac{\sin \theta_2}{r_2} \left(\sin \theta_2 - \frac{r_2}{R} \right) \quad \text{with} \quad D_1 > D_2 \quad . \quad (\text{C.39})$$

The inequality follows since r_1 has been defined to be less than r_2 , and $\sin \theta_1$ exceeds $\sin \theta_2$. The exponents $X(\sigma)$ and $Y(\sigma)$ are approximated without the 2nd and higher order terms of Eq. (C.38):

$$X(\sigma) \approx \frac{\sigma}{r_2 - r_1} \left(\cos \theta_1 + \frac{\sigma}{2} D_1 \right) \quad \text{and} \quad Y(\sigma) \approx \frac{\sigma}{r_2 - r_1} \left(\cos \theta_2 - \frac{\sigma}{2} D_2 \right) \quad . \quad (\text{C.40})$$

Substituting into Eq. (C.33), the column density has the following form:

$$\begin{aligned} u &\equiv v(a s, \rho_1, \cos \theta_1 / h, D_1 / h) + v(b s, \rho_2, -\cos \theta_2 / h, D_2 / h) \\ &= \rho_1 \int_0^{a s} (\rho_2 / \rho_1)^\wedge \left[\frac{\sigma}{r_2 - r_1} \left(\cos \theta_1 + \frac{\sigma}{2} D_1 \right) \right] d\sigma + \rho_2 \int_0^{b s} (\rho_1 / \rho_2)^\wedge \left[\frac{\sigma}{r_2 - r_1} \left(\cos \theta_2 - \frac{\sigma}{2} D_2 \right) \right] d\sigma \\ &= \rho_1 \int_0^{a s} \exp \left[-\sigma \left(\cos \theta_1 + \frac{\sigma}{2} D_1 \right) \frac{\ln(\rho_1 / \rho_2)}{r_2 - r_1} \right] d\sigma + \rho_2 \int_0^{b s} \exp \left[\sigma \left(\cos \theta_2 - \frac{\sigma}{2} D_2 \right) \frac{\ln(\rho_1 / \rho_2)}{r_2 - r_1} \right] d\sigma \quad (\text{C.41}) \\ &= \rho_1 \int_0^{a s} \exp \left[-\frac{\sigma}{h} \left(\cos \theta_1 + \frac{\sigma}{2} D_1 \right) \right] d\sigma + \rho_2 \int_0^{b s} \exp \left[\frac{\sigma}{h} \left(\cos \theta_2 - \frac{\sigma}{2} D_2 \right) \right] d\sigma \quad , \end{aligned}$$

where

$$h \equiv \frac{r_2 - r_1}{\ln(\rho_1 / \rho_2)} \quad \text{and} \quad v \equiv v(S, \rho, \cos \theta / h, D / h) \equiv \rho \int_0^S \exp \left[-\frac{\sigma}{h} \left(\cos \theta + \frac{\sigma}{2} D \right) \right] d\sigma \quad . \quad (\text{C.42})$$

The variable h is the density scale height, and v is a sub-segment column density. Since the density scale height is, by definition, a constant within a given atmospheric layer (or spherical shell), it does not need to be recomputed when the layer refractive path is partitioned into multiple circular arcs using the bi-section algorithm described above.

The parameter $a = 1 - b$ should be set between 0 and 1. It would be desirable to have the density variation with altitude be a continuous function. Mathematically, this would require that

$$\rho_1 \left(\frac{\rho_2}{\rho_1} \right)^{X(a s)} = \rho_2 \left(\frac{\rho_1}{\rho_2} \right)^{Y(b s)} = \rho_1 \left(\frac{\rho_2}{\rho_1} \right) \left(\frac{\rho_2}{\rho_1} \right)^{-Y(b s)} = \rho_1 \left(\frac{\rho_2}{\rho_1} \right)^{1-Y(s-a s)} \Rightarrow X(a s) = 1 - Y(s - a s) . \quad (\text{C.43})$$

Interestingly, this condition does not depend on the density values themselves, but is only a function of the refractive path. With $X(as)$ and $Y(s-as)$ both approximated as quadratic functions of their argument, Eq. (C.40), an expression for the product as can be derived that satisfies Eq. (C.43), as shown in Appendix δ . Unfortunately, it is demonstrated that, for straight line paths, there is no real value of a that satisfies Eq. (C.43); this result has also been empirically demonstrated for refractive paths. Instead, a and b are both simply set to $1/2$.

With the value of a and b set, the column density integrals can be evaluated in terms of special functions. For most atmospheric species, densities decrease with altitude so that ρ_1 exceeds ρ_2 when r_1 is less than r_2 , and the scale height h is positive. The parameters D_1 and D_2 will also generally be positive since both r_1 and r_2 are generally much smaller than the radius of curvature,

R ; even when both the $\sin\theta_1$ and $\sin\theta_2$ approach zero (a vertical path), the reciprocal of R also approaches zero, and D_1 and D_2 both tend to remain positive. Under these circumstances, the argument of the exponential in Eq. (C.42) has a negative quadratic term in σ , and the column density integral can be expressed in terms of error functions:

$$v = \rho \sqrt{\frac{\pi h}{2D}} \exp\left(\frac{\cos^2 \theta}{2hD}\right) \left\{ \operatorname{erf}\left(\frac{\cos \theta}{h\sqrt{2D/h}} + \frac{S}{2} \sqrt{\frac{2D}{h}}\right) - \operatorname{erf}\left(\frac{\cos \theta}{h\sqrt{2D/h}}\right) \right\} \quad \text{for } \frac{D}{h} > 0 \quad (\text{C.44})$$

In this equation and throughout this section, all square roots are defined to produce non-negative real numbers.

If the magnitude of the σ^2 term in the exponential is very small, the column density integral is most easily evaluated by expanding the exponential as follows:

$$\begin{aligned} v &= \rho \int_0^S \left(1 - \frac{\sigma^2 D}{2h} + \frac{\sigma^4 D^2}{8h^2} - \dots \right) \exp\left(-\frac{\sigma \cos \theta}{h}\right) d\sigma \quad \text{for } \left| \frac{S^2 D}{2h} \right| \text{ small} \\ &= \pm \rho_i \left\{ \begin{aligned} &\frac{h}{\cos \theta} \left[1 - \exp\left(-\frac{\cos \theta}{h/S}\right) \right] - \frac{Dh^2}{\cos^3 \theta} \left[1 - \left(\frac{\cos^2 \theta}{2(h/S)^2} + \frac{\cos \theta}{h/S} + 1 \right) \exp\left(-\frac{\cos \theta}{h/S}\right) \right] \\ &+ \frac{3D^2 h^3}{\cos^5 \theta} \left[1 - \left(\frac{\cos^4 \theta}{24(h/S)^4} + \frac{\cos^3 \theta}{6(h/S)^3} + \frac{\cos^2 \theta}{2(h/S)^2} + \frac{\cos \theta}{h/S} + 1 \right) \exp\left(-\frac{\cos \theta}{h/S}\right) \right] - \dots \end{aligned} \right\} \quad (\text{C.45}) \end{aligned}$$

If both the σ linear and quadratic terms are small, then the entire exponential term can be expanded in a Taylor series:

$$\begin{aligned} v &= \rho \int_0^S \left[1 - \frac{\sigma}{h} \left(\cos \theta + \frac{\sigma}{2} D \right) + \frac{\sigma^2}{2h^2} \left(\cos \theta + \frac{\sigma}{2} D \right)^2 - \dots \right] d\sigma \\ &= \rho S \left[1 - \frac{S}{2h} \left(\cos \theta + \frac{SD}{3} \right) + \frac{S^2}{2h^2} \left(\frac{\cos^2 \theta}{3} + \frac{SD \cos \theta}{4} + \frac{S^2 D^2}{20} \right) - \dots \right] \quad (\text{C.46}) \\ &\quad \text{for both } \left| \frac{S \cos \theta}{h} \right| \text{ and } \left| \frac{S^2 D}{2h} \right| \text{ small} . \end{aligned}$$

The column density integrals are somewhat more difficult to evaluate for the least common case, i.e., when the σ^2 term in the exponential is positive. In this case, the column density is evaluated in terms of a Dawson integral,

$$F(x) \equiv e^{-x^2} \int_0^x e^{y^2} dy \quad , \quad x \geq 0 \quad . \quad (\text{C.47})$$

The v integral is expanded as follows:

$$\begin{aligned}
v &= \rho \int_0^S \exp\left(-\frac{\sigma \cos \theta}{h} - \frac{\sigma^2 D}{2h}\right) d\sigma \quad \text{for } \frac{D}{h} < 0 \\
&= \rho \int_0^S \exp\left[\left(\sigma \sqrt{\frac{-D}{2h}} - \frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right)^2 + \frac{\cos^2 \theta}{2hD}\right] d\sigma \\
&= \rho \sqrt{\frac{-2h}{D}} \exp\left(\frac{\cos^2 \theta}{2hD}\right) \int_{\Sigma(0)}^{\Sigma(S)} \exp(\Sigma^2) d\Sigma
\end{aligned} \tag{C.48}$$

where

$$\begin{aligned}
\Sigma(\sigma) &\equiv \sigma \sqrt{\frac{-D}{2h}} - \frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}} ; \\
\left(\sqrt{\frac{-D}{2h}}\right)^2 &= \frac{-D}{2h} ; \quad 2\sqrt{\frac{-D}{2h}} \sqrt{\frac{-h}{2D}} = 1 ; \quad \left(\sqrt{\frac{-h}{2D}}\right)^2 = \frac{-h}{2D} .
\end{aligned} \tag{C.49}$$

Three cases must be distinguished depending on the sign of the integration limits.

Case a: $\Sigma(0) \geq 0$ [equivalently, $-\cos \theta / h \geq 0$]

$$v = \rho \sqrt{\frac{-2h}{D}} \left[\exp\left(-\frac{S^2 D}{2h} - \frac{S \cos \theta}{h}\right) F\left(S \sqrt{\frac{-D}{2h}} - \frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) - F\left(-\frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) \right] \tag{C.50a}$$

Case b: $\Sigma(0) < 0$ and $\Sigma(S) \geq 0$ [equivalently, $-\cos \theta / h < 0$ and $S + \cos \theta / D \geq 0$]

$$v = \rho \sqrt{\frac{-2h}{D}} \left[\exp\left(-\frac{S^2 D}{2h} - \frac{S \cos \theta}{h}\right) F\left(S \sqrt{\frac{-D}{2h}} - \frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) + F\left(\frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) \right] \tag{C.50b}$$

Case c: $\Sigma_{\pm}(S) < 0$ [equivalently, $S + \cos \theta / D < 0$]

$$v = \rho \sqrt{\frac{-2h}{D}} \left[F\left(\frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) - \exp\left(-\frac{S^2 D}{2h} - \frac{S \cos \theta}{h}\right) F\left(-S \sqrt{\frac{-D}{2h}} - \frac{\cos \theta}{h} \sqrt{\frac{-h}{2D}}\right) \right] \tag{C.50c}$$

The column density integration algorithm has been implemented and tested for the 0 to 1 km, half-tangent path defined in Table I. For this >132 km path, 14 bisections are required to provide 1 mm accuracy as is illustrated in Figure C.8. The algorithm was run for a 0 to 1 km altitude density ratio, ρ_1/ρ_2 , domain of 0.01 to 100. The column amount, u , normalized by the higher altitude density, ρ_2 , is plotted on the left in Figure C.10. It follows from Eqs. (C.41) and (C.42) that this normalized column amount depends only on the ρ_1/ρ_2 ratio, not on the individual densities. Eqs. (C.44), (C.45), (C.46), (C.50a) and (C.50c) were all used in the generation of the curves in Figure C.10. The polynomial expansion, Eq. (C.46), is invoked when both $|S \cos \theta / h|$ and $|\frac{1}{2} S^2 D / h|$ are less than 0.0005, and the integral representation of Eq. (C.45) is invoked when only $|\frac{1}{2} S^2 D / h|$ is less than 0.0005.

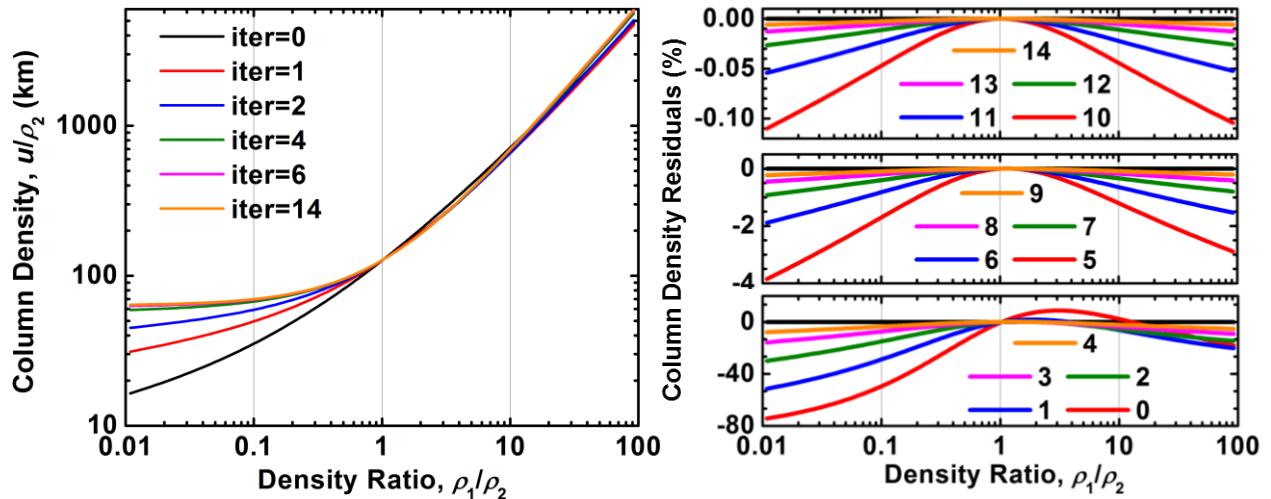


Figure C.10. Column Density Plotted as a Function of the 0 to 1 km Density Ratio for the MLW Refracted Path Defined in Table I with Curves are Labeled by the Number of Altitude Bisections

Results are presented for levels of bisection up to iteration 14. Iteration 0 corresponds to a single circular arc from 0 to 1km, while iteration 14 corresponds to 2^{14} (= 16,384) sub-segment circular arcs with a vertical step size of ~61mm. The right-hand side plot in Figure C.10 contains curves of the column amount percent error plotted as a function of the density ratio for all 15 levels of bisection. These curves were generated with the converged result represented by 17 bisections, a vertical step size of less than 8 mm. By iteration 7, the magnitude of the column amount residual is less than 1% for the entire domain; at iteration 14, the magnitude has dropped to below 0.006%. All the curves appear to intersect at ρ_1/ρ_2 equal to one, but they do differ. The density normalized column amount, u/ρ_2 , equals the path length in this case. As indicated in Table I, the iteration 0 path length is too short by 0.82%. The column density errors tend to increase when the density ratio deviates from unity. In practice, the domain of greatest interest for typical molecular profiles in the terrestrial atmosphere is $1 \leq \rho_1/\rho_2 \leq 2$; at iterations 7 and 14, the magnitude of the column density residuals over this domain is less than 0.04 and 0.0003%, respectively.

11. Summary and Future Work

Traditionally, spherical refractive geometry path lengths have been computed by numerically integrating the path length differential equation. However, numerical problems arise near the tangent point even if the dependent variable is modified to remove the singularity. Here, a new approach for solving the terrestrial spherical refractive geometry problem was introduced based on defining the unique circular arc that satisfies the spherical medium refractive path constant constraint at the arc end points. The circular arc construct leads to very simple analytic expressions for path length, bend angle and Earth center angle. The circular arcs are known to be the exact solution to the spherical refractive geometry problem if the reciprocal of the index of refraction varies linearly with the square of the Earth center distance between altitude levels. For the terrestrial atmosphere, this index of refraction profile essentially translates to requiring that the refractivity vary linearly with altitude. In actuality, the refractivity varies close to exponentially with altitude in the ambient atmosphere. To correct for this inconsistency, altitude bisection is used to converge to the true path length. Within MODTRAN, the new algorithm is used to compute >100 km half-tangent path segments to better than 1 mm accuracy. Finally, a method is defined for computing circular arc column densities in terms of Error and Dawson

functions. When the lower altitude density, ρ_1 , exceeds the upper altitude density, ρ_2 , but is less than twice ρ_2 , then the column density error is less than 0.0003%.

The new spherical refractive geometry algorithm has been tested under conditions of normal refractivity, the purview of MODTRAN. In principle, the technique of approximating path segments as circular arcs is also applicable to anomalous refraction, which includes the sub-refraction, super-refraction and ducting regimes. With sub-refraction, paths curve upward. This would require the circle in Figure C.2 to be flipped from below to above location P_1 , the arc length end point at Earth center distance r_1 . A direct application of radius of curvature formula, Eq. (C.10), would produce a negative value. This indicates that the path length and angle formulas need to be tailored for the sub-refraction case. Super-refraction refers to the situation in which the paths bend downward to such an extent that a maximum tangent point can occur. In terms of the current work, this would imply that the radius of curvature is less than the Earth radius. The path length, bend angle, and Earth center angle formulas provided here are directly applicable to this scenario; however, MODTRAN explicitly assumes that a path with a non-negative elevation angle will either terminate within the current layer or exit out the top. Under conditions of super-refraction in which a maximum tangent height does occur, MODTRAN prints a warning that it cannot handle super-refraction and exits. Ducting is the case in which super-refraction occurs at an upper altitude region and either normal or sub-refraction occurs at a lower altitude region, the result being that radiation is trapped. Future work is needed to determine the accuracy of the circular arc approach for all three anomalous refraction regimes.

12. Acknowledgements

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Appendix α : Deriving the Radius of Curvature

An expression for coordinate y as a function of x and R is determined from Eq. (C.8):

$$\begin{aligned} Rr_2 \cos \phi &= x(y + r_1 \sin \theta - R) - y(x + r_1 \cos \theta) = x(r_1 \sin \theta - R) - yr_1 \cos \theta \\ \Rightarrow y &= \frac{x(r_1 \sin \theta - R) - Rr_2 \cos \phi}{r_1 \cos \theta} \end{aligned} \quad (\text{C.}\alpha 1)$$

The $x^2 + y^2$ term in the expansion of Eq. (C.7) can be replaced by R^2 [Eq. (C.6)], and y is eliminated using Eq. (C.1):

$$\begin{aligned} 2y(r_1 \sin \theta - R) &= r_2^2 - r_1^2 - 2R^2 + 2r_1(R \sin \theta - x \cos \theta) \\ 2(r_1 \sin \theta - R)[x(r_1 \sin \theta - R) - Rr_2 \cos \phi] &= r_1 \cos \theta [r_2^2 - r_1^2 - 2R^2 + 2r_1(R \sin \theta - x \cos \theta)] \end{aligned} \quad (\text{C.}\alpha 2)$$

Solving for x as a function of R results in the following expression:

$$x = \frac{R(r_1 \sin \theta - R)(r_2 \cos \phi + r_1 \cos \theta) + \frac{1}{2} r_1(r_2^2 - r_1^2) \cos \theta}{r_1^2 - 2Rr_1 \sin \theta + R^2} \quad (\text{C.}\alpha 3)$$

Returning to Eq. (C.7), Eq. (C.α1) is used to eliminate y and the remaining terms are sorted in powers of x :

$$\begin{aligned} \left[x(r_1 \sin \theta - R) - R r_2 \cos \phi \right]^2 &= (R^2 - x^2) r_1^2 \cos^2 \theta \\ x^2(r_1^2 - 2Rr_1 \sin \theta + R^2) - 2xRr_2 \cos \phi(r_1 \sin \theta - R) - R^2(r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi) &= 0 \end{aligned} \quad (\text{C.}\alpha 4)$$

A linear equation in x is obtained by substituting for $x(r_1^2 - 2Rr_1 \sin \theta + R^2)$ using Eq. (C.α3):

$$\begin{aligned} x \left[R(r_1 \sin \theta - R)(r_2 \cos \phi + r_1 \cos \theta) + \frac{1}{2} r_1(r_2^2 - r_1^2) \cos \theta \right] \\ = 2xRr_2 \cos \phi(r_1 \sin \theta - R) + R^2(r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi) \end{aligned} \quad (\text{C.}\alpha 5)$$

$$x = \frac{R^2(r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi)}{R(r_1 \sin \theta - R)(r_1 \cos \theta - r_2 \cos \phi) + \frac{1}{2} r_1(r_2^2 - r_1^2) \cos \theta} \quad (\text{C.}\alpha 6)$$

Combine Eqs. (C.α3) and (C.α6) to obtain an equation for R :

$$\begin{aligned} (r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi)(r_1^2 - 2Rr_1 \sin \theta + R^2)R^2 \\ = \left[r_1 \cos \theta(r_1 \sin \theta - R)R + \frac{1}{2}(r_2^2 - r_1^2)r_1 \cos \theta \right]^2 - r_2^2 \cos^2 \phi(r_1 \sin \theta - R)^2 R^2 \\ = (r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi)(r_1 \sin \theta - R)^2 R^2 \\ + R_1^2(R_2^2 - R_1^2) \cos^2 \theta (R_1 \sin \theta - R)R + \frac{1}{4}(R_2^2 - R_1^2)^2 R_1^2 \cos^2 \theta \end{aligned} \quad (\text{C.}\alpha 7)$$

Both the quartic and cubic terms in R exactly cancel when first term on the right-hand side is subtracted from the left-hand side:

$$\begin{aligned} r_1^2 \cos^2 \theta (r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi)R^2 \\ = r_1^2 \cos^2 \theta (r_2^2 - r_1^2)(r_1 \sin \theta - R)R + \frac{1}{4} r_1^2 \cos^2 \theta (r_2^2 - r_1^2)^2 \end{aligned} \quad (\text{C.}\alpha 8)$$

The term $(r_1 \cos \theta)^2$ factors

$$\begin{aligned} 2(r_1^2 \cos^2 \theta - r_2^2 \cos^2 \phi)R^2 - 2(r_2^2 - r_1^2)(r_1 \sin \theta - R)R - \frac{1}{2}(r_2^2 - r_1^2)^2 \\ = 2(r_2^2 \sin^2 \phi - r_1^2 \sin^2 \theta)R^2 - 2(r_2^2 - r_1^2)r_1 \sin \theta R - \frac{1}{2}(r_2^2 - r_1^2)^2 = 0 \end{aligned} \quad (\text{C.}\alpha 9)$$

The discriminant of the quadratic equation in R is a perfect square, so the expression for R simplifies:

$$\begin{aligned}
R &= \frac{(r_2^2 - r_1^2)r_1 \sin \theta \pm \sqrt{(r_2^2 - r_1^2)^2 r_1^2 \sin^2 \theta + (r_2^2 - r_1^2)^2 (r_2^2 \sin^2 \phi - r_1^2 \sin^2 \theta)}}{2 (r_2^2 \sin^2 \phi - r_1^2 \sin^2 \theta)} \\
&= \frac{r_2^2 - r_1^2}{r_2^2 \sin^2 \phi - r_1^2 \sin^2 \theta} \left(\frac{r_1 \sin \theta \pm r_2 \sin \phi}{2} \right) \\
&= \frac{(r_1 - r_2)(r_1 + r_2)/2}{r_1 \sin \theta \mp r_2 \sin \phi} .
\end{aligned} \tag{C.α10}$$

Now consider the straight line (no refraction) spherical geometry case. Without refraction, the terms $r_1 \sin \theta$ and $r_2 \sin \phi$ are equal, and the radius of curvature is infinite. Thus, the upper sign must prevail in Eq. (C.α10):

$$R = \frac{(r_2^2 - r_1^2)/2}{r_2 \sin \phi - r_1 \sin \theta} . \tag{C.α11}$$

Appendix β: Deriving the Cartesian Distance

An expression for the Cartesian distance d between points P_1 and P_2 , Figure C.3, in terms of Earth center distances, r_1 and r_2 , and zenith angles, θ and ϕ , is desired. The derivation starts with Eq. (C.10):

$$d^2 = x^2 + (R - y)^2 = 2R(R - y) . \tag{C.β1}$$

Coordinate y is eliminated from quantity $R - y$ by substituting Eq. (C.α1):

$$\begin{aligned}
R - y &= R - \frac{x (r_1 \sin \theta - R) - R r_2 \cos \phi}{r_1 \cos \theta} \\
&= \frac{R (r_2 \cos \phi + r_1 \cos \theta) + x (R - r_1 \sin \theta)}{r_1 \cos \theta} .
\end{aligned} \tag{C.β2}$$

Substituting Eq. (C.α6) for x , the term $R (r_2 \cos \phi - r_1 \cos \theta)$ factors out

$$\begin{aligned}
R - y &= \frac{R (r_2 \cos \phi + r_1 \cos \theta)}{r_1 \cos \theta} \left[1 + \frac{R (R - r_1 \sin \theta) (r_1 \cos \theta - r_2 \cos \phi)}{\frac{1}{2} r_1 (r_2^2 - r_1^2) \cos \theta - R (R - r_1 \sin \theta) (r_1 \cos \theta - r_2 \cos \phi)} \right] \\
&= \frac{R (r_2 \cos \phi + r_1 \cos \theta) (r_2^2 - r_1^2)}{r_1 (r_2^2 - r_1^2) \cos \theta - 2R (R - r_1 \sin \theta) (r_1 \cos \theta - r_2 \cos \phi)} .
\end{aligned} \tag{C.β3}$$

Returning to Eq. (C.β1), the Cartesian distance is expressed as a function of the radius of curvature:

$$d^2 = \frac{2R^2 (r_2 \cos \phi + r_1 \cos \theta) (r_2^2 - r_1^2)}{r_1 (r_2^2 - r_1^2) \cos \theta - 2R (R - r_1 \sin \theta) (r_1 \cos \theta - r_2 \cos \phi)} . \tag{C.β4}$$

After Eq. (C.54) is used to eliminate the radius of curvature; a factor to $r_2 \cos \phi + r_1 \cos \theta$ factors from the numerator and denominator:

$$d^2 = \frac{(r_2 \cos \phi + r_1 \cos \theta)(r_2^2 - r_1^2)^2}{2r_1(r_2 \sin \phi - r_1 \sin \theta)^2 \cos \theta - [r_2^2 - r_1^2 - 2r_1 \sin \theta(r_2 \sin \phi - r_1 \sin \theta)](r_1 \cos \theta - r_2 \cos \phi)} \quad (\text{C.}\beta 5)$$

$$= \frac{(r_2 \cos \phi + r_1 \cos \theta)(r_2^2 - r_1^2)^2}{\left\{ \begin{aligned} &r_2^3 \cos \phi - r_2^2 r_1 (\cos \theta - 2 \sin^2 \phi \cos \theta + 2 \sin \phi \cos \phi \sin \theta) \\ &- r_2 r_1^2 (\cos \phi - 2 \sin^2 \theta \cos \phi + 2 \sin \phi \sin \theta \cos \theta) + r_1^3 \cos \theta \end{aligned} \right\}} .$$

The $r_2 \cos \phi + r_1 \cos \theta$ term in the numerator factors out of the denominator resulting in the following expression for the square of the Cartesian distance between points P_1 and P_2 :

$$d^2 = \frac{(r_2^2 - r_1^2)^2}{r_2^2 - 2r_2 r_1 \cos(\phi - \theta) + r_1^2} . \quad (\text{C.}\beta 6)$$

Appendix γ: Deriving the Path Bending Angle

The sine of the angle subtended by the circular arc ψ , as illustrated in Figure C.3, can be expressed as a function of the radius of curvature, R , and the Cartesian distance, d :

$$\sin \psi = 2 \sin \frac{\psi}{2} \cos \frac{\psi}{2} = \frac{d}{R} \sqrt{1 - \frac{d^2}{4R^2}} . \quad (\text{C.}\gamma 1)$$

From Eqs. (C.54) and (C.56), $\frac{1}{2}(d/R)$ equals

$$\frac{d}{2R} = \frac{r_2 \sin \phi - r_1 \sin \theta}{\sqrt{r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2}} . \quad (\text{C.}\gamma 2)$$

Substituting this expression into Eq. (C.γ1) eliminates all the square root terms

$$\sin \psi = \frac{2(r_2 \sin \phi - r_1 \sin \theta)}{\sqrt{r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2}} \sqrt{\frac{r_2^2 \cos^2 \phi - 2r_2 r_1 \cos \theta \cos \phi + r_1^2 \cos^2 \theta}{r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2}} \quad (\text{C.}\gamma 3)$$

$$= \frac{2(r_2 \sin \phi - r_1 \sin \theta)(r_1 \cos \theta - r_2 \cos \phi)}{r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2} .$$

For computational purposes, it is convenient to rewrite the denominator using the terms in the numerator:

$$r_2^2 - 2r_2 r_1 \cos(\theta - \phi) + r_1^2 = (r_2 \sin \phi - r_1 \sin \theta)^2 + (r_1 \cos \theta - r_2 \cos \phi)^2 . \quad (\text{C.}\gamma 4)$$

Making this substitution, the circular arc bending and subtended angle is given by

$$\psi = \arcsin \left(\frac{2(r_2 \sin \phi - r_1 \sin \theta)(r_1 \cos \theta - r_2 \cos \phi)}{(r_2 \sin \phi - r_1 \sin \theta)^2 + (r_1 \cos \theta - r_2 \cos \phi)^2} \right) . \quad (\text{C.}\gamma 5)$$

Appendix δ: Determining Path Lengths at which Densities Expanded about the Arc End Points Equal

An equation for the path length as , used in the path length integral of Eq. (C.35), is obtained by inserting Eq. (C.40) into the second line of Eq. (C.43):

$$\begin{aligned} \frac{as}{r_2 - r_1} \left(\cos \theta_1 + \frac{as}{2} D_1 \right) &= 1 - \frac{s - as}{r_2 - r_1} \left(\cos \theta_2 - \frac{s - as}{2} D_2 \right) \\ \Rightarrow 0 &= r_2 - r_1 - as \left(\cos \theta_1 + \frac{as}{2} D_1 \right) - (s - as) \left(\cos \theta_2 - \frac{s - as}{2} D_2 \right) . \end{aligned} \quad (\text{C.81})$$

This expression can be rewritten as a quadratic polynomial in as :

$$0 = r_2 - r_1 - s \left(\cos \theta_2 - \frac{s}{2} D_2 \right) + as (\cos \theta_2 - \cos \theta_1 - s D_2) - \frac{a^2 s^2}{2} (D_1 - D_2) . \quad (\text{C.82})$$

The discriminant of this quadratic equation can itself be expanding as a quadratic in s :

$$\begin{aligned} B^2 - 4AC &= (\cos \theta_2 - \cos \theta_1 - s D_2)^2 + [2(r_2 - r_1) - s(2 \cos \theta_2 - s D_2)](D_1 - D_2) \\ &= (\cos \theta_2 - \cos \theta_1)^2 + 2(r_2 - r_1)(D_1 - D_2) \\ &\quad - 2s[(\cos \theta_2 - \cos \theta_1)D_2 + \cos \theta_2(D_1 - D_2)] + s^2[D_2^2 + D_2(D_1 - D_2)] \\ &= (\cos \theta_2 - \cos \theta_1)^2 + 2(r_2 - r_1)(D_1 - D_2) - 2s(D_1 \cos \theta_2 - D_2 \cos \theta_1) + s^2 D_1 D_2 . \end{aligned} \quad (\text{C.83})$$

The solution to the quadratic equation for as becomes:

$$\begin{aligned} as &= \frac{1}{D_1 - D_2} \left(\cos \theta_2 - \cos \theta_1 - s D_2 \pm \sqrt{2(r_2 - r_1)(D_1 - D_2) + (\cos \theta_2 - \cos \theta_1)^2 - 2s(D_1 \cos \theta_2 - D_2 \cos \theta_1) + s^2 D_1 D_2} \right) \\ &= \frac{2[s \cos \theta_2 - (r_2 - r_1)] - s^2 D_2}{\cos \theta_2 - \cos \theta_1 - s D_2 \mp \sqrt{2(r_2 - r_1)(D_1 - D_2) + (\cos \theta_2 - \cos \theta_1)^2 - 2s(D_1 \cos \theta_2 - D_2 \cos \theta_1) + s^2 D_1 D_2}} . \end{aligned} \quad (\text{C.84})$$

Unfortunately, it turns out that the discriminant is generally negative. This can be analytically demonstrated for the no refraction (constant refractivity, straight path) case. For such a path, the following relationships hold true:

$$\begin{aligned} r_1 &= r_2 \frac{\sin \theta_2}{\sin \theta_1} ; \quad s = r_2 \cos \theta_2 - r_1 \cos \theta_1 = r_2 \left(\cos \theta_2 - \frac{\cos \theta_1 \sin \theta_2}{\sin \theta_1} \right) ; \quad \frac{1}{R} = 0 \\ D_1 &= \frac{\sin^2 \theta_1}{r_1} = \frac{\sin^3 \theta_1}{r_2 \sin \theta_2} ; \quad \text{and} \quad D_2 = \frac{\sin^2 \theta_2}{r_2} . \end{aligned} \quad (\text{C.85})$$

Substituting into the expression for the discriminant and simplifying, one can derive the following expression:

$$B^2 - 4AC = 2 - 3\sin^2 \theta_1 + 3\sin^3 \theta_1 \sin \theta_2 - 3\sin^2 \theta_2 \\ + (3 - 2\sin^2 \theta_1) \sin \theta_1 \sin^2 \theta_2 - 2\cos^3 \theta_1 \cos^3 \theta_2 \quad . \quad (\text{C.86})$$

Note that this equation is symmetric with regards to interchange of the zenith angles, θ_1 and θ_2 . For short, non-vertical paths, the zenith angle at r_2 ($> r_1$) will be slightly less than the value at r_1 :

$$\theta_2 = \theta_1 - \varepsilon \quad . \quad (\text{C.87})$$

Here ε is the decrease in zenith angle. Inserting this expression for θ_2 into Eq. (C.86) and expanding about $\varepsilon = 0$, the following result is obtained

$$B^2 - 4AC = -3\varepsilon^2 \cos^6 \theta_1 + O(\varepsilon^3) \quad . \quad (\text{C.88})$$

Thus, the short path discriminant is negative whenever $\cos \theta_1$ is non-zero. When the $\cos \theta_1$ is zero (the half-tangent case), direct substitution of $\theta_1 = \frac{1}{2}\pi$ into Eq. (C.86) also yields a negative discriminant:

$$B^2 - 4AC = -(1 - \sin \theta_2)^3 \quad \text{for} \quad \theta_1 = \pi/2 \quad . \quad (\text{C.89})$$

APPENDIX D: AN IMPROVED MODTRAN SINGLE SCATTER SOLAR MODEL

A. Berk and R. Panfili

An effort is underway to improve the MODTRAN single scatter solar calculations. From its inception, MODTRAN has computed the solar illumination path molecular transmittances by modeling the path as a single homogeneous layer with an effective temperature and pressure. By contrast, line-of-sight (LOS) paths are modeled within the band model using Curtis-Godson (CG) path averaging. The computationally more intensive and quantitatively more accurate CG approach requires (1) determining spectral molecular band model data for each segment along the line-of-sight, (2) summing molecular total optical depths and optical depth weighted averages for line widths and line densities, and (3) computing full path spectral transmittances from the summed data. These calculations must be performed at every spectral grid point.

The goal of this effort is to upgrade the solar illumination path transmittance calculations using the CG formulation, but to do so without adding appreciably to run time. For a LOS path containing N segments, the GC sums involve N terms. A direct implementation of CG for solar illumination paths would require $\sum_{i=0,N} m_i$ segment calculations, where m_i is the number of segments between LOS level i and the sun. Essentially, the number of segment calculations would grow proportional to the square of the number of line-of-sight segments.

To reduce the dimensionality of the CG problem for solar illumination, the CG sums from level $i - 1$ and over m_{i-1} path segments are reused or scaled to compute the sums from level i ($i > 0$) over m_i path segments. For each level after the first, new band model data need only be calculated for one segment. Thus, only $m_0 + N$ illumination path segments are processed. The two sub-sections below contain (1) the notes compiled on the effort and (2) viewgraphs that were prepared to describe the effort and its status.

D.1 Project Notes

Modtran currently considers the solar path to consist of a single, density averaged, segment. The proposed solution is to separate the solar path into two parts: a local layer and a scaling of the previous solar illumination path. This is illustrated in Figure D.1.

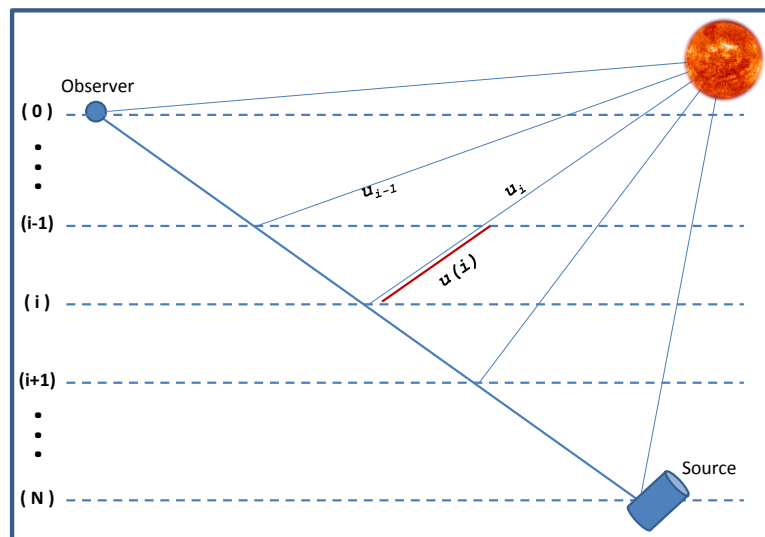


Figure D.1: Schematic illustration of solar path computation.

Rather than reconstruct the Curtis-Godson averaging within Modtran, an approximation to the actual solar path, albeit one that is much improved from the existing approach, is used.

$$\left(\frac{Su}{d}\right)_i = \left(\frac{Su}{d}\right)_{i-1} \cdot \left(\frac{u_i - u(i)}{u_{i-1}}\right) + \frac{S}{d}(i) \cdot u(i) \quad (\text{D.1})$$

Here, $(Su/d)_{i-1}$ and $(Su/d)_i$ are the weak-line molecular absorption optical depths for the solar paths ending at levels $i - 1$ and i , respectively; u_{i-1} and u_i contain pre-computed column amounts for the solar paths ending at levels $i - 1$ and i , respectively; and $u(i)$ contain the single segment column amount between levels i and $i - 1$. Essentially, the previous (Su/d) value is being corrected to account for the slightly different column amount between the nearly parallel paths cover a common set of layers, and then the contribution from the new layer's segment is added. Implementation of this new capability involved 8 steps:

Step 1: Allocate space for the new arrays (sg2Dta_mod.f90).

The SEG2 variable is part of the sg2Dta FORTRAN derived type. The derived type is being extended to hold several additional variables. The new arrays are allocated and deallocated in the same location as SEG2. The new arrays are created as follows:

```
type :: sg2Dta
  real, dimension(:,:), pointer :: SEG2 => NULL()
  real, dimension(:,:), pointer :: SEG2N => NULL()
  real, dimension(:,:), pointer :: SEG2F => NULL()
  real, dimension(:,:), pointer :: SSTLM1 => NULL()
  real, dimension(:,:), pointer :: DPTM1 => NULL()
  real, dimension(:,:), pointer :: ODBM1 => NULL()
  real, dimension(:,:), pointer :: ADBM1 => NULL()
  real, dimension(:,:), pointer :: ACBM1 => NULL()
  real, dimension(:,:), pointer :: AC2M1 => NULL()
end type sg2Dta
```

These arrays contain the following information:

- SEG2N contains the column density of the “near” segment. This can be an individual layer or a density-weighted average of a small number of layers, depending on the geometry of the problem. It is a two dimensional array with one dimension to hold segments along the LOS and another to hold unique values for each molecular radiator.
- SEG2F contains the segment quantities for the first solar path, defined as originating at the observer and extending to the sun. This path – the first in the sequence – is performed to a higher level of fidelity and so each segment must be retained.
- SSTLM1 holds the band model path-averaged line strength for the previous solar path along the LOS. This is continuously updated as new segments along the LOS are added. It is a one dimensional array holding path-weighted line strength values for each radiator.

Then, memory is allocated in sg2Set

```

subroutine sg2Set(pSG2D,MSEGL,MSEGU,MXL,MXU)
integer, parameter :: NTLSUB = 4 ! copied from mbmDta_mod
integer, parameter :: NPRMX =2   ! Num Band Model Pairs (1 or 2)
...
ALLOCATE (pSG2D%SEG2 (MXL:MXU,MSEGL:MSEGU),STAT=istate)
ALLOCATE (pSG2D%SEG2N (MXL:MXU,MSEGL:MSEGU),STAT=istate)
ALLOCATE (pSG2D%SEG2F (MXL:MXU,MSEGL:MSEGU),STAT=istate)
ALLOCATE (pSG2D%SSTLM1 (0:NTLSUB,MXL:MXU),STAT=istate)
ALLOCATE (pSG2D%DPTM1 (1:NPRMX,MXL:MXU),STAT=istate)
ALLOCATE (pSG2D%ODBM1 (1:NPRMX,MXL:MXU),STAT=istate)
ALLOCATE (pSG2D%ADBM1 (1:NPRMX,MXL:MXU),STAT=istate)
ALLOCATE (pSG2D%ACBM1 (1:NPRMX,MXL:MXU),STAT=istate)
ALLOCATE (pSG2D%AC2M1 (1:NPRMX,MXL:MXU),STAT=istate)
...
end subroutine sg2Set

```

These arrays will be used in precisely the same set of subroutines as SEG2, the existing solar path array variable. Both allocation and de-allocation are coupled together.

Finally, the routines necessary to free up memory are modified once these arrays are no longer required.

```

subroutine sg2Clean(pSG2D)
...
if (ASSOCIATED(pSG2D%SEG2 )) DEALLOCATE(pSG2D%SEG2 )
if (ASSOCIATED(pSG2D%SEG2N)) DEALLOCATE(pSG2D%SEG2N)
if (ASSOCIATED(pSG2D%SEG2F)) DEALLOCATE(pSG2D%SEG2F)
if (ASSOCIATED(pSG2D%SSTLM1)) DEALLOCATE(pSG2D%SSTLM1)
if (ASSOCIATED(pSG2D%DPTM1)) DEALLOCATE(pSG2D%DPTM1)
if (ASSOCIATED(pSG2D%ODBM1)) DEALLOCATE(pSG2D%ODBM1)
if (ASSOCIATED(pSG2D%ADBM1)) DEALLOCATE(pSG2D%ADBM1)
if (ASSOCIATED(pSG2D%ACBM1)) DEALLOCATE(pSG2D%ACBM1)
if (ASSOCIATED(pSG2D%AC2M1)) DEALLOCATE(pSG2D%AC2M1)
...
end subroutine sg2Clean

```

These arrays do not need to be initialized to zero. These arrays – if they need to be used – will be specifically set to a value in MODTRAN routine ssgeo().

Step 2: Allocate space for temperature and pressures of first solar path (tisDta_mod.f90).

Space needs to be created to store temperature and pressure variables. Three informational variables, in addition to the temperatures and pressures of the segments along the first path, are defined: (1) to total number of layers in the first solar path, (2) whether the first solar path begins on a vertex or is mid-layer, and (3) whether the LOS looks up or down.

```

type :: tisDta
integer :: NSEGF
logical :: MIDLYR ! mid-layer seg
logical :: ULOOK ! looking up
real, dimension(:), pointer :: TSSEGF => NULL() ! First solar
real, dimension(:), pointer :: PSATMF => NULL() ! First solar
real, dimension(:), pointer :: T5SF => NULL()
real, dimension(:), pointer :: PTSF => NULL()
type(tpsDta), dimension(:), pointer :: LTSA => NULL()
end type tisDta

```


Memory is allocated in tisSet:

```
...
integer iseg

ALLOCATE (pTISD%TSSEGF (MSEGL:MSEGU) )
ALLOCATE (pTISD%PSATMF (MSEGL:MSEGU) )
ALLOCATE (pTISD%LTSA (MSEGL:MSEGU) )
ALLOCATE (pTISD%T5SF (MSEGL:MSEGU) )
ALLOCATE (pTISD%PTSF (MSEGL:MSEGU) )
...
```

De-allocation occurs in tisClean:

```
...
do iseg = LBOUND(pTISD%LTSA,1),UBOUND(pTISD%LTSA,1)
  call tpsCleanup(pTISD%LTSA(iseg))
enddo
DEALLOCATE (pTISD%LTSA)
endif
IF (ASSOCIATED(pTISD%TSSEGF)) DEALLOCATE (pTISD%TSSEGF)
IF (ASSOCIATED(pTISD%PSATMF)) DEALLOCATE (pTISD%PSATMF)
IF (ASSOCIATED(pTISD%PTSF )) DEALLOCATE (pTISD%PTSF )
IF (ASSOCIATED(pTISD%T5SF )) DEALLOCATE (pTISD%T5SF )
...
```

These arrays do not need to be initialized to zero. When necessary, they will be specifically set to a value in MODTRAN routine ssgeo().

Step 3: Allocate space for temperature, pressure, and interpolating factors (tpsDta_mod.f90).

Interpolation values are defined. The process is similar to that for segment lengths. Space is created for (1) temperature and pressure data of “near” segments along the solar paths and (2) interpolating factors for both the near segments and first solar path. Two new variables define temperature interpolating indices (JTS*) and their weighting factors (FTS*). Similar values are defined for pressure (P2S* and iLoP2S*, respectively).

```
type :: tpsDta
...
real(k_DOUBLE), dimension(:), pointer :: DT5SH => NULL(),      &
                                         DC2TSH => NULL(),      &
                                         DPTRtH => NULL()

integer, dimension(:), pointer :: JTSF => NULL()
integer, dimension(:), pointer :: JTSN => NULL()
integer, dimension(:), pointer :: iLoP2SF => NULL()
integer, dimension(:), pointer :: iLoP2SN => NULL()
real, dimension(:), pointer :: FTSF => NULL()
real, dimension(:), pointer :: FTSN => NULL()
real, dimension(:), pointer :: P2SF => NULL()
real, dimension(:), pointer :: P2SN => NULL()
real :: TSSEGN
real :: PSATMN
end type tpsDta
```

Memory is allocated in tpsSetup. For the time being, the original values are used for both band model and correlated-*k* computations.

```

...
ALLOCATE (pTPSD%JTS (ISIZE), pTPSD%iLoP2S (ISIZE), pTPSD%FTS (ISIZE), &
          pTPSD%P2S (ISIZE), pTPSD%PTS (ISIZE), pTPSD%T5S (ISIZE), &
          pTPSD%PSATM (ISIZE), pTPSD%TSSEG (ISIZE))
ALLOCATE (pTPSD%JTSF (ISIZE))
ALLOCATE (pTPSD%JTSN (ISIZE))
ALLOCATE (pTPSD%FTSF (ISIZE))
ALLOCATE (pTPSD%FTSN (ISIZE))
ALLOCATE (pTPSD%iLoP2SF (ISIZE))
ALLOCATE (pTPSD%iLoP2SN (ISIZE))
ALLOCATE (pTPSD%P2SF (ISIZE))
ALLOCATE (pTPSD%P2SN (ISIZE))
...

```

De-allocate occurs in tpsCleanup

```

...
if (ASSOCIATED (pTPSD%JTS)) DEALLOCATE (pTPSD%JTS)
if (ASSOCIATED (pTPSD%JTSF)) DEALLOCATE (pTPSD%JTSF)
if (ASSOCIATED (pTPSD%JTSN)) DEALLOCATE (pTPSD%JTSN)
if (ASSOCIATED (pTPSD%iLoP2S)) DEALLOCATE (pTPSD%iLoP2S)
if (ASSOCIATED (pTPSD%iLoP2SF)) DEALLOCATE (pTPSD%iLoP2SF)
if (ASSOCIATED (pTPSD%iLoP2SN)) DEALLOCATE (pTPSD%iLoP2SN)
if (ASSOCIATED (pTPSD%FTS)) DEALLOCATE (pTPSD%FTS)
if (ASSOCIATED (pTPSD%FTSF)) DEALLOCATE (pTPSD%FTSF)
if (ASSOCIATED (pTPSD%FTSN)) DEALLOCATE (pTPSD%FTSN)
if (ASSOCIATED (pTPSD%P2S)) DEALLOCATE (pTPSD%P2S)
if (ASSOCIATED (pTPSD%P2SF)) DEALLOCATE (pTPSD%P2SF)
if (ASSOCIATED (pTPSD%P2SN)) DEALLOCATE (pTPSD%P2SN)
if (ASSOCIATED (pTPSD%PTS)) DEALLOCATE (pTPSD%PTS)
if (ASSOCIATED (pTPSD%T5S)) DEALLOCATE (pTPSD%T5S)
if (ASSOCIATED (pTPSD%PSATM)) DEALLOCATE (pTPSD%PSATM)
...

```

Step 4: Fill arrays with proper values (ssgeo.f)

Required variables are computed in the MODTRAN geo() subroutine. These values must be stored once defined.

Several module variables are defined:

```

SUBROUTINE SS GEO (...)
...
USE aernsm_mod, ONLY: COSSAP, GFAC, IPH, LSAP, LYDA, ML, NANSAP, NWVSAP, &
&                      ZTOL
...
INTEGER :: IMIN
REAL :: RMIN
...

```

There are a number important features to be aware of as this data is recorded.

- The call to geo() uses an atmospheric path structure (pWPTH). This data is transferred from that structure into the solar path structure (pWSPTH).
- The solar path loop runs through $N+1$ endpoints for a LOS consisting of N segments. Consequently, when the loop provides values for segment *ISEG*, they are stored in an increment one level lower meaning array runs through values 0 to N .

- The first endpoint along the LOS is modeled differently from the rest, and all solar illumination path segment values are retained. The interpretation of segments along the *SEG2F* array is therefore different than the interpretation of line segments along other arrays beginning with the *SEG* name. The *SEG2F* array contains segments along a specific path (the observer to the sun).

```

SUBROUTINE SSGEO(...)
...
10    CONTINUE
!    LOAD CORRECTIONS TO SOLAR PATH LENGTH
    pWSPTH(ILOS)%SEG2N(:, ISEGM1) = pWPTH(0)%SEG2(:, ISEG)
    if (ISEG.eq.1) pWSPTH(ILOS)%SEG2F(:, 1:) = pWPTH(0)%SEG2(:, :)
...
END SUBROUTINE SSGEO

```

Step 5: Record segment temperature, pressure, and logical values (ssgeo.f)

Local temperature and pressure are required for determining Curtis-Godson values. They are present immediately upon computing the solar path and must be recorded segment by segment. The number of segments in the first solar path is stored.

```

SUBROUTINE SSGEO(...)
...
!    @RECORD NEAR LAYER TEMPERATURE AND PRESSURE FOR SOLAR PATH
!    @NOTE 1: ILOS = NLOS+1 IS THE MS PATH. THERE MUST BE A TEST
!            OF ILOS TO MAKE SURE YOU ARE NOT OUT-OF-BOUNDS
!    @NOTE 2: ACCUMULATION MUST BE AGAINST ISEG, NOT ISEGM1
    IF (ILOS.le.size(pTPSA)) THEN
        pWSPTH(ILOS)%SEG2N(:, ISEGM1)=pWPTH(0)%SEG2(:, 1)
        IF (ISEG.EQ.1) THEN
            pWSPTH(ILOS)%SEG2F(:, 1:)=pWPTH(0)%SEG2(:, 1:)
            pTPSA(ILOS)%NSEGF      = NSEG(0)
            pTPSA(ILOS)%TSSEGF(:)  = pPALS(0)%TSEG(:)
            pTPSA(ILOS)%PSATMF(:)  = pPALS(0)%PATM(:)
            pTSLA(ILOS)%NSEGF      = NSEG(0)
            pTSLA(ILOS)%TSSEGF(:)  = pPALS(0)%TSEG(:)
            pTSLA(ILOS)%PSATMF(:)  = pPALS(0)%PATM(:)
        ELSE
            pTPSA(ILOS)%LTSA(ISEGM1)%TSSEGN = pPALS(0)%TSEG(1)
            pTPSA(ILOS)%LTSA(ISEGM1)%PSATMN = pPALS(0)%PATM(1)
            pTSLA(ILOS)%LTSA(ISEGM1)%TSSEGN = pPAPL(0)%TSEG(1)
            pTSLA(ILOS)%LTSA(ISEGM1)%PSATMN = pPAPL(0)%PATM(1)
        ENDIF
    ELSE
        ! MS VERTICAL PATH
        IF (ISEG.EQ.1) THEN
            pTMSS%NSEGF      = NSEG(0)
            IK = ubound(pTMSS%TSSEGF(:), 1)
            pTMSS%TSSEGF(:)  = pPALS(0)%TSEG(:IK)
            pTMSS%PSATMF(:)  = pPALS(0)%PATM(:IK)
        ELSE
            pTMSS%LTSA(ISEG)%TSSEGN = pPALS(0)%TSEG(1)
            pTMSS%LTSA(ISEG)%PSATMN = pPALS(0)%PATM(1)
        ENDIF
    ENDIF
...
END SUBROUTINE SSGEO

```

Next, details of the type of solar paths are recorded.

```

SUBROUTINE SS GEO (...)
...
!
! @RECORD SALIENT DETAILS REGARDING SOLAR LOS
! @ (1) DOES LOS BEGIN AT A LAYER BOUNDARY?
! @ (2) WHAT DIRECTION IS THE LOS POINTING
IF (ISEG.EQ.1) THEN
    IMIN = 1
    RMIN = ABS (LYDA (1) %ZM-H1ALTX)
    DO IK=1,size (LYDA)
        IF (ABS (LYDA (IK) %ZM-H1ALTX) .lt. RMIN) IMIN = IK
        RMIN = min (RMIN,abs (LYDA (IK) %ZM-H1ALTX))
    ENDDO
    IF (ILOS.le.size (pTPSA)) THEN                ! LOS PATH
        IF (RMIN.le.ZTOL) then
            pTPSA (ILOS) %MIDLYR = .false.
            pTSLA (ILOS) %MIDLYR = .false.
        ELSE
            pTPSA (ILOS) %MIDLYR = .true.
            pTSLA (ILOS) %MIDLYR = .true.
        ENDIF
        !
        IF (OBSZEN.LE.90.D0) THEN
            pTPSA (ILOS) %UPLOOK = .true.
            pTSLA (ILOS) %UPLOOK = .true.
        ELSE
            pTPSA (ILOS) %UPLOOK = .false.
            pTSLA (ILOS) %UPLOOK = .false.
        ENDIF
    ELSE                                           ! MS VERTICAL PATH
        IF (RMIN.le.ZTOL) then
            pTMSS %MIDLYR = .false.
        ELSE
            pTMSS %MIDLYR = .true.
        ENDIF
        !
        IF (OBSZEN.LE.90.D0) THEN
            pTMSS %UPLOOK = .true.
        ELSE
            pTMSS %UPLOOK = .false.
        ENDIF
    ENDIF
ENDIF
...
END SUBROUTINE SS GEO

```

Step 6: Perform temperature/pressure interpolation (pt_int.f90)

There are four separate blocks of code that need to be considered when performing the interpolation (ambient radiators, plume radiators, ambient cross-section species, and plume cross-section species). Interpolations are grouped by capability (improved single scattering approximation) rather than computing these quantities within the pre-existing code blocks. The new interpolation values for the “near” segment of the solar path appear in one block of code.

```

...
!=====
!!****im* mbmDta_mod/PNTERP
!!
!! COPYRIGHT
!!   Copyright 2015 Spectral Sciences, Inc.
!!   Neither Spectral Sciences, Inc. nor its employees makes any
!!   warranty, express or implied, regarding the uses, accuracy,
!!   or completeness of this software.
!!
!! DESCRIPTION
!!   PNTERP determines pressure interpolation values
!!
!! CALLED BY
!!   PT_INT
!!
!! INPUTS
!!   PP           Local pressure [atm]
!!
!! OUTPUTS
!!   P2SIDX       Interpolation lower index.
!!   P2SFRC       Interpolation fraction (for upper index)
!!
!! SOURCE
!=====
      SUBROUTINE PNTERP (PP,P2SIDX,P2SFRC)

      IMPLICIT NONE

!      INPUT ARGUMENTS:
      real, intent(in) :: PP
      integer, intent(out) :: P2SIDX
      real, intent(out) :: P2SFRC

      IF (PP.GE.PBAND(1)) THEN
        P2SIDX=1
        P2SFRC=0.
      ELSEIF (PP.LE.PBAND(NPRESS)) THEN
        P2SIDX=NPRESS-1
        P2SFRC=1.
      ELSE
        P2SIDX=1+INT((PBAND2(1)-PP**2)/P2_Den)
        P2SFRC=(PBAND2(P2SIDX)-PP**2)/P2_Den
      ENDIF

      END SUBROUTINE PNTERP

```

Now, all necessary interpolations can be performed.

```

...
!      @NEAR SEGMENT SINGLE SCATTERING INTERPOLATIONS
      DO ILOS=1,NLOS
!      @TEMPERATURE INTERPOLATION (FIRST SOLAR PATH)
        DO ISEG=1,pTPSA(ILOS)%NSEGF
          pSDLD=>pTPSA(ILOS)%LTSA(ISEG)
          IF (ASSOCIATED(pSDLD%FTSF)) THEN
            TT=>pTPSA(ILOS)%TSSEGF(ISEG)
            PP=>pTPSA(ILOS)%PSATMF(ISEG)
            TT0=SQRT(TT/TZERO)
            pTPSA(ILOS)%T5SF(ISEG)=TT0
            pTPSA(ILOS)%PTSF(ISEG)=PP/(SQRT(TT0)*TT0)
          ENDIF
        END DO
      ENDDO

```

```

!      @TEMPERATURE INTERPOLATION (NEAR SEGMENT FOR SOLAR PATHS)
DO ISEG=1,NSEG(ILOS)+1
    PSDLD=>pTPSA(ILOS)%LTSA(ISEG)
    IF (ASSOCIATED(PSDLD%FTSN)) THEN
        TT=>pTPSA(ILOS)%LTSA(ISEG)%TSSEGN
        PP=>pTPSA(ILOS)%LTSA(ISEG)%PSATMN
        TT0=SQRT(TT/TZERO)
        PSDLD%T5SN=TT0
        PSDLD%FTSN=PP/(SQRT(TT0)*TT0)
    ENDIF
ENDDO

!      @BUILT-IN BAND MODEL SPECIES:
DO IMOL=1,NMOL
    !
    DO ISEG=1,pTPSA(ILOS)%NSEGF
        PSDLD=>pTPSA(ILOS)%LTSA(ISEG)
        IF (ASSOCIATED(PSDLD%FTSF)) THEN
            CALL INTERP(PSDLD%JTSF(IMOL),PSDLD%FTSF(IMOL),      &
&                        pTPSA(ILOS)%TSSEGF(ISEG),NTEMP,TBAND)
            CALL PNTERP(pTPSA(ILOS)%PSATMF(ISEG),      &
&                        PSDLD%iLoP2SF(IMOL),PSDLD%P2SF(IMOL))
        ENDIF
    ENDDO
    !
    DO ISEG=1,pTSLA(ILOS)%NSEGF
        pSLLD=>pTSLA(ILOS)%LTSA(ISEG)
        IF (ASSOCIATED(pSLLD%FTSF)) THEN
            CALL INTERP(pSLLD%JTSF(IMOL),pSLLD%FTSF(IMOL),      &
&                        pTSLA(ILOS)%TSSEGF(ISEG),NTEMP,TBAND)
            CALL PNTERP(pTSLA(ILOS)%PSATMF(ISEG),      &
&                        pSLLD%iLoP2SF(IMOL),pSLLD%P2SF(IMOL))
        ENDIF
    ENDDO
    !
    DO ISEG=1,NSEG(ILOS)+1
        PSDLD=>pTPSA(ILOS)%LTSA(ISEG)
        IF (ASSOCIATED(PSDLD%FTSN)) THEN
            CALL INTERP(PSDLD%JTSN(IMOL),PSDLD%FTSN(IMOL),      &
&                        PSDLD%TSSEGN,NTEMP,TBAND)
            CALL PNTERP(PSDLD%PSATMN,PSDLD%iLoP2SN(IMOL),      &
&                        PSDLD%P2SN(IMOL))
        ENDIF
    ENDDO
    DO ISEG=1,NSEG(ILOS)+1
        pSLLD=>pTSLA(ILOS)%LTSA(ISEG)
        IF (ASSOCIATED(pSLLD%FTSN)) THEN
            CALL INTERP(pSLLD%JTSN(IMOL),pSLLD%FTSN(IMOL),      &
&                        pSLLD%TSSEGN,NTEMP,TBAND)
            CALL PNTERP(pSLLD%PSATMN,pSLLD%iLoP2SN(IMOL),      &
&                        pSLLD%P2SN(IMOL))
        ENDIF
    ENDDO
ENDDO

!      @BUILT-IN CROSS-SECTION (X) SPECIES:
DO IMOL=NMOL+1,NMOLXT
    !
    DO ISEG=1,pTPSA(ILOS)%NSEGF
        PSDLD=>pTPSA(ILOS)%LTSA(ISEG)
        IF (ASSOCIATED(PSDLD%FTSF)) THEN
            CALL INTERP(PSDLD%JTSF(IMOL),PSDLD%FTSF(IMOL),      &

```



```

&          TEMP_Y (:, IMOLY) )
&          CALL PNTERP (pSDLD%PSATMN, pSDLD%iLoP2SN (IMOL), &
&          pSDLD%P2SN (IMOL) )
&          ENDDO
&          !
&          DO ISEG=1, NSEG (ILOS) +1
&          pSLLD=>pTSLA (ILOS) %LTSA (ISEG)
&          IF (ASSOCIATED (pSLLD%FTSN) ) THEN
&          CALL INTERP (pSLLD%JTSN (IMOL), pSLLD%FTSN (IMOL), &
&          pSLLD%TSSEGN, NTEMPY (IMOLY), &
&          TEMP_Y (:, IMOLY) )
&          CALL PNTERP (pSLLD%PSATMN, pSLLD%iLoP2SN (IMOL), &
&          pSLLD%P2SN (IMOL) )
&          ENDDO
&          !
&          ENDDO
&          !
! @CROSS-SECTION Y-SPECIES:
DO IMOLY=NMOLYS+1, NMOLY
  IMOL=NMOLXT+IMOLY
  DO ISEG=1, pTPSA (ILOS) %NSEGF
    pSDLD=>pTPSA (ILOS) %LTSA (ISEG)
    CALL INTERP (pSDLD%JTSF (IMOL), pSDLD%FTSF (IMOL), &
    pTPSA (ILOS) %TSSEGF (ISEG), NTEMPY (IMOLY), &
    TEMP_Y (:, IMOLY) )
    ENDDO
    DO ISEG=1, pTSLA (ILOS) %NSEGF
      pSLLD=>pTSLA (ILOS) %LTSA (ISEG)
      CALL INTERP (pSLLD%JTSF (IMOL), pSLLD%FTSF (IMOL), &
      pTSLA (ILOS) %TSSEGF (ISEG), NTEMPY (IMOLY), &
      TEMP_Y (:, IMOLY) )
      ENDDO
      DO ISEG=1, NSEG (ILOS) +1
        pSDLD=>pTPSA (ILOS) %LTSA (ISEG)
        CALL INTERP (pSDLD%JTSN (IMOL), pSDLD%FTSN (IMOL), &
        pSDLD%TSSEGN, NTEMPY (IMOLY), TEMP_Y (:, IMOLY) )
        ENDDO
        DO ISEG=1, NSEG (ILOS) +1
          pSLLD=>pTSLA (ILOS) %LTSA (ISEG)
          CALL INTERP (pSLLD%JTSN (IMOL), pSLLD%FTSN (IMOL), &
          pSLLD%TSSEGN, NTEMPY (IMOLY), TEMP_Y (:, IMOLY) )
          ENDDO
        ENDDO
      ENDDO
    END LOOP OVER ILOS
  ENDDO
  ...

```


Step 7: Pass values to the spectral loop subroutine (bmod.f)

The routine which makes use of the solar segments is bmloop(). It is called under a variety of scenarios through the bmod() driver routine. A series of optional arguments are given to bmloop() for cases in which a solar path is present. In all those instances, two new variables are passed along.

```

...
CALL BMLOOP (ISEG, ISEG, .FALSE., pTINA (ILOS) %LTNA,      &
&      pWPTH (ILOS) %SEG2, pPALS (ILOS) %PATM, WDSA, TX,    &
&      pWSPTH (ILOS) %SEG2 (0:, LSOLAR), pTPSA (ILOS) %LTSA (LSOLAR+1), &
&      pTPSA (ILOS), pWSPTH (ILOS), LSOLAR)
...

```

This must be accompanied by a corresponding change to bmloop() as both new variables must be included in the argument list and declared.

```

SUBROUTINE BMLOOP (ISEG, LMAX, MSPATH, pTPNA, WPTH, PATMLC, WDSD, TX,      &
&      WSPTH, pTPSD, pTPSA, PTHSS, LSOLAR)
...
REAL, DIMENSION(0:), INTENT(IN), OPTIONAL :: WSPTH
TYPE (tisDta), INTENT(IN), OPTIONAL :: pTPSA
TYPE (sg2Dta), INTENT(INOUT), OPTIONAL, TARGET :: PTHSS
INTEGER, INTENT(IN), OPTIONAL :: LSOLAR
...

```

At the completion of this process it is likely that the WSPTH derived type will be redundant. Its properties can be rolled into the LOS derived type and the current 2D arrays will become 1D arrays. This cleanup will be deferred until all appropriate changes have been made to the code.

Step 8: Make use of the new quantities for cross-section species (bmloop.f90)

Cross-section species make use of a database of absorption cross-sections pre-computed for a temperature grid. The total cross-section of the tail contribution is a linear interpolation along that grid. Here, this is denoted as being the original method by the “ $F+N$ ” nomenclature in combination.

$$k_{tail}(F+N) = \chi_{J(F+N)} + \alpha_{J(F+N)} \cdot (\chi_{J(F+N)-1} - \chi_{J(F+N)}) \quad (\text{D.2})$$

Here, the index “ J ” refers to the interpolation index. The weighting term α ranges from 0 to 1 and the cross-section values (χ) have already been computed. The interpolation was done using a density-weighted average temperature for the entire solar path. Finally, the optical depth (τ) prior to the introduction of this new method was given as:

$$\tau = f_{mc} \cdot \rho(F+N) \cdot k_{tail}(F+N) \quad (\text{D.3})$$

The term f is a user-supplied dimensionless scaling factor whose default value is unity and ρ is the column depth of the entire path.

Our improved solar path computation treats the segment within the local layer with a higher degree of accuracy than the rest of the path. Consequently, the absorption cross-section for this segment needs to be determined using the local temperature and associated interpolation factor. This new computation is denoted with the index “ N ” as it only applies to the “near” segment.

$$k_{tail}(N) = \chi_{J(N)} + \alpha_{J(N)} \cdot (\chi_{J(N)-1} - \chi_{J(N)}) \quad (\text{D.4})$$

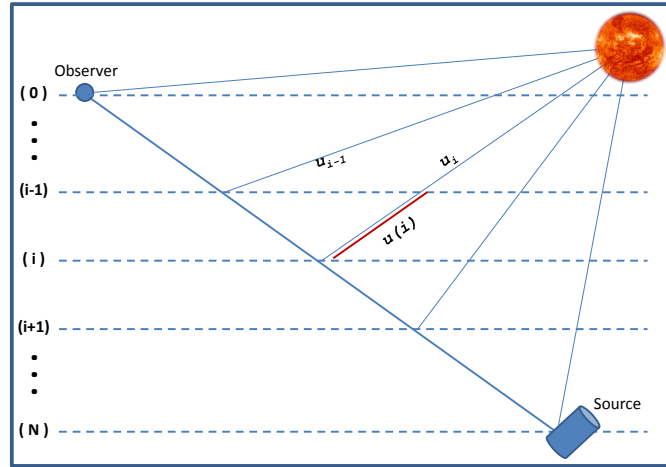
Now, our new optical depth is the sum of two path contributions: the newly-computed near term segment plus the previously computed term modified to account for changes in solar path column densities.

$$\begin{aligned}\tau &\approx f_{mc} \cdot \left[\rho(N) \cdot k_{tail}(N) + \left(\frac{\rho(F+N) - \rho(N)}{\rho(F+N)} \right) \cdot \rho(F+N) \cdot k_{tail}(F+N) \right] \\ &= f_{mc} \cdot [\rho(N) \cdot k_{tail}(N) + (\rho(F+N) - \rho(N)) \cdot k_{tail}(F+N)]\end{aligned}\quad (D.5)$$

Two changes are made to the existing approach. First, the initial solar path (originating at the observer) is computed by explicitly summing over segment contributions.

$$\left(\frac{Su}{d} \right)_{i=0} = \sum_{j=1}^N \frac{S(j)}{\Delta v} \cdot u(j) \quad (D.6)$$

Here, S is a line strength element and u is a column density.



For subsequent solar illumination paths, the algorithm used varies depending on the relative position of the observer, source, and sun. The illustration above shows the case in which an observer is looking down with an overhead sun.

In this case, each subsequent solar path will contain *more* segments than the previous path. The improved solar path computation treats the segment within the local layer with a higher degree of accuracy than the rest of the path. The new, local layer (denoted with an N in the above figure) will be treated as precisely as possible while considering subsequent layers to be a perturbation on the previously computed solar path. Using subscript p to denote the entire solar path, then the improved Curtis-Godson approximation takes the following form for this specific scenario:

$$\left(\frac{Su}{d} \right)_i = \left\{ \frac{u_i(i) - u(i)}{u_{i-1}} \right\} \cdot \left(\frac{Su}{d} \right)_{i-1} + \frac{S(i)}{\Delta v} \cdot u(i) \quad (D.7)$$

The scenario when looking up is meaningfully different. That scenario is illustrated in the image below. Critically, when the sun is overhead, the nearest segment along the previous solar path is completely eliminated from the LOS calculation.

To compensate, then, our ratio of previous to current solar path computation should consider a previous solar path which removes the nearest segment. So, a more appropriate ratio would take a form that looks like:

$$\frac{u_i}{u_{i-1} - u(i-1)} \quad (\text{D.8})$$

This is not the only modification required. The above ratio eliminates the segment missing from the current solar path but it is still present in the path averaged value. This is done when solar path $i - 1$ is considered. Finally, both the current and previous solar path quantities contain a segment associated with the *near* solar path layer. Since it is present in both paths, it can be kept in the column amount ratio. It will, however, need to be eliminated in the path-averaged variable.

$$\left(\frac{Su}{d}\right)_i = \left\{ \frac{u_i}{u_{i-1} - u(i-1)} \right\} \cdot \left\{ \left(\frac{Su}{d}\right)_{i-1} - \frac{S(i-1)}{\Delta v} u(i-1) - \frac{S(i)}{\Delta v} u(i) \right\} + \frac{S(i)}{\Delta v} \cdot u(i) \quad (\text{D.9})$$

So, the first step in this process is to compute – without approximation – the column amount along the first solar LOS. Both the total column amount of the first LOS as well as the first segment of the first LOS is computed.

```

      ....
!      @INITIALIZE SOLAR PATH SEGMENT CURTIS-GODSON VALUES
      IF ((PRESENT(pTPSD)).and.(ISEG.eq.1).and.(LMAX.eq.0)) THEN
        TAILF=0.
        TAILFS = 0.
        DO LLAY = 1, pTPSA%NSEGF
          JUPTMP => pTPSA%LTSA(LLAY)%JTSF(pIMOL)
          pTAILU => pSDDL(JUPTMP)
          pTAILL => pSDDL(JUPTMP-1)
          TAIL = pTAILU+pTPSD%FTSF(pIMOL)*(pTAILL-pTAILU)
          TAILF = TAILF + SEG2F(MOLPNT,LLAY)*TAIL
          IF (LLAY.eq.1) TAILFS = FACMC*SEG2F(MOLPNT,LLAY)*TAIL
        ENDDO
        SSTLM1(MOLPNT) = FACMC*TAILF
      ENDIF
      ....

```

Next, the computation of the solar path for the current layer is performed. Up-looking and down-looking paths will need slightly different approximations, as discussed previously. Finally, the path averaged quantities are updated in preparation for the next solar path in the sequence

```

      ....
!      IF SOLAR PATH, CALCULATE ADDITIONAL LAYER:
      IF (PRESENT(pTPSD)) THEN

        COLDEN=FACMC*WSPTH(MOLPNT)
        IF (COLDEN.GT.0.) THEN

!          SOLAR PATH ONLY:
          IF (MSPATH) THEN
            TDEPTH(0)=0.
          ENDIF

!          LINE TAILS ARE SCALED BY PRESSURE;
!          EXTINCTION CROSS-SECTIONS ARE NOT.
          JUPTMP=>pTPSD%JTS(pIMOL)
          pTAILU=>pSDDL(JUPTMP)
          pTAILL=>pSDDL(JUPTMP-1)

```

```

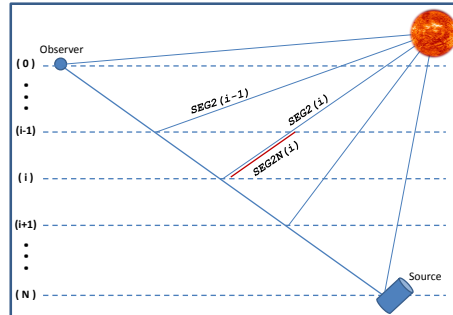
F=>pTPSD%FTS (pIMOL)
TAIL=pTAILU+F*(pTAILL-pTAILU)

!
BOTH SOLAR AND L-SHAPED PATH:
IF (PRESENT (pTPSA) .AND. (LMAX.EQ.0) ) THEN
    TAILF=0.
    DO LLAY = 1, pTPSA%NSEGF
        COLSEG = FACMC*PTHSS%SEG2F (MOLPNT,LLAY)
        JUPTMP => pTPSA%LTSA (LLAY) %JTSF (pIMOL)
        pTAILU => pSDDL (JUPTMP )
        pTAILL => pSDDL (JUPTMP-1)
        F      => pTPSA%LTSA (LLAY) %FTSF (pIMOL)
        TAILSEG = pTAILU+F*(pTAILL-pTAILU)
        TAILF = TAILF + COLSEG*TAILSEG
        IF (LLAY.eq.1) TAILFS = COLSEG*TAILSEG
    ENDDO
    TDEPTH(0)=TDEPTH(0)+TAILF
    IF ((pTPSA%MIDLYR).and.(.not.pTPSA%UPLOOK)) THEN
        PTHSS%SSTLM1(0,MOLPNT) = TAILF - TAILFS
    ELSEIF (pTPSA%UPLOOK) THEN
        PTHSS%SSTLM1(0,MOLPNT) = TAILF - TAILFS
    ELSE
        PTHSS%SSTLM1(0,MOLPNT) = TAILF
    ENDIF
ELSEIF (PRESENT (pTPSA) .AND. (LMAX.NE.0) ) THEN
    COLSEG = FACMC*PTHSS%SEG2N (MOLPNT,LSOLAR)
    JUPTMP => pTPSD%JTSN (pIMOL)
    pTAILU => pSDDL (JUPTMP )
    pTAILL => pSDDL (JUPTMP-1)
    F      => pTPSD%FTSN (pIMOL)
    TAILSEG = pTAILU+F*(pTAILL-pTAILU)
    TAILX = COLSEG*TAILSEG      ! (Su/d) for this segment
    !
    UI      => PTHSS%SEG2 (MOLPNT,LSOLAR )
    UIM1    => PTHSS%SEG2 (MOLPNT,LSOLAR-1)
    USEG    => PTHSS%SEG2N (MOLPNT,LSOLAR )
    USEGM1  => PTHSS%SEG2N (MOLPNT,LSOLAR-1)
    IF (pTPSA%UPLOOK) THEN
        SCL2F = UI/(UIM1-USEGM1)
        TAILN = SCL2F*(PTHSS%SSTLM1(0,MOLPNT)-TAILX)+TAILX
    ELSE
        SCL2F = (UI-USEG)/UIM1
        TAILN = SCL2F*PTHSS%SSTLM1(0,MOLPNT) + TAILX
    ENDIF
    !
    ! Get ready for next call
    IF (pTPSA%UPLOOK) THEN
        PTHSS%SSTLM1(0,MOLPNT)=PTHSS%SSTLM1(0,MOLPNT)-TAILX
    ELSE
        PTHSS%SSTLM1(0,MOLPNT)=PTHSS%SSTLM1(0,MOLPNT)+TAILX
    ENDIF
    TDEPTH(0)=TDEPTH(0)+TAILN
ELSE
    TDEPTH(0)=TDEPTH(0)+COLDEN*TAIL
ENDIF
ELSEIF (MSPATH) THEN
    ....
ENDIF
....

```

D.2 Viewgraph Project Summary

Single Scatter Update: Lex's Approach

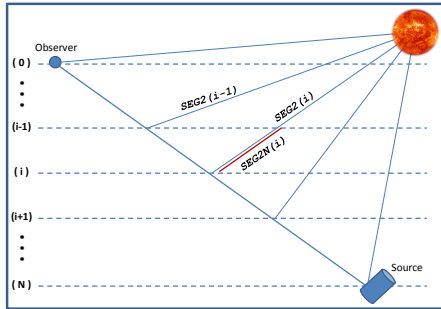


- MODTRAN's Traditional Single Scatter Approach
 - Approximate solar illumination path as a homogeneous segment with effective temperature and pressure
- New Approach
 - Introduce short cut to properly model illumination paths
 - Solar path at level i along the line-of-sight (LOS) defined by
 - DOWN LOOK: the $i-1$ path *scaled* plus a new segment
 - UPLOOK: the $i-1$ path minus the last segment, the quantity scaled

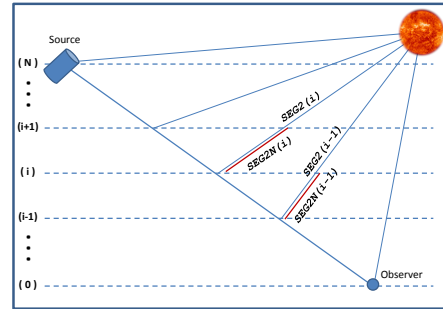
Three Code Blocks

- Solar path computed in three code blocks, each is slightly different
 - Line Center
 - Line Tail
 - Cross-section
- Nearly completed
 - Cross-section = coded + tested
 - Line Tail = coded + tested
 - Line Center = currently coding
- No code commit yet because
 - Current code failed with gfortran
 - Code failed mysteriously in de-allocate step
 - Rolled back earlier commit so everyone else can keep working
 - Have a simple approach to force the error
 - Will re-introduce code (locally) and confirm error is gone
 - Plume MS paths – possible source of issue?

Down vs. Up



$$\left(\frac{Su}{d}\right)_i = \left\{ \frac{u_i - u(i)}{u_{i-1}} \right\} \cdot \left(\frac{Su}{d}\right)_{i-1} + \frac{S}{d}(i) \cdot u(i)$$



$$\left(\frac{Su}{d}\right)_i = \left\{ \frac{u_i}{u_{i-1} - u(i)} \right\} \cdot \left\{ \left(\frac{Su}{d}\right)_{i-1} - \frac{S}{d}(i) \cdot u(i) \right\}$$

- Same concept, but different form if up or down look
 - Subscripts refer to levels; Arguments refer to segments
 - Levels go from 0 to N; Segments from 1 to N; Segment i is bounded by levels $i - 1$ and i
 - u used for all column densities; S/d used for all absorption coefficients

Reorganizing the MS Code Block

ORIGINAL

NEW

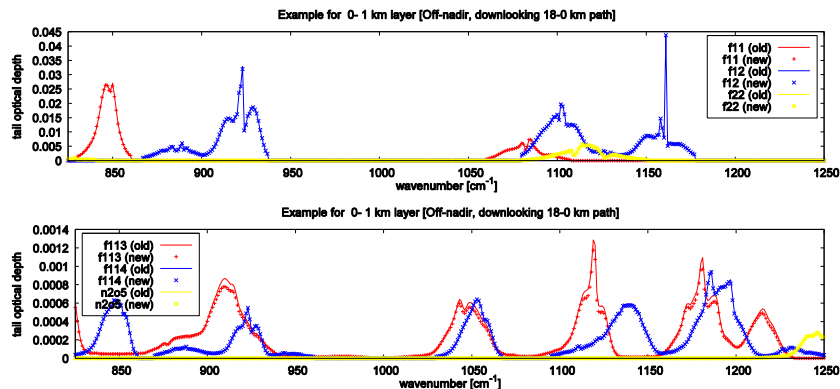
<pre> IF (PRESENT (pTPSD)) THEN COLDEN=FACMC*NSPHT (MOLPNT) IF (COLDEN.GT.0.) THEN !LINE TAILS ARE SCALED BY PRESSURE; !EXTINCTION CROSS-SECTIONS ARE NOT. COLDEN=COLDEN*pTPSD*PSATM (IMOL) JUTMP => pTPSD*JTS (IMOL) JLOTMP = JUTMP-1 F=>pTPSD*F (IMOL) IF (MSPATH) THEN !SOLAR PATH ONLY: DO ITLSUB=0,NTLSUB pTAIL=>SDTL (ITLSUB,JUTMP, 6 pTPSD*ILOP2S (IMOL),IMOL) TAIL=pTAIL+pTPSD*P2S (IMOL)* (SDTL (ITLSUB, 6 JUTMP,pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) JUTMP=pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) pTAIL=>SDTL (ITLSUB,JLOTMP, 6 pTPSD*ILOP2S (IMOL),IMOL) TAILMI=pTAIL+pTPSD*P2S (IMOL)* (SDTL (ITLSUB, 6 JLOTMP,pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) TDEPHT (ITLSUB)=TDEPHT (ITLSUB) COLDEN* (TAIL+F* (TAILMI-TAIL)) ENDDO ELSE !L-SHAPED PATH: DO ITLSUB=0,NTLSUB pTAIL=>SDTL (ITLSUB,JUTMP, 6 pTPSD*ILOP2S (IMOL),IMOL) TAIL=pTAIL+pTPSD*P2S (IMOL)* (SDTL (ITLSUB, 6 JUTMP,pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) JUTMP=pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) pTAIL=>SDTL (ITLSUB,JLOTMP, 6 pTPSD*ILOP2S (IMOL),IMOL) TAILMI=pTAIL+pTPSD*P2S (IMOL)* (SDTL (ITLSUB, 6 JLOTMP,pTPSD*ILOP2S (IMOL)+1,IMOL)-pTAIL) TDEPHT (ITLSUB)=TDEPHT (ITLSUB) COLDEN* (TAIL+F* (TAILMI-TAIL)) ENDDO ENDIF ELSEIF (MSPATH) THEN TX (MOLPNT)=1. CYCLE LOOP1 ENDIF ENDIF </pre>	<pre> IF (PRESENT (pTPSD)) THEN COLDEN=FACMC*NSPHT (MOLPNT) !SPECIAL CASE - NO OPTICAL DEPTH + SOLAR PATH IF ((COLDEN.LE.0.) .AND. (MSPATH)) THEN TX (MOLPNT)=1. CYCLE LOOP1 ENDIF !SOLAR OUT L-SHAPE FOR SOLAR-ONLY PATH IF (MSPATH) THEN DO ITLSUB=0,NTLSUB TDEPHT (ITLSUB) = 0. ENDDO !SOLAR FOUR-POINT INTERPOLATION (TEMPERATURE + PRESSURE) COLDEN = COLDEN*pTPSD*PSATM (IMOL) JUTMP => pTPSD*JTS (IMOL) JLOTMP = JUTMP-1 JUPFRS = pTPSD*ILOP2S (IMOL)+1 JLOFRS => pTPSD*ILOP2S (IMOL) F => pTPSD*F (IMOL) DO ITLSUB=0,NTLSUB pTaillo=>SDTL (ITLSUB,JUTMP,JLOFRS,IMOL) pTaillo=>SDTL (ITLSUB,JUTMP,JUPFRS,IMOL) Taillo=pTaillo+pTPSD*P2S (IMOL)* (pTaillo-pTaillo) pTaillo=>SDTL (ITLSUB,JLOTMP,JLOFRS,IMOL) pTaillo=>SDTL (ITLSUB,JLOTMP,JUPFRS,IMOL) Taillo=pTaillo+pTPSD*P2S (IMOL)* (pTaillo-pTaillo) TAIL=COLDEN* (Taillo+F* (Taillo-Taillo)) TDEPHT (ITLSUB)=TDEPHT (ITLSUB)+TAIL ENDDO ENDIF </pre>
---	--

- Simplify logic before adding new code
- One issue: exiting code block if OD=0.0

Nuances

- How do you know if you're up-looking or down-looking with a limb view?
 - Problem: Illumination path can start as up-looking and become down-looking (or vice-versa)
 - Solution: Record solar zenith at each level
- Do the approximations work if we transition from down-looking to up-looking along a single LOS?
YES!
- What about the tangent point segment?
 - Proper operation must still be verified

Intermediate Examples



- Cross-section species optical depths for a down-looking slant path
- Plotting solar path only
- Modest differences so far
 - line centers may be more dramatic

APPENDIX E: DERIVATION OF LINE TAIL MOLECULAR ABSORPTION PADÉ FIT PARAMETERS

Alexander Berk

- 1
$$K_\nu \equiv \frac{k_0 + k_1 \delta + k_2 \delta^2}{1 + a \delta + b \delta^2} \quad \delta \equiv 2 \left(\frac{\nu - \nu_c}{\Delta \nu} \right)$$
- 2
$$K_0 \equiv K_{\nu=\nu_c} = K_\nu (\delta=0)$$
- 3
$$K_\pm \equiv K_{\nu=\nu_c \pm \frac{\Delta \nu}{2}} = K_\nu (\delta \pm 1) = \frac{k_0 \pm k_1 + k_2}{1 \pm a + b} = \frac{k_0 \pm k_1 + k_2}{\beta \pm a}$$
- 4
$$\beta \equiv 1 + b$$
- 5
$$\frac{\Delta \nu}{2} \frac{\partial K_\nu}{\partial \nu} \equiv K'_0 = \left. \frac{dK_\nu}{d\delta} \right|_{\delta=0} = \left(\frac{k_1 + 2k_2 \delta}{1 + a \delta + b \delta^2} - \frac{(k_0 + k_1 \delta + k_2 \delta^2)(a + 2b \delta)}{1 + a \delta + b \delta^2} \right) \bigg|_{\delta=0} = K_1 - a_0 K_0$$

Solve for k_2, k_1, b , and a (or β) in terms of $K_0, \bar{k}, \Delta k, K'_0$ and β (or a)
- 6
$$\bar{k} \equiv \frac{k_+ + k_-}{2} \quad \Delta k \equiv \frac{k_+ - k_-}{2}$$
- 7
$$\boxed{b = \beta - 1} \quad \text{from (4)}$$
- 8
$$\boxed{K_1 = K'_0 + a_0 K_0} \quad \text{from (5)}$$
- 9
$$(\beta \pm a) K_\pm = k_0 \pm k_1 + k_2 \quad \text{from (3)}$$
- 10
$$(\beta + a) K_+ - (\beta - a) K_- = 2k_1 \quad \text{from (9)}$$
- 11
$$K_1 = \beta \Delta k + a \bar{k} \quad \text{from (10) and (6)}$$
- 12
$$\beta \Delta k + a \bar{k} = K'_0 + a K_0 \quad \text{from (11) and (8)}$$
- 13
$$\boxed{a = \frac{K'_0 - \beta \Delta k}{\bar{k} - K_0}} \quad \text{or} \quad \boxed{\beta = \frac{K'_0 - a(\bar{k} - K_0)}{\Delta k}} \quad \text{from (12)}$$

$$14 \quad (\beta+a)K_+ + (\beta-a)K_- = 2(K_0 + k_2) \quad \text{from (9)}$$

$$15 \quad \boxed{K_2 = \beta \bar{K} + a \Delta k - K_0}$$

Determine β (or a) from the integral condition

$$\langle k \rangle = I \equiv \frac{1}{\Delta v} \int_{K_0 - \frac{\Delta v}{2}}^{K_0 + \frac{\Delta v}{2}} K_v dv = \frac{1}{2} \int_{-1}^1 K_v d\delta = \frac{K_0}{2} \int_{-1}^1 \frac{d\delta}{1+a\delta+b\delta^2} + \frac{K_1}{2} \int_{-1}^1 \frac{\delta d\delta}{1+a\delta+b\delta^2} + \frac{K_2}{2} \int_{-1}^1 \frac{\delta^2 d\delta}{1+a\delta+b\delta^2}$$

$$I = (K_0 I_0 + K_1 I_1 + K_2 I_2) \frac{1}{2}$$

$$I_2 = \left(\frac{\delta}{b} - \frac{a}{2b^2} \ln(1+a\delta+b\delta^2) \right) \Big|_{\delta=-1}^{\delta=1} + \frac{a^2-2b}{2b^2} I_0$$

$$I_2 = \frac{2}{b} - \frac{a}{2b^2} \ln\left(\frac{\beta+a}{\beta-a}\right) + \frac{a^2-2b}{2b^2} I_0$$

$$I_1 = \frac{1}{2b} \ln(1+a\delta+b\delta^2) \Big|_{\delta=-1}^{\delta=1} - \frac{a}{2b} I_0$$

$$I_1 = \frac{1}{2b} \ln\left(\frac{\beta+a}{\beta-a}\right) - \frac{a}{2b} I_0$$

$$I = \left[\frac{2K_2}{b} + \frac{1}{2b} \left(K_1 - \frac{aK_2}{b} \right) \ln\left(\frac{\beta+a}{\beta-a}\right) + \left(K_0 - \frac{aK_1}{2b} + \frac{a^2-2b}{2b^2} K_2 \right) I_0 \right] \frac{1}{2}$$

$$q = b - \frac{a^2}{4}$$

$$q = 4b - a^2$$

$$I_0 = \begin{cases} \frac{2}{\sqrt{q}} \tan^{-1} \frac{2b\delta + a}{\sqrt{q}} \Big|_{\delta=-1}^{\delta=1} & q > 0 \\ -\frac{2}{\sqrt{-q}} \tanh^{-1} \frac{2b\delta + a}{\sqrt{-q}} \Big|_{\delta=-1}^{\delta=1} & q < 0 \end{cases}$$

$$\begin{aligned}
\tan^{-1} \frac{2b\delta + a/2}{\sqrt{g}} \Big|_{\delta=-1}^{\delta=+1} &= \tan^{-1} \frac{2b+a/2}{\sqrt{g}} - \tan^{-1} \frac{a/2-2b}{\sqrt{g}} > 0 \\
&= \tan^{-1} \left\{ \tan \left[\tan^{-1} \frac{a/2+2b}{\sqrt{g}} - \tan^{-1} \frac{a/2-2b}{\sqrt{g}} \right] \right\} > 0 \\
&= \tan^{-1} \left\{ \frac{(a/2+2b)/\sqrt{g} - (a/2-2b)/\sqrt{g}}{1 + (a/2+2b)/\sqrt{g} (a/2-2b)/\sqrt{g}} \right\} > 0 \\
&= \tan^{-1} \left(\frac{42b\sqrt{g}}{g + (a/2+2b)(a/2-2b)} \right) > 0 \\
&= \tan^{-1} \left(\frac{42b\sqrt{g}}{4b - \frac{a^2}{4} + \frac{a^2}{4} - 4b^2} \right) > 0 \\
&= \tan^{-1} \left(\frac{2\sqrt{g}}{1-b} \right) > 0 \\
&= \tan^{-1} \left(\frac{2\sqrt{g}}{2-\beta} \right) > 0 \quad \text{ctn}^{-1} \left(\frac{2-\beta}{2\sqrt{g}} \right)
\end{aligned}$$

$$\begin{aligned}
-\tanh^{-1} \frac{b\delta + a/2}{\sqrt{g}} \Big|_{\delta=-1}^{\delta=+1} &= \tanh^{-1} \frac{a/2-b}{\sqrt{g}} - \tanh^{-1} \frac{a/2+b}{\sqrt{g}} \\
&= \tanh^{-1} \left\{ \frac{(a/2-b)/\sqrt{g} - (a/2+b)/\sqrt{g}}{1 - (a/2-b)/\sqrt{g} (a/2+b)/\sqrt{g}} \right\} \\
&= \tanh^{-1} \left\{ \frac{-2b\sqrt{g}}{-g - (a/2-b)(a/2+b)} \right\} \\
&= \tanh^{-1} \left\{ \frac{2b\sqrt{g}}{(b - \frac{a^2}{4}) + (a/2-b)(a/2+b)} \right\} = \tanh^{-1} \left(\frac{2\sqrt{g}}{1-b} \right) = \tanh^{-1} \left(\frac{2\sqrt{g}}{2-\beta} \right)
\end{aligned}$$

$$I_0 = \begin{cases} \frac{2}{\sqrt{g}} \tan^{-1} \left(\frac{2\sqrt{g}}{2-\beta} \right) & g > 0 \\ \frac{2}{\sqrt{-g}} \tanh^{-1} \left(\frac{2\sqrt{-g}}{2-\beta} \right) & g < 0 \end{cases}$$

- (A) No singularity is a requirement: $D(\delta) = 1 + a\delta + b\delta^2 > 0 \quad |\delta| < 1$
- (B) Necessary condition: $\beta = 1+b > |a|$
- (C) (B) plus $b \leq 0$ is sufficient because $D(\pm 1) > 0$ and $D(\delta)$ has no minimum [i.e. $D''(\delta) = 2b \leq 0$]
- (D) If (B) and $b > 0$, then (A) is true if the location of the minimum $\delta_{\min}$ [defined by $D'(\delta_{\min}) = 0$] is outside $[-1, 1]$
 $D'(\delta_{\min}) = 0 = a + 2b\delta_{\min} \quad \delta_{\min} = -\frac{a}{2b} \quad |\delta_{\min}| > 1 \Rightarrow |a| > 2b$
- (E) If (B) and $b > 0$ and $|a| \leq 2b$, the
 $D(\delta_{\min}) = 1 + a\delta_{\min} + b\delta_{\min}^2 = 1 - \frac{a^2}{2b} + \frac{a^2}{4b} = 1 - \frac{a^2}{4b}$
 must exceed zero. Note, $D(\delta_{\min}) = q/b$ $a^2 < 4b$
 $|a| < 2\sqrt{b}$
- (F) If $0 < b < 1$, then either $|a| > 2b \Rightarrow |\delta_{\min}| > 1$
 or $|a| \leq 2b < 2\sqrt{b} \Rightarrow 1 - \frac{a^2}{4b} > 0$
- (G) If $b \geq 1$, then $|a| \leq 1+b \leq 2b$
 $D(\delta_{\min}) = 1 + \frac{a^2}{4b} = 1 - \frac{a^2}{4(b-1)}$
 Require $1 - \frac{a^2}{4b} > 0 \Rightarrow a^2 - 4b < 0$
 $\Rightarrow \left(\frac{k_0' - \Delta K - b \Delta K}{\bar{K} - K_0} \right)^2 - 4b < 0$
 $\Rightarrow \left(b - \frac{k_0' - \Delta K}{\Delta K} \right)^2 - 4 \left(\frac{\bar{K} - K_0}{\Delta K} \right)^2 b < 0$

Find initial guess for β without singularities,

TRY 1: Ignore integral constraint ($\langle k \rangle = I$) and set k_2 to zero

16 $k_2 = 0$

17 $\beta = \frac{k_0 - a \Delta k}{\bar{k}}$ from (15) and (16)

18 $\frac{k_0 - a \Delta k}{\bar{k}} = \frac{k_0' - a(\bar{k} - k_0)}{\Delta k}$ from (13) and (17)

19 $k_0 \Delta k - k_0' \bar{k} = a \{ (\Delta k)^2 - \bar{k}(\bar{k} - k_0) \}$ from (18)

20 $= a \{ k_0 \bar{k} - k_1 k_2 \}$ from (19) and (6)
 $k_1 k_2 = ?$

21 $a = \frac{k_0 \Delta k - k_0' \bar{k}}{k_0 \bar{k} - k_1 k_2}$

22 $a = \frac{k_0 - \beta \bar{k}}{\Delta k}$ from (17)

23 $\frac{k_0' - \beta \Delta k}{\bar{k} - k_0} = \frac{k_0 - \beta \bar{k}}{\Delta k}$ from (22) and (13)

24 $k_0' \Delta k - k_0(\bar{k} - k_0) = \beta \{ (\Delta k)^2 - \bar{k}(\bar{k} - k_0) \}$ from (23)

25 $\beta = \frac{k_0' \Delta k - k_0(\bar{k} - k_0)}{k_0 \bar{k} - k_1 k_2}$
 $k_1 k_2$

$$\frac{1}{2} \ln \left(\frac{\beta+q}{\beta-q} \right) = \tanh^{-1} \left(\frac{q}{\beta} \right)$$

$$r_1 \equiv \frac{k_1}{b} \quad r_2 \equiv \frac{k_2}{b}$$

$$q = b - \frac{a^2}{4} \quad q = 4b - a^2$$

$$I = \left[2r_2 + \left(r_1 - \frac{ar_2}{b} \right) \tanh^{-1} \left(\frac{q}{\beta} \right) + \left[k_0 - \frac{ar_1}{2} + \left(\frac{a^2}{2b} - 1 \right) r_2 \right] I_0 \right] \frac{1}{2}$$

$$I = \left[2r_2 + \left(r_1 - \frac{ar_2}{b} \right) \tanh^{-1} \left(\frac{q}{\beta} \right) + \left[k_0 - \frac{ar_1}{2} + \left(1 - \frac{a^2}{2b} \right) r_2 \right] I_0 \right] \begin{cases} \frac{2}{1-\beta} \tanh^{-1} \left(\frac{a\sqrt{\beta}}{2-\beta} \right) & q > 0 \\ \frac{2}{1-\beta} \tanh^{-1} \left(\frac{a\sqrt{\beta}}{2-\beta} \right) & q < 0 \end{cases}$$

$$0 < \beta < 2 \quad \beta_{\min} = 0 \quad \beta_{\max} = 2$$

$$\left| \frac{q}{\beta} \right| < 1 \Rightarrow |q| = \left| \frac{k_0' - \beta \Delta k}{\bar{k} - k_0} \right| < \beta \Rightarrow |k_0' - \beta \Delta k| < \beta |\bar{k} - k_0|$$

$$\Rightarrow k_0' - \beta \Delta k < \beta |\bar{k} - k_0| \quad \text{and} \quad \beta \Delta k - k_0' < \beta |\bar{k} - k_0|$$

$$\Rightarrow k_0' < \beta (|\bar{k} - k_0| + \Delta k) \quad \text{and} \quad -k_0' < \beta (|\bar{k} - k_0| - \Delta k)$$

If $(|\bar{k} - k_0| + \Delta k) > 0$, then

$$\tilde{\beta}_{\max} = \beta_{\max} \quad \tilde{\beta}_{\min} = \max \left[\beta_{\min}, \frac{k_0'}{|\bar{k} - k_0| + \Delta k} \right]$$

else if $(|\bar{k} - k_0| + \Delta k) < 0$, then

$$\tilde{\beta}_{\min} = \beta_{\min} \quad \tilde{\beta}_{\max} = \min \left[\beta_{\max}, \frac{k_0'}{|\bar{k} - k_0| + \Delta k} \right]$$

The domain of a and $\beta (=1+b)$

①

The denominator of the Padé approximate is $D(\delta) \equiv 1 + a\delta + b\delta^2$ for $|\delta| \leq 1$. Since $D(0) \equiv 1 > 0$, we require that $D(\delta) > 0$ for $|\delta| \leq 1$ [$D(\delta)$ could actually equal 0 if the numerator also equaled 0, but we ignore that possibility].

A necessary condition is that $D(\pm 1) > 0$. This implies $\beta = 1+b > |a| > 0$. Parameters a and b are related by the equations

$$a = \frac{k_0' - \beta \Delta k}{\bar{k} - k_0} \quad \text{and} \quad \beta = \frac{k_0' - a(\bar{k} - k_0)}{\Delta k}$$

Case 1: For $|\Delta k| > |\bar{k} - k_0|$, let a be the independent variable.

$$\epsilon + |a| \leq \beta = \frac{k_0' - a(\bar{k} - k_0)}{\Delta k} \Rightarrow \epsilon + a \left(\frac{\bar{k} - k_0}{\Delta k} \right) \pm a \leq \frac{k_0'}{\Delta k}$$

$$\Rightarrow a \left[\frac{\bar{k} - k_0}{\Delta k} \pm 1 \right] \leq \frac{k_0'}{\Delta k} - \epsilon$$

$$\text{Since } \left| \frac{\bar{k} - k_0}{\Delta k} \right| < 1, \quad \boxed{\frac{k_0' - \epsilon \Delta k}{\bar{k} - k_0 - \Delta k} \leq a \leq \frac{k_0' - \epsilon \Delta k}{\bar{k} - k_0 + \Delta k}}$$

Case 2: For $|\Delta K| \leq |\bar{k} - k_0|$, let $\beta = (1+b)$ be the (2) independent variable.

$$\beta > |a|^{+e} = \left| \frac{k_0' - \beta \Delta K}{\bar{k} - k_0} \right| + e$$

$$\beta > \pm \left(\frac{k_0'}{\bar{k} - k_0} \right) \mp \beta \left(\frac{\Delta K}{\bar{k} - k_0} \right) + e$$

$$\beta \left[1 \pm \frac{\Delta K}{\bar{k} - k_0} \right] > \pm \frac{k_0'}{\bar{k} - k_0} + e$$

$$\boxed{\beta > \frac{\pm k_0' + e(\bar{k} - k_0)}{\bar{k} - k_0 \pm \Delta K} \text{ for } |\Delta K| < |\bar{k} - k_0|}$$

$$\beta > \frac{k_0'}{2(\bar{k} - k_0)} \text{ for } |\Delta K| = |\bar{k} - k_0|$$

however $\frac{k_0'}{\bar{k} - k_0}$ must be negative? and

we already know that β is positive

$$\beta > \max \left(\frac{k_0' + e(\bar{k} - k_0)}{\bar{k} - k_0 + \Delta K}, \frac{-k_0' + e(\bar{k} - k_0)}{\bar{k} - k_0 - \Delta K} \right) \quad |\Delta K| < |\bar{k} - k_0|$$

Note that

$$|\Delta K| = |\bar{k} - k_0| \Rightarrow \pm \left(\frac{k_+ - k_-}{2} \right) = \left(\frac{k_+ + k_-}{2} \right) - k_0$$

$$\Rightarrow k_0 = k_+ \text{ or } k_0 = k_-$$

Given $D(\pm 1) > 0$, $D(\delta)$ is positive for $|\delta| \leq 1$ (3)
 if $D(\delta)$ has no minimum, equivalently $D''(\delta) = 2b \leq 0$
 $\Rightarrow D(\delta) > 0$ for $|\delta| \leq 1$ if $|a| < b = 1+b \leq 1$

If $D(\delta)$ has a minimum^(b>0), it will be located
 at $0 = D'(\delta_{\min}) = a + 2b \delta_{\min} \Rightarrow \delta_{\min} = -\frac{a}{2b}$. The value
 of $D(\delta)$ at $\delta_{\min}$ is $D(\delta_{\min}) = 1 + a \delta_{\min} + b \delta_{\min}^2 = 1 - \frac{a^2}{4b}$

Given $D(\pm 1) > 0$ and $b < 1$, ~~either~~
 either $|a| > 2b \Rightarrow |\delta_{\min}| > 1$

or $|a| \leq 2b < 2\sqrt{b} \Rightarrow D(\delta_{\min}) > 0$

Finally, if $D(\pm 1) > 0$ and $b \geq 1$ ($b \geq 2$),
 then $|a| < 1+b \leq 2b \Rightarrow \delta_{\min} < 1$. We must
 require that

$$D(\delta_{\min}) = 1 - \frac{a^2}{4b} \geq 0 \quad \text{if } b \geq 1 \text{ assuming } D(\pm 1) > 0$$

$$\Rightarrow a^2 - 4b \leq 0 \quad -4b \leq 0$$

$$\Rightarrow \left(\frac{k_0' - (1+b)\Delta k}{\bar{k} - k_0} \right)^2 - 4b \leq 0 \quad -4b \leq 0$$

$$\Rightarrow \left(b - \frac{k_0' - \Delta k}{\Delta k} \right)^2 - 4 \left(\frac{\bar{k} - k_0}{\Delta k} \right)^2 b \leq 0$$

(4)

Define

$$x \equiv \frac{K_0' - \Delta K}{\Delta K} = \frac{K_0'}{\Delta K} - 1 \quad y \equiv \frac{\bar{K} - K_0}{\Delta K}$$

$$(b-x)^2 - 4y^2 b \leq 0$$

$$b^2 - 2(x+2y^2)b + x^2 \leq 0$$

Define

$$b_{\pm} \equiv x + 2y^2 \pm \left[(x+2y^2)^2 - x^2 \right]^{1/2} = \frac{x^2}{(x+2y^2) \mp \{ [x+2(1-\epsilon)y^2]^2 - x^2 \}^{1/2}} = \frac{(x\Delta K)^2}{\Delta K(x\Delta K) + 2(K\Delta K)^2(1-\epsilon) - \{ []^2 - (x\Delta K)^2 \}}$$

$$(b-b_-)(b-b_+) \leq 0 \quad \text{if } b \geq 1 \quad [\text{given } D(\pm 1) > 0]$$

$$b \not\geq 1 \quad \text{if } |x+2y^2| < |x| \Rightarrow x < -y^2(1-\epsilon)$$

$$\Rightarrow \frac{K_0' - \Delta K}{\Delta K} < - \left(\frac{\bar{K} - K_0}{\Delta K} \right)^2 (1-\epsilon)$$

$$\Rightarrow \Delta K (K_0' - \Delta K) < -(\bar{K} - K_0)^2 (1-\epsilon)$$

$$\Rightarrow K_0' \Delta K < (\Delta K)^2 - (\bar{K} - K_0)^2 (1-\epsilon) = (\Delta K - \bar{K} + K_0)(\Delta K + \bar{K} - K_0) + \epsilon(\bar{K} - K_0)^2$$

$$\Rightarrow K_0' \Delta K < (K_+ - K_0)(K_0 - K_-) + \epsilon(\bar{K} - K_0)^2$$

$$\text{For } K_0' \Delta K \geq (K_+ - K_0)(K_0 - K_-)$$

$$b_- < b < b_+ \Rightarrow b < \max(1, b_+)$$

$$?? \text{ Can } b > 1 \text{ and } \beta = 1+b > 1, \text{ but } b_- \not\leq b ??$$

$$b_{\max} \equiv \begin{cases} 1 & K_0' \Delta K < (K_+ - K_0)(K_0 - K_-) \\ \max(1, b_+) & \text{otherwise} \end{cases}$$

$$b < b_{\max} \Rightarrow \frac{K_0' - a(\bar{K} - K_0)}{\Delta K} - 1 < b_{\max}$$

$$\Rightarrow \left(\frac{K_0' - \Delta K}{\Delta K} \right) - b_{\max} < a \left(\frac{\bar{K} - K_0}{\Delta K} \right)$$

$$\Rightarrow x - b_{\max} < ay$$

$$\text{If } \gamma \equiv \frac{\bar{K} - K_0}{\Delta K} > 0$$

(5)

$$a > \frac{K_0' - \Delta K}{\bar{K} - K_0} - \left(\frac{\Delta K}{\bar{K} - K_0} \right) b_{max}$$

$$\text{If } \frac{\bar{K} - K_0}{\Delta K} < 0$$

$$a < \frac{K_0' - \Delta K}{\bar{K} - K_0} - \left(\frac{\Delta K}{\bar{K} - K_0} \right) b_{max}$$

Two special cases:

$$\text{Case 1: } 4b < 0.0001 \quad \text{and} \quad |a| \leq 0.1$$

$$\text{Case 2: } 4b < 0.0001 \quad \text{and} \quad |a| > 0.1 \quad \left(\frac{4b}{a^2} < .01 \right)$$

Case 1

$$\begin{aligned} \frac{1}{1+a\delta+b\delta^2} &= 1 - (a\delta+b\delta^2) + (a\delta+b\delta^2)^2 - (a\delta+b\delta^2)^3 + (a\delta+b\delta^2)^4 - \dots \\ &= 1 - a\delta - b\delta^2 + a^2\delta^2 + \cancel{2ab\delta^3} + a^3\delta^3 + a^4\delta^4 + O(a^5, ab, b^2) \end{aligned}$$

$$I(C1) = \frac{K_0}{2} \int_{-1}^1 (1 - a\delta - b\delta^2 + a^2\delta^2 - a^3\delta^3 + a^4\delta^4) d\delta$$

$$+ \frac{K_1}{2} \int_{-1}^1 (\quad) \delta d\delta$$

$$+ \frac{K_2}{2} \int_{-1}^1 (\quad) \delta^2 d\delta$$

$$= \frac{K_0}{2} \left(1 - \frac{b}{3} + \frac{a^2}{3} + \frac{a^4}{5} \right) + K_1 \left(-\frac{a}{3} - \frac{a^3}{5} \right) + K_2 \left(\frac{1}{3} - \frac{b}{5} + \frac{a^2}{5} + \frac{a^4}{7} \right)$$

$$I(b) = \frac{k_0}{2} \int_{-1}^1 \frac{d\delta}{1+a\delta+b\delta^2} + \frac{k_1}{2} \int_{-1}^1 \frac{\delta d\delta}{1+a\delta+b\delta^2} + \frac{k_2}{2} \int_{-1}^1 \frac{\delta^2 d\delta}{1+a\delta+b\delta^2}$$

$$I(0) = \frac{k_0}{2} \int_{-1}^1 \frac{d\delta}{1+a\delta} + \frac{k_1}{2} \int_{-1}^1 \frac{\delta d\delta}{1+a\delta} + \frac{k_2}{2} \int_{-1}^1 \frac{\delta^2 d\delta}{1+a\delta}$$

$$= \frac{k_0}{2a} \ln \frac{1+q}{1-q} + \frac{k_1}{2a^2} \left(2a - \ln \frac{1+q}{1-q} \right) + \frac{k_2}{2a^3} \left(-2a + \ln \frac{1+q}{1-q} \right)$$

$$= \frac{k_0}{a} \tanh^{-1} a - \frac{k_1}{a^2} (\tanh^{-1} a - a) + \frac{k_2}{a^3} (\tanh^{-1} a - a)$$

$$= k_0 + (a^2 k_0 - a k_1 + k_2) \left(\frac{\tanh^{-1} a - a}{a^3} \right)$$

$$\left(\frac{\tanh^{-1} a - a}{a^3} \right) = \frac{1}{3} + \frac{a^2}{5} + \frac{a^4}{7} + \dots \quad |a| < 1$$

$$I'(b) = -\frac{k_0}{2} \int_{-1}^1 \frac{\delta^2 d\delta}{(1+a\delta+b\delta^2)^2} - \frac{k_1}{2} \int_{-1}^1 \frac{\delta^3 d\delta}{(1+a\delta+b\delta^2)^2} - \frac{k_2}{2} \int_{-1}^1 \frac{\delta^4 d\delta}{(1+a\delta+b\delta^2)^2}$$

$$I'(0) = -\frac{k_0}{2} \int_{-1}^1 \frac{\delta^2 d\delta}{(1+a\delta)^2} - \frac{k_1}{2} \int_{-1}^1 \frac{\delta^3 d\delta}{(1+a\delta)^2} - \frac{k_2}{2} \int_{-1}^1 \frac{\delta^4 d\delta}{(1+a\delta)^2}$$

$$dv = -\frac{d\delta}{(1+a\delta)^2} \quad u = \delta^n \quad v = \frac{1}{(1+a\delta)a} \quad du = n\delta^{n-1}$$

$$I'(0) = \left[\frac{k_0}{2a} \frac{\delta^2}{1+a\delta} + \frac{k_1}{2a} \frac{\delta^3}{1+a\delta} + \frac{k_2}{2a} \frac{\delta^4}{1+a\delta} \right] \Big|_{\delta=-1}^{\delta=1}$$

$$- \frac{k_0}{a} \int_{-1}^1 \frac{\delta^2 d\delta}{1+a\delta} - \frac{3k_1}{2a} \int_{-1}^1 \frac{\delta^2 d\delta}{1+a\delta} - \frac{2k_2}{a} \int_{-1}^1 \frac{\delta^3 d\delta}{1+a\delta} \quad \frac{\delta^n}{a} = \frac{\delta^{n-1}}{a^2} (1+a\delta - 1)$$

$$= \frac{k_0}{2a} \left(\frac{1}{1+a} - \frac{1}{1-a} \right) + \frac{k_1}{2a} \left(\frac{1}{1+a} + \frac{1}{1-a} \right) + \frac{k_2}{2a} \left(\frac{1}{1+a} - \frac{1}{1-a} \right)$$

$$- \frac{k_0}{a^2} \int_{-1}^1 \left(1 - \frac{1}{1+a\delta} \right) d\delta - \frac{3k_1}{2a^2} \int_{-1}^1 \left(\delta - \frac{\delta}{1+a\delta} \right) d\delta - \frac{2k_2}{a^2} \int_{-1}^1 \left(\delta^2 - \frac{\delta^2}{1+a\delta} \right) d\delta$$

APPENDIX F: A PLANNED UPGRADE TO THE MODTRAN CORRELATED-K ALGORITHM

A. Berk and P.K. Acharya

SSI has undertaken and nearly completed the task of upgrading the correlated- k (Ck) implementation in MODTRAN. This upgrade is not included in the initial MODTRAN6 distribution, but it is anticipated that it will be integrated in a later release of MODTRAN6. Currently MODTRAN contains the statistical Ck implementation described in Chapter 4 and first introduced in 1996 [Bernstein *et al.*, 1996]. The reasons for and the basic methodology of Ck in MODTRAN has not changed.

MODTRAN, being the premier comprehensive atmospheric radiative transfer (RT) code from far IR to UV-VIS, requires that multiple scattering (MS) contributions to radiance be computed accurately. MODTRAN pairs its sophisticated band model (BM) methodology with the state-of-the art DISORT discrete ordinate scattering code for RT computation [Stamnes *et al.*, 1988]. For a line-of-sight (LOS) consisting of nearly homogenous segments, the band model molecular transmittance of the entire LOS is **not** equal to the product of band model molecular transmittances of each segment. That is, Beer's Law is not obeyed for bandpass transmittances. Since the multiple scattering algorithms in MODTRAN, including the DISORT discrete ordinate method, assume layer optical depths are additive, the lack of Beer's Law is problematic.

The Ck method is designed to recast the band model into a Beer's Law compliant method. As illustrated in Figure 1, the optical depth versus spectral frequency relationship is converted to an optical depth versus cumulative probability relationship by sorting optical depths (absorption cross-section, k , and column density, u , products) in narrow vertical strips of equal and infinitesimal widths in non-decreasing order. The cumulative probability distribution of molecular cross-sections, defined from 0 to 1, is referred to as the g -space. For a single segment, the $[\nu_1, \nu_2]$ spectral band transmittance variation with column density u [atm-cm] equals one minus the curve-of-growth (CoG) and is defined by

$$T(u) = \frac{1}{\nu_2 - \nu_1} \int_{\nu_1}^{\nu_2} \exp[-k(\nu)u] d\nu = \int_0^1 \exp[-k(g)u] dg = \sum_{i=1}^N \Delta g_i \exp\left[-u k_{MVT}^u(i)\right] \quad (\text{F.1})$$

The right-most approximation is a discrete sum representation of the integral. The Δg_i weights, which sum to unity, are defined by $g_i - g_{i-1}$ where $0 = g_0 < g_1 < \dots < g_N = 1$; the set of g_i 's are fixed, and the $k_{MVT}^u(i)$ are mean value theorem $k(g)$'s, which depend on u itself. Currently in MODTRAN, N is either 33 (for the Slow speed option) or, more often, 17 (for the Moderate speed option).

In the MODTRAN Ck implementation, great care is taken to ensure that path segment transmittances obtained by the BM method and Ck method are identical. The Ck method is statistical, which enables bypassing of the impractical *ab-initio* generation of $k(g)$ curves. Instead, the MODTRAN $k(g)$ curves are extracted from a small table, generated off-line and only dependent on three band model parameters. The parameters are n , the number of randomly distributed identical lines, and γ_L and γ_D , the Lorentz and Doppler half-widths respectively. The k -table entries are generated using a Monte-Carlo method. In the on-the-fly MODTRAN Ck run, first the total (due to all species) *band model* transmittance is computed for a single segment. A "pseudo-gas" effective line density, line strength and half-widths are then used to obtain the k -values from the Ck table. The Ck values are scaled to exactly match the Ck transmittance to band model transmittance.

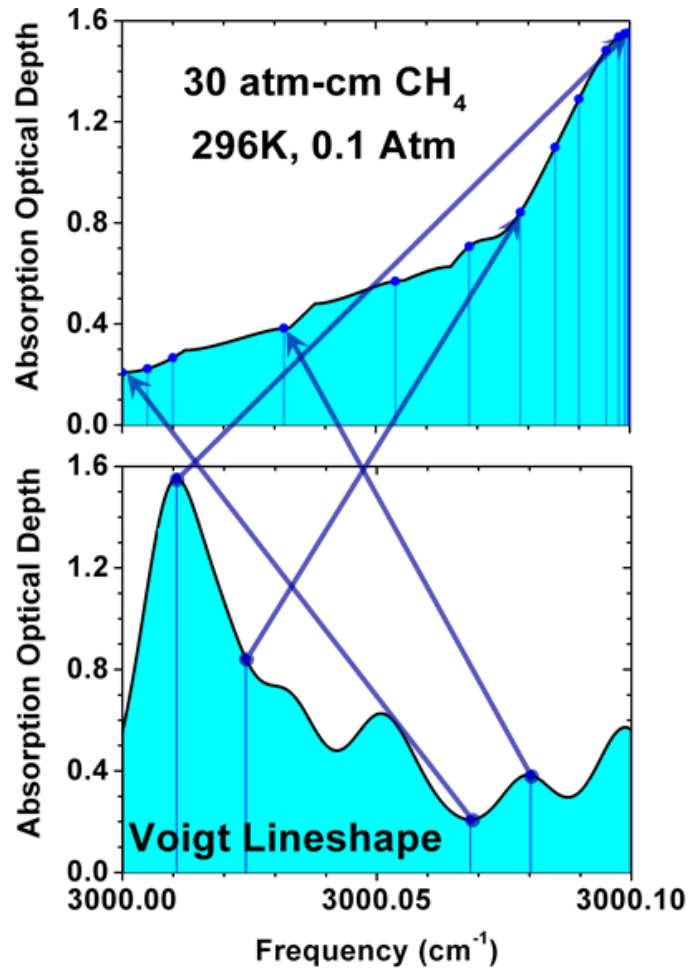


Figure 1. Reordering of the spectral absorption optical depths, i.e., the product of the frequency dependent absorption cross-section, $k(\nu)$, and the column density, u (lower figure). In the upper figure, the monotonically non-decreasing $k(g)u$ absorption optical depths are plotted as a function of the cumulative probability function, g . By definition, the fraction of spectral interval optical depths less than $k(g)$ is g ($0 \leq g \leq 1$).

MODTRAN has two options for discretizing the g -space. In the slow option, g -space is divided into 33 intervals (34 g_i values):

$g_i = \{0.0000000, 0.0500000, 0.1000000, 0.3187070, 0.5358410, 0.6837720, 0.7845560, 0.8532200, 0.9000000, 0.9318710, 0.9535840, 0.9683770, 0.9784560, 0.9853220, 0.9900000, 0.9931870, 0.9953580, 0.9968380, 0.9978460, 0.9985320, 0.9990000, 0.9993190, 0.9995360, 0.9996840, 0.9997850, 0.9998530, 0.9999000, 0.9999320, 0.9999540, 0.9999680, 0.9999780, 0.9999850, 0.9999900, 1.0000000\}$.

In the moderate speed option, the g -space is divided into 17 intervals (18 g_i values) by dropping selected points from the larger set:

$g_i = \{0.0000000, 0.0500000, 0.1000000, 0.3187070, 0.5358410, 0.6837720, 0.7845560, 0.8532200, 0.9000000, 0.9535840, 0.9784560, 0.9900000, 0.9953580, 0.9978460, 0.9990000, 0.9995360, 0.9997850, 1.0000000\}$.

The g -space is chosen so that the $k(g)$ distribution with the largest dynamic range in k -values would have each decade of k -values covered by approximately 3 $[k(g_i), g_i]$ points.

The Ck algorithm is introduced into MODTRAN to provide Beer's Law dependence for the modeling of multiple scatter. It also eliminates the need for Curtis-Godson path averaging when computing multiple segment spectral transmittances. Instead, the LOS segment optical depths at each g_i can be summed if spectral correlation among segments is satisfied, the Ck ansatz. The radiative transfer equation (RTE) can be solved to compute monochromatic spectral radiances and transmittances at the g_i values. Once computed, the g_i -dependent values can be integration over the 0 to 1 range. For example, the formula for the transmittance of a multiple-segment ($j = 1$ to M) path becomes:

$$T(u_1, u_2, \dots, u_M) = \sum_{i=1}^N T_i(u_1, u_2, \dots, u_M) \quad ; \quad T_i = \int_{g_{i-1}}^{g_i} \exp[-\tau_g(u_1, u_2, \dots, u_M)] dg \quad ;$$

$$\tau_g(u_1, u_2, \dots, u_M) \equiv \left[\sum_{j=1}^M u_j k_j(g_{i-1}) \right] \left[\frac{\sum_{j=1}^M u_j k_j(g_i)}{\sum_{j=1}^M u_j k_j(g_{i-1})} \right]^{(g-g_{i-1})/\Delta g_i} \quad . \quad (\text{F.2})$$

Here, $\tau_g(u_1, u_2, \dots, u_M)$ is the full path optical depth at cumulative probability g ; u_j is the column density [atm-cm] of segment j ; $k_j(g)$ is the pseudo-gas absorption cross-section [$\text{cm}^{-1}/\text{atm}$] of segment j , and the total optical depths are modeled as varying exponentially with g between g_{i-1} and g_i . Eq. (F.2) can be expressed in terms of E_1 exponential integrals.

The problem with the current MODTRAN Ck implementation is that Eq. (F.2) is not used. Instead of solving the RTE at the 18 (or 34) monochromatic g_i points, the equations are solved using the 17 (or 33) interval mean value theorem absorption cross-sections, i.e., the $k_{MVT}^u(i)$ values of Eq. (F.1). In contrast to Eq. (F.2), the total path transmittance is computed as

$$T(u_1, u_2, \dots, u_M) = \sum_{i=1}^N T_i(u_1, u_2, \dots, u_M) \quad ; \quad T_i \approx \Delta g_i \exp \left[- \sum_{j=1}^M u_j k_{MVT}^{u_j}(i) \right] \quad ;$$

$$\exp \left[- u_j k_{MVT}^u(i) \right] \equiv \frac{1}{\Delta g_i} \int_{g_{i-1}}^{g_i} \exp \left[- u_j k_j(g_{i-1}) \left(\frac{k_j(g_i)}{k_j(g_{i-1})} \right)^{(g-g_{i-1})/\Delta g_i} \right] dg \quad (\text{F.3})$$

In practice, the Eq. (F.2) and Eq. (F.3) transmittance expressions usually give a similar result. However there are importance differences. Consider the transmittance from the first ($i = 1$) Δg interval for two ($M = 2$) identical segments with $u = u_1 = u_2$ and $k = k_1 = k_2$

$$T_1 = \int_{g_0}^{g_1} \exp \left[- 2u k(g_0) \left(\frac{k(g_1)}{k(g_0)} \right)^{(g-g_0)/\Delta g_1} \right] dg \approx \left\{ \int_{g_0}^{g_1} \exp \left[- u k(g_0) \left(\frac{k(g_1)}{k(g_0)} \right)^{(g-g_0)/\Delta g_1} \right] dg \right\}^2 \quad . \quad (\text{F.4})$$

Eq. (F.2) correctly integrates the total transmittance at each g value [first integral in Eq. (F.4)], while Eq. (3) simply equals the square of the interval averaged transmittance [second integral in Eq. (F.4)]. If $k(g)$ does not vary significantly between g_0 and g_1 , then the two expressions will be similar [identical if $k(g_0) = k(g_1)$]. When the absorption cross-section variations are large, differences will be evident. At the crux of the matter is the issue that the Ck method was introduced to provide a Beer's Law dependence, and this is only realized with the upgraded approach.

REFERENCES

- Acharya, P.K., R. Panfili, A. Berk, S.M. Adler-Golden, M. Fox and A. Wetmore, "A Correlated- k Based Ultra-Fast Radiative Transfer (k URT) Method," SPIE Europe Symposium on Remote Sensing, Bruges, Paper No. 5982-44, Belgium (19-22 September, 2005).
- Acharya, P.K., A. Berk, L.S. Bernstein, D.C. Robertson, G.P. Anderson, J.H. Chetwynd, H.E. Snell and J. Vail, "Recent Upgrades to MODTRAN – New Solar Irradiance and Instrument Scanning Functions," Spectral Sciences, Inc. Technical Report No. 312, Air Force Research Laboratory, Presented as the 20th Annual Atmospheric Transmission Models (ATM) Meeting, (1998).
- Adler-Golden, S.M., personal communication (circa 1997).
- Anderson G.P., J.H. Chetwynd, Jr., J.-M. Theriault, P.K. Acharya, A. Berk, D.C. Robertson, F.X. Kneizys, M.L. Hoke, L.W. Abreu and E.P. Shettle, MODTRAN2: suitability for remote sensing. Proc. SPIE 1968, Atmospheric Propagation and Remote Sensing II, 514 (September 15, 1993); doi:10.1117/12.154854.
- Berk, A., "Voigt equivalent widths and spectral-bin single-line transmittances: Exact expansions and the MODTRAN5 implementation," *J. Quant. Spectrosc. Radiat. Transfer*, **118**, 102 (2013); doi: [10.1016/j.jqsrt.2012.11.026](https://doi.org/10.1016/j.jqsrt.2012.11.026).
- Berk, A., P.K. Acharya, L.S. Bernstein, G.P. Anderson, P. Lewis, J.H. Chetwynd, M.L. Hoke, "Band Model Method for Modeling Atmospheric Propagation at Arbitrarily Fine Spectral Resolution," U.S. Patent #7433806, (October 7, 2008).
- Berk, A., G.P. Anderson, P.K. Acharya, L.S. Bernstein, L. Muratov, J. Lee, M. Fox, S.M. Adler-Golden, J.H. Chetwynd, M.L. Hoke, R.B. Lockwood, J.A. Gardner, T.W. Cooley, C.C. Borel, P.E. Lewis and E.P. Shettle, "MODTRAN5: 2006 Update," Proc. SPIE, Vol. 6233, 62331F, (2006).
- Berk, A., L.S. Bernstein, and D.C. Robertson, "MODTRAN: A Moderate-Resolution Model for LOWTRAN 7," *Rpt. No. GL-TR-89-0122*, U.S. Air Force Geophysics Directorate, Phillips Laboratory, Hanscom AFB, MA (1989).
- Bernstein, L.S., personal communication (1998).
- Bernstein, L.S., A. Berk, P.K. Acharya, G.P. Anderson and J.H. Chetwynd, "Addition of a Correlated- k Capability to MODTRAN," Technical Rpt. No. PL-TR-96, Phillips Laboratory, Geophysics Directorate, Air Force Materiel Command, Hanscom Air Force Base, MA 01731 (1996).
- Bodhaine, B.A., N.B. Wood, E.G. Dutton, and J.R. Slusser, "On Rayleigh Optical Depth Calculations", *J. Atmos. Ocean. Technol.*, **16**, 1854-1861 (1999).
- Chandrasekhar, S. 1960: Radiative Transfer, Dover, New York. (First edition, 1950).
- Clough, S.A., F.X. Kneizys, and R.W. Davies: Line shape and the water vapor continuum, *Atmos. Res.*, **23**, 229-241, (1989).
- Clough, S.A., M.J. Iacono, and J.-L. Moncet, Line-by-line calculation of atmospheric fluxes and cooling rates: Application to water vapor. *J. Geophys. Res.*, **97**, 15761-15785, (1992).
- Clough, S.A., M.W. Shephard, E.J. Mlawer, J.S. Delamere, M.J. Iacono, K. Cady-Pereira, S. Boukabara, and P.D. Brown, Atmospheric radiative transfer modeling: a summary of the AER codes, Short Communication, *J. Quant. Spectrosc. Radiat. Transfer*, **91**, 233-244, (2005).
- Curtis A.R., "Contribution to a discussion of 'A Statistical Model for Water Vapour Absorption', by R.M. Goody," *Quart. J. Roy. Meteorol. Soc.* **78**, 638-640 (1952).

- Delamere, J.S., S.A. Clough, V.H. Payne, E.J. Mlawer, D.D. Turner, and R.R. Gamache, A far-infrared radiative closure study in the Arctic: Application to water vapor, *J. Geophys. Res.*, **115**, D17106, doi: 10.1029/2009JD012968, (2010).
- Fontenla, J.M., P.C. Stancil and E. Landi, "Solar Spectral Irradiance, Solar Activity, and the Near-Ultraviolet," *Ap. J.* **809**:157, 17 pages (2015).
- Godson, W.L., "The Evaluation of Infra-Red Radiative Fluxes Due to Atmospheric Water Vapour," *Quart. J. Roy. Meteorol. Soc.* **79**, 367-379 (1953).
- Goody, R.M., "The Transmission of Radiation Through an Inhomogeneous Atmosphere," *J. Atmos. Sci.* **21**, 575 (1964).
- Goody, R.M., and Y.L. Yung, "Atmospheric Radiation: Theoretical Basis," 2nd Ed. (New York, Oxford University Press, 1989), pp. 1-519.
- Hapke, B.W., Bidirectional reflectance spectroscopy – 1. The extinction coefficient and opposition effect, *Icarus*, **67**, 264-280 (1986).
- Hapke, B.W., Bidirectional reflectance spectroscopy – 1. Theory, *J. Geophys. Res.*, **86**, 3039-3054 (1981).
- Humlíček, J., 1979: The Voigt function and its derivatives. *J. Quant. Spectrosc. Radiat. Transfer*, **21**, 309–313, doi:10.1016/0022-4073(79)90062-1.
- Isaacs, R.G., W.-C. Wang, R.D. Worsham, and S. Goldenberg, "Multiple scattering LOWTRAN and FASCODE models," *Appl. Opt.* **26**, 1272-1281 (1987).
- Isaacs, R.G., W. C. Wang, R. D. Worsham, and S., Goldenberg, "Multiple Scattering Treatment for Use in the LOWTRAN and FASCODE Models," Rpt. No. AFGL-TR-86-0073, Air Force Geophys. Lab., Bedford, MA 01731, pp 1-101, (1986).
- Kneizys, F.X., E.P. Shettle, W.O. Gallery, J.H. Chetwynd, L.W. Abreu, J.E.A. Selby, S.A. Clough, and R.W. Fenn, Atmospheric Transmittance/Radiance: Computer Code LOWTRAN 6, AFGL-TR-83-0187, NTIS ADA137796 (1983).
- Kurucz, R.L., (1) Atomic and Molecular Data for Opacity Calculations; (2) Finding the Missing Solar Ultraviolet Opacity; and (3) Remaining Line Opacity Problems for the Solar Spectrum, all three papers in *Revista Mexicana de Astronomia y Astrofisica*, **23**, (1992).
- Kurucz, R.L., The Solar Irradiance by Computation, "Proceedings of the 17th Annual Review Conference on Atmospheric Transmission Models", edited by Anderson, G.P., Picard, R.H., and Chetwynd, J.H., PL/-TR-95-2060, Special Reports, No. 274, Pl. 332, Phillips Laboratory / Geophysics Directorate, MA, (May 1995).
- Lacis, A.A. and V. Oinas, "A Description of the Correlated-k Distribution Method for Modeling Nongray Gaseous Absorption, Thermal Emission, and Multiple Scattering in Vertically Inhomogeneous Atmospheres", *J. Geophys. Res.*, **96**, 9027 (1991).
- Lucht, W., C.B. Schaaf, and A.H. Strahler, An algorithm for the retrieval of albedo from space using semiempirical BRDF models, *IEEE Trans. Geosci. Remote Sens.*, (1999).
- Meador, W.E. and W.R. Weaver, "Two-stream Approximations to Radiative Transfer in Planetary Atmospheres: A Unified Description of Existing Methods and a New Improvement," *J. Atmos. Sci.* **37**, 630-643 (1980).
- Mlawer, E.J., V.H. Payne, J.-L. Moncet, J.S. Delamere, M.J. Alvarado and D.D. Tobin, Development and recent evaluation of the MT_CKD model of continuum absorption, *Phil. Trans. Roy. Soc. A*, **370**, 1-37, doi:10.1098/rsta.2011.0295, (2012).

- Nakajima, T. and M. Tanaka, "Algorithms for radiative intensity calculations in moderately thick atmospheres using a truncation approximation," *J. Quant. Spectrosc. Radiat. Transfer*, **40**, 51-69 (1988).
- Payne, V.H., E.J. Mlawer, K.E. Cady-Pereira and J-L. Moncet, Water vapor continuum absorption in the microwave, *IEEE Trans. Geosci. Remote Sens*, **49**(6), 2194-2208, doi 10.1109/TGRS.2010.209141, (2011).
- Pierluissi, J.H. and C.E. Maragoudakis, Molecular Transmission Band Models for LOWTRAN, AFGL-TR-86-0272, AD A180655 (1986).
- Pinty, B. and M.M. Verstraete, Extracting Information on surface properties from bidirectional reflectance measurements, *J. Geophys. Res.*, **96**, 2865-2874, (1991).
- Plass, G.N., "Models for Spectral Band Absorption," *J. Opt. Soc. Am.* **48**, 690-703 (1958).
- Rahman, H., B. Pinty, and M.M. Verstraete, Coupled Surface-Atmosphere Reflectance (CSAR) Model 2. Semiempirical Surface Model Usable With NOAA Advanced Very High Resolution Radiometer Data, *J. Geophys. Res.* **98D**, 20,791-20,801 (1993).
- Reichel, A., "The Integral of the n^{th} Power of the Voigt Function," *Mathematics of Computation*, **23**(107):645-649, July 1969. CODEN MCMPAF. ISSN 0025-5718 (print), 1088-6842 (electronic).
- Robertson, D.C., L.S. Bernstein, R. Haimes, J. Wunderlich and L. Vega, "5-cm⁻¹ band model option to LOWTRAN5", *Appl. Opt.*, **20**, 3218-3226 (1981).
- Rodgers, C.D., and A.P. Williams, "Integrated absorption of a spectral line with the Voigt profile," *J. Quant. Spectrosc. Radiat. Transfer*, **14**, 319-323 (1974).
- Ross, V. and D. Dion, "Sea Surface Slope Statistics Derived from Sun Glint Radiance Measurements and Their Apparent Dependence on Sensor Elevation," *J. Geophys. Res.*, **112**, C09015, DOI:10.1029/2007JC004137 (2007).
- Ross, V., D. Dion and G. Potvin, "Detailed Analytical Approach to the Gaussian Surface Bidirectional Reflectance Distribution Function Specular Component Applied to the Sea Surface," *J. Opt. Soc. Am. A*, **22**, 2442-2453 (2005).
- Rothman, L.S., *et al.*, "The HITRAN2012 molecular spectroscopic database," *J. Quant. Spectrosc. Radiat. Transfer*, **130** (HITRAN2012 special issue): 4–50, (November 2013).
- Roujean, J.-L., M. Leroy, P.-Y. Dechamps, A Bidirectional Reflectance Model of the Earth's Surface for the Correction of Remote Sensing Data, *J. Geophys. Res.*, **97D**, 20,455-20,468 (1992).
- Stamnes, K., S.C. Tsay, W.J. Wiscombe, and I. Laszlo, "DISORT: A general-purpose Fortran program for discrete ordinate-method radiative transfer in scattering and emitting layered media," ftp://climate1.gsfc.nasa.gov/wiscombe/Multiple_Scatt/DISORTReport1.1.pdf NASA Report (2000).
- Stamnes, K., S.C. Tsay, W.J. Wiscombe and K. Jayaweera, "Numerically Stable Algorithms for Discrete Ordinate Methods Radiative Transfer in Multiple Scattering and Emitting Layered Media," *Applied Optics*, **27**, 2502 (1988).
- van de Hulst, H.C., "Theory of Absorption Lines in the Atmosphere of the Earth," *Ann. Astrophys.* **8**, 1-11 (1945).
- Walthall, C.L., J.M. Norman, J.M. Welles, G. Campbell, and B.L. Blad, Simple equation to approximate the bidirectional reflectance from vegetative canopies and bare soil surfaces, *Applied Optics*, **24**, 383-387 (1985).

- Wanner, W., A.H. Strahler, B. Hu, X. Li, C.L. Barker Schaaf, P. Lewis, J.-P. Muller, and M.J. Barnsley, Global retrieval of bidirectional reflectance and albedo over land from EOS MODIS and MISR data: theory and algorithm, *J. Geophys. Res.*, 102D, 17,143-17,162 (1997).
- Wanner, W., X. Li, and A.H. Strahler, On the derivation of kernels for kernel-driven models of bidirectional reflectance, *J. Geophys. Res.*, 100D, 21,077-21,090 (1995).
- Wiscombe, W.J., "The Delta-*M* Method: Rapid Yet Accurate Radiative Flux Calculations for Strongly Asymmetric Phase Functions," *J. Atmos. Sci.* **34**, 1408-1422 (1977).